Self-Adaptive Kernels as Evolving Measures: Transport Adaptive Control

Mathematical Foundations of Self-Adaptive Kernels, version 3: a complete mathematical treatment

Delhi Technological University

October 2026

Version 3. This document supersedes version 2 and, before it, version 1, *Mathematical Foundations of Self-Adaptive Kernels Achieving Lyapunov Convergence*. Section 2 states what survived from version 1 and what was withdrawn; version 3 adds Lemma E.0 (explicit constants for Theorem E), Theorems C′, D′, I and J and Proposition K (Table 1); a second increment adds the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2); a third increment adds the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″). This is research in progress and has not been peer reviewed. Code, tests, experiments and review records: https://github.com/Dutta-06/Lyapunov-Convergence-Kernels-Showcase.

Copyright © 2026 Odwitiyo Dutta. All rights reserved.

Abstract

This is a complete mathematical treatment of adaptive tracking control with Gaussian radial-basis-function (RBF) networks whose weights, centres and widths are adapted online. Part I is the projection-based adaptive RBF controller of version 1. An audit found its stability argument to be a correct but standard composition of known results, with two claims wrong as written (a non-convex "projection" for centre separation and an implicit-function-theorem argument that is not a proof); it contains no new theorem. We state it with explicit assumptions, a Hessian-free disturbance bound and an RMS bound, and list the errata to version 1. Part II develops transport adaptive control: the network is an empirical measure on a lifted parameter space, and the adaptive law is its transport. Theorems A–F, C′ and D′ are proved; A, B, C, C′, E and F are also checked by tests. Theorem E is self-contained: an explicit strict Lyapunov function (Lemma E.0) gives the radius, overshoot and rate of the label-free convergence under tangential excitation. Theorem C′ is the OT σ-modification toward a prior with a different number of atoms, through the barycentric projection of the optimal plan (Lloyd's algorithm in continuous time); Theorem D′ is propagation of chaos for the dithered closed loop, at the dimension-free rate $m^{-1/2}$. Proposition G is formal and checked directly. Theorem H shows that the averaged mean-field loop with saturated features, σ-modification and dither is mean-field Langevin dynamics and converges globally and exponentially to the regularised optimum; this is proved for the averaged system, the reduction to it is formal, and the real loop is a conjecture. At finite gain the averaged loop is in general not a gradient flow of any free energy in the $W_{G^{-1}}$ metric: Proposition H.7 (frozen-measure part proved) exhibits a semi-gradient term, whose performative defect is non-reciprocal in the checked instances ($n = 1, 2$), and an orbit-area circulation kernel ($n \geq 2$). With a second-order averaging term they predict a closed-loop effect for which a targeted literature search found no precedent: traversing the reference orbit backwards changes the learned kernels by $O(1/k_e)$, although the occupation measure is unchanged. At $k_e = 40$ the prediction matches to 3% in size and 11% as a vector (cosine 0.994). Proposition H.8, proved assuming the chain rule for the perturbed flow, gives input-to-state stability of the

averaged system. Theorem I characterises the non-reciprocity at order k_e^{-2}: for $n = 1$ the averaged loop is reciprocal exactly when the residual vanishes on the orbit, and no open set of measures is reciprocal for any n. Theorem J proves the deterministic finite-m case of the averaging conjecture: every hyperbolic critical point of the regularised loss has a unique nearby periodic orbit of the real closed loop with the same stability type, whose period mean is the first-order prediction up to $O(k_e^{-2})$. Proposition K shows that a collapsed configuration is a strict local minimum iff the leak gain exceeds a critical value σ_{crit}, and that in a window around σ_{crit} the direction in which the reference is traversed decides whether the kernels separate. A second increment adds the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2). A third increment adds the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″).

Contents

1 Introduction **5**
- 1.1 The idea . . . 5
- 1.2 Why the classical theory needs it . . . 5
- 1.3 What the closed loop actually is . . . 5
- 1.4 Conventions and verification . . . 6
- 1.5 Contributions and their status . . . 6
- 1.6 Where the new mathematics lies . . . 10
- 1.7 Organisation . . . 10

2 Relation to version 1 **11**
- 2.1 What version 1 claimed . . . 11
- 2.2 The audit's verdict . . . 12
- 2.3 Errata to version 1 . . . 14
- 2.4 What survives and what is withdrawn . . . 15
- 2.5 How Part II answers the structural defects . . . 15

I The projection-based controller **17**

3 Setting and standing assumptions **17**

4 The algorithm **18**

5 Lemmas and the boundedness theorem **18**
- 5.1 Lemmas . . . 18
- 5.2 Theorem 1: uniform ultimate boundedness . . . 20

6 The no-bias claim, stated precisely **21**

7 Prior art **21**

8 What the experiments show **22**
- 8.1 Robustness mechanism versus geometry . . . 22
- 8.2 Kernel-regression baselines . . . 23
- 8.3 The bound against data . . . 24
- 8.4 Summary . . . 26

II Transport adaptive control **29**

9 Setting **29**

10 Theorem A: identification **31**

11 Theorem B: the optimal-assignment Lyapunov function **33**

12 Theorem C: optimal-transport σ-modification **37**

13 Theorem C′: the OT leak toward a prior with a different number of atoms **41**

14 Theorem D: the mean-field closed loop **46**

15 Theorem E: label-free local convergence under tangential excitation **49**

16 Theorem F: the transport cost of adaptation **65**

17 Theorem D′: propagation of chaos for the dithered closed loop on finite horizons **67**
Theorem D″: uniform-in-time propagation of chaos in the leak-dominated regime . . . 70

18 Proposition G: the slow geometry **73**

19 Theorem H: Langevin adaptive control and global convergence to the regularised optimum **78**

20 Finite gain: semi-gradient and non-reciprocal structure **84**
20.1 Frozen-measure structure of the exactly averaged loop 84
20.2 The oscillation correction and the orientation of the reference orbit 88
20.3 Reciprocity at order k_e^{-2}: a pairwise criterion and its characterisation 90
20.4 Input-to-state stability of $(\mathfrak{A})$ and the route to Conjecture H.6 93
20.5 Evidence from the real closed loop 94

21 Theorem J: the finite-m closed loop as a perturbed gradient flow **95**

22 Proposition K: collapsed configurations, the critical leak, and the orientation of the orbit as a design input **101**

23 Theorem J′: the stationary law of the dithered finite-m closed loop **104**

III Evidence and outlook **115**

24 Numerical evidence **115**
24.1 Theorem B: the label-free Lyapunov function 115
24.2 Theorem C: OT σ-modification 116
24.3 Theorem C′: the barycentric OT leak 117
24.4 Theorem D: convergence in m 118
24.5 Theorem D′: propagation of chaos for the dithered loop 118
24.6 Theorem E: label-free convergence under excitation 119
24.7 Lemma E.0 and Theorem E with explicit constants 120
24.8 Theorem F: transport cost 122
24.9 Proposition G: slow geometry 122
24.10Theorem H: Langevin adaptive control 124
24.11Remark H.0 and Proposition H.7: the exactly averaged closed loop 125
24.12Remark H.7a: the orientation of the reference orbit 126
24.13Theorem I: reciprocity at order k_e^{-2} 126
24.14Theorem J: periodic orbits of the finite-m closed loop 127
24.15Proposition K: collapsed configurations and the orientation of the orbit 128
24.16Theorem J′: the stationary law of the dithered single-kernel loop 129

25 Research program **129**

A Verification map **133**

1 Introduction

1.1 The idea

An adaptive approximator with nonlinearly parametrised features can be read as an *empirical measure* $\mu = \frac{1}{m}\sum_i \delta_{\vartheta_i}$ on a lifted parameter space. Here the features are Gaussian RBF kernels with adaptive weights, centres and widths. The adaptive law transports this measure, and the control output $\hat{f} = \int \Phi \, \mathrm{d}\mu$ is linear in μ. Stability, robustness modifications, excitation, the many-kernel limit and the slow evolution of kernel geometry all acquire formulations on Wasserstein space. So does the global optimality of kernel placement. These formulations do not depend on how the kernels are labelled.

1.2 Why the classical theory needs it

This work identifies two structural defects of the classical labelled analysis, that of version 1 of this manuscript and of the literature it builds on.

- The classical Lyapunov analyses depend on labels. A network that is a relabelling of the ideal has zero function error but a large parameter error.
- There is no object that describes where the kernels go, or whether their placement is good.

A third apparent defect was that constants grow with the number of kernels. It turned out to be an artefact of the normalisation. The mean-field normalisation removes it for labelled and label-free analyses alike (Remark A.1).

1.3 What the closed loop actually is

- At infinite feedback gain, the averaged closed loop is a Wasserstein gradient flow. With dither it is a mean-field Langevin dynamics (Theorem H).
- At finite gain it is in general *not* a gradient flow of any free energy in the $W_{G^{-1}}$ metric. This already holds for $n = 1$, so it is not a two-dimensional effect (Proposition H.7). The first correction has two parts at a frozen measure, and a third along the real loop.
  - The first part is a *semi-gradient* term. The adaptive law differentiates the loss on the closed-loop state distribution while ignoring that this distribution depends on the parameters, as in performative prediction. Like semi-gradient TD, the resulting update is not the gradient of any objective. Its performative defect is present for every n; for $n = 1$ it is non-reciprocal exactly when the residual is non-zero on the orbit (Theorem I(iii)), and it is non-reciprocal in the checked $n = 2$ instance.
  - The second part is an antisymmetric interaction between kernels ($n \geq 2$). Its kernel is an integral of $\sum_k \det[\nabla\tilde{\Phi}_k(\vartheta), \nabla\tilde{\Phi}_k(\vartheta')]$ over the region that the reference orbit encloses (Stokes).
  - Along the real loop, a third part comes from the parameters' oscillation within each period. It is a second-order averaging term (Remark H.7a).
- Together these predict a new closed-loop effect: traversing the reference orbit backwards changes the learned kernels by $O(1/k_e)$, although the occupation measure is unchanged. At $k_e = 40$ the prediction matches to 3% in size and 11% as a vector (cosine 0.994). The vector error halves with each doubling of k_e (`experiments/exp_orbit_orientation.py`).
- The gradient structure survives as an input-to-state Lyapunov function (Proposition H.8). This reduces the global conjecture for the real loop to an averaging estimate.

1.4 Conventions and verification

Status labels. Every result of Part II is labelled *proved*, *formal* or *conjectural*, and the label is given in a status line after the result.

- "Proved" means proved here. "Formal" means a derivation without error control.
- "Checked" means a test evaluates the statement, or every inequality in it, along simulated trajectories or on sampled states.
- The novelty column of Table 1 follows the verdicts in `LITERATURE_TRANSPORT.md`.

In Part I, every step of the stability argument is either proved or verified numerically by `tests/test_theorem_to_code.py`.

Verification and review.

- Every inequality the tests can evaluate is checked numerically along simulated trajectories, and every number in this document is produced by a script in the accompanying repository (the experiments are in `experiments/`); no table is edited by hand.
- The theory went through three rounds of automated multi-agent adversarial review, the manuscript through a fourth that compared it line by line with the theory, the results added in version 3 (Lemma E.0, Theorems C′, D′, I and J, Proposition K) through a fifth, and the second increment — the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2) — through a sixth, and the third increment — the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) — through a seventh. Every confirmed issue and its fix is logged in `review/REVIEW_LOG.md`, and the raw reports of rounds 1–3, 5 and 6 are kept beside it.
- The claim-by-claim novelty audit is in `LITERATURE_TRANSPORT.md`.

Appendix A maps each checked statement to the test or experiment that checks it.

1.5 Contributions and their status

Table 1 lists the results of Part II with their status, in the conventions of Section 1.4, and with the novelty verdict of the literature audit.

Table 1: Contributions of Part II and their honest status. Status conventions: Section 1.4.

result	statement	status	novelty (audit)
A	The adaptive law is the particle form of the Wasserstein gradient flow of a μ-linear potential. Projected rates coincide only for a per-kernel projection ($m \geq 2$).	proved and checked	interpretive; ingredients known [10], [16], [42], [57]
B	$V_{\mathrm{OT}} = \frac{1}{2}\|e\|^2 + \frac{1}{2}W^2$ satisfies a Dini inequality with a two-regime matched remainder. It gives conditional RMS bounds along (L), and along (L_σ) with a leak term.	proved and checked, including a moving ideal, $\varepsilon \neq 0$, switches, an active projection, and the failure of the plain bound along (L_σ)	packaging new; ingredients known [70], [1], [56], [12]

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
C	OT σ-modification: exact dissipation for selection solutions; a gap-protected strict inequality with $\sqrt{C} \le b + 2\eta$; a local exponential estimate, m-uniform for a single-atom prior and $O(m^{-1/2})$ otherwise; reshuffling by the labelled leak	proved and checked; the C(iv) instance satisfies every hypothesis	novel as a control construction; C(i) and the obstruction are known [66], [6]
C′	The OT leak toward a prior with any number of atoms, through the barycentric projection of the optimal plan: dissipation of $W(\mu,\nu)$ up to the within-cell variance of the plan; the fixed points of the pure leak are the equal-mass centroidal configurations of the prior (Lloyd's algorithm in continuous time, the G-gradient flow of the quantisation energy); the induced-coupling bound $\sqrt{C_{\mathrm{bar}}} \le b + 2\eta$; a Lyapunov inequality and a local exponential estimate with the bias b_{bar} := distance of the ideal from the centroids of its prior cells ($\le b$)	proved for vertex-selection solutions (existence assumed) and for generalised solutions (which exist); checked (dissipation at every step, convergence to a Lloyd fixed point, the coupling bound and plan stability, the Lyapunov inequality along the closed loop)	the law is the continuous-time Lloyd descent of coverage control [15] with equal-mass (capacity-constrained) OT plans [4], used as a σ-modification leak; new here are the induced-coupling bound, the Lyapunov inequality with the bias b_{bar} and the transfer reading; the quantisation side is Lloyd/CVT theory [47]; it reduces to Theorem C for $m_0 = m$
D	Local well-posedness of the mean-field loop; a Dobrushin estimate; a tube bound and global existence under a self-consistent growth condition; the mean-field limit on the existence interval	proved; consistency in m checked numerically (the constant is not checked)	technique anticipated [22, 60, 61]
E	Label-free local exponential convergence under tangential PE, in a uniform tube, at fixed m, with explicit radius, overshoot and rate (Lemma E.0); exponential merging of solutions. In the slow-adaptation regime a second strict Lyapunov function (Lemma E.0′: second-order slaving and a second-order averaging correction, with the peak factors of the regressor as data) has the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, and under parameter dither bounds the stationary parameter spread by $(1+O(\varepsilon))k\|\Sigma\|_F^2/(2(1-\delta)\lambda_{\min})$ (Corollary E.0′.3); for non-periodic PE regressors the same construction with an exponential average over a window gives the rate $(1-\delta-\delta_2)\lambda_T/k$, λ_T the weighted PE level of the window (Lemma E.0″)	proved, self-contained; linearisation checked; the strict Lyapunov inequalities checked pointwise (E.0′ by a certificate over all states) and the explicit envelope checked on the linearised and on the implemented loop; convergence observed. The rate of E.0 is conservative in a quantified way (Remark E.0a); the rate of E.0′ is within 12% of the exact rate above its threshold	partially anticipated [12], [27]; the strict Lyapunov function is of the Mazenc–Malisoff type
F	Along (L), the Wasserstein length of learning is $\le \lambda_G^{1/2} L_1 \int\|e\|$; leak versions	proved and checked, including the leak clause, a tight leak instance with $e \equiv 0$ and a tight velocity instance	elementary

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
G	Averaged joint descent on the occupation loss. The slow geometry follows the gradient flow of the profile loss in the metric induced on the slow manifold, which is plain variable projection only when that manifold is flat ($S \ll 1$).	formal; checked directly, including a counterexample to plain VarPro at $S = O(1)$	a fixed-m, closed-loop analogue of the two-timescale/VarPro training of [48], [5]
H	The averaged mean-field loop with saturated features, σ-modification and dither is mean-field Langevin dynamics. It converges globally and exponentially to the regularised optimum, with the log-Sobolev constant given in control quantities.	proved for ($\mathfrak{A}$); the reduction to ($\mathfrak{A}$) is formal (Remark H.0); real loop: Conjecture H.6	a bridge to [11], [52]; new in adaptive control
H.7	At a frozen measure, the exactly averaged loop is ($\mathfrak{A}$) plus an explicit $O(1/k_e)$ correction: a semi-gradient term, whose performative defect is present for every n, and an orbit-area circulation kernel ($n \geq 2$). Along the real loop, a second-order averaging term is added (Remark H.7a).	Frozen-measure part proved. Stokes identity and parity hold to machine precision. The expansion is checked through its $O(1/k_e)$ convergence. Non-reciprocity of the exact second variation is checked to 3% (asserted; 1–1.5% measured) and characterised by Theorem I. The full first-order theory predicts the orientation dependence of the learned kernels in the real closed loop to 3% in size and 11% as a vector (cosine 0.994, $k_e = 40$).	no precedent found (targeted search in the third review round); ingredients known (higher-order averaging; performative prediction [59]; semi-gradient TD [73], [71])
I	Reciprocity at order k_e^{-2} is a pairwise property: the antisymmetric kernel $A_{\text{even}} + 2K$ is a coboundary iff it vanishes. For $n = 1$, $A_\mu \equiv 0$ iff the residual vanishes on the orbit, and the performative non-reciprocity is bounded by the residual. For $n \geq 2$ at perfect fit, $A_\mu \equiv 0$ iff the oriented orbit carries no circulation ($n = 2$: zero winding function) and the residual's Jacobian is symmetric along the orbit; $A_\mu = 2K$ when the residual vanishes near the orbit. A_{even} is the curl of the residual paired with the features plus the residual paired with the Lie bracket of the feature fields. No open set of measures is reciprocal, for any n: the averaged loop is nowhere a gradient flow at order k_e^{-2}.	proved; every formula checked against the generic second-variation code and, by the round-5 referee, against the exact second variation; the characterisations checked on perfect-fit, curl and zero-current instances	new; the analytic tools (injectivity of Gaussian convolution; Stokes with winding numbers) are classical

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D′	Finite-horizon propagation of chaos for the dithered closed loop (L_τ): by synchronous coupling, $\mathbb{E}\sup_{t\le T}\big(\|x^m-\bar{x}\|+\frac{1}{m}\sum_i\|\vartheta_i-\bar{\vartheta}_i\|\big)\le TK_\Phi e^{\Lambda T}m^{-1/2}$, with Λ independent of m and τ, and a per-particle sup bound of the same order; the plant and the learned function converge at the rate $m^{-1/2}$, whose exponent is dimension-free, and only $W_1(\mu^m,\bar{\mu})$ carries $m^{-1/p}$; under a dominant leak (small-gain condition $\kappa_e\kappa_\vartheta > L_{b'}L_\vartheta$) the bound is uniform in time (Theorem D″), a leak-dominated regime in which the network does not learn (at $\tau=0$ it collapses to one kernel; at $\tau>0$ it stays in the dither cloud of the prior)	proved; checked: the coupled distance to a 4096-particle reference decays like $m^{-0.47\pm0.06}$ (24 seeds, $\tau=0$; -0.46 ± 0.05 with dither) and like $(1/m-1/M)^{0.94\pm0.08}$, the law of the shared-particle coupling	the technique is Sznitman's, and the dimension-free exponent for the state is its standard output; new in that the plant is in the loop, with constants in control quantities
J	At finite m, without dither, in the scaling $\sigma=\sigma_0/k_e$: every hyperbolic critical point of the m-particle regularised loss J_m has a unique nearby T_0-periodic orbit of the real closed loop, at distance $O(1/k_e)$, with the same stability type; the parameter Floquet exponents are $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})+O(k_e^{-2})$; the basin of a minimum is uniform in k_e; the period mean is $\Theta^*-J^{-1}(b_{\mathrm{frozen}}+b_{\mathrm{osc}})+O(k_e^{-2})$	proved (tube and slaving, periodic Riccati block-triangularisation, second-order averaging, Poincaré map); checked by Newton shooting for the exact orbit, its monodromy matrix and its period mean against the averaged Hessian and the first-order prediction, at $k_e=10$–80	the first step of Section 25, branch 1; the tools (two-time-scale averaging under attraction) are classical [40, Ch. 10–11], [64, Ch. 5], [67, Ch. 4]
K	At a collapsed critical point (all kernels equal) the Hessian of J_m splits by permutation symmetry into a separation block $\sigma_0G^{-1}-R_2$, with R_2 the Hessian of the first-variation potential at the atom (independent of m), and a collective block; the collapsed configuration is a strict local minimum iff the atom minimises the single-kernel objective and $\sigma_0>\sigma_{\mathrm{crit}}:=\lambda_{\max}(G^{1/2}R_2G^{1/2})$. The orientation-odd k_e^{-2} terms of the real loop (U_{odd}, b_{osc} and the third derivative of J_m against the first-order offset) shift the separation Floquet exponents of the collapsed orbit by $O(k_e^{-2})$, identically for the $(m-1)$-fold multiplet, and in a window of width $O(1/k_e)$ around σ_{crit} the direction in which the reference is traversed decides whether the kernels separate	proved ((iii)–(iv) given the second-order normal form, Section 22); checked by shooting on instance A: the orientation difference of the softest separation exponent scales like $k_e^{-2.18}$ and matches the prediction from the averaged data to 3% at $k_e=80$; at a leak gain computed from that prediction, one traversal leaves the collapsed configuration stable and the other makes it unstable ($k_e=10$ and 20)	new as a design statement; the splitting is the isotypic decomposition of the permutation action, and the stability criterion is the local Chizat–Bach condition at a Dirac mass

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
J′	At finite m with dither $\tau = \tau_0/k_e$ and leak $\sigma = \sigma_0/k_e$: the closed loop has an invariant law (unique under a smoothness hypothesis) whose parameter marginal is the Gibbs measure $\exp(-J_m/\tau_0)$ up to $O(1/k_e)$, with the explicit first-order correction $\varepsilon \int \varphi g \, \mathrm{d}\pi$ driven by the first-order drift b of Theorem J as a vector field on parameter space (g solves $\tau_0 \nabla \cdot (\pi G \nabla g) = \nabla \cdot (\pi b)$); the stationary tracking error is $(2/k_e^2)\pi(\bar{L}_m) + O(k_e^{-3})$; the stationary parameter process is not reversible, with a circulation $\varepsilon[\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle] + O(\varepsilon^2)$, where $j_1 = \pi(b - \tau_0 G \nabla g)$ is the first-order stationary current, nonzero exactly when b is not a G-gradient (Proposition J′.2: the finite-m form of Theorem I's reciprocity defect plus a Hessian commutator from second-order averaging; never a gradient for $m \geq 3$)	proved (the Poisson-equation expansion of the invariant measure with the fast error and phase eliminated; three standard theorems imported and named); checked on a single-kernel instance: the first-order prediction removes the $O(1/k_e)$ discrepancy between the real loop's stationary means and the Gibbs measure, the orientation-odd part of the means (the dither version of Remark H.7a) matches its prediction, and the current machinery is checked on the first-order averaged SDE	the technique is the Pavliotis–Stuart [58] / Mattingly–Stuart–Tretyakov [49] expansion; new in its object (a hypoelliptic learning loop with a plant) and in the identification of the first-order non-equilibrium current with the reciprocity defect of Theorem I
H.8	$(\mathfrak{A})$ is input-to-state stable under drift perturbations, with gain $1/(\tau\rho_\tau)$	proved, assuming the chain rule for the perturbed flow	elementary given [11]

1.6 Where the new mathematics lies

The new mathematics this viewpoint requires lies in the open problems of Section 25.

1. Infinite-horizon averaging for McKean–Vlasov closed loops. Propositions H.7 and H.8 reduce it to controlling a near-identity averaging transformation along the moving measure. Theorem J settles the deterministic finite-m case.
2. Uniform-in-time propagation of chaos with a plant in the loop.
3. The theory of the non-reciprocal, semi-gradient mean-field dynamics that the closed loop actually produces. Theorem I characterises its non-reciprocity (completely for $n = 1$ and at perfect fit; by an explicit bilinear form otherwise, and nowhere reciprocal on an open set); its long-time behaviour is open. Proposition K turns one of its effects into a design rule: near the critical leak, the orientation of the reference orbit decides whether a collapsed network separates.
4. Global convergence without dither.

1.7 Organisation

Section 2 relates this document to version 1: what version 1 claimed, the audit's verdict, the errata, what survives and what is withdrawn, and how Part II answers the structural defects of Section 1.2. Part I treats the projection-based controller: the setting and standing assumptions (Section 3), the algorithm (Section 4), the lemmas and the uniform-ultimate-boundedness theorem (Section 5), the "no bias" claim stated precisely (Section 6), prior art and the honest novelty statement (Section 7), and the experiments (Section 8). Part II develops transport adaptive control in Sections 9–22: the setting (Section 9), Theorems A–F together with Theorem C′ and Theorem D′ (Sections 10–17), Proposition G (Section 18), Theorem H (Section 19), the finite-gain structure of Propositions H.7 and H.8 and the reciprocity criterion of Theorem I (Section 20),

Theorem J (Section 21) and Proposition K (Section 22). Part III collects the numerical evidence (Section 24) and the research program (Section 25). Appendix A is the verification map.

2 Relation to version 1

Version 1 of this document, *Mathematical Foundations of Self-Adaptive Kernels Achieving Lyapunov Convergence* (14 pages), presented the controller of Part I. An audit (`ASSESSMENT.md`) re-examined its theory, its code and its experiments; every numerical statement of the audit was regenerated by running the code, and the scripts and raw files are named next to each claim. A revised theory (`THEORY_REVISED.md`, Part I of this document) replaced its stability argument. This section records what version 1 claimed, the audit's verdict, the errata, what survives and what is withdrawn, and how Part II answers the structural defects of Section 1.2.

2.1 What version 1 claimed

Version 1 was subtitled *Uniform Ultimate Boundedness via Projection-Based Adaptive RBF Control Without Persistent Bias of Leaky Modification*. It treated the plant $\dot{x} = f(x,t) + u$ on a compact domain with the certainty-equivalence control $u = \dot{x}_m - k_e e - \hat{f}(x,t)$ and the network $\hat{f} = W^\top \phi(x, c, \sigma)$, adapted weights, centres and widths by projected gradient laws, and used the Lyapunov candidate $V = \frac{1}{2} e^\top e + \frac{1}{2}\,\mathrm{tr}(\tilde{W}^\top \Gamma_W^{-1} \tilde{W}) + \frac{1}{2}\tilde{c}^\top \Gamma_c^{-1} \tilde{c} + \frac{1}{2}\tilde{\sigma}^\top \Gamma_\sigma^{-1} \tilde{\sigma}$. Its claims were the following.

- *Central contribution* (abstract): "a rigorous proof that a fully shape-shifting kernel architecture — with online adaptation of weights, centers, and bandwidths — achieves Uniform Ultimate Boundedness (UUB) of the tracking error without the persistent steady-state bias introduced by classical leaky modification schemes (σ-modification, e-modification)".
- *Key mechanism* (abstract, §1.3): "the strict decoupling of offline Tikhonov regularization, which guarantees the well-posedness of the ideal parameter target, from the online smooth projection operator, which provides hard parameter boundedness", with an implicit-function-theorem (IFT) argument giving finite velocity bounds V_W, V_c, V_σ for the ideal parameters.
- *A "critical and non-trivial" result* (abstract, §7.2): the projection's hard cap on parameter energy "converts the standard boundary-only Lyapunov descent argument into a global interior argument", proving convergence to the infimal invariant set without LaSalle's invariance principle or Barbalat's lemma, with the ultimate bound

$$\limsup_{t\to\infty} \|e(t)\| \le \sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}.$$

- Centre separation $\|c_i - c_j\| \ge \delta > 0$, "maintained via the projection set construction" (§1.2); and a corollary that a standard persistent-excitation condition on the regressor is insufficient to guarantee parameter convergence, so that alternative mechanisms, such as integral concurrent learning (ICL), are required (§8).
- §9 summarised five architectural choices and the failure mode each was said to close: $\sigma_{\min} > 0$ (Hessian blow-up near $\sigma \to 0$); offline Tikhonov regularisation (circular self-consistency of the projection geometry); a smooth convex projection operator (a cancellation gap when the projection is active); the projection's hard energy cap (the LaSalle and Barbalat requirements); and stability from the Lyapunov function and the projection alone (the circularity of ICL, which needs bounded states).
- §10 stated that "the theoretical core of this manuscript is mathematically complete and internally consistent", and that the remaining items were "completeness additions and

Table 2: Novelty of the ingredients of version 1 (audit).

ingredient	prior source	status of the claim of version 1
Adaptive RBF control with projection-type robustification	Sanner & Slotine 1992 [65]; Ioannou & Sun 1996 [39]	known
Adapting centres/widths by first-order Taylor expansion with bounded remainder	Lewis, Yesildirek & Liu 1996 [45]; Polycarpou 1996 [62]; Fabri & Kadirkamanathan 1996 [21]; Ge & Wang 2002 [26]	known; version 1 cites none of these
Smooth projection with ε boundary layer and inequality (P)	Pomet & Praly 1992 [63]; Lavretsky & Gibson 2011 [43]; Lavretsky & Wise 2013 [44]	known; the showcase README called it "the core theoretical contribution"
Projection introduces no bias in $\dot{V}$ when $\theta^* \in \Omega$	Ioannou & Sun 1996 [39, §4.4]	known
Level-set ("global interior") UUB argument	Khalil [40, Thm. 4.18]; Lewis et al. 1999 [46]	known; called "critical and non-trivial"
Simultaneous adaptation of W, c, σ for Gaussian RBFs with all three projected	—	the only combinational novelty; minor

empirical validation requirements . . . , not theoretical repairs": a constructive feasibility theorem; empirical validation (Lorenz-driven tracking against σ- and e-modification "to prove bias elimination", a Monte Carlo sweep "verifying R_{UUB} bound is not violated", an ablation of fixed against adaptive centres); and a novelty framing in which the architecture "eliminates this tradeoff entirely", the projection introduces "zero bias into the Lyapunov derivative", and the result is "not a marginal improvement but a structural result enabled by the strict decoupling of offline regularization from online projection".

2.2 The audit's verdict

The theory is a correct but standard composition of known results (Taylor-linearised inner-layer adaptation, boundary-layer projection, level-set UUB), presented with language that overstates its novelty and its conclusion. Two of its claims are wrong as written (non-convex "projection" for centre separation; an IFT argument that is not a proof) and one was not implemented (the Γ-weighted projection). The experiments contained a hand-chosen "theoretical" threshold, a confounded headline comparison against baselines that failed representationally, GP baselines handicapped by stale predictions and a mis-placed inducing grid, invalid significance tests on autocorrelated samples, and the public figure omitted the one benchmark ($n = 3$) where the method is weak and the one baseline (NTK) that beats it. After revision, the implemented algorithm is exactly the proved one and is machine-checked; the honest experiments show a real, consistent advantage of projection over σ- and e-modification on identical geometry (5/5 seeds in every cell) and a modest but consistent gain from adaptive geometry, while the a-priori bound is shown to be $915\times$ to $10\,954\times$ loose.

Novelty. Table 2 compares the ingredients of version 1 with prior work. Net: no new theorem. What can honestly be claimed after revision (Section 7) is a machine-checked theorem-to-code map, an explicit Hessian-free disturbance bound, the RMS form of the bound, and a controlled mechanism-versus-geometry experiment.

Integrity findings. Table 3 lists the integrity findings of the audit, each verified by running code, with the evidence and the fix. File names refer to the accompanying repository.

Table 3: Integrity findings of the audit of version 1 and its code (each verified by running code).

#	finding	evidence	fix
I1	Implemented centre update $\neq$ proved update: a pairwise repulsion term broke the projection inequality the proof needs	`tests/test_theorem_to_code.py` (oracle plant): inequality violated at 6054 of 20000 samples with the legacy term, 0 of 20000 without; the term also failed its own purpose (min separation 0.07 vs $\delta = 0.5$)	removed from the algorithm (`enforce_separation=False` default); tracking unchanged on the headline run (mean steady-state error $0.0767 \to 0.0774$, max $0.46 \to 0.40$)
I2	Projection applied the isotropic formula to any Γ; the proof needs the Γ-weighted operator	`tests/test_controller_paths.py`	Γ-weighted projection implemented; fast/dense paths agree to 10^{-12}
I3	"R_{UUB} threshold" in the Monte Carlo figure was the constant 0.5	`tier1_monte_carlo.py` line 69: "A chosen arbitrary bound"	`exp_monte_carlo_honest.py`: only exactly checkable statements (200/200 hard-bound and ISS checks)
I4	"Theoretical R_{UUB}" in the k_e sweep used invented constants and an underived formula	`tier2_ruub_sweep.py`, former `estimate_D_max`	`exp_oracle_ruub.py`: all constants computable; a-priori bound 915×–10 954× loose; RMS bound with measured disturbance holds 48/48
I5	Headline figure: fixed-grid baselines with a grid not covering the reference; their error (≈ 15) exceeded open loop, so the figure showed representational failure, not "bias"	`results/tier1_lorenz_tracking.png` vs ablation figure (projection W-only fails identically)	`exp_bias_factorial.py`: 3 mechanisms × 3 geometries, covering grid, 5 seeds
I6	Wilcoxon tests on ∼12 000 time samples with lag-1 autocorrelation 0.9999; the README's "$p > 0.3$ vs Exact GP" is contradicted by its own data ($p = 1.6 \times 10^{-11}$)	recomputed from `results/*.npy`	seed-level statistics only
I7	$n = 3$ benchmark and NTK baseline removed from the public figure (commits "exclude n=3 and NTK")	`results/n3/*.npy`, `results/*/ntk_*.npy`; `plot_comparison_hd_complete.py`	complete table regenerated: proposed ranks 3/5, 2/5, 3/5, 2/5
I8	GP baselines held a stale prediction for 50–100 steps; sparse GP inducing points on a diagonal line over $x \in [-3, 3]$ while the reference spans ± 20	`sim1`–`sim3`	`kernel_baselines_fair.py`: per-step prediction, 9×3 covering grid, seeds
I9	"$O(mn)$ per step" claim false with the $O(m^2 n)$ repulsion	`projection.py`	true after I1
I10	"Concurrent Learning" hook used the current residual only	former `controller.py`	removed
I11	Lyapunov monitor requires θ^*; never used by any experiment, so "Lyapunov descent verification" was never run	`integrator.py`	oracle module and tests perform it on a representable plant

Recommendation. The audit recommended that version 1 not be submitted in its form, but re-framed as an empirical-plus-verified-implementation paper: the factorial result is a clean, reproducible finding, and the theory should be the revised one (Sections 3–6) with the prior work cited.

2.3 Errata to version 1

Table 4 lists the errata to version 1. Section numbers in the second column refer to version 1; "README" refers to the public showcase README that accompanied it.

Table 4: Errata to version 1.

#	section	statement in version 1	status
E1	1.2, 9	Centre separation $\|c_i - c_j\| \geq \delta$ "maintained via the projection set construction" with "convex projection boundaries"	**Wrong.** The separation set is non-convex; no projection onto it has property (P). The implemented repulsion term violated (P) in 30% of samples and did not keep the separation (min distance 0.07 vs $\delta = 0.5$). Removed from the algorithm; not needed by the proof.
E2	1.3	Offline Tikhonov regularisation of the ideal target: "This strict separation eliminates the circularity"	**Irrelevant to the proof.** Optimality of θ^* is never used (Section 3, remarks on the assumptions).
E3	1.3	IFT gives finite velocity bounds V_W, V_c, V_σ	**Not a proof.** The Hessian of a Tikhonov-regularised non-convex loss at a minimiser is PSD, not necessarily PD; the IFT hypothesis is asserted, not established. Replaced by Assumption A4.
E4	2.1	$D_{\max}$ from the Taylor remainder, "a finite geometric scalar"	**Valid but not explicit.** Replaced by the explicit, Hessian-free Lemma 2. The code's former `estimate_D_max` $= \varepsilon(1 + 2W_{\max}c_{\max})$ had no derivation and is deleted.
E5	4	$\mathrm{tr}(\tilde{W}^\top \Gamma_W^{-1} \tilde{W})$ with Γ_W "positive-definite learning rate matrices"	**Type error** for $\Gamma_W \in \mathbb{R}^{mn \times mn}$; the correct form is $\mathrm{vec}(\tilde{W})^\top \Gamma_W^{-1} \mathrm{vec}(\tilde{W})$. Harmless for scalar Γ.
E6	5	Projection inequality "holds as an absolute mathematical certainty — the cancellation gap does not exist"	**True only under A3** (θ^* in the inner set) and only for the Γ-weighted operator; the previous code applied the isotropic formula to any Γ. Fixed and tested.
E7	7.2	"A critical and non-trivial consequence of the projection operator"; its hard energy cap "converts what would be a boundary-only argument … into a global interior argument"	**Standard** level-set / comparison argument (Khalil [40, Thm. 4.18]). Correct, not new; also "strict monotone descent through every intermediate level set until the infimal set is penetrated" overstates: entry into every level above $c_{\inf}$ is finite-time, entry into $c_{\inf}$ itself only asymptotic.
E8	abstract, 7.3, 10.3	Projection gives UUB "without the persistent steady-state bias" (abstract); "no modification bias is introduced" (7.3); "zero bias into the Lyapunov derivative", "not a marginal improvement but a structural result" (10.3)	**Overclaimed.** The ultimate bound $\sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}$ contains the parameter-energy term; see Section 6. The tighter, bias-free statement is the RMS bound (iii).
E9	8	"a standard Persistent Excitation (PE) condition on the regressor ϕ is insufficient to guarantee $\tilde{\Theta}(t) \to 0$"; convergence "requires alternative mechanisms", such as ICL	Parameter convergence is not treated; the remark is speculation and the code's former "Concurrent Learning" hook used only the current residual (not recorded data) and is removed.
E10	README	"100% of the independent simulation trials converged strictly below the calculated R_{UUB} threshold"	**False.** The threshold 0.5 was a hand-chosen constant (`tier1_monte_carlo.py`: "A chosen arbitrary bound"). Replaced by `exp_monte_carlo_honest.py` / `exp_oracle_ruub.py`.
E11	README	"$O(mn)$ time"	**Was false** with the $O(m^2 n)$ pairwise repulsion; true now that it is off by default.

2.4 What survives and what is withdrawn

What survives.

- *The corrected theorem.* Theorem 1 of Section 5 holds under the explicit Assumptions A1–A5, for an algorithm identical to the implemented one. Its pointwise ultimate bound (ii) is the conclusion of version 1; it is the textbook level-set argument, and its radius contains the parameter-energy term $V_{\max,\Theta}$. Its RMS bound (iii) does not contain that term and is the form of the result that can be confronted with data; its reference bound (iv) holds whatever the adaptation does. The ingredients are the projection property (P) (Lemma 1), the explicit Hessian-free disturbance bound (Lemma 2) and the Lyapunov derivative (Lemma 3), verified numerically along simulated trajectories. Nothing in the stability theorem is new (Section 7). The "no bias" claim survives only in the precise form of Section 6: projection removes the *Lyapunov-derivative* bias of leakage, not the ultimate *tracking* bound's dependence on the parameter set.
- *The projection-versus-leakage finding* (Section 8). On identical geometry, projection beats both σ- and e-modification in every cell of a mechanism-by-geometry factorial, 5/5 seeds, with paired confidence intervals excluding zero. Within projection, adaptive centres improve on the fixed grid by 0.012 (5/5) and adaptive widths by a further 0.035 (5/5). So the method's real advantage is the robustness mechanism, with geometry adaptation a smaller second-order gain, close to the reverse of the story told by the original figure.
- *The comparison with kernel regression, with its limits* (Section 8). Against kernel-regression baselines the method is never the most accurate. Its edge is cost: it is cheaper per step than every GP baseline at every n and than NTK except at $n = 3$, and in the fair $n = 1$ rerun, with per-step prediction for every method, it has the cheapest step. Every kernel-regression baseline (GP and NTK, including the fair $n = 1$ rerun) is trained on noise-free samples of the true $f(x, t)$, which is equivalent to measuring the state derivative exactly, while the adaptive law uses only the tracking error.

What is withdrawn.

- Earlier versions of the public README stated that the method's steady-state error was "statistically indistinguishable (Wilcoxon signed-rank $p > 0.3$) from both the Exact GP and the Sparse GP", that "100% of the independent simulation trials converged strictly below the calculated R_{UUB} threshold", and announced "Curse of Dimensionality Solved". Those statements are withdrawn: the significance test was invalid, the threshold was an arbitrary constant, and the omitted $n = 3$ and NTK results contradict them.
- The "zenith" language is dropped: "absolute mathematical certainty", "curse of dimensionality solved", "structural result". None of it survives the data. Claims of a "structural" advantage over leaky modifications are withdrawn.
- The claim that centre separation is "maintained via the projection set construction" is withdrawn (erratum E1); no centre-separation constraint is needed by the proof.

2.5 How Part II answers the structural defects

Section 1.2 named two structural defects of the classical labelled analysis, that of version 1 and of the literature it builds on: its Lyapunov analyses depend on labels, and it has no object that describes where the kernels go or whether their placement is good. Part II addresses both by working with the measure $\mu = \frac{1}{m}\sum_i \delta_{\vartheta_i}$, which does not see the labels, and with its transport.

- *Labels.* The Lyapunov function of Theorem B, $V_{\mathrm{OT}} = \frac{1}{2}\|e\|^2 + \frac{1}{2}W(\mu_t, \mu_t^*)^2$, measures the parameter error by the optimal-assignment (Wasserstein) distance W, a minimum

over all assignments of kernels to ideal atoms (Section 9). The labelled σ-modification is not a functional of μ; Theorem C replaces it by an optimal-transport σ-modification, Theorem C$'$ extends the OT leak to priors with a different number of atoms through the barycentric projection of the optimal plan, and Proposition C.5 shows that the labelled leak reshuffles kernels. Theorem E gives label-free local exponential convergence under tangential persistent excitation, with explicit radius, overshoot and rate (Lemma E.0).

- *Where the kernels go.* The measure μ_t is that object. By Theorem A the adaptive law is the particle form of the Wasserstein gradient flow of a μ-linear potential; Theorems D and D$'$ treat the mean-field (many-kernel) loop, without and with dither; Theorem F bounds the Wasserstein length of learning by the integrated tracking error; Proposition G describes the slow geometry (formal). Whether the placement is good is the subject of Theorem H: with saturated features, σ-modification and dither, the averaged mean-field loop converges globally and exponentially to the regularised optimum. This is proved for ($\mathfrak{A}$); the reduction to ($\mathfrak{A}$) is formal, and the real loop is Conjecture H.6, which Propositions H.7 and H.8 reduce to an averaging estimate. Theorem J proves its deterministic finite-m case, Theorem I characterises the non-reciprocity of the averaged loop at finite gain, and Proposition K turns the orientation of the reference orbit into a design input for collapsed configurations.
- *Constants growing with m.* The third apparent defect is an artefact of the normalisation; the mean-field normalisation removes it for labelled and label-free analyses alike (Remark A.1).

The status of each of these results is that of Table 1.

Part I
The projection-based controller

3 Setting and standing assumptions

This part replaces the stability argument of version 1 with a statement whose assumptions are explicit, whose algorithm is exactly the one in `src/core/` of the accompanying repository, and whose every step is either proved here or verified numerically by `tests/test_theorem_to_code.py`. The errata to version 1 are listed in Section 2; Section 7 positions the result against prior work.

Notation. $\|\cdot\|$ is the Euclidean norm for vectors and the Frobenius norm for matrices; $\mathrm{vec}(\cdot)$ is the row-major vectorisation used throughout the code; there are m kernels and n states.

Plant, reference and error.

$$\dot{x} = f(x,t) + u, \qquad e = x - x_m(t), \qquad u = \dot{x}_m - k_e e - \hat{f}(x), \tag{3.1}$$

$$\hat{f}(x) = W^\top \varphi(x, c, \sigma), \qquad \varphi_i(x) = \exp\Big(-\frac{\|x - c_i\|^2}{2\sigma_i^2}\Big). \tag{3.2}$$

Assumption A1 (operation on a compact set)**.** There is a compact set $X \subset \mathbb{R}^n$ with $x(t) \in X$ for all $t \geq 0$, and $F_{\max} := \sup_{x \in X,\, t \geq 0} \|f(x,t)\| < \infty$. *(Version 1 assumes "a compact spatial domain X" in the same way. For globally bounded f the assumption is discharged by Theorem 1(iv); for the Lorenz plant it is a genuine assumption.)*

Assumption A2 (reference)**.** x_m is C^1 with x_m, $\dot{x}_m$ bounded, and $\dot{x}_m$ is available to the controller.

Assumption A3 (admissible ideal parameters)**.** There is a curve $\theta^*(t) = (W^*(t), c^*(t), \sigma^*(t))$ lying for all t in the *inner* projection sets

$$\|W^*\| \leq W_{\max} - \varepsilon, \qquad \|c^*\| \leq c_{\max} - \varepsilon, \qquad \sigma_{\min} + \varepsilon \leq \sigma_i^* \leq \sigma_{\max} - \varepsilon.$$

Assumption A4 (ideal-parameter regularity)**.** $\theta^*(\cdot)$ is C^1 with $\|\dot{W}^*\| \leq V_W$, $\|\dot{c}^*\| \leq V_c$, $\|\dot{\sigma}^*\| \leq V_\sigma$. (For a time-invariant plant one may take θ^* constant, $V_W = V_c = V_\sigma = 0$.)

Assumption A5 (gains and geometry)**.** $k_e > 0$; Γ_W, Γ_c symmetric positive definite on $\mathbb{R}^{mn}$ (a zero Γ freezes that group, see Remark 3); Γ_σ diagonal positive; $0 < \varepsilon < \min\big(W_{\max}, c_{\max}, (\sigma_{\max} - \sigma_{\min})/2\big)$; $\sigma_{\min} > 0$.

Remarks on the assumptions.

- **The proof never uses optimality of θ^*.** θ^* is an analytic device; any admissible C^1 curve works, and the approximation quality of the network enters the result only through the *actual* size of the disturbance $d(t)$ defined below, not through the bound. The Tikhonov-regularised definition of θ^* in version 1 and its implicit-function-theorem (IFT) argument for A4 are therefore unnecessary — and, as written, not a proof (erratum E3, Section 2).
- **No centre-separation constraint is needed.** The set $\{c : \|c_i - c_j\| \geq \delta \ \forall\, i \neq j\}$ is not convex, so no projection onto it has property (P) below; the claim of version 1 that separation is "maintained via the projection set construction" is withdrawn (erratum E1, Section 2). The bounded-Jacobian property that the proof needs follows from $\sigma \geq \sigma_{\min}$ alone.

4 The algorithm

The algorithm is identical to the one implemented in `src/core/controller.py`:

$$\begin{aligned} \dot{W} &= \mathrm{Proj}_W\big(W,\ \Gamma_W\,\varphi e^\top\big), \\ \dot{c} &= \mathrm{Proj}_c\big(c,\ \Gamma_c\,\varphi_c'^\top We\big), \\ \dot{\sigma} &= \mathrm{Proj}_\sigma\big(\sigma,\ \Gamma_\sigma\,\varphi_\sigma'^\top We\big), \end{aligned} \tag{4.1}$$

with $\varphi_c' = \partial\varphi/\partial\,\mathrm{vec}(c) \in \mathbb{R}^{m\times mn}$ (row i has the single non-zero block $\varphi_i(x-c_i)^\top/\sigma_i^2$) and $\varphi_\sigma' = \mathrm{diag}\big(\varphi_i\|x-c_i\|^2/\sigma_i^3\big)$.

Norm-ball projection (W and c), boundary layer of width ε.

$$\mathrm{Proj}(\theta,\Gamma y) = \begin{cases} \Gamma y - h(\theta)\,\Gamma\theta\,\dfrac{\theta^\top\Gamma y}{\theta^\top\Gamma\theta} & \text{if } h(\theta)>0 \text{ and } \theta^\top\Gamma y>0, \\ \Gamma y & \text{otherwise,} \end{cases} \tag{4.2}$$

$$h(\theta) = \begin{cases} 0 & \text{for } \|\theta\| \le \theta_{\max}-\varepsilon, \\ \big((\|\theta\|-\theta_{\max}+\varepsilon)/\varepsilon\big)^2 & \text{in between,} \\ 1 & \text{for } \|\theta\| \ge \theta_{\max}. \end{cases} \tag{4.3}$$

Box projection (σ, component-wise). The update component is multiplied by $\mathrm{clip}\big((\sigma_i-\sigma_{\min})/\varepsilon, 0, 1\big)$ when it points below the lower face and by $\mathrm{clip}\big((\sigma_{\max}-\sigma_i)/\varepsilon, 0, 1\big)$ when it points above the upper face.

Discrete time. The simulations integrate with fixed-step RK4 and re-clip the parameters to their sets after every stage (`src/core/integrator.py`). The theorem is a continuous-time statement; the clip is a safety net that is inactive if the step is small enough (property (P) already removes the outward component on the boundary).

5 Lemmas and the boundedness theorem

5.1 Lemmas

Lemma 1 (projection property (P))**.** *Under A3 and A5, for every θ in its set and every y,*

$$\tilde\theta^\top\Gamma^{-1}\,\mathrm{Proj}(\theta,\Gamma y) \le \tilde\theta^\top y, \qquad \tilde\theta := \theta-\theta^*. \tag{P}$$

Proof. If the projection is inactive the two sides are equal. Otherwise

$$\tilde\theta^\top\Gamma^{-1}\big(\mathrm{Proj}\ -\Gamma y\big) = -h\,\frac{\theta^\top\Gamma y}{\theta^\top\Gamma\theta}\,\tilde\theta^\top\theta,$$

with $h>0$ and $\theta^\top\Gamma y>0$, and $\tilde\theta^\top\theta = \|\theta\|^2 - \theta^{*\top}\theta \ge \|\theta\|\big(\|\theta\|-\|\theta^*\|\big) \ge 0$ because $h>0$ forces $\|\theta\| \ge \theta_{\max}-\varepsilon \ge \|\theta^*\|$. For the box, component i, with s_i the clip factor applied to that component: $\tilde\sigma_i(s_i-1)y_i \le 0$ because $s_i \le 1$, and on the lower face $y_i<0$ while $\tilde\sigma_i = \sigma_i-\sigma_i^* \le \sigma_{\min}+\varepsilon-\sigma_i^* \le 0$ (symmetrically on the upper face). □

Status. Proved.

Checked by: `tests/test_projection.py` (randomised; isotropic and non-isotropic Γ; vector, matrix and box), together with the counterexample showing that (P) fails when θ^* is outside the inner set.

Lemma 2 (lumped disturbance)**.** *Define*

$$d := f - \hat{f} + \tilde{W}^\top \varphi + W^\top \varphi_c' \,\tilde{c} + W^\top \varphi_\sigma' \,\tilde{\sigma}. \tag{5.1}$$

Then identically

$$\dot{e} = -k_e e - \tilde{W}^\top \varphi - W^\top \varphi_c' \,\tilde{c} - W^\top \varphi_\sigma' \,\tilde{\sigma} + d, \tag{5.2}$$

and under A1, A3,

$$\|d\| \le D_{\max} := F_{\max} + 3W_{\max}\sqrt{m} + W_{\max}\big(2c_{\max}J_c + \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})J_\sigma\big), \tag{5.3}$$
$$J_c = \mathrm{e}^{-1/2}/\sigma_{\min}, \qquad J_\sigma = 2\mathrm{e}^{-1}/\sigma_{\min}.$$

Proof. The identity is the definition of d substituted into $\dot{e} = -k_e e + f - \hat{f}$. For the bound: $\|f\| \le F_{\max}$; $\|\hat{f}\| \le \|W\|\|\varphi\| \le W_{\max}\sqrt{m}$ $(0 < \varphi_i \le 1)$; $\|\tilde{W}\| \le 2W_{\max}$; $\|\tilde{c}\| \le 2c_{\max}$; $\|\tilde{\sigma}\| \le \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})$; the rows of φ_c' have disjoint supports, so

$$\|\varphi_c'\|_2 = \max_i \frac{\varphi_i \|x - c_i\|}{\sigma_i^2} \le \sup_{r \ge 0} \frac{r\,\mathrm{e}^{-r^2/(2\sigma^2)}}{\sigma^2} = \frac{\mathrm{e}^{-1/2}}{\sigma} \le J_c,$$

and $\|\varphi_\sigma'\|_2 = \max_i \varphi_i \|x - c_i\|^2/\sigma_i^3 \le 2\mathrm{e}^{-1}/\sigma \le J_\sigma$. □

Status. Proved.

Checked by: `tests/test_rbf_kernel.py` checks the two suprema by brute force.

Remark 1 (local form)**.** Writing $\varphi^* = \varphi(x; c^*, \sigma^*)$ and expanding to first order in $(\tilde{c}, \tilde{\sigma})$ gives

$$d = \varepsilon + W^{*\top} O_{\mathrm{ho}} + \tilde{W}^\top \varphi_c' \,\tilde{c} + \tilde{W}^\top \varphi_\sigma' \,\tilde{\sigma}, \qquad \varepsilon := f - W^{*\top}\varphi^*,$$

with O_{ho} the Taylor remainder — the decomposition of version 1. Near convergence ($\tilde{\theta} \to 0$) this shows $\|d\| \lesssim \|\varepsilon\| + O(\|\tilde{\theta}\|^2)$, which is why the controller performs far better than $D_{\max}$ suggests; but it does not yield a smaller global constant than Lemma 2, which needs no Hessian and no Taylor expansion.

Lemma 3 (Lyapunov derivative)**.** *Let*

$$V = \tfrac{1}{2} e^\top e + \tfrac{1}{2}\,\mathrm{vec}(\tilde{W})^\top \Gamma_W^{-1}\,\mathrm{vec}(\tilde{W}) + \tfrac{1}{2}\tilde{c}^\top \Gamma_c^{-1}\tilde{c} + \tfrac{1}{2}\tilde{\sigma}^\top \Gamma_\sigma^{-1}\tilde{\sigma}.$$

Along trajectories of the algorithm,

$$\begin{aligned}
\dot{V} &= -k_e\|e\|^2 + e^\top d + \mathrm{cross} - \Omega^*, \\
\mathrm{cross} &= \langle \tilde{W},\, \Gamma_W^{-1}\dot{W} - \varphi e^\top \rangle + \tilde{c}^\top\big(\Gamma_c^{-1}\dot{c} - \varphi_c'^\top W e\big) + \tilde{\sigma}^\top\big(\Gamma_\sigma^{-1}\dot{\sigma} - \varphi_\sigma'^\top W e\big) \le 0, \\
\Omega^* &= \langle \tilde{W},\, \Gamma_W^{-1}\dot{W}^* \rangle + \tilde{c}^\top \Gamma_c^{-1}\dot{c}^* + \tilde{\sigma}^\top \Gamma_\sigma^{-1}\dot{\sigma}^*, \qquad |\Omega^*| \le C_\Omega, \\
C_\Omega &:= 2W_{\max}\|\Gamma_W^{-1}\|_2 V_W + 2c_{\max}\|\Gamma_c^{-1}\|_2 V_c + \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})\|\Gamma_\sigma^{-1}\|_2 V_\sigma.
\end{aligned} \tag{5.4}$$

Proof. Differentiate V, substitute the identity of Lemma 2 for $\dot{e}$, and use $e^\top \tilde{W}^\top \varphi = \langle \tilde{W},\, \varphi e^\top \rangle$, $e^\top W^\top \varphi_c' \,\tilde{c} = \tilde{c}^\top \varphi_c'^\top W e$, $e^\top W^\top \varphi_\sigma' \,\tilde{\sigma} = \tilde{\sigma}^\top \varphi_\sigma'^\top W e$. That cross ≤ 0 is Lemma 1 applied to the three groups; the bound on Ω^* is Cauchy–Schwarz with A3, A4. □

Status. Proved.

Checked by: `tests/test_theorem_to_code.py`: the identity (to machine precision) and cross ≤ 0 (at every sample) are verified along simulated trajectories, on a plant that is exactly representable with known constant θ^*.

5.2 Theorem 1: uniform ultimate boundedness

Theorem 1 (uniform ultimate boundedness)**.** *Assume A1–A5, let*

$$R_{\mathrm{UUB}} := \frac{D_{\max}}{2k_e} + \sqrt{\frac{D_{\max}^2}{4k_e^2} + \frac{C_\Omega}{k_e}},$$

and let $V_{\max,\Theta}$ be the supremum over the projection sets of the parameter part of V.

(i) ***Descent outside a ball.*** *$\dot V \le -k_e\|e\|^2 + D_{\max}\|e\| + C_\Omega$, hence $\dot V < 0$ whenever $\|e\| > R_{\mathrm{UUB}}$.*

(ii) ***Pointwise ultimate bound.*** *Every solution satisfies $V(t) \le \max\{V(0), c_{\inf}\}$ with $c_{\inf} := \frac{1}{2}R_{\mathrm{UUB}}^2 + V_{\max,\Theta}$, every level $c > c_{\inf}$ is reached in finite time, and*

$$\limsup_{t\to\infty}\|e(t)\| \le \sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}.$$

(iii) ***RMS bound (the quantitatively meaningful one).*** *For every $T_1 < T_2$,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{D_{\max}}{2k_e} + \sqrt{\frac{D_{\max}^2}{4k_e^2} + \frac{C_\Omega}{k_e} + \frac{V(T_1)}{k_e(T_2-T_1)}},$$

so $\limsup_{T_2\to\infty}\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le R_{\mathrm{UUB}}$ — without the parameter-energy term of (ii). The same inequality holds with $D_{\max}$ and C_Ω replaced by the trajectory-wise suprema of $\|d(t)\|$ and $|\Omega^*(t)|$ on $[T_1,T_2]$.

(iv) ***Reference bound (adaptation-free).*** *Under A1 and the hard parameter bounds alone, $\dot e = -k_e e + (f - \hat f)$ with $\|f - \hat f\| \le F_{\max} + W_{\max}\sqrt{m} =: \tilde F$, so*

$$\|e(t)\| \le \mathrm{e}^{-k_e t}\|e(0)\| + \tilde F/k_e \qquad \text{for all } t,$$

whatever the adaptation does.

Proof. (i) Lemma 3 with Cauchy–Schwarz on $e^\top d$.

(ii) If $V > c_{\inf}$ then, since the parameter part of V is at most $V_{\max,\Theta}$, $\frac{1}{2}\|e\|^2 > \frac{1}{2}R_{\mathrm{UUB}}^2$, so $\|e\| > R_{\mathrm{UUB}}$ and $\dot V < 0$; more precisely, with $\Delta := V - c_{\inf} > 0$ one has $\|e\|^2 \ge R_{\mathrm{UUB}}^2 + 2\Delta$ and therefore

$$\dot V \le -g\Big(\sqrt{R_{\mathrm{UUB}}^2 + 2\Delta}\Big) =: -\psi(\Delta), \qquad g(r) = k_e r^2 - D_{\max} r - C_\Omega,$$

with g increasing for $r \ge R_{\mathrm{UUB}}$ and ψ of class $\mathcal{K}$. The comparison lemma gives $V(t) \le \max\{V(0), c_{\inf}\}$, finite-time entry into every sublevel set $\{V \le c\}$ with $c > c_{\inf}$, and $\limsup V \le c_{\inf}$, i.e. the stated bound on $\|e\|$ ($\|e\|^2 \le 2V$).

(iii) Integrate (i):

$$\begin{aligned} k_e\int_{T_1}^{T_2}\|e\|^2\,\mathrm{d}t &\le V(T_1) - V(T_2) + D_{\max}\int_{T_1}^{T_2}\|e\|\,\mathrm{d}t + C_\Omega(T_2-T_1) \\ &\le V(T_1) + D_{\max}\sqrt{T_2-T_1}\,\sqrt{\int_{T_1}^{T_2}\|e\|^2\,\mathrm{d}t} + C_\Omega(T_2-T_1) \end{aligned}$$

by Cauchy–Schwarz; with $S := \mathrm{RMS}_{[T_1,T_2]}(\|e\|)$ this reads

$$k_e S^2 - D_{\max} S - \big(C_\Omega + V(T_1)/(T_2-T_1)\big) \le 0,$$

and S is below the positive root.

(iv) Variation of constants. □

Status. Proved.

What (ii)–(iv) say together. Part (ii) is the conclusion of version 1; it is the textbook level-set argument (Khalil [40, Thm. 4.18]; Lewis et al. [46]) and its radius contains the parameter-energy reservoir $V_{\max,\Theta}$ — a bias-like term — so it does **not** establish that projection yields a tighter tracking bound than σ-modification. Part (iii) is tighter and is the statement that can be checked without unknown constants: replace $D_{\max}$ by the measured $\sup\|d\|$, see `experiments/exp_oracle_ruub.py`, where it is satisfied in 48/48 runs (Section 8). Part (iv) shows that mere boundedness of e is already a consequence of projection plus a stable error channel, independent of the adaptation law; the content of adaptive control is in making the *actual* $d(t)$ small, which no a-priori constant captures. In the oracle experiments the a-priori R_{UUB} exceeds the observed RMS error by factors of 915 to 10954.

Remark 2 (no LaSalle / Barbalat)**.** Version 1's claim that the argument needs neither LaSalle's invariance principle nor Barbalat's lemma is correct, and standard: UUB statements never need them. Nothing here claims asymptotic convergence of e or of $\tilde{\theta}$.

Remark 3 (frozen groups)**.** If $\Gamma_c = 0$ (fixed centres), drop the $\tilde{c}$ term from V; the identity of Lemma 2 still holds with $-W^\top \varphi_c' \tilde{c}$ absorbed into d, with the same bound. Likewise for σ.

Remark 4 (what is not proved)**.** Parameter convergence (it requires persistent excitation of a nonlinear-in-parameter regressor — not addressed); exact tracking; any statement about the size of the ideal approximation error ε.

6 The no-bias claim, stated precisely

With σ-modification,

$$\dot{V} \le -k_e\|e\|^2 - \tfrac{1}{2}\sigma_{\mathrm{mod}}\|\tilde{\theta}\|_\Gamma^2 + \tfrac{1}{2}\sigma_{\mathrm{mod}}\|\theta^*\|_\Gamma^2 + D\|e\| + C_\Omega:$$

the leak adds a positive constant and removes the parameter-energy reservoir, which gives exponential convergence of V to a ball whose radius grows with $\sigma_{\mathrm{mod}}\|\theta^*\|^2$. With projection,

$$\dot{V} \le -k_e\|e\|^2 + D\|e\| + C_\Omega:$$

no positive constant is added, but nothing dissipates parameter energy, which is why the pointwise bound (ii) of Theorem 1 keeps $V_{\max,\Theta}$. Projection therefore removes the *Lyapunov-derivative* bias of leakage (true, and known since Pomet–Praly [63] and Ioannou–Sun [39]), not the dependence of the ultimate *tracking* bound on the parameter set (false as stated in Sections 7.3 and 10.3 of version 1). Empirically the two mechanisms are compared on identical geometry in `experiments/exp_bias_factorial.py` (Section 8).

7 Prior art

- Adaptive Gaussian-network control with robust modification: Sanner and Slotine [65].
- Adaptation of inner-layer parameters via first-order Taylor expansion of the feature map with bounded remainder — the exact device of Sections 4 and 5: Lewis, Yesildirek and Liu [45]; Polycarpou [62]; Ge and Wang [26]; dynamic-structure RBF networks with tuned centres and widths: Fabri and Kadirkamanathan [21].
- Smooth projection with a boundary layer and the inequality (P), including the "no bias when $\theta^* \in \Omega$" observation: Pomet and Praly [63]; Ioannou and Sun [39, §4.4]; Lavretsky and Gibson [43]; Lavretsky and Wise [44, Ch. 11].
- σ- and e-modification and their bias: Ioannou and Kokotovic [38]; Narendra and Annaswamy [51].

- UUB via level sets: Khalil [40, Thm. 4.18]; Lewis, Jagannathan and Yesildirek [46].
- Concurrent learning (recorded-data stacks, finite excitation): Chowdhary and Johnson [13, 14].

Table 5 sets the ingredients of version 1 against these sources.

Table 5: Novelty of the ingredients of the stability argument.

Ingredient	Prior source	Status of the claim of version 1
Adaptive RBF control with projection-type robustification	Sanner and Slotine [65]; Ioannou and Sun [39]	known
Adapting centres/widths by first-order Taylor expansion with bounded remainder	Lewis, Yesildirek and Liu [45]; Polycarpou [62]; Fabri and Kadirkamanathan [21]; Ge and Wang [26]	known; version 1 cites none of these
Smooth projection with ε boundary layer and inequality (P)	Pomet and Praly [63]; Lavretsky and Gibson [43]; Lavretsky and Wise [44]	known; the public showcase README calls it "the core theoretical contribution"
Projection introduces no bias in $\dot{V}$ when $\theta^* \in \Omega$	Ioannou and Sun [39, §4.4]	known
Level-set ("global interior") UUB argument	Khalil [40, Thm. 4.18]; Lewis et al. [46]	known; called "critical and non-trivial"
Simultaneous adaptation of W, c, σ for Gaussian RBFs with all three projected	—	the only combinational novelty; minor

Net: no new theorem.

What is new here. Nothing in the stability theorem: Theorem 1(i)–(ii) is the composition of known pieces (Taylor-linearised inner-layer adaptation, boundary-layer projection and level-set UUB), applied to Gaussian RBFs with simultaneously adaptive weights, centres and widths. The contribution of the accompanying repository, after the revision of version 1, is (a) a theorem-to-code map in which the implemented and the proved algorithm coincide and are machine-checked (Lemmas 1–3 verified numerically along trajectories), (b) the explicit Hessian-free disturbance bound of Lemma 2 and the RMS bound (iii) of Theorem 1, which is the form of the result that can actually be confronted with data, and (c) a controlled empirical separation of "projection vs leakage" from "adaptive vs fixed geometry" (Section 6; `results/factorial/`). Claims of a "structural" advantage over leaky modifications are withdrawn.

8 What the experiments show

Every numerical statement in this section was regenerated from the scripts of the accompanying repository; the result file and the generating script are named with each table. Statistics are computed over seeds, not over autocorrelated time samples.

8.1 Robustness mechanism versus geometry

The factorial experiment `experiments/exp_bias_factorial.py` separates the two factors that the original headline figure (`results/tier1_lorenz_tracking.png`) confounded: there, the fixed-grid baselines used a grid not covering the reference, and their error (≈ 15) exceeded open loop, so the figure showed representational failure, not "bias". Here three robustness mechanisms (projection, σ-modification, e-modification, the leaky ones implemented on the same architecture

in `src/baselines/leaky_geometry.py`) are crossed with three geometries (fixed grid with only W adapted; adaptive W, c; adaptive W, c, σ) on Lorenz tracking with a covering grid, $m = 27$, $k_e = 10$ and 5 seeds (Tables 6 and 7, Figure 1).

Table 6: Robustness mechanism × geometry (Lorenz tracking, $m = 27$, $k_e = 10$). Seeds: 5; metric = steady-state $\|e\|$ over the last 40% of a 30 s run; mean ± 95% CI over seeds; max SS = maximum over the window, mean over seeds; "div" = diverged runs. Source: `results/factorial/table.md`.

geometry	mechanism	mean SS $\|e\|$	median SS	max SS	max $\|W\|_F$	min sep	div
fixed (W)	projection	0.1584 ± 0.0072	0.1288	0.559	47.5	1.365	0
fixed (W)	σ-mod	0.7327 ± 0.0223	0.5647	2.720	38.5	1.469	0
fixed (W)	e-mod	0.5619 ± 0.0306	0.4264	2.192	26.7	1.469	0
adaptive (W, c)	projection	0.1468 ± 0.0038	0.1020	0.751	40.9	0.039	0
adaptive (W, c)	σ-mod	0.4004 ± 0.0050	0.3468	1.863	32.3	0.078	0
adaptive (W, c)	e-mod	0.2327 ± 0.0024	0.1993	0.718	28.1	0.099	0
adaptive (W, c, σ)	projection	0.1119 ± 0.0043	0.0852	0.512	38.5	0.172	0
adaptive (W, c, σ)	σ-mod	0.3862 ± 0.0057	0.3422	1.655	30.9	0.215	0
adaptive (W, c, σ)	e-mod	0.2385 ± 0.0029	0.2103	0.820	27.7	0.239	0

Table 7: Paired (same seed) differences of mean steady-state $\|e\|$ for the factorial experiment. Top: projection minus baseline. Bottom: geometry effect within the projection mechanism. Source: `results/factorial/table.md`.

geometry	baseline	mean diff	95% CI	projection better in
fixed (W)	σ-mod	−0.5743	±0.0232	5/5 seeds
fixed (W)	e-mod	−0.4035	±0.0310	5/5 seeds
adaptive (W, c)	σ-mod	−0.2536	±0.0021	5/5 seeds
adaptive (W, c)	e-mod	−0.0858	±0.0025	5/5 seeds
adaptive (W, c, σ)	σ-mod	−0.2743	±0.0043	5/5 seeds
adaptive (W, c, σ)	e-mod	−0.1266	±0.0031	5/5 seeds
comparison		mean diff	95% CI	adaptive better in
adaptive (W, c) − fixed (W)		−0.0116	±0.0056	5/5
adaptive (W, c, σ) − adaptive (W, c)		−0.0349	±0.0026	5/5

Projection beats both leaky baselines in every cell, in 5/5 seeds, with paired confidence intervals excluding zero. Within projection, adaptive centres improve on the fixed grid by 0.0116 ± 0.0056 (5/5 seeds) and adaptive widths by a further 0.0349 ± 0.0026 (5/5 seeds). So the method's real advantage is the robustness mechanism, with geometry adaptation a smaller second-order gain — close to the reverse of the story told by the original headline figure.

8.2 Kernel-regression baselines

Table 8 is the comparison against kernel-regression baselines from saved single runs, with nothing excluded (in particular the benchmark $n = 3$ and the NTK baseline are included). The proposed method is never first: it ranks 3/5, 2/5, 3/5 and 2/5 at $n = 1, 2, 3, 6$. NTK is first at $n = 2$ and $n = 6$ (mean steady-state $\|e\|$ 0.0015 and 0.0013, against 0.0372 and 0.0044 for the proposed method), and the sparse GP is first at $n = 3$ (3.2729), where the error of

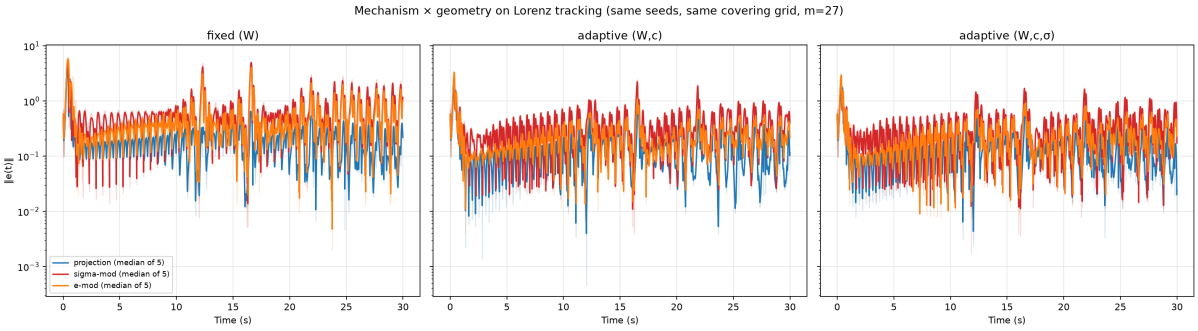


Figure 1: Mechanism × geometry on Lorenz tracking (same seeds, same covering grid, $m = 27$): $\|e(t)\|$ against time, logarithmic scale, median of the 5 seeds for projection, σ-modification and e-modification; shaded bands: minimum to maximum over the 5 seeds. Panels: fixed (W), adaptive (W, c), adaptive (W, c, σ). Generated by `experiments/exp_bias_factorial.py`.

the proposed method is 7.7190. Its genuine edge is cost against the GP baselines: the mean update time of the proposed method is 0.10, 0.22 and 0.60 ms per step at $n = 1, 2, 6$ and 4.11 ms at $n = 3$, against 52.92–1888.80 ms per update for the GP baselines, which update every 50–100 steps. NTK takes 1.59–13.22 ms per update; it updates every 50 steps at $n = 1$ and every step at $n = 2, 3, 6$ (`experiments/sim4_ntk_control.py`, `experiments/sim4_ntk_n2.py`, `experiments/sim4_ntk_n3.py`, `experiments/sim4_ntk_n6.py`), so it is cheaper than the proposed method only at $n = 3$ (2.35 against 4.11 ms per step). The comparison is not information-matched: every baseline, here and in the fair re-run below, is trained on noise-free samples of the true $f(x, t)$, which amounts to measuring the state derivative exactly, whereas the adaptive law uses only the tracking error e. With one run per cell no valid significance test is available.

In the original baseline scripts (`sim1`–`sim3`) the GP baselines held a stale prediction for 50–100 steps, and the inducing points of the sparse GP lay on a diagonal line over $x \in [-3, 3]$ while the reference spans ± 20. The fair re-run at $n = 1$ (`experiments/kernel_baselines_fair.py`, `experiments/run_fair_n1.py`, `experiments/plot_fair_n1.py`) uses per-step prediction for all methods, a covering inducing grid and 3 seeds (Table 9, Figure 2). The mean steady-state $\|e\|$ is 0.0591 for the exact GP, 0.1867 for the sparse GP, 0.1978 for the proposed method, 0.3087 for the deep GP and 0.4700 for NTK; the median online step is 6.309, 1.353, 0.060, 4.461 and 0.212 ms respectively. Once it is allowed to predict every step, the exact GP is 3–4× more accurate than the proposed method (mean steady-state $\|e\|$ 0.0591 against 0.1978), and more accurate in every seed (the proposed method has the lower mean in 0/3 seeds). The proposed method is on par with a fixed-grid sparse GP (the proposed method has the lower mean in 2/3 seeds) at 22.6× lower cost (median step 0.060 ms against 1.353 ms); its step is the cheapest of the five methods. This is the version of the comparison to quote.

8.3 The bound against data

Oracle bound study. The oracle experiment (`experiments/exp_oracle_ruub.py`) uses a plant that is exactly representable with constant θ^* ($m = 8$, $n = 2$), so every constant of Theorem 1 is computable; with θ^* constant, $C_\Omega = 0$ and $R_{\mathrm{UUB}} = D_{\max}/k_e$, reported as D_{theory}/k_e with $D_{\max}$ from Lemma 2. The "full-state bound" is $\sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}$ of Theorem 1(ii), and D_{meas} is the measured $\sup_{t\in[10,30]}\|d(t)\|$ over the evaluation window. Table 10 and Figure 3 cover $k_e \in \{2, 5, 10, 20\}$ and four learning-rate scales, 3 seeds each. The Lyapunov identity of Lemma 3 holds to 3.55×10^{-15} (maximal $|\dot{V} - (-k_e\|e\|^2 + e^\top d + \mathrm{cross})|$), property (P) is never violated, the hard bounds held, and the RMS bound of Theorem 1(iii) with the measured disturbance holds in 48/48 runs and tracks $1/k_e$. The a-priori R_{UUB} is 26.1–261.1 against observed RMS

Table 8: Higher-dimensional comparison, complete (nothing excluded). Single trajectory per cell; steady state = last 40% of 30 s. Step time = mean wall-clock time of the method's update call (GP: per update, every 50–100 steps; NTK: per update, every 50 steps at $n = 1$ and every step at $n = 2, 3, 6$; proposed: per step). No valid significance test is available (one run per cell). Generated by `experiments/plot_comparison_hd_complete.py`; source: `results/comparison_complete_table.md`.

benchmark	method	mean SS $\|e\|$	median SS	max SS	rank (mean SS)	mean update time (ms)
$n = 1$ (1-D plant, Lorenz ref.)	Proposed (Proj)	0.2342	0.1496	1.290	3/5	0.10
	Exact GP	0.2206	0.1415	1.016	1/5	273.78
	Sparse GP	0.2285	0.1914	0.874	2/5	52.92
	Deep GP	0.3963	0.2994	1.436	4/5	94.34
	NTK	0.4286	0.3318	1.324	5/5	13.22
$n = 2$ (Van der Pol)	Proposed (Proj)	0.0372	0.0137	0.310	2/5	0.22
	Exact GP	0.0555	0.0370	0.278	3/5	498.50
	Sparse GP	0.1771	0.0929	0.886	5/5	150.30
	Deep GP	0.1758	0.1169	1.097	4/5	415.37
	NTK	0.0015	0.0009	0.006	1/5	1.59
$n = 3$ (Lorenz plant)	Proposed (Proj)	7.7190	5.7197	30.476	3/5	4.11
	Exact GP	10.2774	8.8437	34.467	5/5	1888.80
	Sparse GP	3.2729	2.5099	19.176	1/5	264.64
	Deep GP	9.5344	8.3305	33.755	4/5	802.93
	NTK	5.7642	4.2944	18.240	2/5	2.35
$n = 6$ (coupled Duffing)	Proposed (Proj)	0.0044	0.0013	0.026	2/5	0.60
	Exact GP	0.0841	0.0695	0.279	3/5	668.01
	Sparse GP	0.1444	0.0872	0.670	5/5	523.83
	Deep GP	0.0953	0.0553	0.342	4/5	1607.48
	NTK	0.0013	0.0010	0.004	1/5	4.75

Table 9: Fair 1-D kernel comparison: per-step prediction for every method, covering inducing grid, seed-level statistics. Seeds: 3 (different reference and plant initial conditions). Steady state = last 40% of 30 s. Step time = median full online step (prediction + amortised training) after warm-up, same process type, 2 BLAS threads. Source: `results/fair_n1/table.md`.

method	mean SS $\|e\|$ (mean $\pm$ 95% CI over seeds)	max SS $\|e\|$ (mean)	median step (ms)	mean step (ms)	seeds where proposed has lower mean SS
Proposed (Proj)	0.1978 ± 0.0503	1.147	0.060	0.069	—
Exact GP	0.0591 ± 0.0061	0.780	6.309	13.936	0/3
Sparse GP (9×3 grid)	0.1867 ± 0.0210	0.699	1.353	2.732	2/3
Deep GP	0.3087 ± 0.1121	1.262	4.461	5.495	2/3
NTK	0.4700 ± 0.0250	1.438	0.212	0.503	3/3

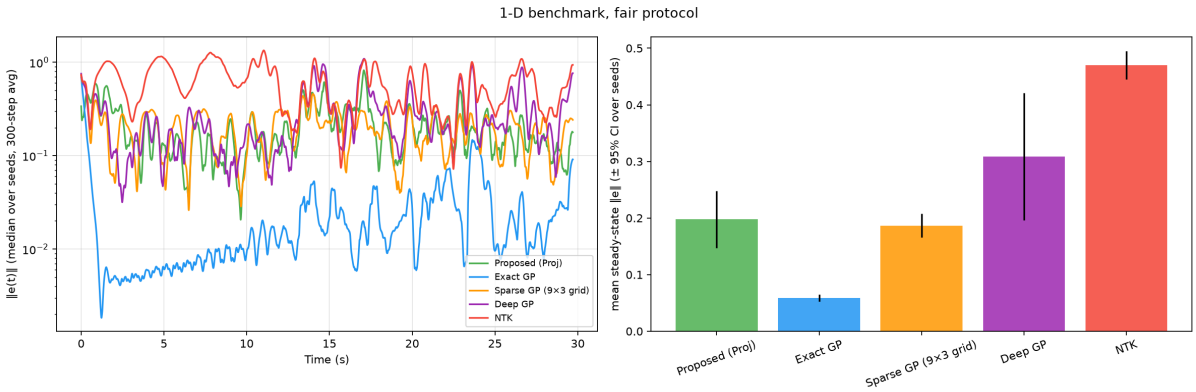


Figure 2: 1-D benchmark, fair protocol. Left: $\|e(t)\|$ (median over seeds, 300-step average), logarithmic scale, for the proposed method (Proj), the exact GP, the sparse GP (9×3 grid), the deep GP and NTK. Right: mean steady-state $\|e\|$ ($\pm$ 95% CI over seeds). Generated by `experiments/plot_fair_n1.py`; numbers in Table 9.

errors of 0.0087–0.0908; the ratio of the a-priori bound to the observed RMS error ranges from 915× to 10954×.

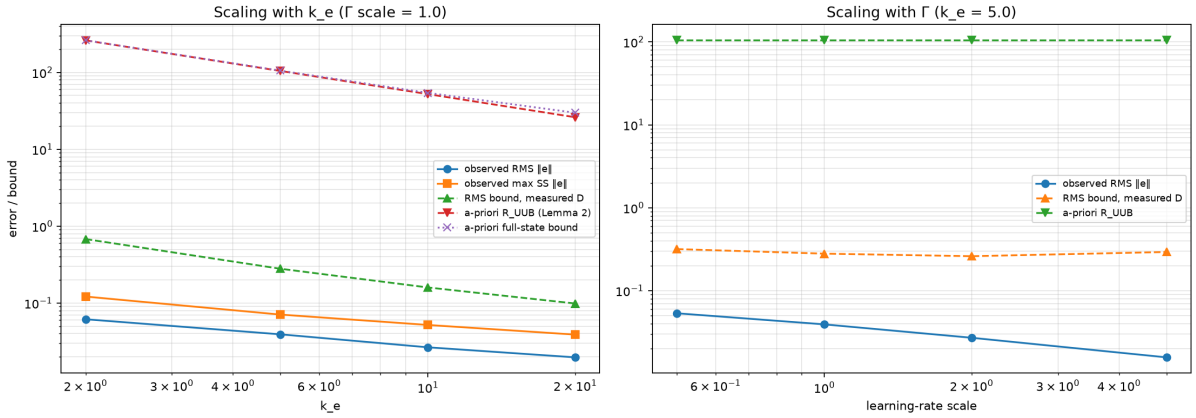


Figure 3: Oracle bound study, logarithmic axes. Left: scaling with k_e at learning-rate scale 1.0 — observed RMS $\|e\|$, observed maximal steady-state $\|e\|$, the RMS bound with the measured disturbance, the a-priori R_{UUB} (Lemma 2) and the a-priori full-state bound. Right: scaling with the learning-rate scale at $k_e = 5$ — observed RMS $\|e\|$, the RMS bound with the measured disturbance and the a-priori R_{UUB}. Generated by `experiments/exp_oracle_ruub.py`; numbers in Table 10.

Monte Carlo over initial conditions. On a plant that is not exactly representable only exactly checkable statements are tested (`experiments/exp_monte_carlo_honest.py`, Table 11): all 200 trajectories are finite, the hard parameter bounds held at every recorded step in 200/200 trials, and the a-posteriori ISS bound held in 200/200 trials (maximal ratio 0.979). No a-priori R_{UUB} threshold is drawn: $D_{\max}$ and C_Ω are not computable for this plant (the threshold 0.5 of the original figure was an arbitrary constant, not a calculated bound).

8.4 Summary

- On identical geometry, projection beats σ- and e-modification in every cell and in every seed (5/5), with paired confidence intervals excluding zero.

Table 10: Oracle R_{UUB} study (exactly representable plant, constant θ^*, $m = 8$, $n = 2$). Window $[10, 30]$ s; mean over 3 seeds. All bounds in the same units as $\|e\|$. "P viol" = samples with cross > 0 (must be 0); "inside" = hard bounds held. RMS bound (measured disturbance) satisfied in all 48 runs: True. Maximal identity error $|\dot{V} - (-k_e\|e\|^2 + e^\top d + \mathrm{cross})|$: 3.55×10^{-15}. Source: `results/oracle_ruub/table.md`.

k_e	Γ scale	RMS $\|e\|$	max SS $\|e\|$	RMS bound (D_{meas}, finite window)	D_{meas}/k_e	D_{theory}/k_e	full-state bound	ratio theory/RMS	P viol	inside
2	0.5	0.0908	0.1704	0.6791	0.5435	261.1	261.9	2875×	0	True
2	1	0.0616	0.1215	0.6801	0.6123	261.1	261.5	4239×	0	True
2	2	0.0383	0.0796	0.7139	0.6808	261.1	261.3	6824×	0	True
2	5	0.0238	0.0476	1.1766	1.1649	261.1	261.2	10954×	0	True
5	0.5	0.0533	0.1054	0.3170	0.1979	104.4	106.4	1958×	0	True
5	1	0.0392	0.0708	0.2793	0.2123	104.4	105.4	2664×	0	True
5	2	0.0270	0.0539	0.2601	0.2240	104.4	104.9	3866×	0	True
5	5	0.0157	0.0329	0.2925	0.2780	104.4	104.6	6662×	0	True
10	0.5	0.0398	0.0793	0.1970	0.0981	52.2	56.1	1313×	0	True
10	1	0.0266	0.0519	0.1596	0.1003	52.2	54.2	1965×	0	True
10	2	0.0194	0.0369	0.1395	0.1057	52.2	53.2	2688×	0	True
10	5	0.0117	0.0240	0.1335	0.1188	52.2	52.6	4466×	0	True
20	0.5	0.0285	0.0522	0.1290	0.0501	26.1	33.2	915×	0	True
20	1	0.0197	0.0389	0.0987	0.0494	26.1	29.9	1328×	0	True
20	2	0.0132	0.0260	0.0799	0.0503	26.1	28.0	1971×	0	True
20	5	0.0087	0.0168	0.0676	0.0534	26.1	26.9	3011×	0	True

Table 11: Monte Carlo over random initial conditions (Van der Pol, $m = 16$, $k_e = 5.0$, 200 trials). Source: `results/monte_carlo_honest/summary.md`.

check	result
all trajectories finite	200/200
hard parameter bounds held at every recorded step	200/200
a-posteriori ISS bound $\|e(t)\| \le \mathrm{e}^{-k_e(t-T_1)}\|e(T_1)\| + \sup\|f - \hat{f}\|/k_e$ held	200/200 (max ratio 0.979)
steady-state max $\|e\|$ (last 25%): median / 95th pct / max	0.0568 / 0.0938 / 0.1075
steady-state mean $\|e\|$: median / 95th pct / max	0.0302 / 0.0484 / 0.0593
$\sup\|f - \hat{f}\|$ in steady state / k_e: median / max	0.0627 / 0.1475
transient max $\|e\|$: max over trials	4.903

- Adaptive geometry is a smaller but consistent gain: 0.0116 for adaptive centres and a further 0.0349 for adaptive widths in mean steady-state $\|e\|$, 5/5 seeds each.
- Against kernel-regression baselines the proposed method is never the most accurate. Its edge is cost against the GP baselines (NTK is cheaper only at $n = 3$), and in the fair re-run at $n = 1$ it has the cheapest step of the five methods.
- In the fair re-run at $n = 1$ the exact GP is 3–4$\times$ more accurate than the proposed method.
- In the oracle study the a-priori bound exceeds the observed RMS error by factors of 915 to 10954, while the RMS bound with the measured disturbance holds in 48/48 runs.

Part II
Transport adaptive control

9 Setting

Plant, reference, control, error.

$$\dot{x} = f(x,t) + u, \qquad e = x - x_m(t), \qquad u = \dot{x}_m - k_e e - \hat{f}(x).$$

Lifted parameter. $\vartheta = (a, c, s) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}_{>0}$, and $p := 2n + 1$. The feature is $\Phi(x;\vartheta) := a\,\varphi(x;c,s) \in \mathbb{R}^n$, where $\varphi(x;c,s) = \exp\bigl(-\|x-c\|^2/(2s^2)\bigr)$.

Network as a measure.

$$\hat{f}(x) = \frac{1}{m}\sum_{i=1}^{m} \Phi(x;\vartheta_i) = \int \Phi(x;\vartheta)\,\mathrm{d}\mu(\vartheta), \qquad \mu := \frac{1}{m}\sum_{i=1}^{m} \delta_{\vartheta_i}.$$

The map $\mu \mapsto \hat{f}$ is linear.

Metric. $G = \mathrm{diag}(g_a I_n, g_c I_n, g_s) \succ 0$ is the gain matrix. Distances are measured in $\|v\|_{G^{-1}}^2 = v^\top G^{-1} v$, and $\lambda_G := \lambda_{\max}(G)$.

Admissible box and projection.

- **The box.** $\Theta_K := \{\|a\| \le a_{\max}\} \times \{\|c\| \le c_{\max}\} \times [s_{\min}, s_{\max}]$.
- **Layer and inner set.** ε_K is the width of the boundary layer. Shrinking each bound of Θ_K by ε_K gives the inner set Θ_K^ε, with interior $\operatorname{int}\Theta_K^\varepsilon$.
- **The operator.** $\mathrm{Proj}_K(\vartheta, y)$ is the per-particle, factor-wise projection (implemented in `src/core/projection.py` in the accompanying repository). It uses Pomet–Praly norm-ball operators [63] for a and c, with a quadratic ramp $h \in [0,1]$ (defined in Section 4), and a box operator for s, with a linear ramp.
- **Regularity.** The operator is locally Lipschitz in (ϑ, y). The norm-ball factors are C^1 at the inner edge of their layers. The box factor, and every factor at the outer edge, is only Lipschitz. The proofs use only Lipschitz continuity and ramp values in $[0,1]$.

Property (P). For $\vartheta \in \Theta_K$, $\vartheta' \in \Theta_K^\varepsilon$ and all y,

$$\langle \vartheta - \vartheta',\, G^{-1}\,\mathrm{Proj}_K(\vartheta, Gy)\rangle \le \langle \vartheta - \vartheta',\, y\rangle. \tag{P}$$

On $\partial\Theta_K$ the projected velocity has no outward component, so Θ_K is forward invariant (Nagumo; Pomet–Praly 1992 [63]).

Lemma P.1 (Proj_K is non-expansive)**.** *For every $\vartheta = (a,c,s) \in \Theta_K$ and $y = (y_a, y_c, y_s)$,*

$$\|\mathrm{Proj}_K(\vartheta, Gy)\|_{G^{-1}}^2 = \|y\|_G^2 - \sum_{b\in\{a,c,s\}} h_b(2-h_b)\,\frac{\bigl[(n_b^\top G_b y_b)_+\bigr]^2}{n_b^\top G_b n_b} \;\le\; \|y\|_G^2.$$

The terms are as follows.

- *For the balls, $n_a = a$ and $n_c = c$, with the quadratic ramps $h_a = h(\|a\|; a_{\max}, \varepsilon_K)$ and $h_c = h(\|c\|; c_{\max}, \varepsilon_K)$.*

- *For $s < s_{\min} + \varepsilon_K$, $n_s = -1$ and $h_s = 1 - \mathrm{clip}\big((s - s_{\min})/\varepsilon_K, 0, 1\big)$.*
- *For $s > s_{\max} - \varepsilon_K$, $n_s = +1$ and $h_s = 1 - \mathrm{clip}\big((s_{\max} - s)/\varepsilon_K, 0, 1\big)$.*
- *Otherwise $h_s = 0$.*

Proof. G is block-diagonal, so both squared norms split over the blocks. On an active ball block, $\mathrm{Proj}_b = G_b q_b$ with $q_b = y_b - h_b n_b\,(n_b^\top G_b y_b)/(n_b^\top G_b n_b)$. Expanding $q_b^\top G_b q_b$ gives the subtracted term. On an inactive block ($h_b = 0$ or $n_b^\top G_b y_b \le 0$), $\mathrm{Proj}_b = G_b y_b$. The s-block is the one-dimensional case. Finally $h_b \in [0,1]$, so $h_b(2 - h_b) \in [0,1]$. □

Checked by: `tests/test_transport.py`: `test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound`.

Potential. $g_t(\vartheta) := e(t)^\top \Phi(x(t);\vartheta)$, and $F_t(\mu) := -\int g_t \,\mathrm{d}\mu = -e(t)^\top \hat{f}(x(t))$.

Adaptive law (the algorithm).

$$\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\, G\nabla_\vartheta g_t(\vartheta_i)\big), \qquad i = 1, \dots, m. \tag{L}$$

Ideal measure. For $j = 1, \dots, m$, $\vartheta_j^*(t) \in \Theta_K^\varepsilon$ is C^1 in t, and $\mu_t^* := \frac{1}{m}\sum_j \delta_{\vartheta_j^*(t)}$. Write $\varepsilon(x,t) := f(x,t) - \int \Phi(x;\cdot)\,\mathrm{d}\mu_t^*$, with $\|\varepsilon\| \le \varepsilon_{\max}$ on the operating set X, and set

$$V_*(t) := \Big(\frac{1}{m}\sum_j \|\dot{\vartheta}_j^*\|_{G^{-1}}^2\Big)^{1/2}, \qquad V_* := \sup_t V_*(t).$$

The proofs never use optimality of μ^*. It is an analytic device, and any admissible C^1 curve may be used.

Optimal-transport distance. For uniform atomic $\mu = \frac{1}{m}\sum \delta_{\vartheta_i}$ and $\nu = \frac{1}{m}\sum \delta_{\vartheta'_j}$,

$$W(\mu,\nu)^2 := \min_{\pi \in S_m} \frac{1}{m}\sum_i \|\vartheta_i - \vartheta'_{\pi(i)}\|_{G^{-1}}^2.$$

This is the Kantorovich 2-Wasserstein distance with ground cost $\|\cdot\|_{G^{-1}}^2$. For equal masses the optimum is attained at a permutation (Birkhoff). It is computed by linear assignment in $O(m^3)$. For general measures on Θ_K, W denotes the same Kantorovich distance.

Label-free Lyapunov function.

$$V_{\mathrm{OT}}(t) := \tfrac{1}{2}\|e(t)\|^2 + \tfrac{1}{2}W(\mu_t, \mu_t^*)^2 = \min_\pi V_\pi(t),$$
$$V_\pi := \tfrac{1}{2}\|e\|^2 + \frac{1}{2m}\sum_i \|\vartheta_i - \vartheta^*_{\pi(i)}\|_{G^{-1}}^2.$$

Standing hypotheses (S).

(S0) $k_e > 0$; $G = \mathrm{diag}(g_a I_n, g_c I_n, g_s) \succ 0$; $s_{\min} > 0$; and $0 < \varepsilon_K < \min\{a_{\max}, c_{\max}, (s_{\max} - s_{\min})/2\}$ (`src/core/projection.py` enforces the last condition).

(S1) f is continuous in (x,t) and locally Lipschitz in x, uniformly in t on bounded time intervals.

(S2) $x_m \in C^1$, and x_m and $\dot{x}_m$ are bounded.

(S3) $\vartheta_i(0) \in \Theta_K$.

(S4) The ϑ_j^* are C^1 with values in Θ_K^ε, and $V_* < \infty$.

(S5) $x(t) \in X$ on the interval considered. For bounded f, take $X := \bigcup_t \{x : \|x - x_m(t)\| \le r(F)\}$ with $F = \sup\|f\|$ (Theorem D(b)). Then $\varepsilon_{\max}$ is the supremum over this X, and it depends on $\|e(0)\|$.

Under (S0)–(S3) the right-hand side of (L) is continuous in t and locally Lipschitz in (x, ϑ). Solutions are classical (C^1) and unique on their maximal interval. Under (S4) as well, each V_π is C^1.

Notation.

- Times are written $t_0 \le t_1$ (or $T_1 < T_2$), and r is the integration variable in time.
- $\mathrm{RMS}_{[T_1,T_2]}\|e\| := \Big((T_2 - T_1)^{-1} \int_{T_1}^{T_2} \|e\|^2 \,\mathrm{d}r\Big)^{1/2}$.

Constants of the Gaussian. Closed forms (`gaussian_constants` in `src/core/transport.py` of the accompanying repository). With $u := \|x - c\|^2/s^2$ and $\varphi = e^{-u/2}$, on $s \ge s_{\min}$:

$$\|\nabla_c \varphi\| \le J_c := e^{-1/2}/s_{\min}, \qquad |\partial_s \varphi| \le J_s := 2e^{-1}/s_{\min}, \qquad J := (J_c^2 + J_s^2)^{1/2},$$
$$\|\nabla^2_{(c,s)} \varphi\| \le H_\varphi := \big(\max(h_{cc}, h_{ss}) + h_{cs}\big)/s_{\min}^2,$$
$$h_{cc} = 1, \qquad h_{ss} = 2e^{-1/2}, \qquad h_{cs} = \sqrt{u}\, e^{-u/2}|u - 2| \ \text{ evaluated at } u = (5 - \sqrt{17})/2.$$

J and H_φ are upper bounds. The true supremum of $\|\nabla_{(c,s)}\varphi\|$ is about $0.92/s_{\min}$, and that of $\|\nabla^2_{(c,s)}\varphi\|$ is about $1.57/s_{\min}^2$. The Hessian of the lifted potential is the block matrix

$$\begin{pmatrix} 0 & u\,\nabla\varphi^\top \\ \nabla\varphi\, u^\top & (u^\top a)\,\nabla^2\varphi \end{pmatrix}.$$

Hence

$$\sup_{x \in \mathbb{R}^n,\ \vartheta \in \Theta_K,\ \|u\|=1} \|\nabla^2_\vartheta \big(u^\top \Phi(x;\vartheta)\big)\| \le H_K := J + a_{\max} H_\varphi, \qquad K := \tfrac{1}{2} H_K. \tag{1}$$

A brute-force numerical check over the full lifted Hessian gives sup $\approx 0.76\, H_K$.

10 Theorem A: identification

Theorem A. *Assume* (S0).

(i) Assume scalar classical gains $\Gamma_W = \gamma_W I$, $\Gamma_c = \gamma_c I$ and $\Gamma_\sigma = \gamma_\sigma I$, and use the dictionary $W_i = a_i/m$, $\gamma_W = g_a/m$, $\gamma_c = m g_c$, $\gamma_\sigma = m g_s$.

- *The* raw (*unprojected*) *rates of* (L) *and of the classical law of Section 4 coincide.*
- *The* projected *rates coincide when the classical joint projections are replaced by per-kernel ones. The classical law projects W onto one Frobenius ball and* $\mathrm{vec}(c)$ *onto one Euclidean ball. The per-kernel version projects:*
  - *W_i onto $\|W_i\| \le a_{\max}/m$, with layer ε_K/m;*
  - *c_i onto $\|c_i\| \le c_{\max}$, with layer ε_K;*
  - *σ_i onto $[s_{\min}, s_{\max}]$, with layer ε_K.*
- *For $m \ge 2$, no joint ball, of any radius $W_{\max}$ and layer ε_W, reproduces the per-kernel field. For $m = 1$ the two coincide, with $W_{\max} = a_{\max}$ and layer ε_K.*

(ii) Fix a closed-loop trajectory $(x(\cdot), e(\cdot))$. Then:

- *The field* $v_t(\vartheta) := \mathrm{Proj}_K(\vartheta, G\nabla g_t(\vartheta))$ *is Lipschitz in* ϑ *on* Θ_K*, and* Θ_K *is forward invariant.*
- *For every* $\mu_0 \in \mathcal{P}(\Theta_K)$*,* $\mu_t := (X_t)_{\#}\mu_0$ *is a distributional solution of* $\partial_t\mu + \nabla\cdot(\mu v_t) = 0$*.*
- *On* $\operatorname{int}\Theta_K^\varepsilon$*,* $v_t = -G\nabla(\delta F_t/\delta\mu)$*. So at frozen time,* μ_t *is the Wasserstein gradient flow (metric* G^{-1}*) of the linear functional* F_t*, and* (L) *is its particle form.*
- F_t *is generated by the closed loop, so the coupled system is the one studied in Theorem D.*

(iii) ***Dictionary****, all in the* G^{-1} *metric:*

classical device	measure-level object
projection onto Θ_K	A velocity-level (Pomet–Praly) regularisation of the support constraint $\mu_t \in \mathcal{P}(\Theta_K)$. For $\varepsilon_K > 0$ it is not the gradient flow of any functional of μ. Formally, as $\varepsilon_K \to 0$, the field tends to the G^{-1}-projection of $G\nabla g_t$ onto the tangent cone of Θ_K, which is the velocity of the $W_{G^{-1}}$-gradient flow of $F_t + \iota_{\mathcal{P}(\Theta_K)}$ ([42, Prop. 3.1] for the exact W_2 projection).
σ-modification in the G^{-1} metric (leak outside the gain, $\dot\vartheta = G\nabla g - \sigma(\vartheta - \vartheta^0)$, as in ($\mathrm{L}_\sigma$) and Theorem H)	the potential energy $(\sigma/2)\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$, which is geodesically σ-convex
classical σ-modification $\Gamma(y - \sigma\theta)$ (leak inside the gain; `src/baselines/sigma_mod.py` implements its W-block, with c and σ frozen)	For the full-vector leak, under (i), the object is $(\sigma/2)\int(\vartheta - \vartheta^0)^\top D(\vartheta - \vartheta^0)\,\mathrm{d}\mu$ with $D = \mathrm{diag}(I_n/m, mI_n, m)$. In the G^{-1} metric it is λ-convex with $\lambda = \sigma\cdot\min(g_a/m, mg_c, mg_s)$, which depends on m. For the W-block alone it is $(\sigma/2)\int(a - a^0)^\top(I_n/m)(a - a^0)\,\mathrm{d}\mu$.
σ-modification toward a labelled (ϑ_i^0)	not a functional of μ; it is a functional on the labelled lift $\mathbb{R}^{mp}$
OT σ-modification (Theorem C)	$(\sigma/2)W(\mu, \mu^0)^2$. For atomic μ with a unique optimal assignment, its W-gradient at atom i is $\sigma(\vartheta_i - \vartheta^0_{\tau(i)})$. At ties, ($\mathrm{L}_\sigma$) uses a measurable selection from the superdifferential $\{\sigma(\vartheta_i - \vartheta^0_{\tau(i)}) : \tau \text{ optimal}\}$.
e-modification	the same functionals with coefficient $\sigma\|e\|$
parameter dither (Theorem H)	the entropy $\tau\int\mu\log\mu$

Status. Proved and checked. Novelty (literature audit): interpretive; ingredients known [10], [16], [42], [57].

Checked by: `tests/test_transport.py`: `test_identification_with_classical_law` (raw rates in (i)); `test_identification_projected_rates_per_kernel` (projected rates and the only-if construction in (i)).

Proof. **(i) Gain map.** We have $\nabla_a g = \varphi e$, $\nabla_c g = (a^\top e)\nabla_c\varphi$ and $\partial_s g = (a^\top e)\partial_s\varphi$. Matching these with the classical rates gives the gain map.

Scaling identity. $(1/m)\,\mathrm{Proj}_{r,\varepsilon}(a, y) = \mathrm{Proj}_{r/m,\varepsilon/m}(a/m, y/m)$. Three facts give it:

- $h_{r/m,\varepsilon/m}(\|a\|/m) = h_{r,\varepsilon}(\|a\|)$;
- the direction $a(a^\top y)/\|a\|^2$ is homogeneous of degree 0 in a and of degree 1 in y;
- the activation $a^\top y > 0$ is invariant under scaling.

If the layer is not rescaled, the ramps differ.

Only if $(m \geq 2)$. Let $r_{\text{in}} := (a_{\max} - \varepsilon_K)/m$, and let the joint ball have radius $W_{\max}$, layer ε_W and $\rho := W_{\max} - \varepsilon_W$. In both cases below the driving term points outward.

- If $\rho < \sqrt{2}\, r_{\text{in}}$, take two rows of equal norm r with $\sqrt{2}\, r \in \big(\rho, \min(\sqrt{2}\, r_{\text{in}}, W_{\max})\big]$, parallel to e. The per-kernel field is inactive and the joint field is active.
- If $\rho \geq \sqrt{2}\, r_{\text{in}}$, take one row with norm in $\big(r_{\text{in}}, \min(a_{\max}/m, \rho)\big)$ and the others zero. The per-kernel field is active and the joint field is inactive.

The same argument applies to $\mathrm{vec}(c)$. For $m = 1$ the two operators coincide.

The test `test_identification_projected_rates_per_kernel` checks three things: the identity with active projections; that the layer must be rescaled; and the only-if construction against `ProjectionOperator.project_matrix` on five joint balls. Four are active against an inactive per-kernel field, and one is inactive against an active per-kernel field.

(ii) Proj_K is Lipschitz in ϑ. The ramp vanishes for $\|\vartheta\| \leq \theta_{\max} - \varepsilon_K$, so the direction $\vartheta/\|\vartheta\|$ is evaluated only away from 0. Also, ∇g_t is C^1 on $s \geq s_{\min}$.

- **Invariance.** Forward invariance follows from Nagumo's theorem, since $n_b^\top \mathrm{Proj}_b \leq 0$ on the boundary, with the outward normals n_b of Lemma P.1.
- **Continuity equation.** The push-forward of μ_0 by a Lipschitz flow solves the continuity equation (Golse [30]; Ambrosio–Gigli–Savaré [2, §8.1]).
- **Gradient form.** On $\operatorname{int} \Theta_K^\varepsilon$ the projection is inactive, so $v_t = G\nabla g_t = -G\nabla(\delta F_t/\delta\mu)$.

(iii) Potential-energy rows. These follow from the computation in the metric G^{-1}: for any V, $\nabla_{G^{-1}} V = G\nabla V$. For example, $V = (\sigma/2)\|\vartheta - \vartheta^0\|^2_{G^{-1}}$ gives $G\nabla V = \sigma(\vartheta - \vartheta^0)$.

Classical-leak row. Applying the gain map of (i) to the classical leak $\Gamma(y - \sigma\theta)$ gives the velocity $-\sigma GD(\vartheta - \vartheta^0)$.

Projection row. The G^{-1}-rescaled correction field has an asymmetric Jacobian. Take the example $m = 1$, $n = 1$, $G = I$, $x = 0.7$, $e = 1$, $\vartheta = (0.93, 0.2, 1.5)$, $a_{\max} = 1$, $\varepsilon_K = 0.15$ (so $h(0.93) = 0.284$) and $c_{\max} = 2$, with a in the layer. A numerical evaluation gives:

- the c-derivative of the a-component is -0.060;
- the a-derivative of the c-component is 0.

So the field is not a gradient. □

Remark A.1 (normalisation)**.** The algorithm's raw rates are unchanged. Two things do change: the admissible set becomes a product of per-kernel balls, and $a_{\max}$, not $\|W\|_F$, is held fixed in m.

- ***m*-free constants.** In this normalisation the structural constants k_e, K, L_0, L_1, λ_G and Δ are free of m for fixed G and $a_{\max}$. So are the constants of Theorems B, C(i)–(iii), D and F.
- ***m*-dependent data.** $\varepsilon_{\max}$ and V_* are data of the chosen m-atom ideal. The local constants of C(iv) and of Theorem E depend on m.
- **Classical convention.** Fixed classical γ's correspond to $g_a = m\gamma_W \propto m$ and $g_c, g_s \propto 1/m$. Weights speed up, centres and widths freeze, $\lambda_G \propto m$ and $\Delta \propto \sqrt{m}$.

11 Theorem B: the optimal-assignment Lyapunov function

Lemma B.1 (Dini derivative of a minimum)**.** *Suppose each V_π is C^1 and $V_{\mathrm{OT}} = \min_\pi V_\pi$. Then V_{OT} is locally Lipschitz, and $D^+ V_{\mathrm{OT}}(t) \leq \dot{V}_\pi(t)$ for every π active at t.*

Proof. For $h > 0$, $V_{\mathrm{OT}}(t + h) \leq V_\pi(t + h)$, and $V_{\mathrm{OT}}(t) = V_\pi(t)$. □

Lemma B.2 (matched remainder in two regimes; gradient bound)**.** *Let* $L_0 := 2a_{\max}$ *and* $L_1 := (1 + a_{\max}^2 J^2)^{1/2}$*. For* $\vartheta, \vartheta' \in \Theta_K$*:*

(a) the remainder $R := g_t(\vartheta') - g_t(\vartheta) - \langle \nabla g_t(\vartheta),\, \vartheta' - \vartheta \rangle$ *satisfies*

$$|R| \le \|e(t)\| \cdot \min\bigl\{ K\|\vartheta - \vartheta'\|^2,\ L_0 + L_1\|\vartheta - \vartheta'\| \bigr\};$$

(b) $\|\nabla_\vartheta g_t(\vartheta)\| \le L_1 \|e(t)\|$.

Proof. (b) We have $\nabla_a g = \varphi e$ and $\nabla_{(c,s)} g = (a^\top e)\nabla_{(c,s)}\varphi$, with $\varphi \le 1$ and $\|\nabla_{(c,s)}\varphi\| \le J$ on $s \ge s_{\min}$. So $\|\nabla g\|^2 \le \|e\|^2(1 + a_{\max}^2 J^2)$.

(a) If $e(t) = 0$, both sides vanish. Otherwise set $h := \vartheta' - \vartheta$.

- *Quadratic branch.* Θ_K is convex, so the Lagrange remainder of $\tau \mapsto g_t(\vartheta + \tau h)$ gives $R = \frac{1}{2} h^\top \nabla^2 g_t(\vartheta + \tau_0 h) h$ for some $\tau_0 \in (0,1)$. Since $g_t = \|e\| \cdot (u^\top \Phi)$ with $u = e/\|e\|$, the Hessian bound (1) of Section 9 gives this branch.
- *Linear branch.* It follows from $|g_t| \le \|e\| a_{\max}$ and (b). □

Theorem B. *Under (S), along every solution of (L), for every* t*, with* $W := W(\mu_t, \mu_t^*)$*:*

$$D^+ V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\Bigl(\varepsilon_{\max} + \min\bigl\{K\lambda_G W^2,\ L_0 + L_1 \lambda_G^{1/2} W\bigr\}\Bigr) + V_*(t)\, W. \tag{B}$$

Status. Proved and checked, including a moving ideal, $\varepsilon \neq 0$, switches, an active projection, and the failure of the plain bound along (L_σ). Novelty (literature audit): packaging new; ingredients known [70], [1], [56], [12].

Checked by: `tests/test_transport.py`: `test_theorem_B_inequality_along_trajectories`, `test_theorem_B_moving_ideal_nonrepresentable_switches_and_active_projection`, `test_dini_derivative_of_V_OT_bounded_by_active_branch`, `test_permuted_ideal_labelled_vs_OT`; `experiments/exp_transport_permutation.py`.

Proof. Fix t and an optimal π, and write $\tilde\vartheta_i := \vartheta_i - \vartheta^*_{\pi(i)}$. Then

$$\dot V_\pi = e^\top \dot e + \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta_i\rangle - \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta^*_{\pi(i)}\rangle.$$

(a) $e^\top \dot e = -k_e\|e\|^2 + e^\top \varepsilon + \frac{1}{m}\sum_i \bigl[g_t(\vartheta^*_{\pi(i)}) - g_t(\vartheta_i)\bigr]$.

(b) By (P) with $\vartheta' = \vartheta^*_{\pi(i)}$, $\frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta_i\rangle \le \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, \nabla g_t(\vartheta_i)\rangle$.

(c) Since π is a bijection and is optimal,

$$-\frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta^*_{\pi(i)}\rangle \le \Bigl(\frac{1}{m}\sum_i \|\tilde\vartheta_i\|^2_{G^{-1}}\Bigr)^{1/2}\Bigl(\frac{1}{m}\sum_i \|\dot\vartheta^*_{\pi(i)}\|^2_{G^{-1}}\Bigr)^{1/2} = W \cdot V_*(t).$$

Summing gives

$$\dot V_\pi \le -k_e\|e\|^2 + \|e\|\varepsilon_{\max} + \frac{1}{m}\sum_i R_i + W\, V_*(t),$$

where each R_i is a matched remainder. Average Lemma B.2(a), using three facts:

- $\frac{1}{m}\sum_i \min\{A_i, B_i\} \le \min\{\mathrm{avg}\, A, \mathrm{avg}\, B\}$;
- $\frac{1}{m}\sum_i \|\tilde\vartheta_i\|^2 \le \lambda_G W^2$;
- $\frac{1}{m}\sum_i \|\tilde\vartheta_i\| \le \lambda_G^{1/2} W$.

Lemma B.1 then completes the proof. □

Corollary B.3 (ultimate bounds)**.** *Let* $\Delta := \operatorname{diam}_{G^{-1}}(\Theta_K)$*, so* $W \le \Delta$*. Define*

$$D^{\mathrm{OT}} := \varepsilon_{\max} + \min\{K\lambda_G\Delta^2,\ L_0 + L_1\lambda_G^{1/2}\Delta\}, \qquad C^{\mathrm{OT}} := V_*\Delta,$$
$$R^{\mathrm{OT}} := \frac{D^{\mathrm{OT}}}{2k_e} + \Big(\frac{(D^{\mathrm{OT}})^2}{4k_e^2} + \frac{C^{\mathrm{OT}}}{k_e}\Big)^{1/2}.$$

Then:

- $D^+V_{\mathrm{OT}} < 0$ *whenever* $\|e\| > R^{\mathrm{OT}}$*.*
- *Under (S5) on* $[0,\infty)$*,* $\limsup\|e\| \le \big((R^{\mathrm{OT}})^2 + \Delta^2\big)^{1/2}$*. This uses* $\sup \frac{1}{2}W^2 \le \Delta^2/2$*.*
- $\mathrm{RMS}_{[T_1,T_2]}\|e\| \le \dfrac{D^{\mathrm{OT}}}{2k_e} + \Big(\dfrac{(D^{\mathrm{OT}})^2}{4k_e^2} + \dfrac{C^{\mathrm{OT}}}{k_e} + \dfrac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}$.

Proof. As in Theorem 1 (Section 5). The Dini comparison lemma, applied with the continuous, V-independent right-hand side, gives the integrated form. □

Remark B.4 (what OT buys and what it does not)**.**

- **Globally, the OT constant is not smaller than the classical one.** Take the test box: $a_{\max} = 8$, $c_{\max} = 4$, $s_{\min} = 0.3$, $s_{\max} = 3$, $G = \mathrm{diag}(2, 1, 0.5)$ and $F_{\max} := a_{\max}$. Compare constants on the same per-kernel admissible set.

constant	value
classical Lemma 2 (Section 5), $F_{\max} + 3a_{\max} + 2a_{\max}c_{\max}J_c + a_{\max}(s_{\max} - s_{\min})J_s$	214, for every m
OT linear branch, crude form $L_0 + L_1\lambda_G^{1/2}\Delta$	533
OT linear branch, block split through W	382
OT linear branch, bounded per particle by $\|e\|\big(2a_{\max} + \|\tilde{a}_i\| + a_{\max}(J_c\|\tilde{c}_i\| + J_s\|\tilde{s}_i\|)\big)$	214

  Bounded per particle, the OT constant equals the classical one. An earlier comparison figure of "101–138" used a joint centre ball of radius $c_{\max}$. That radius is $\sqrt{m}$ times smaller than the radius $\sqrt{m}\,c_{\max}$ of the ball that circumscribes the per-kernel product of Theorem A.

- **Locally, the excess disturbance is quadratic in the matched error** (Corollary B.5).
- **Labels.** A labelled theory can make the same local statement only for a labelling fixed in advance, and only while the optimal assignment does not switch. (B) holds uniformly, with the minimum over π taken at each t.
- m**-uniformity** is a property of the normalisation (Remark A.1), not of OT.

Corollary B.5 (conditional bound)**.** *Along a solution of (L), suppose* $\sup_{[T_1,T_2]} W(\mu_t, \mu_t^*) \le \bar{\eta}$*. This quantity is available in oracle simulations, since it requires* μ_t^**, or it can be controlled a priori by Theorem E. With* $\bar{D} := \varepsilon_{\max} + \min\{K\lambda_G\bar{\eta}^2,\ L_0 + L_1\lambda_G^{1/2}\bar{\eta}\}$*,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{\bar{D}}{2k_e} + \Big(\frac{\bar{D}^2}{4k_e^2} + \frac{V_*\bar{\eta}}{k_e} + \frac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}. \tag{11.1}$$

As $\bar{\eta} \to 0$ *and* $T_2 \to \infty$*, this tends to* $\varepsilon_{\max}/k_e$*, the value of the frozen-parameter error channel.*

Variant for (L_σ). *Take a solution of* (L_σ) *in the sense of Section 12 with* $\sup_{[T_1,T_2]} W(\mu_t, \mu_t^*) \le \bar{\eta}$ *and* $4\bar{\eta}(b + \bar{\eta}) < \Delta_{\mathrm{gap}}$*. For example,* $\bar{\eta}$ *can come from Theorem C(iv). Then, with* $\bar{D} := \varepsilon_{\max} + K\lambda_G\bar{\eta}^2$*,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{\bar{D}}{2k_e} + \Big(\frac{\bar{D}^2}{4k_e^2} + \frac{V_*\bar{\eta} + \sigma_m b^2/4}{k_e} + \frac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}. \tag{11.2}$$

Proof of the variant. Apply Theorem C(iii) with $C = b(t)^2 \le b^2$, and use $-\sigma_m\eta^2 + \sigma_m\eta b \le \sigma_m b^2/4$. Without the gap condition, replace b^2 by $(b + 2\bar\eta)^2$ (from $\sqrt{C} \le b + 2\eta$ in C(ii)), or by $b^2 + 4\bar\eta R$, with $R := \Delta$ as in Theorem C. □

As $\bar\eta \to 0$ and $T_2 \to \infty$ the variant tends formally to $\varepsilon_{\max}/(2k_e) + \big(\varepsilon_{\max}^2/(4k_e^2) + \sigma_m b^2/(4k_e)\big)^{1/2}$. Along ($\mathrm{L}_\sigma$) with $b > 0$, the measured $\bar\eta$ stays of order b (about $0.9b$ in the test instance), so the bound is first order in b. The plain bound of Corollary B.5 is false along (L_σ). In the test instance, every hypothesis of C(iv) is asserted and $\sup W \le \bar\eta$ is measured. There, the bound of Corollary B.5 is exceeded by more than 3×, while the variant holds (`tests/test_transport.py::test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`). Independent numerical checks found violations of 3.8–8.3×.

Status. The bound along (L) is not checked in the repository; its failure along (L_σ) and the (L_σ) variant are checked (Appendix A).

Corollary B.6 (approximation error as quantisation error)**.** *Suppose $f = \int \Phi \,\mathrm{d}\mu_\infty^* + \varepsilon_\infty$ for a probability measure μ_∞^* on Θ_K^ε, with $\|\varepsilon_\infty\| \le \bar\varepsilon_\infty$. Then for every m-atom μ_m^* supported in Θ_K^ε,*

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + L_1 \lambda_G^{1/2}\, W(\mu_m^*, \mu_\infty^*).$$

Status. Not checked in the repository (Appendix A).

Proof. For any coupling γ,

$$\begin{aligned}\Big\| \int \Phi \,\mathrm{d}(\mu_m^* - \mu_\infty^*) \Big\| &\le \int \|\Phi(\vartheta) - \Phi(\vartheta')\| \,\mathrm{d}\gamma \le L_1 \lambda_G^{1/2} \int \|\vartheta - \vartheta'\|_{G^{-1}} \,\mathrm{d}\gamma \\ &\le L_1 \lambda_G^{1/2} \Big(\int \|\vartheta - \vartheta'\|_{G^{-1}}^2 \,\mathrm{d}\gamma \Big)^{1/2}.\end{aligned}$$

Minimise over γ. The block constant $\big(g_a + a_{\max}^2 (J_c^2 g_c + J_s^2 g_s)\big)^{1/2}$ may replace $L_1 \lambda_G^{1/2}$. □

Rates.

(a) **First order.** Let μ_∞^* be compactly supported and $p \ge 3$. Then there are equal-weight μ_m^*, with atoms (cell barycentres) in Θ_K^ε, such that $W(\mu_m^*, \mu_\infty^*) = O(m^{-1/p})$. This follows from an equal-mass dyadic decomposition (Chevallier 2018 [9]). For $p > 4$, i.i.d. atoms also achieve the rate in expectation (Fournier–Guillin 2015 [23]). If μ_∞^* has a non-trivial absolutely continuous part, the order cannot be improved, even with free weights (Zador; Graf–Luschgy [31, Thm. 6.2 with Lemma 3.4]).

(b) **Second order.** Suppose μ_m^* is a centroid quantiser for an optimal plan; a best equal-weight approximation is one. Then the first-order term cancels, and by the Hessian bound (1) of Section 9 and convexity of Θ_K^ε,

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + \min\big\{ K \lambda_G W^2,\ L_1 \lambda_G^{1/2} W \big\} = \bar\varepsilon_\infty + O(m^{-2/p}).$$

(c) **Free of p.** I.i.d. atoms drawn from μ_∞^* give, with positive probability, an m-atom ideal with

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + O\Big(a_{\max} \big(n \log(m \cdot \operatorname{diam} X)/m \big)^{1/2} + a_{\max}/(s_{\min}\sqrt{m}) \Big).$$

This is Maurey's empirical method; the constants are not optimised.

The exponent $1/p$ is therefore a property of controlling $\varepsilon_{\max}$ through W, as Theorem D(e) requires. It is not a property of the approximation error itself.

Remark B.7 (continuum form, via couplings)**.** Let $\mu_0 \in \mathcal{P}(\Theta_K)$ and $\mu_t = (X_t)_{\#}\mu_0$, and let μ_t^* be an absolutely continuous ideal curve in $(\mathcal{P}(\Theta_K^\varepsilon), W)$ with velocity field w_t. Set $V_* := \operatorname{ess\,sup}_t \|w_t\|_{L^2(\mu_t^*;G^{-1})}$. Its smallest admissible value is the metric derivative $|\dot\mu_t^*|_W$ (AGS [2, Thm. 8.3.1], after the isometry $\vartheta \mapsto G^{-1/2}\vartheta$). Then (B) holds for a.e. t.

Status. Not checked in the repository (Appendix A).

Sketch.

1. Push an optimal plan γ_t forward by $(X_{t,t+h}, Y_{t,t+h})$. Here Y is the flow of w, or the superposition measure of AGS [2, Thm. 8.2.1] when w is not Lipschitz.
2. This couples (μ_{t+h}, μ_{t+h}^*), so $D^+\big(\frac12 W^2\big) \le \int \langle \vartheta - \vartheta',\, v_t(\vartheta) - w_t(\vartheta')\rangle_{G^{-1}}\, \mathrm{d}\gamma_t$.
3. Integrate (P) and Lemma B.2 against γ_t. □

12 Theorem C: optimal-transport σ-modification

Let $\mu^0 = \frac1m \sum_k \delta_{\vartheta_k^0}$ be a prior configuration with $\vartheta_k^0 \in \Theta_K^\varepsilon$. The *OT σ-modified law* is

$$\begin{aligned} \dot\vartheta_i &= \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \vartheta^0_{\tau(i)})\big), \\ \tau(t) &\in \operatorname*{argmin}_{\rho}\ \frac1m \sum_i \|\vartheta_i - \vartheta^0_{\rho(i)}\|_{G^{-1}}^2. \end{aligned} \tag{L_σ}$$

Solution concept. The right-hand side of (L_σ) is discontinuous at assignment ties.

- **Selection solutions.** A solution is an absolutely continuous (x, ϑ) such that, for a.e. t, $\dot\vartheta_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \vartheta^0_{\tau(t)(i)})\big)$ for some $\tau(t)$ optimal at $\vartheta(t)$. This is what `src/core/transport.py` in the accompanying repository integrates.
- **Filippov solutions.** These exist by Filippov's theorem, since the right-hand side is measurable and locally bounded. But they include sliding motions on repelling ties, along which (i) below fails. For example, take ideal = prior = $\{a = 0,\ a = 1\}$, both particles at $a = \frac12$ and $e \equiv 0$: this is a stationary Filippov solution.
- **What holds for which class.**
  - Parts (ii)–(iv) hold for every Filippov solution, with C taken as the maximum over optimal τ.
  - Part (i) and the exact decay hold for selection solutions.
  - For Filippov solutions, (i) weakens as stated below.
- **Regularity and existence.** V_{OT} and Z are absolutely continuous along solutions, and every inequality holds for a.e. t. Existence of selection solutions is proved for the pure leak (Lemma C.3). In general it is assumed: two-way ties are never attracting, by monotonicity of Proj_K, but multi-way ties, the degenerate ramp $h = 1$ and accumulating switch times are not handled.

Notation.

- $c_\rho(\vartheta) := \frac1m \sum_i \|\vartheta_i - \vartheta^0_{\rho(i)}\|_{G^{-1}}^2$.
- $\eta := W(\mu_t, \mu_t^*)$, and $R := \Delta = \mathrm{diam}_{G^{-1}}(\Theta_K)$.
- $b(t) := W(\mu_t^*, \mu^0)$, and $b := \sup_t b(t)$.

- $\Delta_{\mathrm{gap}}(t) := \min\{c_\rho(\vartheta^*(t)) - b(t)^2 : c_\rho(\vartheta^*(t)) > b(t)^2\}$, with $\min\emptyset := +\infty$; and $\Delta_{\mathrm{gap}} := \inf_t \Delta_{\mathrm{gap}}(t)$.
- $A_{\tau\tau'} := \frac{1}{m}\sum_i \|\vartheta^0_{\tau(i)} - \vartheta^0_{\tau'(i)}\|^2_{G^{-1}}$.

Theorem C. *Under* (S)*, along every solution of* (L_σ) *in the sense above:*

(i) Dissipation toward the prior (selection solutions). *Let* $Z := \frac{1}{2}W(\mu_t,\mu^0)^2$ *and* $\Gamma := \big(\frac{1}{m}\sum_i \|G\nabla g_t(\vartheta_i)\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e\|$*. Then:*

- $D^+Z \le -2\sigma_m Z + (2Z)^{1/2}\Gamma(t)$*;*
- *hence* $W(\mu_t,\mu^0) \le \max\{W(\mu_0,\mu^0),\ \sup\Gamma/\sigma_m\}$*;*
- *for the pure-leak flow* ($\Gamma \equiv 0$)*, the decay* $W(\mu_t,\mu^0) = e^{-\sigma_m t}W(\mu_0,\mu^0)$ *is exact for every* $G \succ 0$ *(Lemma C.3).*

For a Filippov solution with weights θ_k *on tied optimal assignments* τ_k*, the bound weakens to*

$$D^+Z \le -2\sigma_m Z + \frac{\sigma_m}{2}\min_\tau \sum_k \theta_k A_{\tau\tau_k} + (2Z)^{1/2}\Gamma,$$

with the minimum over assignments τ *optimal at* t*.*

(ii) Induced coupling. *Let* π^* *be optimal from* μ *to* μ^**, and let* τ *be the optimal assignment from* μ *to* μ^0 *that the law uses at* t *(for Filippov solutions, any optimal* τ*). Set* $C := \frac{1}{m}\sum_i \|\vartheta^*_{\pi^*(i)} - \vartheta^0_{\tau(i)}\|^2_{G^{-1}}$*. Then*

$$C \le b(t)^2 + 4\eta R \qquad \text{and} \qquad \sqrt{C} \le b(t) + 2\eta,$$

and $C = b(t)^2$ *whenever* $4\eta(b(t)+\eta) < \Delta_{\mathrm{gap}}$*.*

(iii) Lyapunov inequality. *With* C *as in* (ii) *(for Filippov solutions, its maximum over optimal* τ*),*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \sigma_m\eta^2 + \sigma_m\eta C^{1/2}.$$

Under $4\eta(b+\eta) < \Delta_{\mathrm{gap}}$ *this becomes*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \frac{\sigma_m}{2}\eta^2 + \frac{\sigma_m}{2}b^2.$$

(iv) Local exponential estimate. *Suppose* $4\bar\eta(b+\bar\eta) < \Delta_{\mathrm{gap}}$ *and* $8K^2\lambda_G^2\bar\eta^2 \le k_e\sigma_m$*. Set* $\alpha := \min(k_e, \sigma_m/2)$ *and* $\beta := \varepsilon^2_{\max}/k_e + (\sigma_m/2)b^2 + 2V_*^2/\sigma_m$*, and suppose* $\max\{V_{\mathrm{OT}}(0), \beta/\alpha\} < \bar\eta^2/2$*. Then:*

- $\{V_{\mathrm{OT}} \le c\}$ *is forward invariant for every* $c \in \big[\max\{V_{\mathrm{OT}}(0), \beta/\alpha\},\ \bar\eta^2/2\big)$*;*
- $V_{\mathrm{OT}}(t) \le e^{-\alpha t}V_{\mathrm{OT}}(0) + \beta/\alpha$*;*
- *on this set* $\|e\| < \bar\eta$*, so* (S5) *holds with* $X := \bigcup_t\{x : \|x - x_m(t)\| \le \bar\eta\}$*.*

(v) The labelled leak. *Define* $b_{\mathrm{lab}}(\pi)(t) := \big(\frac{1}{m}\sum_i \|\vartheta^*_{\pi(i)}(t) - \vartheta^0_i\|^2_{G^{-1}}\big)^{1/2}$ *and* $b_{\mathrm{lab}}(\pi) := \sup_t b_{\mathrm{lab}}(\pi)(t)$*.*

- *For any fixed* π*, the labelled* σ*-modification (leak toward* ϑ^0_i*) satisfies the second display of* (iii) *at every state, with no gap condition. Here* b *is replaced by* $b_{\mathrm{lab}}(\pi)$*,* V_{OT} *by* V_π*, and* η *by the* π*-labelled error.*
- *Choosing* π *to minimise* $b_{\mathrm{lab}}(\pi)$ *gives the bias* $\min_\pi \sup_t b_{\mathrm{lab}}(\pi)(t) \ge b$*. Equality holds when one assignment is optimal for* (μ^*_t, μ^0) *at every* t*, for example a static ideal, or whenever* $\Delta_{\mathrm{gap}} > 0$*.*

- *The labelled leak does* ***not*** *satisfy* (iii) *for* V_{OT} (*Proposition C.5*).

Status. Proved and checked; the instance used to check (iv) satisfies every hypothesis of (iv). Not checked in the repository: the Filippov form of (i) and the inequality of (v). Novelty (literature audit): novel as a control construction; C(i) and the obstruction are known [66], [6].

Checked by: `tests/test_transport.py`:
(i) and Lemma C.3 (unprojected pure leak): `test_theorem_C_i_exact_dissipation_pure_leak`;
(ii): `test_lemma_C_ii_induced_coupling_bound`;
(iv): `test_theorem_C_iv_exponential_estimate`.
(v) and Proposition C.5, in `experiments/`: `exp_ot_sigma_mod.py` and `exp_sigma_mod_transient.py`.

Scope.

- **Finite gap.** If $\Delta_{\mathrm{gap}} < \infty$, that is, not all ideal–prior assignments tie, then $\Delta_{\mathrm{gap}} \le 2R^2/m$ always. Indeed, if every transposition of an optimal ρ_0 were cost-neutral, the matrix $S(i,j) := B(i,j) - (B(i,i) + B(j,j))/2$ built from $B(i,j) := \langle \vartheta_i^*, \vartheta^0_{\rho_0(j)} \rangle_{G^{-1}}$ would be antisymmetric, and every assignment would tie.
  - Hence (iv) needs $\bar\eta < \sqrt{\Delta_{\mathrm{gap}}}/2 \le R/\sqrt{2m}$.
  - $\beta/\alpha < \bar\eta^2/2$ forces $b < \bar\eta/\sqrt{2}$.
  - So (iv) is a near-prior local result of size $O(m^{-1/2})$ in W, and $O(1/m)$ in V_{OT}.
- **Infinite gap.** If $\Delta_{\mathrm{gap}} = +\infty$, for example for a single-atom prior (the classical σ-modification), the gap condition is void. Then (iv) is m-uniform; $\bar\eta$ is then fixed by $8K^2\lambda_G^2\bar\eta^2 \le k_e\sigma_m$ alone.
- **Switching.** $\Delta_{\mathrm{gap}} > 0$ forces the set of optimal ideal–prior assignments to be constant in t. If that set switches along a moving ideal, then $\Delta_{\mathrm{gap}} = 0$, and the second clause of (ii), the second display of (iii) and all of (iv) are vacuous.

Proof. (i) For a selection solution, at a.e. t the law uses an optimal $\tau(t)$, and $Z \le Z_\tau := \frac{1}{2}c_\tau$ with equality at t. By Lemma B.1 it suffices to bound $\dot Z_\tau$. Apply (P) with $\vartheta' = \vartheta^0_{\tau(i)}$ and Cauchy–Schwarz in the G^{-1} inner product: $\dot Z_\tau \le (2Z)^{1/2}\Gamma - 2\sigma_m Z$. Write $y := (2Z)^{1/2}$.

- Where $Z > 0$, $D^+ y \le -\sigma_m y + \Gamma$.
- At $Z = 0$, $D^+ y \le \big(\frac{1}{m}\sum \|\dot\vartheta_i\|^2_{G^{-1}}\big)^{1/2} \le \Gamma$, by Lemma P.1 (the leak term vanishes there).
- $\Gamma \le \lambda_G^{1/2} L_1 \|e\|$ by Lemma B.2(b).

The pure-leak case is Lemma C.3, and the projection is inactive there. In the layer, $a^\top(a^0 - a) \le \|a\|(\|a^0\| - \|a\|) \le 0$, and similarly for c. For s, the leak points away from the nearby face.

For Filippov velocities, the leak part averaged over the tie is

$$-\sigma_m \frac{1}{m} \sum_i \sum_k \theta_k \langle \vartheta_i - \vartheta^0_{\tau(i)},\, \vartheta_i - \vartheta^0_{\tau_k(i)} \rangle_{G^{-1}}.$$

By polarisation, and since $c_\tau = c_{\tau_k} = 2Z$ for tied optimal assignments, this equals $-2\sigma_m Z + (\sigma_m/2)\sum_k \theta_k A_{\tau\tau_k}$. Property (P) holds termwise for the convex combination. The bound is tight at the tie equilibrium of the solution-concept example ($0 = -\sigma_m/4 + \sigma_m/4$).

(ii) Relabel so that $\frac{1}{m}\sum \|\vartheta_i - \vartheta_i^*\|^2_{G^{-1}} = \eta^2$. For any ρ:

- $|c_\rho(\vartheta) - c_\rho(\vartheta^*)| \le 2\eta R$;
- $|\sqrt{c_\rho(\vartheta)} - \sqrt{c_\rho(\vartheta^*)}| \le \eta$, by the reverse triangle inequality for the $(1/m)$-weighted G^{-1} norm.

Let ρ_0 be optimal for $(\vartheta^*(t), \vartheta^0)$. Then

$$C = c_\tau(\vartheta^*) \le c_\tau(\vartheta) + 2\eta R \le c_{\rho_0}(\vartheta) + 2\eta R \le b(t)^2 + 4\eta R,$$
$$\sqrt{C} \le \sqrt{c_\tau(\vartheta)} + \eta \le \sqrt{c_{\rho_0}(\vartheta)} + \eta \le b(t) + 2\eta.$$

So $C - b(t)^2 \le 4\eta(b(t) + \eta) < \Delta_{\mathrm{gap}} \le \Delta_{\mathrm{gap}}(t)$. No assignment cost at time t lies strictly between $b(t)^2$ and $b(t)^2 + \Delta_{\mathrm{gap}}(t)$, so $C = b(t)^2$.

(iii) Apply Theorem B with $y = \nabla g_t - \sigma_m G^{-1}(\vartheta_i - \vartheta^0_{\tau(i)})$ in (P). With $\tilde{\vartheta}_i := \vartheta_i - \vartheta^*_{\pi^*(i)}$ as in the proof of Theorem B, the extra term is

$$-\sigma_m \frac{1}{m} \sum \langle \tilde{\vartheta}_i,\, \tilde{\vartheta}_i + \vartheta^*_{\pi^*(i)} - \vartheta^0_{\tau(i)} \rangle_{G^{-1}} \le -\sigma_m \eta^2 + \sigma_m \eta C^{1/2}.$$

Under the gap condition, (ii) gives $C^{1/2} = b(t) \le b$, and $\sigma_m \eta b \le (\sigma_m/2)(\eta^2 + b^2)$.

(iv) Apply Young's inequality three times:

- $\|e\|\varepsilon_{\max} \le (k_e/4)\|e\|^2 + \varepsilon^2_{\max}/k_e$;
- $\|e\| K \lambda_G \eta^2 \le (k_e/4)\|e\|^2 + (\sigma_m/8)\eta^2$ on $\{\eta \le \bar{\eta}\}$;
- $V_* \eta \le (\sigma_m/8)\eta^2 + 2V_*^2/\sigma_m$.

This gives $D^+ V_{\mathrm{OT}} \le -\alpha V_{\mathrm{OT}} + \beta$. Each set $\{V_{\mathrm{OT}} \le c\}$ with $\beta/\alpha < c < \bar{\eta}^2/2$ lies inside $\{\eta \le \bar{\eta}\}$ and has a negative boundary derivative. Intersect over $c > \beta/\alpha$ for the endpoint, and apply the comparison lemma.

(v) Steps (a)–(c) of Theorem B hold for any fixed π, so use the classical completion of squares for V_π.

- *Equality case.* If $\Delta_{\mathrm{gap}} > 0$, the costs minus $b(t)^2$ take values in $\{0\} \cup [\Delta_{\mathrm{gap}}, \infty)$ and are continuous in t, so the optimal set is constant.
- *Strict case.* If the optimal ideal–prior assignment changes along the ideal, the inequality can be strict. For example, $a^*_1(t) = t$ and $a^*_2(t) = 1 - t$ on $[0, 1]$, with prior $a \in \{0, 1\}$, give $b = 0.5$ against 1.0.
- *Not always strict.* For $a^*(t) = (u(t),\, 1 - u(t))$ with $u(t) = -0.025 + 0.575 \sin t$, the optimal assignment alternates between the identity and the swap, yet $\min_\pi \sup_t b_{\mathrm{lab}}(\pi)(t) = \sup_t b_{\mathrm{lab}}(\mathrm{id})(t) = 0.6 = b$ (computed numerically).

The negative statement is Proposition C.5. □

Lemma C.3 (no switching along the pure leak; exact decay for any $G \succ 0$)**.** *Let τ be optimal for $(\mu_{\mathrm{init}}, \mu^0)$ and $\dot{\vartheta}_i = -\sigma_m(\vartheta_i - \vartheta^0_{\tau(i)})$. Then τ stays optimal and $W(\mu_t, \mu^0) = e^{-\sigma_m t} W(\mu_{\mathrm{init}}, \mu^0)$.*

Proof. Put $d_i := \vartheta_i(0) - \vartheta^0_{\tau(i)}$, so $\vartheta_i(t) - \vartheta^0_{\tau(i)} = \lambda d_i$ with $\lambda = e^{-\sigma_m t}$. For any ρ, with $q_i := \vartheta^0_{\tau(i)} - \vartheta^0_{\rho(i)}$,

$$c_\rho(t) - c_\tau(t) = \frac{2\lambda}{m} \sum \langle d_i,\, q_i \rangle_{G^{-1}} + \frac{1}{m} \sum \|q_i\|^2_{G^{-1}}.$$

This is affine in λ. It is non-negative at $\lambda = 1$ by optimality and at $\lambda = 0$ trivially, hence on all of $[0, 1]$.

At an exact tie at $t = 0$, the chosen assignment becomes strictly optimal immediately against every ρ with $A_\rho := \frac{1}{m} \sum \|q_i\|^2 > 0$. Assignments with $A_\rho = 0$ induce the same velocity. So no selection solution of the pure leak switches. Filippov sliding on the tie is a different solution, excluded by the solution concept. □

Checked by: `test_theorem_C_i_exact_dissipation_pure_leak` in `tests/test_transport.py`: zero switches, and exact decay to integrator accuracy.

Proposition C.5 (the labelled leak reshuffles)**.** *There are configurations with* $b = b_{\mathrm{lab}}(\mathrm{id}) = 0$ *and* $e = 0$ *at which the labelled leak gives* $D^+V_{\mathrm{OT}} > 0$*.*

Example. Take $n = 1$, $G = I$ and $a_{\max} - \varepsilon_K \geq 1$. Let ideal = prior = $\{a = 0,\ a = 1\}$, with the other coordinates equal and in the inner set, and let the network weights be $(1 - \delta, \delta)$ with $0 < \delta < \frac{1}{2}$.

- The optimal assignment to the ideal is the swap.
- The labelled leak moves each particle toward the *other* ideal atom.
- So $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2) > 0$. □

From a relabelled start, the labelled leak moves every kernel in a non-trivial cycle of the relabelling to its prior position. In the reshuffling experiment (`results/ot_sigma_mod/transient.md` in the accompanying repository):

- the steady states coincide;
- the labelled leak's particle path is 5.8× longer (2.26 against 0.39);
- the early RMS$\|e\|$ is only about 4% larger (0.0771 against 0.0743 on $[0, 5]$).

Theorem F does not price this motion. With a leak, Theorem F bounds transport from above by tracking error plus a leak term, and the leak can pay for all of it. With $\delta = 0$ above, $e \equiv 0$ while every particle crosses the domain.

Checked by: the reshuffling transient, `experiments/exp_sigma_mod_transient.py`. The value $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2)$ is not checked in the repository.

Remark C.6 (the obstruction is known)**.**

- **Non-convexity.** For some ν, for example a two-atom ν in dimension ≥ 2, $\frac{1}{2}W^2(\cdot, \nu)$ is not λ-convex along geodesics for any λ (Ambrosio–Gigli–Savaré [2, Example 9.1.5]; Santambrogio [66, §4.5]). It is 1-semiconcave in every dimension [2, Thm. 7.3.2]. For $\nu = \delta_{\vartheta^0}$ it is a potential energy, and 1-convex.
- **Generalised geodesics.** $\frac{1}{2}W^2(\cdot, \nu)$ is 1-convex along generalised geodesics with base ν ([2, Lemma 9.2.1]; Carlen–Craig [6]).
- **Label-free leaks.**
  - A label-free leak that is λ-convex ($\lambda > 0$) on the lift $\mathbb{R}^{mp}$ has a unique, permutation-invariant minimiser, at which all particles coincide. So it cannot vanish at a μ^0 with distinct atoms.
  - 0-convex leaks that vanish at μ^0 exist, for example toward $\operatorname{conv}\operatorname{supp}\mu^0$. They do not single out μ^0.

So the trade-off reflects this geometry. On one side is a global inequality for a labelled V_π, which is large at relabelled starts (hence the reshuffling of Proposition C.5). On the other is a local, gap-protected inequality for the label-free V_{OT}.

13 Theorem C′: the OT leak toward a prior with a different number of atoms

Theorem C needs a prior with as many atoms as the network. Here the prior is any $\nu \in \mathcal{P}_2(\Theta_K^\varepsilon)$, for example the empirical measure of a network of a different size (the transfer problem of Section 25, branch 8), and the leak points at the barycentre of the prior mass that the optimal plan assigns to each particle. The dissipation toward the prior survives up to the quantisation

error, the fixed points of the pure leak are the equal-mass centroidal configurations of ν (the leak is Lloyd's algorithm in continuous time), and the Lyapunov bias is measured by the distance of the ideal from the *centroids* of its prior cells, not by $W(\mu^*, \nu)$.

The law. Let $\nu \in \mathcal{P}_2(\Theta_K^\varepsilon)$, for instance $\nu = \frac{1}{m_0}\sum_{j=1}^{m_0} \delta_{\vartheta_j^0}$ with any $m_0 \geq 1$. For $\mu = \frac{1}{m}\sum \delta_{\vartheta_i}$, let γ be a vertex optimal plan from μ to ν for the cost $\|\cdot\|_{G^{-1}}^2$ (for atomic ν a vertex of the transportation polytope; for general ν any optimal plan), and let

$$\bar{T}_i := m \int \vartheta' \, \mathrm{d}\gamma(\vartheta_i, \vartheta')$$

(the barycentric projection of γ at ϑ_i; for atomic ν, $\bar{T}_i = m\sum_j \gamma_{ij}\vartheta_j^0$),

$$V_{\mathrm{bar}}^2 := \frac{1}{m}\sum_i \|\vartheta_i - \bar{T}_i\|_{G^{-1}}^2, \qquad \mathrm{Var}_\gamma := \int \|\vartheta' - \bar{T}(\vartheta)\|_{G^{-1}}^2 \, \mathrm{d}\gamma(\vartheta, \vartheta'),$$
$$\text{so that} \qquad W(\mu,\nu)^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma.$$

The **barycentric OT leak** is

$$\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \bar{T}_i)\big). \tag{$\mathrm{L}_{\sigma,\mathrm{bar}}$}$$

For atomic ν the plan is a vertex of the transportation polytope (row sums $1/m$, column sums $1/m_0$), computed by linear programming (`transport.optimal_plan`, `transport.barycentric_projection`); for $m_0 = m$ it is a permutation and ($\mathrm{L}_{\sigma,\mathrm{bar}}$) is ($\mathrm{L}_\sigma$). The right-hand side is discontinuous where the optimal plan is not unique, and here, unlike Section 12 (Lemma C.3), the plan does switch along solutions (numerics below), so two solution concepts are used. *Vertex-selection solutions* are absolutely continuous $\vartheta(t)$ with $\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t - \sigma_m(\vartheta_i - \bar{T}_i^{\gamma(t)})\big)$ for a.e. t and $\gamma(t)$ a vertex optimal plan at $\vartheta(t)$; the time-stepped implementation produces these (switches at step times), and their existence for all time is assumed. *Generalised solutions* allow $\gamma(t)$ to be any measurable selection of optimal plans. The optimal face $F(\vartheta)$ of the transportation LP is a compact convex polytope, upper semicontinuous in ϑ (finitely many vertices, continuous costs), and $\gamma \mapsto \bar{T}^\gamma$ is linear, so the set of generalised velocities is compact, convex and upper semicontinuous in ϑ: it contains the Filippov regularisation of the vertex law, and generalised solutions exist on $[0,\infty)$ by the standard theorem for such differential inclusions (the projection is handled as in Section 12). Every statement below holds for both concepts, with $\bar{T}$ and V_{bar} computed from the plan the solution uses. Since Θ_K^ε is convex, $\bar{T}_i \in \Theta_K^\varepsilon$, so property (P) applies with $\vartheta' = \bar{T}_i$.

Notation. $\eta := W(\mu_t, \mu_t^*)$, $b := \sup_t W(\mu_t^*, \nu)$, and

$$b_{\mathrm{bar}}(t)^2 := \frac{1}{m}\sum_j \|\vartheta_j^*(t) - \bar{T}_j^*(t)\|_{G^{-1}}^2 = W(\mu_t^*,\nu)^2 - \mathrm{Var}_{\gamma^*(t)}, \qquad b_{\mathrm{bar}} := \sup_t b_{\mathrm{bar}}(t),$$

the *centroid defect of the ideal*, computed with an optimal plan $\gamma^*(t)$ from μ_t^* to ν. $\Delta_{\mathrm{gap}}(t)$ is the gap between the optimal and the next vertex cost of the transportation polytope at μ_t^* ($\min\emptyset := +\infty$), and $\Delta_{\mathrm{gap}} := \inf_t \Delta_{\mathrm{gap}}(t)$.

Theorem C′. *Under* (S)*, along every vertex-selection or generalised solution of* ($\mathrm{L}_{\sigma,\mathrm{bar}}$)*:*

(i) Dissipation toward the centroidal configurations. *With* $Z := \frac{1}{2}W(\mu_t,\nu)^2$ *and* Γ *as in Theorem C(i),*

$$D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2 + V_{\mathrm{bar}}\Gamma \leq -2\sigma_m\big(Z - \tfrac{1}{2}\mathrm{Var}_\gamma\big) + (2Z)^{1/2}\Gamma \qquad \textit{for a.e. } t.$$

For the pure leak ($\Gamma \equiv 0$)*,* $W(\mu_t,\nu)$ *is non-increasing, strictly decreasing wherever the plan in use is not centroidal* ($V_{\mathrm{bar}} > 0$)*, and at every* ω*-limit point* ϑ_∞ *of a solution on the compact* Θ_K *some optimal plan* γ_∞ *at* ϑ_∞ *is centroidal:* $\vartheta_{\infty,i} = \bar{T}_i^{\gamma_\infty}$ *for all* i *— a vertex plan for a vertex-selection solution. Such configurations are the fixed points of the equal-mass Lloyd iteration* $\vartheta \leftarrow \bar{T}^{\gamma}(\vartheta)$ *with an optimal (vertex) plan; where the optimal plan is unique they are the critical points of the quantisation energy* $\vartheta \mapsto \frac{1}{2}W^2\big(\frac{1}{m}\sum \delta_{\vartheta_i}, \nu\big)$*, which is differentiable there, and the pure leak is its* G*-gradient flow with the factor* m*:* $\dot{\vartheta}_i = -\sigma_m m\, G\, \partial_{\vartheta_i}\big[\frac{1}{2}W^2\big]$*. They include degenerate configurations with coincident particles. For* $m_0 = m$*,* $\mathrm{Var}_\gamma = 0$ *and* (i) *is Theorem C(i).*

(ii) Induced coupling. *Let* π^* *be optimal from* μ *to* μ^**, and let* γ *be the plan the law uses at* t*. Put* $C_{\mathrm{bar}} := \frac{1}{m}\sum_i \|\vartheta^*_{\pi^*(i)} - \bar{T}_i\|^2_{G^{-1}}$*. Then*

$$\sqrt{C_{\mathrm{bar}}} \le b(t) + 2\eta \quad \textit{always},$$

and, for atomic ν*,* $C_{\mathrm{bar}} = b_{\mathrm{bar}}(t)^2$ *whenever* $4\eta(b(t)+\eta) < \Delta_{\mathrm{gap}}(t)$ *and* $\gamma^*(t)$ *is unique.*

(iii) Lyapunov inequality. *With* C_{bar} *as in* (ii)*,*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \sigma_m\eta^2 + \sigma_m\eta\sqrt{C_{\mathrm{bar}}},$$

and under the gap condition of (ii) *the leak terms are at most* $-(\sigma_m/2)\eta^2 + (\sigma_m/2)b^2_{\mathrm{bar}}$*.*

(iv) Local exponential estimate. *For atomic* ν*, Theorem C(iv) holds verbatim with* b *replaced by* b_{bar}*: under* $4\bar{\eta}(b+\bar{\eta}) < \Delta_{\mathrm{gap}}$*, uniqueness of* $\gamma^*(t)$*, and* $8K^2\lambda_G^2\bar{\eta}^2 \le k_e\sigma_m$*, with* $\alpha := \min(k_e, \sigma_m/2)$ *and* $\beta := \varepsilon^2_{\max}/k_e + (\sigma_m/2)b^2_{\mathrm{bar}} + 2V_*^2/\sigma_m$*, if* $\max\{V_{\mathrm{OT}}(0), \beta/\alpha\} < \bar{\eta}^2/2$ *then* $V_{\mathrm{OT}}(t) \le e^{-\alpha t}V_{\mathrm{OT}}(0) + \beta/\alpha$ *and* $\{V_{\mathrm{OT}} \le c\}$ *is forward invariant for* $c \in \big[\max\{V_{\mathrm{OT}}(0), \beta/\alpha\},\ \bar{\eta}^2/2\big)$*.*

(v) What the bias measures. $b_{\mathrm{bar}} \le b$*, with* $b_{\mathrm{bar}} = 0$ *exactly when the ideal is itself an equal-mass centroidal configuration of* ν*. So a prior with many atoms whose centroidal* m*-quantisation coincides with the ideal anchors the network without bias, although* $W(\mu^*,\nu) > 0$*; a prior with the right number of atoms but the wrong positions has* $b_{\mathrm{bar}} = b$*.*

Status. Proved and checked: dissipation at every step and convergence of the pure leak to an equal-mass Lloyd fixed point, the coupling bound and plan stability, the Lyapunov inequality along the closed loop (`tests/test_barycentric.py`, `exp_barycentric_leak.py`). Novelty (literature audit): new as a control construction; the quantisation side is Lloyd/CVT theory; it reduces to Theorem C for $m_0 = m$.

Scope of (ii)–(iv). The gap condition $4\eta(b+\eta) < \Delta_{\mathrm{gap}}(t)$ asks the network to be closer to the ideal than the optimal vertex of the LP is to the next one. For a prior with many atoms, Δ_{gap} is at most the cost of exchanging two atoms between two cells, which is $O(1/m_0)$, so the condition holds only in a small neighbourhood of the ideal; in the closed-loop experiment below $4\eta(0)(b+\eta(0))$ exceeds that bound by factors 83–175, and (iv) applies to no tested instance (the table reports the exchange bound on Δ_{gap}, $\eta(0)$ and $4\eta(0)(b+\eta(0))$ for each run). What the experiment confirms is the fixed-point structure, (i) and (v): the loop settles at the centroidal configuration of its prior, whose distance from the ideal is b_{bar}.

Proof. (i) At a.e. t the law uses an optimal plan γ, and for $h > 0$ the same plan couples μ_{t+h} and ν, so $W(\mu_{t+h},\nu)^2 \le \sum_{ij}\gamma_{ij}\|\vartheta_i(t+h) - \vartheta^0_j\|^2_{G^{-1}}$ (for general ν, $\int\|\vartheta_i(t+h) - \vartheta'\|^2\,\mathrm{d}\gamma$). Differentiating the right-hand side at $h = 0$,

$$D^+Z \le \sum_{ij}\gamma_{ij}\langle\vartheta_i - \vartheta^0_j,\, G^{-1}\dot{\vartheta}_i\rangle = \frac{1}{m}\sum_i\langle\vartheta_i - \bar{T}_i,\, G^{-1}\dot{\vartheta}_i\rangle,$$

because $\sum_j \gamma_{ij}(\vartheta_i-\vartheta_j^0)=\frac{1}{m}(\vartheta_i-\bar T_i)$. Apply (P) with $\vartheta'=\bar T_i\in\Theta_K^\varepsilon$ and $y=\nabla g_t-\sigma_m G^{-1}(\vartheta_i-\bar T_i)$: $\langle\vartheta_i-\bar T_i,\,G^{-1}\dot\vartheta_i\rangle\le\langle\vartheta_i-\bar T_i,\,\nabla g_t(\vartheta_i)\rangle-\sigma_m\|\vartheta_i-\bar T_i\|^2_{G^{-1}}$. Average and use Cauchy–Schwarz for the first term: $\frac{1}{m}\sum\langle\vartheta_i-\bar T_i,\,\nabla g_t\rangle\le V_{\mathrm{bar}}\Gamma\le(2Z)^{1/2}\Gamma$. The Pythagorean identity $W^2=V_{\mathrm{bar}}^2+\mathrm{Var}_\gamma$ follows from $\sum_j\gamma_{ij}(\vartheta_j^0-\bar T_i)=0$: for each i, $\sum_j\gamma_{ij}\|\vartheta_i-\vartheta_j^0\|^2=\sum_j\gamma_{ij}\big(\|\vartheta_i-\bar T_i\|^2+\|\bar T_i-\vartheta_j^0\|^2\big)$. For the pure leak, Z is non-increasing and $D^+Z\le-\sigma_m V_{\mathrm{bar}}(t)^2$, with V_{bar} computed from the plan in use (the coupling inequality above holds for every optimal plan, vertex or not), so Z converges and $\int_0^\infty V_{\mathrm{bar}}^2\,\mathrm{d}t\le Z(0)/\sigma_m<\infty$. V_{bar} itself is neither monotone nor continuous — it jumps at plan switches, upward as well as downward (numerics below) — so the ω-limit statement rests on the stability of the LP instead. Let ϑ_∞ be an ω-limit point and F_∞ the set of optimal plans at ϑ_∞ (for a vertex-selection solution, its finitely many vertices). For a fixed plan γ the barycentres $\bar T_i^\gamma$ do not depend on ϑ, so $V_{\mathrm{bar}}^\gamma(\vartheta):=\big(\frac{1}{m}\sum\|\vartheta_i-\bar T_i^\gamma\|^2_{G^{-1}}\big)^{1/2}$ is 1-Lipschitz in ϑ for the norm $\big(\frac{1}{m}\sum\|\cdot\|^2_{G^{-1}}\big)^{1/2}$; and the optimal face is upper semicontinuous — the plan costs are continuous in ϑ and the polytope has finitely many vertices — so every plan optimal at a ϑ close enough to ϑ_∞ lies in F_∞. Suppose no plan of F_∞ is centroidal; then $\delta:=\min_{\gamma\in F_\infty}V_{\mathrm{bar}}^\gamma(\vartheta_\infty)>0$ (a minimum over a compact set), and for $\|\vartheta-\vartheta_\infty\|<\delta/2$, $V_{\mathrm{bar}}(\vartheta)\ge\delta/2$ whatever optimal plan is in use. The speed is bounded by $\sigma_m\cdot\mathrm{diam}_{G^{-1}}(\Theta_K)$ (Lemma P.1), so the solution, which enters the ball of radius $\delta/4$ around ϑ_∞ at arbitrarily large times, either stays in the ball of radius $\delta/2$ from some time on or spends a time $\ge\delta/(4\sigma_m\,\mathrm{diam})$ in it on each of infinitely many visits; either way $\int V_{\mathrm{bar}}^2\,\mathrm{d}t=\infty$, a contradiction. Hence some plan of F_∞ — a vertex for vertex-selection solutions — is centroidal at ϑ_∞. (A two-way tie between vertex plans γ and ρ is not a sliding mode of the vertex law unless $\bar T^\gamma=\bar T^\rho$: the rate of change of $c_\rho-c_\gamma$ when following γ, plus that of $c_\gamma-c_\rho$ when following ρ, equals $(2\sigma_m/m)\sum_i\|\bar T_i^\gamma-\bar T_i^\rho\|^2_{G^{-1}}\ge0$, so at least one of the two plans stays optimal when followed. Ties between three or more plans are not analysed.) The gradient identity is the envelope theorem for the transportation LP at a configuration with a unique optimal plan: $\partial_{\vartheta_i}\big[\frac12\sum\gamma_{ij}\|\vartheta_i-\vartheta_j^0\|^2_{G^{-1}}\big]=\sum_j\gamma_{ij}G^{-1}(\vartheta_i-\vartheta_j^0)=\frac{1}{m}G^{-1}(\vartheta_i-\bar T_i)$, so $-\sigma_m(\vartheta_i-\bar T_i)=-\sigma_m m\,G\,\partial_{\vartheta_i}\big[\frac12W^2\big]$. (Where the plan is not unique, $\frac12W^2(\cdot,\nu)$ is not differentiable; the generalised velocities are then $-\sigma_m m\,G\,\xi$ with ξ ranging over the barycentric projections of the optimal plans — the elements of the (sub)differential of $\frac12W^2(\cdot,\nu)$ at an atomic μ, cf. Ambrosio–Gigli–Savaré [2, §10.4].)

(ii) Relabel so that $\frac{1}{m}\sum\|\vartheta_i-\vartheta_i^*\|^2_{G^{-1}}=\eta^2$. For any plan ρ with row sums $1/m$, the plan cost $c_\rho(\vartheta):=\sum\rho_{ij}\|\vartheta_i-\vartheta_j^0\|^2_{G^{-1}}$ is the squared norm of $(\vartheta_i-\vartheta_j^0)$ in $L^2(\rho)$, so $|\sqrt{c_\rho(\vartheta)}-\sqrt{c_\rho(\vartheta^*)}|\le\eta$ by the triangle inequality (the labels of ϑ and ϑ^* are matched). The barycentric part is dominated by the plan cost: $C_{\mathrm{bar}}=\frac{1}{m}\sum\|\vartheta_i^*-\bar T_i\|^2\le c_\gamma(\vartheta^*)$ because $\sum_j\gamma_{ij}\|\vartheta_i^*-\vartheta_j^0\|^2=\frac{1}{m}\|\vartheta_i^*-\bar T_i\|^2+\sum_j\gamma_{ij}\|\bar T_i-\vartheta_j^0\|^2$. Let γ^* be optimal for (ϑ^*,ν). Then

$$\sqrt{C_{\mathrm{bar}}}\le\sqrt{c_\gamma(\vartheta^*)}\le\sqrt{c_\gamma(\vartheta)}+\eta\le\sqrt{c_{\gamma^*}(\vartheta)}+\eta\le\sqrt{c_{\gamma^*}(\vartheta^*)}+2\eta=b(t)+2\eta.$$

For the equality case: $c_\gamma(\vartheta^*)-b(t)^2\le(b+2\eta)^2-b^2=4\eta(b+\eta)<\Delta_{\mathrm{gap}}(t)$; the vertex γ (a vertex by the choice of the law) has cost at ϑ^* strictly below the next vertex cost, so it is optimal at ϑ^*; if $\gamma^*(t)$ is unique, $\gamma=\gamma^*(t)$ and $C_{\mathrm{bar}}=\frac{1}{m}\sum\|\vartheta_i^*-\bar T_i^*\|^2=b_{\mathrm{bar}}(t)^2$. [For a generalised solution whose plan at t is not a vertex, the same chain applies to each vertex of the optimal face at ϑ, so every such vertex is optimal at ϑ^*, and uniqueness of $\gamma^*(t)$ again gives $\gamma=\gamma^*(t)$.]

(iii) Apply Theorem B with $y=\nabla g_t-\sigma_m G^{-1}(\vartheta_i-\bar T_i)$ in (P). The extra term is

$$-\sigma_m\frac{1}{m}\sum\langle\tilde\vartheta_i,\,\tilde\vartheta_i+\vartheta^*_{\pi^*(i)}-\bar T_i\rangle_{G^{-1}}\le-\sigma_m\eta^2+\sigma_m\eta\sqrt{C_{\mathrm{bar}}}.$$

Under the gap condition $\sqrt{C_{\mathrm{bar}}}=b_{\mathrm{bar}}(t)\le b_{\mathrm{bar}}$ and $\sigma_m\eta b_{\mathrm{bar}}\le(\sigma_m/2)(\eta^2+b_{\mathrm{bar}}^2)$.

(iv) is the proof of Theorem C(iv) with b_{bar} in place of b.

(v) $b_{\mathrm{bar}}(t)^2=W(\mu_t^*,\nu)^2-\mathrm{Var}_{\gamma^*(t)}\le W(\mu_t^*,\nu)^2$, with equality iff $\mathrm{Var}_{\gamma^*}=0$, that is, iff every cell of γ^* consists of a single atom of ν (coincident atoms counted once) — the case $m_0=m$,

but also $m_0 \mid m$ and priors with repeated atoms — and $b_{\mathrm{bar}} = 0$ iff $\vartheta_j^* = \bar{T}_j^*$ for all j (at a tie, with the plan $\gamma^*(t)$ fixed by the selection). □

Remark C′.1 (reading)**.** The barycentric leak replaces "pull each kernel to its prior kernel" by "pull each kernel to the centre of mass of the prior it is responsible for". Three consequences.

- *Transfer across sizes.* A network of m kernels can be anchored to a learned measure of $m_0 \neq m$ kernels. The bias in the Lyapunov inequality is b_{bar}, the distance of the ideal from the centroids of its cells, not the quantisation error $W(\mu^*, \nu)$ itself.
- *Quantisation on the fly.* With $e \equiv 0$ the law is Lloyd's algorithm for the equal-mass quantisation of ν in the G^{-1} metric, in continuous time — the continuous-time Lloyd descent of coverage control [15], with the equal-mass constraint of capacity-constrained quantisation [4]; so the controller's leak and the quantisation theory of Corollary B.6 (Section 25, branch 9) are the same object. Like Lloyd's algorithm it converges to a critical point of the quantisation energy, not necessarily the global minimiser.
- *Theorem F.* The leak clauses (i′)–(iii′) of Theorem F hold with $W(\mu_r, \nu)$ replaced by $V_{\mathrm{bar}}(r) \leq W(\mu_r, \nu)$, since $\|\dot{\vartheta}_i\|_{G^{-1}} \leq \|\nabla g_t\|_G + \sigma_m \|\vartheta_i - \bar{T}_i\|_{G^{-1}}$ by Lemma P.1 and Minkowski.

Checked (`tests/test_barycentric.py`). The plan reduces to the assignment for $m_0 = m$ (10^{-10}) and the Pythagorean split $W^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma$ holds (10^{-10}); along the pure leak W^2 is non-increasing at every step, the Dini bound $D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2$ holds at every state (within 5% of the first-order Euler quotient), the flow reaches $V_{\mathrm{bar}}^2 < 10^{-8}$, and the end point is a fixed point of the equal-mass Lloyd iteration with the same W^2; $\sqrt{C_{\mathrm{bar}}} \leq b + 2\eta$ on 90 random configurations, with $C_{\mathrm{bar}} = b_{\mathrm{bar}}^2$ at small η; the Lyapunov inequality (iii) holds at every recorded state of a closed loop with the barycentric leak toward a 24-atom prior. The numbers are in `results/barycentric_leak/table.md` (`exp_barycentric_leak.py`).

Numerics (`results/barycentric_leak/table.md`).

- *Pure leak toward a 48-atom prior* ($m = 6$, lifted dimension 3, $G = \mathrm{diag}(2, 1, 0.5)$, $\sigma_m = 1$, three random starts). $W^2(\mu_t, \nu)$ is non-increasing at every one of the 3000 steps; the flow stops where the centroid defect is at the rounding level ($V_{\mathrm{bar}}^2 \leq 2.2 \cdot 10^{-25}$) and $W^2 = \mathrm{Var}_\gamma$ (1.164, 1.213, 1.203 from 2.55, 2.30, 2.58), after 2–6 plan switches, at which V_{bar} jumps (upward as well as downward). The end points are fixed points of the equal-mass Lloyd iteration; the best of 20 Lloyd runs from random starts reaches 1.171, below two of the three end points and above the third: like Lloyd's algorithm, the leak finds a local minimiser of the quantisation energy, not always the global one. With $m_0 = m$ the same code reproduces the exact decay of Theorem C(i) to $5 \cdot 10^{-16}$.
- *Closed loop* (representable plant with an $m^* = 4$ ideal, $m = 4$, $k_e = 4$, $\sigma_m = 0.5$, three seeds). Three leaks of the same gain: (A) the OT leak toward the ideal itself, (B) the barycentric leak toward a 32-atom cloud of jittered copies of the ideal atoms, (C) the labelled leak toward a random 4-subset of the cloud. The network converges, in each case, to the configuration its leak anchors: (A) to the ideal ($W(\mu_T, \mu^*)$ and RMS$\|e\|$ below the printed precision); (B) to the equal-mass centroids of the cloud, at $W(\mu_T, \mu^*) = 0.167$, 0.097, 0.157, which equal the centroid defects b_{bar} of the ideal (0.168, 0.097, 0.159) although $W(\mu^*, \nu) = 0.45$, 0.33, 0.40; (C) to the subset, at $W(\mu_T, \mu^*) = 0.61$, 1.02, 0.35 against $W(\mu^*, \mathrm{subset}) = 0.65$, 1.03, 0.40. The steady tracking RMS is 0.002–0.007 for (B) and 0.03–0.04 for (C). So the loop settles at the centroidal configuration of its prior, at distance b_{bar} from the ideal, as the fixed-point statements (i) and (v) describe; the Lyapunov bound (iv) does not apply to these runs (scope paragraph: the exchange bound on Δ_{gap} and $4\eta(0)(b + \eta(0))$ are in the table). The comparison with (C) shows the cost of a prior of the

right size at the wrong positions, not an advantage of barycentric over labelled leaks as such: a labelled leak toward the cloud's own equal-mass m-quantiser would have the same bias b_{bar} by (v).

14 Theorem D: the mean-field closed loop

The state is $(x, \mu) \in \mathbb{R}^n \times \mathcal{P}(\Theta_K)$, with

$$\begin{gathered}\dot{x} = f(x,t) + \dot{x}_m - k_e(x - x_m) - \int \Phi(x;\vartheta)\,\mathrm{d}\mu_t, \qquad \partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0, \\ v_t = \mathrm{Proj}_K\big(\vartheta,\, G\nabla_\vartheta\big[(x - x_m)^\top \Phi(x;\vartheta)\big]\big).\end{gathered} \tag{14.1}$$

Solutions are Lagrangian: $\mu_t = (\Xi_t)_{\#}\mu_0$ for the flow Ξ_t of v_t. This is law (L). The OT leak is multivalued at ties and is excluded here.

Theorem D. *Assume* (S0)–(S2).

(a) Local well-posedness. *For every $(x_0, \mu_0) \in \mathbb{R}^n \times \mathcal{P}(\Theta_K)$ there is a unique Lagrangian solution. It continues for as long as x stays bounded.*

(b) Adaptation-free tube (m-uniform, μ^*-free). *For every $\mu \in \mathcal{P}(\Theta_K)$, $\|\int \Phi(x;\cdot)\,\mathrm{d}\mu\| \le a_{\max}$. For $F \ge 0$ let*

$$r(F) := \|e(0)\| + \frac{F + a_{\max}}{k_e}, \qquad S(F) := \sup\big\{\|f(x,t)\| : t \ge 0,\ \|x - x_m(t)\| \le r(F)\big\}.$$

Suppose some $F \ge 0$ satisfies $S(F) \le F$. Then every solution, for every m and every μ_0:

- *exists for all $t \ge 0$;*
- *satisfies $\|x(t) - x_m(t)\| \le r(F)$;*
- *satisfies $\|e(t)\| \le e^{-k_e t}\|e(0)\| + (1 - e^{-k_e t})(F + a_{\max})/k_e$.*

Examples:

- *For bounded f, $F = \sup\|f\|$ works for every $e(0)$.*
- *If $\|f(x,t)\| \le \alpha + \beta\|x\|$ with $\beta < k_e$, then*

$$F = \frac{\alpha + \beta\big(\sup\|x_m\| + \|e(0)\| + a_{\max}/k_e\big)}{1 - \beta/k_e}$$

  works.
- *The condition can fail for superlinear f. For the Van der Pol plant of the numerics, $S(F) - F \ge 51.6$ for every F.*

(c) Dobrushin estimate. *Let two solutions of the same loop (same f, x_m, k_e, G, Θ_K and ε_K) remain in a compact X on $[0,T]$. Then*

$$\begin{gathered}\|x_t - x'_t\| + W(\mu_t, \mu'_t) \le e^{\Lambda t}\big(\|x_0 - x'_0\| + W(\mu_0, \mu'_0)\big), \\ \Lambda := \max(L_\vartheta + L_2,\, L_x + M_x),\end{gathered} \tag{14.2}$$

and Λ is independent of m. The constants are:

- *L_ϑ: a Lipschitz constant of $\vartheta \mapsto v$ in the G^{-1} norm, uniform over $x \in X$ and $t \le T$. It is proportional to $\bar{E} := \sup_{X\times[0,T]}\|x - x_m(t)\|$, and it is $O(\bar{E}/\varepsilon_K)$ because of the ramp ($h' \le 2/\varepsilon_K$).*

- *L_x: a Lipschitz constant, Euclidean to G^{-1}, of $x \mapsto v(\vartheta; x, x - x_m(t))$, uniform over $\vartheta \in \Theta_K$, $t \le T$ and $x \in X$.*
- *M_x: $M_x := \sup_{t \le T} \operatorname{Lip}_X f(\cdot, t) + k_e + a_{\max} J_c$. Here $a_{\max} J_c$ bounds $\|\nabla_x \int \Phi(x; \cdot)\, \mathrm{d}\mu\|$ over $x \in \mathbb{R}^n$ and $\mu \in \mathcal{P}(\Theta_K)$.*
- *L_2: $L_2 := \lambda_G^{1/2} L_1$, with L_1 from Lemma B.2, is a W-Lipschitz constant of $\mu \mapsto \int \Phi(x; \cdot)\, \mathrm{d}\mu$. For any coupling, $\|\int \Phi(x; \cdot)\, \mathrm{d}(\mu - \mu')\| \le L_2 \big(\int \|\vartheta - \vartheta'\|_{G^{-1}}^2 \, \mathrm{d}\gamma\big)^{1/2}$.*

(d) Mean-field limit. *Let (x, μ) be the solution from (x_0, μ_0), and let T be smaller than its existence time. Put $X_1 := \{y : \operatorname{dist}(y, x([0,T])) \le 1\}$, and let Λ_1 be the Λ of (c) computed on $X_1 \times [0,T]$. Suppose $x_0^m \to x_0$ and $\mu_0^m \to \mu_0$ in W, and set $\delta_m := \|x_0^m - x_0\| + W(\mu_0^m, \mu_0)$. Then, for all large m:*

- *the m-particle loop exists on $[0,T]$ and stays in X_1;*
- *$\sup_{t \le T} \big(\|x_t^m - x_t\| + W(\mu_t^m, \mu_t)\big) \le e^{\Lambda_1 T} \delta_m \to 0$.*

Under the hypothesis of (b), for example for bounded f, the limit is global and T is arbitrary.

(e) Passage of (B) to the limit. *Work in the setting of (d), with $X := X_1$. Let $\mu_t^{*m} = \frac{1}{m} \sum_j \delta_{\vartheta_j^{*m}(t)}$ satisfy* (S4), *and let $\varepsilon_{\max}^m \ge \sup_{X \times [0,T]} \|f - \int \Phi \, \mathrm{d}\mu_t^{*m}\|$. Assume $\mu_t^{*m} \to \mu_t^*$ uniformly in W on $[0,T]$, $\sup_m \varepsilon_{\max}^m \le \varepsilon_{\max}$ and $\sup_m V_*^m \le V_*$. Only tails matter, so $\sup_m$ may be replaced by $\limsup_m$. Then:*

- *μ_t^* is V_*-Lipschitz in W, with values in $\mathcal{P}(\Theta_K^\varepsilon)$;*
- *the limit satisfies* (B) *in integrated form, and in Dini form for every $t \in [0,T)$, with constants $(k_e, \varepsilon_{\max}, K, L_0, L_1, \lambda_G, V_*)$.*

V_* depends on the approximating ideals; the intrinsic version is Remark B.7.

Status. Proved; consistency in m checked numerically (the constant is not checked). Not checked in the repository: (b), the constant of (c), and the one-sided constant of Remark D.1. Novelty (literature audit): technique anticipated [22, 60, 61].

Proof. (a) Write the loop as an ODE for $(x, \Xi) \in \mathbb{R}^n \times C(\Theta_K; \mathbb{R}^p)$, with the sup norm (and G^{-1} on $\mathbb{R}^p$):

$$\dot{x} = f(x,t) + \dot{x}_m - k_e(x - x_m) - \int \Phi(x; \Xi(\vartheta))\, \mathrm{d}\mu_0(\vartheta), \qquad \partial_t \Xi(\vartheta) = v(\Xi(\vartheta); x, t), \qquad \Xi_0 = \mathrm{id},$$

so that $\mu_t = \Xi_{t\#}\mu_0$. Extend v Lipschitz to a neighbourhood of Θ_K inside $\{s > s_{\min}/2\}$ (McShane). On $\{\|x\| \le R\}$, with the constants of (c), the right-hand side is Lipschitz:

- $\|v(\Xi(\cdot); x, t) - v(\Xi'(\cdot); x', t)\|_\infty \le L_\vartheta \|\Xi - \Xi'\|_\infty + L_x \|x - x'\|$;
- the x-field changes by at most $M_x \|x - x'\| + L_2 \|\Xi - \Xi'\|_\infty$, because $W(\Xi_\# \mu_0, \Xi'_\# \mu_0) \le \|\Xi - \Xi'\|_{L^2(\mu_0; G^{-1})}$.

Picard–Lindelöf gives a unique local solution. Each characteristic stays in Θ_K (Nagumo, as in A(ii)), so the solution continues while x is bounded. Golse [30, Thms. 1.3.2 and 1.4.3], and [22, 60, 61], are structural analogues (pairwise C^1 kernels, W_1), not the proof.

(b) While $\|e\| \le r(F)$, $\dot{e} = -k_e e + (f - \hat{f})$ with $\|f - \hat{f}\| \le F + a_{\max}$. Variation of constants gives

$$\|e(t)\| \le e^{-k_e t} \|e(0)\| + (1 - e^{-k_e t})(F + a_{\max})/k_e < r(F)$$

for every finite t, since $a_{\max} > 0$. So x never reaches the boundary of the tube (first exit), stays bounded, and (a) continues the solution.

(c) Let γ_0 be an optimal plan for (μ_0, μ_0'), let Ξ_t and Ξ_t' be the two flows, and set $D(t)^2 := \int \|\Xi_t(\vartheta) - \Xi_t'(\vartheta')\|_{G^{-1}}^2 \,\mathrm{d}\gamma_0$. Then $(\Xi_t, \Xi_t')_\# \gamma_0$ couples μ_t and μ_t', so $W(\mu_t, \mu_t') \le D(t)$. Split

$$v_t(\Xi) - v_t'(\Xi') = \big[v_t(\Xi) - v_t(\Xi')\big] + \big[v_t(\Xi') - v_t'(\Xi')\big]$$

and apply Cauchy–Schwarz:

$$\frac{\mathrm{d}}{\mathrm{d}t} D^2 \le 2L_\vartheta D^2 + 2L_x \|x - x'\| D, \qquad D^+ \|x - x'\| \le M_x \|x - x'\| + L_2 D.$$

Apply Grönwall to $D_\delta + \|x - x'\|$, with $D_\delta := (D^2 + \delta^2)^{1/2}$, and let $\delta \downarrow 0$.

(d) Let $\tau_m \le T$ be the first time x^m leaves X_1, with $\tau_m = T$ if it never leaves. On $[0, \tau_m]$ both solutions lie in X_1, so (c) gives $\|x_t^m - x_t\| \le e^{\Lambda_1 T} \delta_m < 1$ once m is large. Then $x^m(\tau_m)$ is interior to X_1, so $\tau_m = T$. By (a), the m-solution continues while it is bounded.

(e) Theorem B for each m gives $V^m(t_2) - V^m(t_1) \le \int R^m \le \int \bar{R}^m$. Here $\bar{R}^m$ is R^m with $\varepsilon_{\max}^m$ and $V_*^m(t)$ replaced by $\varepsilon_{\max}$ and V_*. By (d) and the triangle inequality for W, $e^m \to e$ and $W(\mu_t^m, \mu_t^{*m}) \to W(\mu_t, \mu_t^*)$ uniformly. So $V^m \to V$ and $\bar{R}^m \to \bar{R}$ uniformly, and $V(t_2) - V(t_1) \le \int \bar{R}$. $\bar{R}$ is continuous, which gives the Dini form. The identity coupling gives $W(\mu_t^{*m}, \mu_s^{*m}) \le V_* |t - s|$. This passes to the uniform limit, and Θ_K^ε is closed. □

Remark D.1 (a one-sided constant free of ε_K)**.** The coupling step uses v only through $\langle \Xi - \Xi',\, G^{-1}(v(\Xi) - v(\Xi'))\rangle$. So the one-sided constant

$$L_\vartheta^+ := \sup \frac{\langle \vartheta - \vartheta',\, G^{-1}(v(\vartheta) - v(\vartheta'))\rangle}{\|\vartheta - \vartheta'\|_{G^{-1}}^2}$$

may replace L_ϑ. Let $N(\theta, y) := h(\|\theta\|)(\hat{\theta}^\top y)_+ \hat{\theta}$ be the norm-ball correction, with $r = \|\theta\|$, $c = \hat{\theta}^\top \hat{\theta}'$ and $\alpha = (\hat{\theta}^\top y)_+$. Then

$$\langle \theta - \theta',\, N(\theta, y) - N(\theta', y)\rangle \ge -\frac{\|y\| \|\theta - \theta'\|^2}{2(\theta_{\max} - \varepsilon_K)}.$$

This follows from three facts:

- h is monotone;
- $|\alpha - \alpha'| \le \|y\| \sqrt{2(1 - c)}$;
- $\|\theta - \theta'\|^2 \ge 2\sqrt{rr'}\, |r - r'| \sqrt{2(1 - c)}$.

The box correction is monotone. Hence L_ϑ^+ stays bounded as $\varepsilon_K \to 0$, and so does $\Lambda := \max(L_\vartheta^+ + L_2,\, L_x + M_x)$. On the experiment's box, a numerical measurement gives $L_\vartheta^+ \approx 88$–98 for ε_K from 0.2 to 0.005. Over the same range the two-sided constant grows like $30/\varepsilon_K$. This measurement is not repeated in the repository's tests.

Numerics (`results/mean_field_m`). Halving $W(\mu_0^m, \mu_0^{2m})$ halves $W(\mu_T^m, \mu_T^{2m})$, with a nearly constant factor (1.64–1.69 over the four refinements). $\sup_t \|e^m - e^{2m}\|$ falls from 4.6×10^{-3} (16 against 32 particles) to below 10^{-4} (128 against 256).

This is a consistency check only.

- On the experiment's box the guaranteed factor $e^{\Lambda T}$ is at least e^{496}, from $L_2 = 33.08$ alone over $T = 15$.
- The condition of (b) fails for this Van der Pol plant, so the run is covered by the localised (d) through its observed boundedness, not by (b).

Checked by: `experiments/exp_mean_field_m.py` (consistency in m; not the constant $e^{\Lambda T}$).

15 Theorem E: label-free local convergence under tangential excitation

Assumptions.

- (S0) holds, in particular $k_e > 0$.
- **Uniform tube.** There is $\rho > 0$ such that $\varepsilon(x,t) = 0$ whenever $\|x - x_m(t)\| < \rho$, for all $t \geq 0$.
- **Interior ideal.** μ^* is constant, with $\vartheta_j^* \in \operatorname{int}\Theta_K^\varepsilon$. Put $\delta := \min_j \operatorname{dist}(\vartheta_j^*, \partial\Theta_K^\varepsilon) > 0$.
- $x_m \in C^1$, and x_m and $\dot{x}_m$ are bounded.
- Define $\Psi(t) := \big[\nabla_\vartheta \Phi(x_m(t);\vartheta_1^*), \dots, \nabla_\vartheta \Phi(x_m(t);\vartheta_m^*)\big] \in \mathbb{R}^{n\times mp}$.
- **(PE)** There are $T, \alpha > 0$ with $\int_t^{t+T} \Psi^\top \Psi \,\mathrm{d}r \succeq \alpha I_{mp}$ for all $t \geq 0$.

(PE) forces $a_j^* \neq 0$ and pairwise distinct pairs (c_j^*, s_j^*).

- If $a_j^* = 0$, the (c,s)-columns of block j vanish.
- If $(c_j^*, s_j^*) = (c_k^*, s_k^*)$, the kernel of $\int \Psi^\top \Psi \,\mathrm{d}r$ contains the vector with v in the a-slot of block j, $-v$ in the a-slot of block k, and zeros elsewhere.

In particular the ϑ_j^* are pairwise distinct, and

$$d^2 := \min_{\pi\neq\pi'} \frac{1}{m}\sum_i \|\vartheta_{\pi(i)}^* - \vartheta_{\pi'(i)}^*\|_{G^{-1}}^2 = \frac{2}{m}\cdot \min_{i\neq j} \|\vartheta_i^* - \vartheta_j^*\|_{G^{-1}}^2 > 0, \tag{15.1}$$

with $d^2 := +\infty$ for $m = 1$. The closed form holds because a non-identity permutation moves at least two indices, and the transposition of the G^{-1}-closest pair attains the bound.

Lemma E.0 (explicit strict Lyapunov function for the skew-symmetric mixed class)**.** *Let $\Psi : [0,\infty) \to \mathbb{R}^{n\times q}$ be C^1 with $\|\Psi(t)\| \leq \Psi_M$ and $\|\dot\Psi(t)\| \leq \Psi_D$ for all t, and persistently exciting: there are $T, \alpha > 0$ with $\int_t^{t+T} \Psi(r)^\top \Psi(r)\,\mathrm{d}r \succeq \alpha I_q$ for all $t \geq 0$. Let $k > 0$ and consider*

$$\dot e = -ke - \Psi(t)\zeta, \qquad \dot\zeta = \Psi(t)^\top e, \qquad w := (e,\zeta) \in \mathbb{R}^{n+q}. \tag{LTV}$$

Put $M := \Psi^\top\Psi$ and

$$Q(t) := \int_t^{t+T} (t+T-r)M(r)\,\mathrm{d}r, \qquad Q_M := \frac{T^2\Psi_M^2}{2}, \qquad c_1 := T^2\Psi_M^3 + Tk\Psi_M + T\Psi_D,$$
$$V(t,e,\zeta) := \tfrac12\|e\|^2 + \tfrac12\|\zeta\|^2 - \varepsilon\,\zeta^\top Q(t)\zeta + \varepsilon T\, e^\top \Psi(t)\zeta.$$

(i) Every admissible ε. *For every $\varepsilon > 0$ with $\varepsilon(Q_M + T\Psi_M/2) < \frac12$, V is C^1 and, along* (LTV),

$$p_{\mathrm{lo}}\|w\|^2 \leq V \leq p_{\mathrm{hi}}\|w\|^2, \qquad p_{\mathrm{lo/hi}} := \tfrac12 \mp \varepsilon(Q_M + T\Psi_M/2),$$
$$\dot V \leq -(k - \varepsilon T\Psi_M^2)\|e\|^2 - \varepsilon\alpha\|\zeta\|^2 + \varepsilon c_1\|e\|\|\zeta\|.$$

Hence $\dot V \leq -\gamma(\varepsilon)\|w\|^2$ with $\gamma(\varepsilon) := \lambda_{\min}\left[\begin{smallmatrix} a & -c/2 \\ -c/2 & b\end{smallmatrix}\right]$, $a := k - \varepsilon T\Psi_M^2$, $b := \varepsilon\alpha$, $c := \varepsilon c_1$, whenever this is positive, and then $\|w(t)\| \leq (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-(\gamma/(2p_{\mathrm{hi}}))(t-t_0)}\|w(t_0)\|$ for all $t \geq t_0 \geq 0$.

(ii) Closed form. *With $\varepsilon := \min\{1/(4(Q_M + T\Psi_M/2)),\ k/(4T\Psi_M^2),\ k\alpha/c_1^2\}$, one has $p_{\mathrm{lo}} \geq \frac14$, $p_{\mathrm{hi}} \leq \frac34$ and $\dot V \leq -\gamma\|w\|^2$ with $\gamma := \min\{k/4, \varepsilon\alpha/2\}$. Hence*

$$\|w(t)\| \leq \sqrt{3}\cdot e^{-(2\gamma/3)(t-t_0)}\cdot\|w(t_0)\| \qquad \text{for all } t \geq t_0 \geq 0.$$

(iii) Dependence on Ψ_D and on time rescaling. *The rate tends to* 0 *as* $\Psi_D \to \infty$*, and it must:* $\Psi = \sin t^2$ *is bounded and PE with* $\Psi_D = \infty$*, and* (LTV) *is then not uniformly exponentially stable (part (f) of the proof). Under the time rescaling* $\Psi_\omega(t) := \Psi(\omega t)$*, with* $\Psi_D > 0$*, the data transform as* $(\alpha, T, \Psi_M, \Psi_D) \mapsto (\alpha/\omega, T/\omega, \Psi_M, \omega\Psi_D)$*, so* $c_1 = T\Psi_D \cdot (1 + O(1/\omega))$ *and the closed-form rate is* $k\alpha^2/(3\omega^2 T^2 \Psi_D^2) \cdot (1 + O(1/\omega))$ *as* $\omega \to \infty$*. This is the* ω^{-2} *law of Remark E.1.*

Proof. (a) The Gramian weight. $0 \preceq Q(t) \preceq \int_t^{t+T} (t + T - r)\Psi_M^2 \,\mathrm{d}r \cdot I = Q_M I$. Since M is continuous, Q is C^1 with $\dot{Q}(t) = \int_t^{t+T} M(r)\,\mathrm{d}r - TM(t)$: the upper limit contributes $(t + T - (t + T))M(t + T) = 0$, the lower limit contributes $-TM(t)$, and differentiating the weight under the integral gives $\int_t^{t+T} M$.
(b) Bounds on V. $|\zeta^\top Q\zeta| \le Q_M\|\zeta\|^2$ and $|e^\top\Psi\zeta| \le \Psi_M\|e\|\|\zeta\| \le (\Psi_M/2)(\|e\|^2 + \|\zeta\|^2)$.
(c) Three derivatives along (LTV).

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\tfrac{1}{2}(\|e\|^2 + \|\zeta\|^2) &= -k\|e\|^2 - e^\top\Psi\zeta + \zeta^\top\Psi^\top e = -k\|e\|^2,\\
\frac{\mathrm{d}}{\mathrm{d}t}(-\zeta^\top Q\zeta) &= -\zeta^\top\dot{Q}\zeta - 2\zeta^\top Q\dot{\zeta} = -\zeta^\top\Big(\int_t^{t+T} M\Big)\zeta + T\|\Psi\zeta\|^2 - 2\zeta^\top Q\Psi^\top e\\
&\le -\alpha\|\zeta\|^2 + T\|\Psi\zeta\|^2 - 2\zeta^\top Q\Psi^\top e,\\
\frac{\mathrm{d}}{\mathrm{d}t}(e^\top\Psi\zeta) &= \dot{e}^\top\Psi\zeta + e^\top\dot{\Psi}\zeta + e^\top\Psi\dot{\zeta} = -k\,e^\top\Psi\zeta - \|\Psi\zeta\|^2 + e^\top\dot{\Psi}\zeta + \|\Psi^\top e\|^2.
\end{aligned}$$

Adding them with weights 1, ε and εT, the $\|\Psi\zeta\|^2$ terms cancel:

$$\dot{V} \le -k\|e\|^2 - \varepsilon\alpha\|\zeta\|^2 - 2\varepsilon\zeta^\top Q\Psi^\top e - \varepsilon Tk\,e^\top\Psi\zeta + \varepsilon T\,e^\top\dot{\Psi}\zeta + \varepsilon T\|\Psi^\top e\|^2.$$

The last four terms are at most $2\varepsilon Q_M\Psi_M\|e\|\|\zeta\|$, $\varepsilon Tk\Psi_M\|e\|\|\zeta\|$, $\varepsilon T\Psi_D\|e\|\|\zeta\|$ and $\varepsilon T\Psi_M^2\|e\|^2$. This is (i); the quadratic form $-a\|e\|^2 - b\|\zeta\|^2 + c\|e\|\|\zeta\|$ is at most

$$-\lambda_{\min}\begin{bmatrix} a & -c/2 \\ -c/2 & b \end{bmatrix} \cdot \|w\|^2,$$

and $V \le p_{\mathrm{hi}}\|w\|^2$, $\|w\|^2 \le V/p_{\mathrm{lo}}$ give the estimate.
(d) The closed form. Young's inequality gives $\varepsilon c_1\|e\|\|\zeta\| \le (\varepsilon\alpha/2)\|\zeta\|^2 + (\varepsilon c_1^2/(2\alpha))\|e\|^2$. With $\varepsilon \le k\alpha/c_1^2$ the second coefficient is at most $k/2$, and with $\varepsilon \le k/(4T\Psi_M^2)$ the coefficient of $\|e\|^2$ in (i) is at least $k - k/4$. So $\dot{V} \le -(k/4)\|e\|^2 - (\varepsilon\alpha/2)\|\zeta\|^2 \le -\gamma\|w\|^2$. Since $V \le (3/4)\|w\|^2$, $\dot{V} \le -(4\gamma/3)V$, and $\|w\|^2 \le 4V$ gives

$$\|w(t)\|^2 \le 4V(t) \le 4e^{-(4\gamma/3)(t-t_0)}V(t_0) \le 3e^{-(4\gamma/3)(t-t_0)}\|w(t_0)\|^2.$$

(e) Rescaling. Ψ_ω has sup norm Ψ_M, derivative bound $\omega\Psi_D$, and

$$\int_t^{t+T/\omega} \Psi_\omega^\top\Psi_\omega \,\mathrm{d}r = \frac{1}{\omega}\int_{\omega t}^{\omega t+T} \Psi^\top\Psi \,\mathrm{d}r \succeq \frac{\alpha}{\omega}I.$$

Substitute into c_1 and the closed form.
(f) $\Psi = \sin t^2$ *is not uniformly exponentially stabilising* ($n = q = 1$, $k > 0$). Let $F(s) := \int_s^\infty \sin r^2\,\mathrm{d}r$, so that $\sin s^2 = -F'(s)$ and $|F(s)| \le C/s$ (Fresnel). From $w(t_0) = (0, \zeta_0)$, $e(t) = -\int_{t_0}^t e^{-k(t-s)}\sin(s^2)\zeta(s)\,\mathrm{d}s$; integrating by parts and using $|\zeta| \le \|w(t_0)\|$ (the energy $\frac{1}{2}\|w\|^2$ is non-increasing) and $|\dot{\zeta}| = |\Psi e| \le |e|$,

$$|e(t)| \le \frac{C}{t_0}(2 + 1)\|w(t_0)\| + \frac{C}{t_0}\int_{t_0}^t e^{-k(t-s)}|e(s)|\,\mathrm{d}s,$$

so $\sup_{t\ge t_0}|e| \le 3C\|w(t_0)\|/\big(t_0(1-C/(kt_0))\big)$ for $t_0 > C/k$. Hence

$$\tfrac{1}{2}\|w(t_0+\Delta)\|^2 \ge \tfrac{1}{2}\|w(t_0)\|^2 - k\Delta\Big(\frac{3C}{t_0}\Big)^2\|w(t_0)\|^2(1+o(1)) :$$

the decay over any fixed Δ tends to 0 as $t_0 \to \infty$, which excludes a uniform exponential rate. The three ratios quoted in step 3 of the proof of Theorem E (0.9715, 0.9942, 0.9980 for $t_0 = 5$, 20, 50 and $\Delta = 200$) match $1 - \frac{1}{8}\big(1/t_0 - 1/(t_0+200)\big)$ to three digits. □

Checked by: `tests/test_excitation.py`: the two bounds on V and $\dot V \le -\gamma\|w\|^2$ at random (t, w) for four regressors of different shapes and gains, with the closed-form ε and with the ε maximising the rate (`excitation.optimise_eps`; every admissible ε gives a valid Lyapunov function, so the maximum over the grid is a valid bound for the given PE data; the data Ψ_M, Ψ_D, α are themselves grid maxima and a refinement-checked quadrature—the PE level over the 20π window is computed with 800 Gauss–Legendre nodes, and `excitation.pe_level` raises if a 2× refinement disagrees—so the ordering against the exact rate is numerical); the envelope along trajectories; the ordering explicit rate ≤ exact Floquet rate; the ω^{-2} law; the slow-adaptation limit (a no-growth check of the ratio, not a saturation check); and the explicit nonlinear envelope of Theorem E on the implemented closed loop. The round-5 referee verified the pointwise form $\lambda_{\max}(N(t)) \le -\gamma$ of $\dot V \le -\gamma\|w\|^2$ over all w, with an independent quadrature, for ten regressor families ($q > n$, $q < n$, k from 10^{-3} to 10^4, Ψ_D/Ψ_M up to 100, non-periodic PE, near-degenerate PE) and eight choices of ε, without finding a violation.

Remark E.0a (how conservative the explicit rate is)**.** The construction is of the Mazenc–Malisoff type, a backward–forward integral of the PE Gramian as a cross term; its content here is the explicit, self-contained statement for the mixed class (LTV), which removes the dependence of Theorem E on [50]. The price is conservativeness, which the table of `exp_explicit_rate.py` → `results/explicit_rate/table.md` quantifies against the exact Floquet rate of the linearised loop.

- *Slow adaptation* ($T\Psi_M^2 \ll k$). To leading order in the gains, $c_1 \approx T(k\Psi_M + \Psi_D)$ and $\alpha = T\lambda_{\min}\langle\Psi^\top\Psi\rangle$ for a window equal to the period, so the closed-form bound $\frac{2}{3}\gamma_{\mathrm{cf}}$ of (ii) is $k\alpha^2/(3c_1^2)$, and the rate with the optimal ε of the same quadratic form is $k\alpha^2/c_1^2$ (maximise $\varepsilon\alpha - \varepsilon^2 c_1^2/(4k)$ over ε). The exact rate is $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ when k is also large compared with the bandwidth of the reference (Remark E.1). Hence, in that doubly asymptotic regime,

$$\frac{\text{exact}}{\text{explicit}} \approx 3\kappa_\Psi\Big(1+\frac{\Psi_D}{k\Psi_M}\Big)^2 \text{ for (ii)}, \qquad \approx \kappa_\Psi\Big(1+\frac{\Psi_D}{k\Psi_M}\Big)^2 \text{ for the optimised } \varepsilon,$$

  with $\kappa_\Psi := \Psi_M^2/\lambda_{\min}\langle\Psi^\top\Psi\rangle$ the condition number of the averaged Gramian. The exact rate pays for ill-conditioning once, through $\lambda_{\min}$; the bound pays twice, because the Young step that absorbs the cross term $\varepsilon c_1\|e\|\|\zeta\|$ compares $c_1 \propto \Psi_M$ with $\alpha \propto \lambda_{\min}$ (Lemma E.0′ below removes this by slaving the error to the slow manifold before the Young step). On the ideal of the Theorem E tests ($m = 3$, $n = 2$, $k_e = 6$), driven by a two-tone reference of period 20π, with $G = g\cdot\mathrm{diag}(2,2,1)$: $\kappa_\Psi = 1.27\cdot10^5$, $(1+\Psi_D/(k_e\Psi_M))^2 = 3.84$, so the two formula values are $1.47\cdot10^6$ and $4.9\cdot10^5$. The measured ratio with the optimised ε falls from $6.56\cdot10^{10}$ at $g = 30$ to $4.22\cdot10^5$ at $g = 0.01$ ($5.63\cdot10^5$ with the closed-form ε, which the table evaluates with the exact 2×2 form and the exact p_{hi} rather than the bound (ii)). Two corrections account for the difference from $4.9\cdot10^5$: on this instance the exact rate is 0.66 of $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ at every g ($k_e = 6$ is not large against the reference's bandwidth; the two last columns of the table), and the $T^2\Psi_M^3$ term is still 13% of c_1 at $g = 0.01$ ($(c_1/c_1^\infty)^2 = 1.31$); their product, $0.66\times1.31 = 0.86$, brings $4.9\cdot10^5$ to $4.2\cdot10^5$. The ratio is still decreasing at $g = 0.01$ and tends to about $0.66\cdot\kappa_\Psi(1+\Psi_D/(k_e\Psi_M))^2 \approx 3.2\cdot10^5$ as $g \to 0$. In the mean-field normalisation $\Psi_G \propto m^{-1/2}$, so large networks sit in this regime at fixed geometry.

- *Fast adaptation* ($T\Psi_M^2 \gtrsim k$). The ratio grows further with the excitation strength, through the $T^2\Psi_M^3$ term of c_1 and the Q_M term of ε.
- *Remark E.1's instance* ($m = n = 1$, $G = I$, $k_e = 3$, $x_m = 1.5\sin\omega t$). The ratio exact/explicit decreases from 954 at $\omega = 1$ to 17.6 at $\omega = 64$. Over $\omega \geq 4$ the log-log slopes are -1.97 (exact) and -1.56 (explicit); over the last octave, $\omega = 32 \to 64$, the explicit slope is -1.92, approaching the asymptotic ω^{-2} of (iii).
- *The window.* Any window length T is admissible with its own $\alpha(T)$. On the test ideal the best window is half a period.
- *Answered below.* Lemma E.0′ is a strict Lyapunov function whose rate is $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, above a threshold in k set by the peak factors of the regressor: in the slow regime it pays for the ill-conditioning once.

Lemma E.0′ (the slow-adaptation regime: a strict Lyapunov function whose rate is the averaged rate)**.** *Let Ψ be T-periodic and C^2, $M := \Psi^\top\Psi$, $\bar M := \frac{1}{T}\int_0^T M\,\mathrm{d}t \succ 0$, $\|v\|_{\bar M} := (v^\top \bar M v)^{1/2}$, $\lambda_{\min} := \lambda_{\min}(\bar M)$, and*

$$\Psi_M := \sup\|\Psi\|, \qquad \Psi_D := \sup\|\dot\Psi\|, \qquad \Psi_{DD} := \sup\|\ddot\Psi\|,$$
$$\rho := \sup_{t,\,v\neq 0}\frac{\|\Psi(t)v\|^2}{\|v\|_{\bar M}^2}, \qquad \rho_D := \sup\frac{\|\dot\Psi(t)v\|^2}{\|v\|_{\bar M}^2}, \qquad \rho_{DD} := \sup\frac{\|\ddot\Psi(t)v\|^2}{\|v\|_{\bar M}^2},$$

the peak factors*: the largest ratio, over directions, between the instantaneous excitation of a direction (by Ψ, $\dot\Psi$, $\ddot\Psi$) and its average excitation; $\rho \geq 1$, and all three are finite because $\bar M \succ 0$. Write $\varepsilon := 1/k$ and let*

$$P_1(t) := -\frac{2}{T}\int_t^{t+T}(t+T-r)\big(M(r)-\bar M\big)\,\mathrm{d}r, \qquad F := P_1M + MP_1,$$
$$S := \text{the zero-mean antiderivative of } F, \qquad P_2 := -(M-\bar M) + S,$$
$$e_2(\zeta,t) := -\varepsilon\Psi\zeta + \varepsilon^2\dot\Psi\zeta - \varepsilon^3\Psi M\zeta, \qquad w := e - e_2(\zeta,t),$$
$$V(e,\zeta,t) := \tfrac12\zeta^\top\big(I + \varepsilon P_1(t) + \varepsilon^2 P_2(t)\big)\zeta + \frac{\beta}{2}\|w\|^2,$$

where P_1 is the zero-mean antiderivative of $2(M-\bar M)$ and e_2 is the slow manifold of the error, expanded through order ε^3. F has zero mean: $\langle P_1M + MP_1\rangle = \langle P_1\rangle\bar M + \bar M\langle P_1\rangle + \frac12\big\langle\frac{\mathrm{d}}{\mathrm{d}t}(P_1^2)\big\rangle = 0$, so S exists and the second-order averaged correction $N_2 := \frac12\langle F\rangle$ vanishes identically. Put

$$\pi_1 := \tfrac{T}{2}\Psi_M(\sqrt\rho+1), \qquad \pi_2 := \Psi_M(\sqrt\rho+1) + \tfrac{T^2}{8}\Psi_M^3(2\sqrt\rho+1),$$
$$p_1 := \tfrac{T}{2}\Psi_M^2, \qquad p_2 := \Psi_M^2 + \tfrac{T^2}{4}\Psi_M^4,$$
$$c_1 := \sqrt\rho + \varepsilon\Psi_M\pi_1 + \varepsilon^2\Psi_M\pi_2,$$
$$c_2 := \varepsilon^2\sqrt{\rho_{DD}} + \varepsilon^3\big(3\Psi_D\Psi_M\sqrt\rho + 2\Psi_M^2\sqrt{\rho_D}\big) + \varepsilon^4\big(2\Psi_M^4\sqrt\rho + \Psi_D\Psi_M\sqrt{\rho_D}\big)$$
$$\quad + \varepsilon^5\big(\Psi_M^4\sqrt{\rho_D} + \Psi_D\Psi_M^3\sqrt\rho\big) + \varepsilon^6\Psi_M^6\sqrt\rho,$$
$$d_w := \varepsilon\Psi_M^2 + \varepsilon^2\Psi_D\Psi_M + \varepsilon^3\Psi_M^4,$$
$$r_\zeta := \varepsilon^3\big(\pi_2\Psi_M\sqrt\rho + \pi_1\Psi_M\sqrt{\rho_D} + \Psi_M^2\rho\big) + \varepsilon^4\big(\pi_2\Psi_M\sqrt{\rho_D} + \pi_1\Psi_M^3\sqrt\rho\big) + \varepsilon^5\pi_2\Psi_M^3\sqrt\rho,$$
$$\beta := 2c_1^2/\delta, \qquad c_\zeta := 1 - \delta - r_\zeta/\varepsilon.$$

Fix $\delta \in (0,1)$. If

$$\text{(A) } c_1c_2 \leq \delta/2, \qquad \text{(B) } d_w \leq k/4, \qquad \text{(C) } c_\zeta > 0, \qquad \text{(D) } \varepsilon p_1 + \varepsilon^2 p_2 < 1,$$

then along (LTV)*, for all (e,ζ,t),*

$$\dot V \leq -\varepsilon c_\zeta\,\zeta^\top\bar M\zeta - \frac{\beta k}{4}\|w\|^2 \leq -\gamma V, \qquad \gamma := \min\Big\{\frac{2c_\zeta\lambda_{\min}}{k(1+\varepsilon p_1+\varepsilon^2 p_2)},\ \frac{k}{2}\Big\},$$

and $\|(e,\zeta)(t)\| \le Ce^{-(\gamma/2)(t-t_0)}\|(e,\zeta)(t_0)\|$ *with*

$$C := \big(1+\varepsilon\Psi_M+\varepsilon^2\Psi_D+\varepsilon^3\Psi_M^3\big)^2\cdot\Big(\frac{\max\{1+\varepsilon p_1+\varepsilon^2 p_2,\ \beta\}}{\min\{1-\varepsilon p_1-\varepsilon^2 p_2,\ \beta\}}\Big)^{1/2}.$$

In the slow-adaptation limit ($k\to\infty$ at fixed Ψ) the conditions hold for every $\delta\in(0,1)$ once $k\ge k_0(\delta)$, with $k_0\approx(4\rho\rho_{DD}/\delta^2)^{1/4}$ when (A) binds (small δ, or $T\Psi_M^2\lesssim 1$; otherwise (C) sets k_0 through r_ζ, as on the sinusoidal regressor below), and the rate is $\gamma/2=(1-\delta)(1-\varepsilon p_1+O(k^{-2}))\lambda_{\min}/k$: the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ of Remark E.1 up to the factor $1-\delta$ and the first-order deficit $\varepsilon p_1=T\Psi_M^2/(2k)$ of the quadratic form. The bound pays for the ill-conditioning of $\bar M$ once, through $\lambda_{\min}$, as the exact rate does; the peak factors enter only the threshold k_0 and the overshoot $C=O((\rho/\delta)^{1/2})$, not the rate.

Status. Proved (below); checked pointwise by a certificate—the supremum of $\dot V/V$ over all states and over a time grid is a generalised eigenvalue problem—on the sinusoidal regressor and on the slow-adaptation ideal of the Theorem E tests, where the guaranteed rate is within 12% ($\delta=0.1$) and 26% ($\delta=0.2$) of the exact rate at $k=400$ while Lemma E.0's is five orders of magnitude below it (`tests/test_excitation.py`, `exp_explicit_rate.py`, Table 12). The round-6 referee re-derived every step and certified the inequality independently on five regressor families without a failure (`review/REVIEW_round6_E0prime.md`).

Proof. Write M, Ψ, … for their values at t. Two facts about zero-mean antiderivatives are used throughout: if f is T-periodic with zero mean and g is its zero-mean antiderivative, then $\|g\|_\infty\le(T/4)\|f\|_\infty$—for the maximiser t^* of g, $g(t^*)=\frac1T\int_0^T(g(t^*)-g(s))\,\mathrm{d}s\le\frac1T\int_0^T\operatorname{dist}(t^*,s)\|f\|_\infty\,\mathrm{d}s=(T/4)\|f\|_\infty$ with dist the distance on the circle, and the same for $-g$—and the same bound holds for every bilinear coordinate $u^\top gv$, hence $\|g(t)v\|\le(T/4)\sup_r\|f(r)v\|$. As $\|M-\bar M\|\le\Psi_M^2$ (two positive semidefinite matrices of norm $\le\Psi_M^2$), this gives $\|P_1\|\le T\Psi_M^2/2=p_1$ and $\|P_1v\|\le(T/2)\sup\|(M-\bar M)v\|$, and $\|S\|\le(T/4)\|F\|_\infty$, $\|Sv\|\le(T/4)\sup\|Fv\|$.
(a) The fast variable. From $\dot e=-ke-\Psi\zeta$ and $ke_2=-\Psi\zeta+\varepsilon\dot\Psi\zeta-\varepsilon^2\Psi M\zeta$,

$$\dot e=-kw-\varepsilon\dot\Psi\zeta+\varepsilon^2\Psi M\zeta.$$

Differentiating e_2 along the flow, with $\dot\zeta=\Psi^\top e=\Psi^\top w-\varepsilon M\zeta+\varepsilon^2\Psi^\top\dot\Psi\zeta-\varepsilon^3M^2\zeta$ and $\dot M=\dot\Psi^\top\Psi+\Psi^\top\dot\Psi$, the terms of order ε and ε^2 in ζ cancel and

$$\begin{aligned}\dot w=&-kw+\varepsilon\Psi\Psi^\top w-\varepsilon^2\dot\Psi\Psi^\top w+\varepsilon^3\Psi M\Psi^\top w-\varepsilon^2\ddot\Psi\zeta+\varepsilon^3\big(2\dot\Psi M+\Psi\dot\Psi^\top\Psi+2\Psi\Psi^\top\dot\Psi\big)\zeta\\&-\varepsilon^4\big(2\Psi M^2+\dot\Psi\Psi^\top\dot\Psi\big)\zeta+\varepsilon^5\big(\Psi M\Psi^\top\dot\Psi+\dot\Psi M^2\big)\zeta-\varepsilon^6\Psi M^3\zeta.\end{aligned}$$

(b) The slow part of $\dot V$. With $H:=I+\varepsilon P_1+\varepsilon^2P_2$ (symmetric), $\frac{\mathrm{d}}{\mathrm{d}t}[\frac12\zeta^\top H\zeta]=\zeta^\top H\dot\zeta+\frac12\zeta^\top\dot H\zeta$ and $\dot H=2\varepsilon(M-\bar M)+\varepsilon^2(F-\dot M)$. Insert $\dot\zeta$ and collect by powers of ε. At order ε: $-\zeta^\top M\zeta+\zeta^\top(M-\bar M)\zeta=-\zeta^\top\bar M\zeta$. At order ε^2:

$$\begin{aligned}-\zeta^\top P_1M\zeta+\zeta^\top\Psi^\top\dot\Psi\zeta+\tfrac12\zeta^\top(F-\dot M)\zeta=&-\tfrac12\zeta^\top(P_1M+MP_1)\zeta+\tfrac12\zeta^\top\dot M\zeta\\&+\tfrac12\zeta^\top(P_1M+MP_1-\dot M)\zeta=0,\end{aligned}$$

using $\zeta^\top\Psi^\top\dot\Psi\zeta=\frac12\zeta^\top\dot M\zeta$: the choice of P_2 removes the whole second-order term, and this is where $N_2=0$ enters (otherwise $-\zeta^\top N_2\zeta$ would remain). The remaining terms are

$$R_\zeta:=\varepsilon^3\big(-\zeta^\top P_2M\zeta+\zeta^\top P_1\Psi^\top\dot\Psi\zeta-\zeta^\top M^2\zeta\big)+\varepsilon^4\big(\zeta^\top P_2\Psi^\top\dot\Psi\zeta-\zeta^\top P_1M^2\zeta\big)-\varepsilon^5\zeta^\top P_2M^2\zeta,$$

plus the coupling $\zeta^\top H\Psi^\top w$, so $\frac{\mathrm{d}}{\mathrm{d}t}[\frac12\zeta^\top H\zeta]=-\varepsilon\zeta^\top\bar M\zeta+R_\zeta+\zeta^\top H\Psi^\top w$.

(c) Per-direction bounds. For every v and t:

$$\|\Psi v\| \le \sqrt{\rho}\|v\|_{\bar M}, \qquad \|\dot\Psi v\| \le \sqrt{\rho_D}\|v\|_{\bar M}, \qquad \|\ddot\Psi v\| \le \sqrt{\rho_{DD}}\|v\|_{\bar M};$$
$$\|Mv\| \le \Psi_M\|\Psi v\| \le \Psi_M\sqrt{\rho}\|v\|_{\bar M}; \qquad \|\bar M v\| \le \lambda_{\max}(\bar M)^{1/2}\|v\|_{\bar M} \le \Psi_M\|v\|_{\bar M};$$
$$\|\Psi^\top\dot\Psi v\| \le \Psi_M\sqrt{\rho_D}\|v\|_{\bar M}; \qquad \|P_1 v\| \le \tfrac{T}{2}\sup_r\|(M(r)-\bar M)v\| \le \pi_1\|v\|_{\bar M};$$
$$\|Fv\| \le \|P_1 Mv\| + \|MP_1 v\| \le \tfrac{T}{2}\Psi_M^3\sqrt{\rho}\|v\|_{\bar M} + \Psi_M^2\pi_1\|v\|_{\bar M} = T\Psi_M^3(\sqrt{\rho}+\tfrac12)\|v\|_{\bar M},$$

so $\|Sv\| \le (T/4)\sup\|Fv\| = (T^2/8)\Psi_M^3(2\sqrt{\rho}+1)\|v\|_{\bar M}$ and $\|P_2 v\| \le \|(M-\bar M)v\| + \|Sv\| \le \pi_2\|v\|_{\bar M}$; also

$$\|P_2\| \le \Psi_M^2 + \tfrac{T}{4}\|F\|_\infty \le \Psi_M^2 + \tfrac{T}{4}\cdot 2\cdot\tfrac{T\Psi_M^2}{2}\Psi_M^2 = p_2.$$

Hence $|R_\zeta| \le r_\zeta\|\zeta\|_{\bar M}^2$, $|\zeta^\top H\Psi^\top w| = |\langle \Psi H\zeta, w\rangle| \le c_1\|\zeta\|_{\bar M}\|w\|$, and from (a), $\beta w^\top\dot w \le \beta\big[-(k-d_w)\|w\|^2 + c_2\|\zeta\|_{\bar M}\|w\|\big]$: each ζ-term of $\dot w$ is bounded through $\|\Psi\zeta\|$, $\|\dot\Psi\zeta\|$, $\|\ddot\Psi\zeta\|$, $\|M\zeta\|$ or $\|M^2\zeta\|$ (the terms of c_2, in the order of (a); the last one is $\|\Psi M^3\zeta\| \le \Psi_M\cdot\Psi_M^4\|M\zeta\| \le \Psi_M^6\sqrt{\rho}\|\zeta\|_{\bar M}$), the w-terms through operator norms (d_w).

(d) Assembly. $\dot V \le -\varepsilon\|\zeta\|_{\bar M}^2 + r_\zeta\|\zeta\|_{\bar M}^2 + (c_1+\beta c_2)\|\zeta\|_{\bar M}\|w\| - \beta(k-d_w)\|w\|^2$. By Young, $(c_1+\beta c_2)\|\zeta\|_{\bar M}\|w\| \le \delta\varepsilon\|\zeta\|_{\bar M}^2 + (c_1+\beta c_2)^2\|w\|^2/(4\delta\varepsilon)$, and $(c_1+\beta c_2)^2 \le 2c_1^2 + 2\beta^2 c_2^2$. With $\beta = 2c_1^2/\delta$ the term $2c_1^2/(4\delta\varepsilon)$ equals $\beta k/4$, and (A) gives $2\beta^2 c_2^2/(4\delta\varepsilon) \le \beta k/4$, so with (B) the coefficient of $\|w\|^2$ is at most $-\beta(k-d_w-k/2) \le -\beta k/4$; the coefficient of $\|\zeta\|_{\bar M}^2$ is $-\varepsilon(1-\delta-r_\zeta/\varepsilon) = -\varepsilon c_\zeta$. This is the first inequality. Since $\frac12(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2 \le V \le \frac12(1+\varepsilon p_1+\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2$ by (D), and $\zeta^\top\bar M\zeta \ge \lambda_{\min}\|\zeta\|^2$, $\dot V \le -\gamma V$ follows, and the envelope with C from $\|e\| \le \|w\| + (\varepsilon\Psi_M + \varepsilon^2\Psi_D + \varepsilon^3\Psi_M^3)\|\zeta\|$ and its converse. □

Remark E.0′.1 (reading)**.**

- *What is sharp and what is not.* The exact rate of (LTV) is $\lambda_{\min}/k\cdot(1+O(k^{-2}))$: the first-order averaging of $\dot\zeta = -(1/k)M\zeta$ after the slaving gives $\lambda_{\min}/k$, and the second-order correction to the symmetric part of the averaged generator is $N_2 = 0$ (Table 12 measures $1-\text{exact}/(\lambda_{\min}/k) \approx 29/k^2$ on the ideal; on $\Psi = (\sin t, \cos t)$ the deviation is of order k^{-4}). Lemma E.0′ recovers this rate up to the factor $c_\zeta/(1+\varepsilon p_1+\varepsilon^2 p_2) \to 1-\delta$, and δ can be taken $\to 0$ as $k\to\infty$ (condition (A) allows $\delta \gtrsim 2\sqrt{\rho\rho_{DD}}/k^2$). The overshoot constant $C = O((\rho/\delta)^{1/2})$ is an artefact of the quadratic form, not of the dynamics.
- *Two regimes, two lemmas.* Lemma E.0 needs only $(\alpha, T, \Psi_M, \Psi_D)$ and gives the ω^{-2} law of the fast regime; its rate is quadratic in $\lambda_{\min}$. Lemma E.0′ needs k above the threshold $k_0 \approx (4\rho\rho_{DD}/\delta^2)^{1/4}$, set by the peak factors of Ψ and its two derivatives, and gives the sharp rate. On the regressors of Theorem E the derivative peak factors are large—the directions that are poorly excited on average are excited in short bursts, so ρ_{DD}/ρ is of the order of the fourth power of the effective bandwidth—and k_0 is in the hundreds there, while on $\Psi = (\sin t, \cos t)$ ($\rho = \rho_D = \rho_{DD} = 2$) it is about 10, set by (C) rather than (A). A third-order slaving would replace ρ_{DD} by the peak factor of $\dddot\Psi$ at one order higher in ε and lower k_0 further; this is not pursued.
- *Hypotheses.* Only the periodicity of Ψ is used (through the zero-mean antiderivatives; a non-periodic PE regressor would need a windowed $\bar M(t)$ and its derivative). Theorem E's x_m is periodic, so Ψ_G is. Theorem E's steps 4–5 go through with this V in place of Lemma E.0's (Corollary E.0′.2 below): the nonlinear rate becomes $\gamma/4 \approx (1-\delta)\lambda_{\min}/(2k)$, while the explicit ball radius carries the anisotropy $p'_{\text{lo}}/p'_{\text{hi}} \approx 1/\beta$ of the quadratic form and can be smaller than Lemma E.0's.

Numerics (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 4; reproduced in Table 12). The slow-adaptation ideal of Remark E.0a (table 2 of that file) at $g = 0.01$

($T = 20\pi$): $\lambda_{\min}(\bar{M}) = 2.16 \cdot 10^{-7}$, $\kappa(\bar{M}) = 7.8 \cdot 10^3$, $\Psi_M = 0.168$, $\Psi_D = 0.957$, $\Psi_{DD} = 8.39$, $T\Psi_M^2 = 1.78$; peak factors $\rho = 69.9$, $\rho_D = 3.7 \cdot 10^3$, $\rho_{DD} = 3.7 \cdot 10^5$ (the poorly excited directions are excited in bursts: $\rho_D/\rho \approx 53$ and $\rho_{DD}/\rho_D \approx 99$ are effective squared bandwidths); $\|N_2\| = 3 \cdot 10^{-21}$. The threshold of (A)–(D) is $k_0 = 226$ for $\delta = 0.2$ and 321 for $\delta = 0.1$, set by (A): $c_1 c_2 = 2.07$, 0.51, 0.13, 0.056, 0.032, 0.014 at $k = 50$, 100, 200, 300, 400, 600, the k^{-2} law of $\varepsilon^2\sqrt{\rho\rho_{DD}}$.

Table 12: Lemma E.0′ in the slow regime on the slow-adaptation ideal of Remark E.0a at $g = 0.01$ (`results/explicit_rate/table.md`, table 4).

k	exact Floquet rate	exact / ($\lambda_{\min}/k$)	guaranteed rate $\gamma/2$, $\delta = 0.2$ ($\delta = 0.1$)	exact / guaranteed	exact / Lemma E.0 (optimised ε)	$\sup \dot{V}/V$ on the grid, $-\gamma$ ($\delta = 0.1$)
300	$7.21 \cdot 10^{-10}$	0.9997	$5.70 \cdot 10^{-10}$ (below threshold)	1.26 (—)	$1.33 \cdot 10^5$	—
400	$5.41 \cdot 10^{-10}$	0.9998	$4.30 \cdot 10^{-10}$ ($4.84 \cdot 10^{-10}$)	1.26 (1.12)	$1.32 \cdot 10^5$	$-1.079 \cdot 10^{-9}$, $-9.68 \cdot 10^{-10}$
600	$3.61 \cdot 10^{-10}$	0.9999	$2.88 \cdot 10^{-10}$ ($3.24 \cdot 10^{-10}$)	1.25 (1.12)	$1.30 \cdot 10^5$	$-7.20 \cdot 10^{-10}$, $-6.47 \cdot 10^{-10}$

Above the threshold the guaranteed rate is within 26% ($\delta = 0.2$) and 12% ($\delta = 0.1$) of the exact rate—the factor $1/(1-\delta)$ of the lemma, with $c_\zeta = 0.80$ and 0.90—while Lemma E.0's optimised rate is $1.3 \cdot 10^5$ times smaller; the certificate $\sup \dot{V}/V \le -\gamma$ holds on the 128-point grid in every row where the conditions hold, with $\sup \dot{V}/V = -1.078 \cdot 10^{-9} \approx -2\lambda_{\min}/k$ at $k = 400$ (attained in the slow direction at $w = 0$). The exact rate deviates from $\lambda_{\min}/k$ by 1.15%, 0.29%, 0.07% at $k = 50$, 100, 200: the k^{-2} law that $N_2 = 0$ predicts. The overshoot constant is $C = \sqrt{\beta} \cdot (1 + O(1/k)) = 27$ ($\delta = 0.2$) and 38 ($\delta = 0.1$). Below k_0 the lemma makes no claim; Lemma E.0 does (Remark E.0a; table 2 of that file).

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos t)$ the computed $\lambda_{\min}$, ρ, ρ_D, ρ_{DD} equal 1/2, 2, 2, 2, $N_2 = 0$ to 10^{-12}, the threshold of (A)–(D) is below 12 for $\delta = 0.2$, the certificate $\sup \dot{V}/V \le -\gamma$ holds at $k = 20$, 50, 200, and the guaranteed rate is below the exact Floquet rate and within a factor 2 of it, the ratio decreasing towards $1/(1-\delta)$ (`test_lemma_E0prime_sinusoidal_regressor_certificate_and_sharpness`); on the slow-adaptation ideal of the Theorem E tests at $g = 0.01$ and $k = 400$, $\delta = 0.2$: $\kappa(\bar{M}) > 10^3$, the conditions hold, the certificate holds on a 128-point grid, the guaranteed rate is within a factor 1.5 of the exact rate and exceeds Lemma E.0's optimised rate by more than 10^4 (`test_lemma_E0prime_pays_for_ill_conditioning_once_on_the_theorem_E_ideal`).

Corollary E.0′.2 (Theorem E in the slow regime)**.** *Under the hypotheses of Theorem E, suppose the regressor Ψ_G of its step 3 satisfies the conditions (A)–(D) of Lemma E.0′ at $k = k_e$ for some δ, with constants $(\beta, \gamma, p_1, p_2)$ and $L_M := \varepsilon\Psi_{G,M} + \varepsilon^2\Psi_{G,D} + \varepsilon^3\Psi_{G,M}^3$ ($\varepsilon = 1/k_e$), and that $\beta L_M^2 < 1 - \varepsilon p_1 - \varepsilon^2 p_2$. Put*

$$p'_{\mathrm{lo}} := \tfrac{1}{2}\min\{\beta/2,\ 1 - \varepsilon p_1 - \varepsilon^2 p_2 - \beta L_M^2\},$$
$$p'_{\mathrm{hi}} := \tfrac{1}{2}\max\{\beta(1+L_M),\ 1 + \varepsilon p_1 + \varepsilon^2 p_2 + \beta L_M(1+L_M)\},$$
$$c'_0 := \min\Big\{\Big[\frac{\gamma p'_{\mathrm{lo}}}{4(p'_{\mathrm{hi}})^{1/2}\tilde{C}_R}\Big]^2,\ \frac{p'_{\mathrm{lo}} r^2}{\|S^{-1}\|^2}\Big\}, \qquad r'_0 := \Big(\frac{c'_0}{p'_{\mathrm{hi}}}\Big)^{1/2}, \qquad \lambda' := \frac{\gamma}{4}, \qquad C' := \Big(\frac{2p'_{\mathrm{hi}}}{p'_{\mathrm{lo}}}\Big)^{1/2},$$

with r, $\tilde{C}_R$, S as in Theorem E. Then the conclusions of Theorem E hold with (c_0, λ, C) replaced by $((r'_0)^2/2, \lambda', C')$. In the slow regime $\lambda' = (1-\delta)(1 - \varepsilon p_1 + O(k_e^{-2}))\lambda_{\min}(\bar{M}_G)/(2k_e)$, $\bar{M}_G := \langle \Psi_G^\top \Psi_G \rangle$: half the exact rate of the linearised loop, up to the factor $1-\delta$ and the first-order deficit εp_1.

Proof. Write $V := \frac{1}{2} w^\top \mathbf{H}(t) w$ for Lemma E.0′'s V in the coordinates $w = (e, \zeta)$ of step 3:

$$\mathbf{H} = \begin{bmatrix} \beta I & -\beta L \\ -\beta L^\top & H_\zeta + \beta L^\top L \end{bmatrix}$$

with $e_2 = L(t)\zeta$, $\|L\| \le L_M$ and $H_\zeta := I + \varepsilon P_1 + \varepsilon^2 P_2$. Then $w^\top \mathbf{H} w = \beta\|e - L\zeta\|^2 + \zeta^\top H_\zeta \zeta$, and $\|e - L\zeta\|^2 \le (1 + L_M)(\|e\|^2 + L_M\|\zeta\|^2)$, $\|e - L\zeta\|^2 \ge \frac{1}{2}\|e\|^2 - L_M^2\|\zeta\|^2$ give $2p'_{\mathrm{lo}}\|w\|^2 \le w^\top \mathbf{H} w \le 2p'_{\mathrm{hi}}\|w\|^2$. Along the nonlinear loop $\dot w = [A_0 + B(t)]w + \tilde R(t, w)$ with $\|\tilde R\| \le \tilde C_R\|w\|^2$ on $\{\|w\| < r/\|S^{-1}\|\}$ (step 4), so with Lemma E.0′,

$$\dot V \le -\gamma V + \langle \mathbf{H} w, \tilde R\rangle \le -\gamma V + \|\mathbf{H}^{1/2} w\|\,\|\mathbf{H}^{1/2}\|\,\tilde C_R\|w\|^2 \le -\gamma V + \frac{2(p'_{\mathrm{hi}})^{1/2}\tilde C_R}{p'_{\mathrm{lo}}} V^{3/2},$$

using $\|\mathbf{H}^{1/2} w\| = (2V)^{1/2}$, $\|\mathbf{H}^{1/2}\| = (2p'_{\mathrm{hi}})^{1/2}$ and $\|w\|^2 \le V/p'_{\mathrm{lo}}$. Hence $\dot V \le -(\gamma/2)V$ wherever $V^{1/2} \le \gamma p'_{\mathrm{lo}}/(4(p'_{\mathrm{hi}})^{1/2}\tilde C_R)$, and the sublevel set $\{V \le c'_0\}$—inside the ball where $\tilde R$ is bounded, by the second term of c'_0—is forward invariant with $V(t) \le e^{-(\gamma/2)(t-t_0)}V(t_0)$ on it. It contains $\{\|w\| \le r'_0\}$ because $V \le p'_{\mathrm{hi}}\|w\|^2$, and $\|w(t)\| \le (p'_{\mathrm{hi}}/p'_{\mathrm{lo}})^{1/2} e^{-(\gamma/4)(t-t_0)}\|w(t_0)\|$ follows. Step 5 of Theorem E is unchanged: $\|w\|^2 = 2V_\pi$, so $V_{\mathrm{OT}}(t_0) \le (r'_0)^2/2$ puts the state in the ball, $\|e\| + W \le \sqrt{2}\|w\|$ gives C', and uniqueness and the limits follow as there. The asymptotics of λ' are those of γ in Lemma E.0′. □

Numerics (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 5; reproduced in Table 13). Remark E.1's single-kernel instance ($m = n = 1$, $\vartheta^* = (1.5, 0.3, 0.8)$, $G = 0.3 \cdot I$, $x_m = 1.5 \sin t$) at $k_e = 40$: $\lambda_{\min}(\bar M_G) = 5.41 \cdot 10^{-2}$, $\kappa(\bar M_G) = 6.7$, peak factors $\rho = 4.8$, $\rho_D = 30.3$, $\rho_{DD} = 461$, threshold $k_0 = 24.4$ ($\delta = 0.2$) and 33.5 ($\delta = 0.1$); $L_J = 4.62$ estimated from samples; exact linear rate $1.35 \cdot 10^{-3}$.

Table 13: Theorem E in the slow regime on Remark E.1's single-kernel instance at $k_e = 40$ (`results/explicit_rate/table.md`, table 5).

Lyapunov function	explicit ball radius r_0 (w-coordinates)	nonlinear rate λ	overshoot C	exact linear rate / λ
Lemma E.0, optimised ε (Theorem E)	$6.15 \cdot 10^{-6}$	$2.63 \cdot 10^{-5}$	1.43	51
Lemma E.0′, $\delta = 0.2$ (this corollary)	$1.53 \cdot 10^{-6}$	$4.53 \cdot 10^{-4}$	11.8	3.0
Lemma E.0′, $\delta = 0.1$ (this corollary)	$8.32 \cdot 10^{-7}$	$5.16 \cdot 10^{-4}$	17.1	2.6

On this well-conditioned regressor ($\kappa \approx 7$) Lemma E.0 is only an order of magnitude conservative, and the corollary trades the certified ball for the rate: λ' is within a factor 3 of the exact linear rate (the factor 2 of the Lyapunov argument times $1/(1-\delta)$), while r'_0 is four times smaller than Theorem E's r_0, carrying the anisotropy $p'_{\mathrm{lo}}/p'_{\mathrm{hi}} \approx 1/\beta$ of the quadratic form. On ill-conditioned regressors the rate improves by the factor of Remark E.0a (Table 12); the radius comparison was not computed there.

Checked by: `tests/test_excitation.py`: on the single-kernel instance at $k_e = 40$, $\delta = 0.2$, from a start with $\|w(0)\| \le r'_0$, the implemented closed loop satisfies $\|w(t)\| \le C' e^{-\lambda' t}\|w(0)\|$ at every sample over 1.5 e-folds of the explicit envelope ($\mathrm{d}t = 0.005$: the fixed-step loop has a discretisation floor $\approx 6 \cdot 10^{-6}$ at $\mathrm{d}t = 0.02$, above the explicit ball), with λ' more than ten times Theorem E's rate from Lemma E.0 and within a factor 4 of the exact linear rate (`test_corollary_E0prime_2_envelope_on_the_implemented_loop`).

Corollary E.0′.3 (the stationary parameter spread of the dithered linearised loop)**.** *Under the hypotheses of Lemma E.0′ (conditions (A)–(D) at (k, δ)), consider* (LTV) *with parameter dither,*

$$\dot e = -ke - \Psi(t)\zeta, \qquad \mathrm{d}\zeta = \Psi(t)^\top e\,\mathrm{d}t + \Sigma\,\mathrm{d}B_t, \tag{LTV$_\tau$}$$

with $\Sigma \in \mathbb{R}^{q\times q'}$ constant and B a standard Brownian motion. Let V be the strict Lyapunov function of Lemma E.0′ and $\mathbf{H}(t)$ its quadratic form ($V = \frac{1}{2}x^\top \mathbf{H}(t)x$, $x = (e,\zeta)$), $L(t) := -\varepsilon\Psi + \varepsilon^2\dot\Psi - \varepsilon^3\Psi M$ the slaving map ($e_2 = L(t)\zeta$), $L_M := \varepsilon\Psi_M + \varepsilon^2\Psi_D + \varepsilon^3\Psi_M^3 \geq \sup\|L\|$, and

$$\nu(t) := \tfrac{1}{2}\operatorname{tr}\big(\Sigma^\top \mathbf{H}_{\zeta\zeta}(t)\Sigma\big) = \tfrac{1}{2}\operatorname{tr}\big(\Sigma^\top(I + \varepsilon P_1 + \varepsilon^2 P_2 + \beta L^\top L)\Sigma\big),$$
$$\bar\nu := \sup_t \nu(t) \leq \tfrac{1}{2}\|\Sigma\|_F^2\big(1 + \varepsilon p_1 + \varepsilon^2 p_2 + \beta L_M^2\big).$$

Then, for every initial law with finite second moment,

$$\limsup_{t\to\infty} \mathbb{E}V(t,x_t) \leq \frac{\bar\nu}{\gamma}, \qquad \limsup_{t\to\infty} \mathbb{E}\|\zeta_t\|^2 \leq \frac{2\bar\nu}{\gamma(1-\varepsilon p_1 - \varepsilon^2 p_2)}, \qquad \limsup_{t\to\infty} \mathbb{E}\|w_t\|^2 \leq \frac{2\bar\nu}{\gamma\beta},$$

with γ the rate of Lemma E.0′; in the slow regime $\gamma = 2c_\zeta\lambda_{\min}(\bar M)/(k(1+\varepsilon p_1 + \varepsilon^2 p_2))$, so that

$$\limsup \mathbb{E}\|\zeta_t\|^2 \leq \frac{k\|\Sigma\|_F^2\,(1+\varepsilon p_1+\varepsilon^2 p_2)(1+\varepsilon p_1+\varepsilon^2 p_2+\beta L_M^2)}{2c_\zeta(1-\varepsilon p_1-\varepsilon^2 p_2)\lambda_{\min}(\bar M)} = (1+O(\varepsilon))\,\frac{k\|\Sigma\|_F^2}{2(1-\delta)\lambda_{\min}(\bar M)}.$$

For comparison, the averaged Langevin dynamics $\dot\zeta = -\frac{1}{k}\bar M\zeta + \Sigma\dot B$ has the exact stationary covariance $\bar C = kX$ with $\bar M X + X\bar M = \Sigma\Sigma^\top$, so $\mathbb{E}\|\zeta\|^2 = k\operatorname{tr}X$, equal to $\frac{k}{2}\operatorname{tr}(\bar M^{-1}\Sigma\Sigma^\top)$ when $\Sigma\Sigma^\top$ commutes with $\bar M$: the bound is within $(1+O(\varepsilon))/(1-\delta)$ of it when $\bar M$ is isotropic, and pays the condition number of $\bar M$ in the trace otherwise (it replaces every $1/\lambda_i$ by $1/\lambda_{\min}$), as a scalar Lyapunov function must.

Status. Proved (Itô's formula and the pointwise inequality of Lemma E.0′). Checked (`exp_explicit_rate.py`, table 6; `tests/test_excitation.py`) against the exact periodic stationary covariance of (LTV_τ), computed from the periodic Lyapunov equation.

Proof. $V(t,x) = \frac{1}{2}x^\top\mathbf{H}(t)x$ is C^1 in t and quadratic in x, so by Itô's formula along (LTV_τ), with $\tilde\Sigma := [0;\Sigma] \in \mathbb{R}^{(n+q)\times q'}$,

$$\begin{aligned}\mathrm{d}V &= \big[\partial_t V + \nabla_x V\cdot A(t)x\big]\,\mathrm{d}t + \nabla_x V\cdot\tilde\Sigma\,\mathrm{d}B_t + \tfrac{1}{2}\operatorname{tr}\big(\tilde\Sigma^\top\mathbf{H}(t)\tilde\Sigma\big)\,\mathrm{d}t\\ &\leq \big[-\gamma V + \nu(t)\big]\,\mathrm{d}t + \nabla_x V\cdot\tilde\Sigma\,\mathrm{d}B_t,\end{aligned}$$

the inequality being Lemma E.0′'s $\dot V \leq -\gamma V$ pointwise in (t,x), and $\tilde\Sigma^\top\mathbf{H}\tilde\Sigma = \Sigma^\top\mathbf{H}_{\zeta\zeta}\Sigma$ with $\mathbf{H}_{\zeta\zeta} = I+\varepsilon P_1+\varepsilon^2P_2+\beta L^\top L$ from the block form of $\mathbf{H}$ (Corollary E.0′.2). The stochastic integral is a martingale because its integrand is linear in x and the linear SDE has bounded second moments (Gaussian with a bounded covariance), so $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}V \leq -\gamma\mathbb{E}V + \bar\nu$ and $\mathbb{E}V(t) \leq e^{-\gamma t}\mathbb{E}V(0) + \bar\nu/\gamma$. The bounds on $\mathbb{E}\|\zeta\|^2$ and $\mathbb{E}\|w\|^2$ follow from $V \geq \frac{1}{2}(1-\varepsilon p_1-\varepsilon^2p_2)\|\zeta\|^2$ and $V \geq \frac{\beta}{2}\|w\|^2$; the estimate of $\bar\nu$ uses $\|P_1\| \leq p_1$, $\|P_2\| \leq p_2$, $\|L\| \leq L_M$. The averaged covariance is the stationary solution of the Lyapunov equation of the time-invariant averaged system. □

Numerics (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 6; reproduced in Table 14). $\Sigma = (2\cdot10^{-3})^{1/2}I$; the exact stationary covariance from the periodic Lyapunov equation (RK4 with $k\cdot\mathrm{d}t \leq 1/4$; the period mean and the maximum of $\mathbb{E}\|\zeta\|^2$ coincide to four digits), the averaged value $k\cdot\operatorname{tr}X$, and the bound:

Table 14: Corollary E.0′.3: the explicit bound on the stationary parameter spread of the dithered (LTV) against the exact periodic stationary covariance and the averaged value (`results/explicit_rate/table.md`, table 6).

regressor ($\kappa(\bar{M})$)	k	δ	exact $\mathbb{E}\|\zeta\|^2$	averaged $k \cdot \operatorname{tr} X$	bound	bound / exact	$1/(1-\delta)$
Theorem E ideal of table 4 ($7.8 \cdot 10^3$)	300	0.2	$1.509 \cdot 10^6$	$1.509 \cdot 10^6$	$2.64 \cdot 10^7$	17.5	1.25
	400	0.2 / 0.1	$2.012 \cdot 10^6$	$2.012 \cdot 10^6$	$3.51 \cdot 10^7$ / $3.11 \cdot 10^7$	17.4 / 15.5	1.25 / 1.11
	600	0.2 / 0.1	$3.018 \cdot 10^6$	$3.018 \cdot 10^6$	$5.23 \cdot 10^7$ / $4.65 \cdot 10^7$	17.3 / 15.4	1.25 / 1.11
$\Psi = (\sin t, \cos t)$ (1)	20	0.2 / 0.1	$8.025 \cdot 10^{-2}$	$8.000 \cdot 10^{-2}$	0.216 / 0.202	2.69 / 2.52	1.25 / 1.11
	50	0.2 / 0.1	$2.001 \cdot 10^{-1}$	$2.000 \cdot 10^{-1}$	0.315 / 0.282	1.58 / 1.41	1.25 / 1.11
	200	0.2 / 0.1	$8.000 \cdot 10^{-1}$	$8.000 \cdot 10^{-1}$	1.051 / 0.935	1.31 / 1.17	1.25 / 1.11

The exact stationary spread is the averaged Langevin value up to $3 \cdot 10^{-4}$ (ideal) and $1.25/k^2$ (sinusoid): on these instances, where Σ is isotropic so that $\Sigma\Sigma^\top$ commutes with $\bar{M}$, the averaging prediction of the dithered loop's stationary covariance is exact to $O(k^{-2})$. When $\Sigma\Sigma^\top$ does not commute with $\bar{M}$ the deviation is $O(1/k)$ (referee round 7 measured $\pm 0.25/k$ on two-dimensional sinusoids with anisotropic Σ, the bound still holding with ratios 2.7–5.1). On the isotropic sinusoid the bound exceeds the exact value by a factor that falls toward $1/(1-\delta)$ (2.7, 1.6, 1.3 at $k = 20$, 50, 200 for $\delta = 0.2$; the excess is the $O(1/k)$ of the factors $(1 \pm \varepsilon p_1) \dots$ and βL_M^2). On the ideal, where one direction is excited $7.8 \cdot 10^3$ times less than the others, the bound is 15–18 times the exact value: it charges every one of the $q = 15$ directions the price of the worst, $(q/\lambda_{\min})/\sum_i (1/\lambda_i) \approx q = 15$ (up to $1/(1-\delta)$), the trace-level price of a scalar Lyapunov function. The exact $\mathbb{E}\|e\|^2$ is $\|\Sigma\|_F^2/(2k) = \operatorname{tr}(\bar{M}X)/k$ to five digits on both instances (the error rides on its slow manifold $e \approx \Psi\zeta/k$, and $\operatorname{tr}(\bar{M}X) = \frac{1}{2}\|\Sigma\|_F^2$ by the trace of the Lyapunov equation), while the bound on $\mathbb{E}\|w\|^2$ is coarse (factor $1/\beta$): on the sinusoid the exact stationary $\mathbb{E}\|w\|^2 = \mathbb{E}\|e - L\zeta\|^2$, computed from the periodic covariance and the slaving map $L = -\varepsilon\Psi + \varepsilon^2\dot{\Psi} - \varepsilon^3\Psi M$, is more than three orders of magnitude below the bound (asserted in the test below).

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos t)$ with $\Sigma = (2 \cdot 10^{-3})^{1/2} I$, the explicit bound dominates the exact period mean and the exact maximum of $\mathbb{E}\|\zeta\|^2$ at $k = 20$, 50, 200 for $\delta = 0.1$ and 0.2, its ratio to the exact value decreases toward $1/(1-\delta)$ (within 6% at $k = 200$), the grid value of $\bar{\nu}$ lies below its explicit bound, and the exact value agrees with the averaged $k \cdot \operatorname{tr} X$ within $2/k^2$ (`test_corollary_E0prime_3_dither_covariance_bound_against_the_periodic_lyapunov_solution`).

Lemma E.0″ (the slow regime for non-periodic persistently exciting regressors)**.** *Let $\Psi\colon [0,\infty) \to \mathbb{R}^{n\times q}$ be C^2 with $\|\Psi\| \le \Psi_M$, $\|\dot{\Psi}\| \le \Psi_D$, $\|\ddot{\Psi}\| \le \Psi_{DD}$, $M := \Psi^\top\Psi$, and fix a window $T > 0$. For a bounded continuous X let*

$$\langle X\rangle_T(t) := \frac{1}{T}\int_0^\infty e^{-s/T} X(t+s)\,\mathrm{d}s$$

(the forward exponential average; $T\frac{\mathrm{d}}{\mathrm{d}t}\langle X\rangle_T = \langle X\rangle_T - X$),

and $\bar{M} := \langle M\rangle_T$. Assume the exponentially weighted PE condition

$$\bar{M}(t) \succeq \lambda I_q \quad \text{for all } t \ge 0, \quad \text{with } \lambda > 0, \tag{PE$_T$}$$

which is equivalent to classical PE ($\int_t^{t+T} M \succeq \alpha I$ for all t gives (PE$_T$) *with $\lambda \ge e^{-1}\alpha/T$, and* (PE$_T$) *gives PE over windows $T' > T\log(\Lambda/\lambda)$ with level $T(\lambda - e^{-T'/T}\Lambda)$; proof below). Let $\Lambda := \sup_t \lambda_{\max}(\bar{M}(t)) \le \Psi_M^2$. Fix symmetric matrices $\hat{M}$, $\hat{F}$ (any; the long-time means of $\bar{M}$*

and $\bar F$ minimise the time-averaged squares of $\|\bar M-\hat M\|$, $\|\bar F-\hat F\|$, not necessarily the suprema below) and define

$$P_1:=-2T(\bar M-\hat M)\ \ (so\ \dot P_1=2(M-\bar M)),\qquad F:=P_1M+MP_1,$$
$$\bar F:=\langle F\rangle_T,\qquad S:=-T(\bar F-\hat F)\ \ (\dot S=F-\bar F),$$
$$P_2:=-(M-\bar M)+S,\qquad \bar N_2:=\tfrac12(\bar F-\dot{\bar M})=\tfrac12\big(\bar F-(\bar M-M)/T\big),$$

and e_2, w, V exactly as in Lemma E.0′ ($V=\frac12\zeta^\top(I+\varepsilon P_1+\varepsilon^2P_2)\zeta+\frac{\beta}{2}\|w\|^2$, $\varepsilon=1/k$). The constants are those of Lemma E.0′ with the period mean replaced by the exponential average, all of them computable from the regressor: the time-local peak factors

$$\rho:=\sup_{t,v}\frac{\|\Psi(t)v\|^2}{v^\top\bar M(t)v},\qquad \rho_D:=\sup\frac{\|\dot\Psi(t)v\|^2}{v^\top\bar M(t)v},\qquad \rho_{DD}:=\sup\frac{\|\ddot\Psi(t)v\|^2}{v^\top\bar M(t)v}$$
$$(all\ \le\Psi_\cdot^2/\lambda),$$

the operator and per-direction norms

$$p_1:=\sup_t\|P_1(t)\|=2T\sup\|\bar M-\hat M\|,\qquad p_2:=\sup_t\|P_2(t)\|\le\Psi_M^2+T\sup\|\bar F-\hat F\|,$$
$$\pi_1:=\sup_{t,v}\frac{\|P_1(t)v\|}{\|v\|_{\bar M(t)}}\le 2T\big(\sqrt\Lambda+\|\hat M\|/\sqrt\lambda\big),$$
$$\pi_2:=\sup_{t,v}\frac{\|P_2(t)v\|}{\|v\|_{\bar M(t)}}\le\Psi_M\sqrt\rho+\sqrt\Lambda+T\big(2p_1\Psi_M^2+\|\hat F\|\big)/\sqrt\lambda,$$
$$n_2:=\sup_t\|\bar N_2(t)\|\le\tfrac12\big(\sup\|F\|+2\Psi_M^2/T\big)\le p_1\Psi_M^2+\Psi_M^2/T,$$

and, with these π_1, π_2, p_1, p_2, the quantities c_1, c_2, d_w, r_ζ and $\beta:=2c_1^2/\delta$ of Lemma E.0′ (same formulas), together with

$$\delta_2:=\varepsilon n_2/\lambda,\qquad c_\zeta:=1-\delta-\delta_2-r_\zeta/\varepsilon.$$

Fix $\delta\in(0,1)$. If (A) $c_1c_2\le\delta/2$, (B) $d_w\le k/4$, (C) $c_\zeta>0$ *and* (D) $\varepsilon p_1+\varepsilon^2p_2<1$, *then along* (LTV), *for all (e,ζ,t),*

$$\dot V\le-\varepsilon c_\zeta\,\zeta^\top\bar M(t)\zeta-\frac{\beta k}{4}\|w\|^2\le-\gamma V,\qquad \gamma:=\min\Big\{\frac{2c_\zeta\lambda}{k(1+\varepsilon p_1+\varepsilon^2p_2)},\ \frac{k}{2}\Big\},$$

and $\|(e(t),\zeta(t))\|\le Ce^{-\gamma(t-t_0)/2}\|(e(t_0),\zeta(t_0))\|$ with the C of Lemma E.0′. In the slow regime the guaranteed rate is

$$\gamma/2=(1-\delta-\delta_2)\big(1-\varepsilon p_1+O(k^{-2})\big)\,\lambda/k,\qquad \lambda=\inf_t\lambda_{\min}\big(\langle M\rangle_T(t)\big):$$

the exponentially weighted PE level of the window divided by k, up to the factor $1-\delta-\delta_2$, where $\delta_2=O(1/k)$ is the price of the residual second-order term (zero for a periodic Ψ with $\hat M=\langle M\rangle$ in the limit $T\to\infty$, where Lemma E.0′ applies directly).

Status. Proved (the proof of Lemma E.0′ with the changes listed below). Checked (`exp_pe_general.py`, `tests/test_excitation.py`) on a quasi-periodic regressor and on a regressor driven by a Lorenz orbit, where no periodic construction applies: the conditions hold from $k=20$ on for the quasi-periodic regressor and from $k=100$ on for the chaotic one (the derivative peak factors of a chaotic orbit are large), the certificate $\sup\dot V/V\le-\gamma$ holds on the time grid wherever they hold, and the guaranteed rate is within a factor 1.3–2.5 of the exact decay rate on the quasi-periodic regressor ($k=50$–200) and, with the window $T=20$, within a factor 3.5–5 on the chaotic one (shorter windows pay more: 7.5–11 at $T=10$ and 14–138 at $T=5$), where the worst exponential window is three times less exciting than the long-time average. Referee round 7 (`review/REVIEW_round7_E0p3_E0dp.md`) confirmed the proof and the code and found no counterexample on an attack set of barely exciting, bursty, near-resonant and chaotic regressors.

Proof. Follow the proof of Lemma E.0′ with $\bar{M}$, P_1, S, P_2 as defined here. (a) is unchanged (it does not involve the averages). In (b), $\dot{H} = 2\varepsilon(M-\bar{M})+\varepsilon^2(F-\bar{F}-\dot{M}+\dot{\bar{M}})$ exactly, by $\dot{P}_1 = 2(M-\bar{M})$ and $\dot{P}_2 = -\dot{M} + \dot{\bar{M}} + \dot{S} = -\dot{M} + \dot{\bar{M}} + F - \bar{F}$; the order-$\varepsilon$ terms give $-\zeta^\top \bar{M}\zeta$ as before, and the order-ε^2 terms give $-\zeta^\top P_1 M\zeta + \frac{1}{2}\zeta^\top \dot{M}\zeta + \frac{1}{2}\zeta^\top(F - \bar{F} - \dot{M} + \dot{\bar{M}})\zeta = \frac{1}{2}\zeta^\top(\dot{\bar{M}} - \bar{F})\zeta = -\zeta^\top \bar{N}_2\zeta$, since $\zeta^\top P_1 M\zeta = \frac{1}{2}\zeta^\top F\zeta$: the periodic cancellation $N_2 = 0$ is replaced by the residual $-\varepsilon^2\zeta^\top\bar{N}_2\zeta$, and R_ζ has the same form, so $\frac{\mathrm{d}}{\mathrm{d}t}[\frac{1}{2}\zeta^\top H\zeta] = -\varepsilon\zeta^\top(\bar{M} + \varepsilon\bar{N}_2)\zeta + R_\zeta + \zeta^\top H\Psi^\top w$. In (c), every per-direction bound is taken in the time-local norm $\|v\|_{\bar{M}(t)}$: $\|\Psi(t)v\| \le \sqrt{\rho}\|v\|_{\bar{M}(t)}$ and likewise for $\dot{\Psi}$, $\ddot{\Psi}$ by the definition of the peak factors; $\|M(t)v\| \le \Psi_M\sqrt{\rho}\|v\|_{\bar{M}(t)}$; $\|\bar{M}(t)v\| \le \sqrt{\Lambda}\|v\|_{\bar{M}(t)}$; $\|P_1(t)v\| = 2T\|(\bar{M}(t) - \hat{M})v\| \le 2T(\sqrt{\Lambda} + \|\hat{M}\|/\sqrt{\lambda})\|v\|_{\bar{M}(t)}$, using $\sqrt{\lambda}\|v\| \le \|v\|_{\bar{M}(t)}$ (this is where the exponential average pays off: $P_1(t)$ is a function of $\bar{M}(t)$ alone, with no integral over other times); for $\bar{F}(t)v$ and $S(t)v$, which do involve other times $r \ge t$, $\|F(r)v\| \le \|P_1(r)\|\|M(r)v\| + \|M(r)\|\|P_1(r)v\| \le 2p_1\Psi_M^2\|v\| \le (2p_1\Psi_M^2/\sqrt{\lambda})\|v\|_{\bar{M}(t)}$, an average of such bounds bounds $\|\bar{F}(t)v\|$, and $\|S(t)v\| \le T(\|\bar{F}(t)v\| + \|\hat{F}v\|)$, whence the bound on π_2; the stated bounds on p_2 and n_2 are $\|P_2\| \le \|M - \bar{M}\| + \|S\|$ and $\|\bar{N}_2\| \le \frac{1}{2}(\|\bar{F}\| + \|\dot{\bar{M}}\|)$ with $\|\dot{\bar{M}}\| = \|\bar{M} - M\|/T \le 2\Psi_M^2/T$ and $\|\bar{F}\| \le \sup\|F\| \le 2p_1\Psi_M^2$. The bounds $|R_\zeta| \le r_\zeta\|\zeta\|_{\bar{M}}^2$, $|\zeta^\top H\Psi^\top w| \le c_1\|\zeta\|_{\bar{M}}\|w\|$ and $\beta w^\top\dot{w} \le \beta[-(k-d_w)\|w\|^2 + c_2\|\zeta\|_{\bar{M}}\|w\|]$ follow with the same formulas (the suprema over v at the current time t of the norms that appear). In (d), the extra term is bounded by $\varepsilon^2|\zeta^\top\bar{N}_2\zeta| \le \varepsilon^2 n_2\|\zeta\|^2 \le (\varepsilon^2 n_2/\lambda)\zeta^\top\bar{M}\zeta = \varepsilon\delta_2\zeta^\top\bar{M}\zeta$, which turns $1-\delta-r_\zeta/\varepsilon$ into $c_\zeta = 1-\delta-\delta_2-r_\zeta/\varepsilon$; the rest of the assembly, the sandwich $\frac{1}{2}(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2 \le V \le \frac{1}{2}(1+\varepsilon p_1+\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2$ and $\zeta^\top\bar{M}(t)\zeta \ge \lambda\|\zeta\|^2$ give $\dot{V} \le -\gamma V$. The equivalence of (PE$_T$) with classical PE: if $\int_t^{t+T} M \succeq \alpha I$ then $\bar{M}(t) \succeq \frac{1}{T}\int_0^T e^{-s/T}M(t+s)\,\mathrm{d}s \succeq \frac{e^{-1}}{T}\int_t^{t+T} M \succeq (e^{-1}\alpha/T)I$; conversely (PE$_T$) gives, for every $T' > 0$,

$$\int_t^{t+T'} M \succeq \int_t^{t+T'} Me^{-(r-t)/T}\,\mathrm{d}r \succeq T\big(\bar{M}(t) - e^{-T'/T}\bar{M}(t+T')\big) \succeq T(\lambda - e^{-T'/T}\Lambda)I,$$

positive for $T' > T\log(\Lambda/\lambda)$ (the identity $\int_t^{t+T'} e^{-(r-t)/T}M(r)\,\mathrm{d}r = T[\bar{M}(t) - e^{-T'/T}\bar{M}(t+T')]$ follows from the definition of the exponential average). □

Remark E.0″.1 (reading)**.**

- *What changes and what does not.* The construction, the conditions and the rate formula are those of Lemma E.0′; the period mean becomes the exponential average over a window, the peak factors become time-local, and one new term appears: the residual $\bar{N}_2$, whose price $\delta_2 = \varepsilon n_2/\lambda$ vanishes like $1/k$. For a periodic regressor the exponential average with a finite window keeps residual oscillations of relative size $O(\text{period}/T)$, so Lemma E.0′ is sharper there; Lemma E.0″ is the version for references that are not periodic — quasi-periodic (several incommensurate frequencies) or chaotic (a reference along a strange attractor, as in the Lorenz-tracking experiments of Part I), where only PE is available.
- *The window is a design parameter.* The lemma holds for every $T > 0$; the guaranteed rate is $(1-\delta-\delta_2)\lambda_T/k$ with $\lambda_T = \inf\lambda_{\min}(\langle M\rangle_T)$, which increases toward the long-time excitation level as T grows, against $p_1 \propto T$, $n_2 \propto T$ and the threshold in k, which grow with T. The numerics show the optimum at a window of a few characteristic times of the regressor.
- *Exact rate.* For a non-periodic regressor the exact rate is the top Lyapunov exponent of (LTV), which the certificate does not need: the lemma is a pointwise inequality in (t, e, ζ), checked on a grid in t by the same generalised eigenvalue problem as Lemma E.0′.

Numerics (`exp_pe_general.py` → `results/explicit_rate/general_pe.md`). Two regressors that are not periodic, on $[0, 300]$ (grid step 0.005; the exponential averages formed from

an extended grid; the certificate $\sup \dot{V}/V$ over 400 times and all states; the exact rate the top Lyapunov exponent of (LTV) over 200 time units), with the measured constants (the grid suprema of $\|P_1\|$, $\|P_2\|$, the per-direction norms and $\|\bar{N}_2\|$, computable constants of the regressor like the peak factors).

Quasi-periodic $\Psi = (\sin t, \cos\sqrt{2}\,t)$ ($q = 2$; the long-time mean of M is $\frac{1}{2}I$, and the exact decay rate is $\lambda_\infty/k = 0.5/k$ to within 1%, 0.6% at $k = 20$ and 0.8% at $k = 400$): the window average has $\lambda_T = 0.297$, 0.391, 0.444 for $T = 2\pi$, 4π, 8π ($\rho = 4.0$, $\rho_D \approx 6.1$–6.4, $\rho_{DD} \approx 10.0$, $p_1 \approx 2.9$, $p_2 = 8.0$–9.2, $\pi_1 = 4.3$–4.7, $\pi_2 = 11.6$–13.0, $\|\bar{N}_2\| = 0.71$, 0.35, 0.17); the conditions hold from $k = 20$ on for both δ, the certificate holds at every row, and (Table 15)

Table 15: Lemma E.0″ on the quasi-periodic regressor $\Psi = (\sin t, \cos\sqrt{2}\,t)$: the guaranteed rate against the exact decay rate for two windows (`results/explicit_rate/general_pe.md`).

T	k	δ_2	guaranteed $\gamma/2$ ($\delta = 0.2$ / 0.1)	exact	exact/guaranteed	λ_T/k
4π	20	0.045	0.0100 / 0.0117	0.0249	2.49 / 2.13	0.0195
4π	50	0.018	0.00557 / 0.00631	0.00995	1.78 / 1.58	0.00781
4π	200	0.004	0.00153 / 0.00172	0.00249	1.63 / 1.45	0.00195
8π	20	0.020	0.0118 / 0.0137	0.0249	2.10 / 1.81	0.0222
8π	200	0.002	0.00174 / 0.00196	0.00249	1.43 / 1.27	0.00222

($T = 2\pi$: 3.69 → 2.16 for $\delta = 0.2$ and 3.10 → 1.92 for $\delta = 0.1$ from $k = 20$ to 200.) The ratio exact/guaranteed is the product of the window loss λ_∞/λ_T (1.69, 1.28, 1.13 for the three windows), the factor $1/(1 - \delta - \delta_2)$ and the $O(1/k)$ factors; the explicit closed-form constants satisfy the conditions from $k = 50$ on and give rates below the measured-constant ones by 22–79% at $k = 50$ (the longer the window, the larger the deficit), 5–19% at $k = 100$, at most 5% at $k = 200$ and about 1% at $k = 400$. All constants are suprema over $[0, 300]$ (the exponential averages are formed on an extended grid): on the quasi-periodic regressor they are identical to four digits on $[0, 10^4]$ (closed form); on the Lorenz regressor below, the suprema over $[0, 1500]$ are larger by up to 14% (p_1 5.31 → 5.67, p_2 19.4 → 20.7, π_2 49.4 → 51.9, $\|\bar{N}_2\|$ 0.42 → 0.48 at $T = 20$), which lowers the guaranteed rate at $k = 100$ by 3% and keeps the certificate (referee round 7): the constants of a chaotic regressor are those of the stretch of the attractor that the run visits.

Chaotic $\Psi = (x_1/20, (x_3 - 25)/20)$ *along a Lorenz orbit* ($\sigma = 10$, $r = 28$, $b = 8/3$, traversed five times slower than standard Lorenz time, $\dot{\Psi}$ by the chain rule through the vector field; $q = 2$): the exact decay rates are 0.00759, 0.00304, 0.00152, 0.00076, 0.00038 at $k = 20$, 50, 100, 200, 400 (an effective excitation level 0.15), while the worst exponential window has $\lambda_T = 0.015$, 0.024, 0.051 for $T = 5$, 10, 20 ($\Lambda = 0.55$, 0.36, 0.24; $\rho \approx 20$–26, $\rho_D \approx 110$–160, $\rho_{DD} \approx 2000$–3100; $\|\bar{N}_2\| = 1.05$, 0.34, 0.42: the residual does not follow a monotone rule in T). The conditions hold from $k = 100$ on (condition (A) fails below: the derivative peak factors of a chaotic orbit are large), the certificate holds at every admissible row, and at $T = 20$ the guaranteed rate is 0.00030 / 0.00035 ($k = 100$), 0.00018 / 0.00021 ($k = 200$) against the exact 0.00152 and 0.00076: exact/guaranteed = 5.0 / 4.3 and 4.2 / 3.7, of which the window loss $\lambda_\infty/\lambda_T \approx 3$ is the dominant factor. For a chaotic reference the uniform-in-time construction pays the least exciting stretch of the attractor; a longer window ($T = 20$ against the slowed Lorenz loop time ≈ 3.75) recovers part of it ($\lambda_T = 0.015 \to 0.051$ from $T = 5$ to 20) at the price of a larger p_2 and π_2 (9.9 → 19.4 and 13.7 → 49.4). An exploratory regressor with components $\tanh(x_i/8)$ traversed at standard Lorenz speed (not in the table) had ρ_{DD} of order 10^6–10^7 and a threshold beyond $k = 400$: the slow regime requires the reference to be slow relative to the loop, and the time scale of a chaotic reference is the designer's choice.

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos\sqrt{2}\,t)$ with $T = 4\pi$, the differential identities $T\dot{\bar{M}} = \bar{M} - M$ and $\dot{P}_1 = 2(M - \bar{M})$ on the grid (10^{-4}), the conditions at $k = 20$, 50, 200 for $\delta = 0.1$, 0.2 with $\delta_2 \le 5\%$, the certificate $\sup \dot{V}/V \le -\gamma$ on 400 times, and the guaranteed rate below the exact

decay rate (top Lyapunov exponent over 200 time units) within a factor 2.6, the ratio improving with k (`test_lemma_E0dprime_quasi_periodic_regressor_certificate_and_rate`).

Theorem E (with explicit constants)**.** *Let $\Psi_G := m^{-1/2}\Psi(I_m \otimes G^{1/2}) \in \mathbb{R}^{n\times mp}$, and let $(\alpha_G, T, \Psi_{G,M}, \Psi_{G,D})$ be PE data for it: $\int_t^{t+T} \Psi_G^\top \Psi_G \,\mathrm{d}r \succeq \alpha_G I$, $\|\Psi_G\| \le \Psi_{G,M}$, $\|\dot\Psi_G\| \le \Psi_{G,D}$. One may take $\alpha_G := \alpha\lambda_{\min}(G)/m$, $\Psi_{G,M} := m^{-1/2}\lambda_G^{1/2}\sup\|\Psi\|$ and $\Psi_{G,D} := m^{-1/2}\lambda_G^{1/2}\sup\|\dot\Psi\|$; $\dot\Psi$ is bounded because $x_m \in C^1$ with bounded derivative and the Gaussian's mixed derivatives are bounded for $s \ge s_{\min}$. Let $(\varepsilon, \gamma, p_{\mathrm{lo}}, p_{\mathrm{hi}}, g_V)$ be the constants of Lemma E.0 for the data $(k_e, \alpha_G, T, \Psi_{G,M}, \Psi_{G,D})$, with any admissible ε (for example the closed form of E.0(ii)), where $g_V := 1 + \varepsilon(2Q_{G,M} + T\Psi_{G,M})$ bounds $\|\partial V/\partial w\|/\|w\|$ (the Hessian of V is I plus a matrix of norm at most $\varepsilon T\Psi_{G,M} + 2\varepsilon Q_{G,M}$) and $Q_{G,M} := T^2\Psi_{G,M}^2/2$. Let $r := \min(\rho, \delta)$, let L_J be a Lipschitz constant, uniform in t, of the Jacobian of the closed-loop field F of step 2 on the ball D, and let $S := \mathrm{diag}\big(I_n,\, m^{-1/2}(I_m \otimes G^{-1/2})\big)$ be the similarity to the coordinates $w := (e, \zeta)$ of step 3. Put*

$$\tilde C_R := \|S\|\,\frac{L_J}{2}\,\|S^{-1}\|^2, \qquad r_w := \min\Big\{\frac{r}{\|S^{-1}\|},\ \frac{\gamma}{2g_V\tilde C_R}\Big\}, \qquad r_0 := r_w\Big(\frac{p_{\mathrm{lo}}}{p_{\mathrm{hi}}}\Big)^{1/2},$$

$$\lambda := \frac{\gamma}{4p_{\mathrm{hi}}}, \qquad c_0 := \frac{r_0^2}{2}, \qquad C := \Big(\frac{2p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} \quad (\le \sqrt{6} \text{ for the closed-form } \varepsilon).$$

If $V_{\mathrm{OT}}(t_0) \le c_0$, then:

- *the optimal assignment π at t_0 is unique, and it is the unique optimal assignment for all large t;*
- *$\|e(t)\| + W(\mu_t, \mu^*) \le Ce^{-\lambda(t-t_0)}\big(\|e(t_0)\| + W(\mu_{t_0}, \mu^*)\big)$ for $t \ge t_0$;*
- *$\vartheta_i(t) \to \vartheta^*_{\pi(i)}$. The labelled error*

$$E_{\mathrm{lab}}(t) := \Big(\frac{1}{m}\sum_i \|\vartheta_i(t) - \vartheta_i^*\|_{G^{-1}}^2\Big)^{1/2} \quad \text{tends to} \quad \Big(\frac{1}{m}\sum_i \|\vartheta^*_{\pi(i)} - \vartheta_i^*\|_{G^{-1}}^2\Big)^{1/2},$$

  which is non-zero for $\pi \ne \mathrm{id}$.

Every constant is an explicit function of $(k_e, \alpha_G, T, \Psi_{G,M}, \Psi_{G,D}, m, G, r, L_J)$, hence uniform over the class of regressors with given PE data $(T, \alpha, \sup\|\Psi\|, \sup\|\dot\Psi\|)$. With the closed-form ε, $\lambda \ge \frac{1}{3}\cdot\min\{k_e/4,\, \varepsilon\alpha_G/2\}$. The constants are conservative, in a quantified way (Remark E.0a); their role is to make the statement self-contained. L_J is defined as a supremum; a closed-form bound follows from the third-derivative suprema of the Gaussian and is not computed here (`excitation. jacobian_ lipschitz_ estimate` estimates it from samples for the tests). The previous version proved this theorem with existence constants, conditional on Morgan–Narendra [50, Thm. 1]; Lemma E.0 removes that citation.

Status. Proved, self-contained (Lemma E.0 replaces the citation of Morgan–Narendra [50]); linearisation checked; the strict Lyapunov inequalities checked pointwise (E.0′ by a certificate over all states) and the explicit envelope checked on the linearised and on the implemented loop; convergence observed. The rate of E.0 is conservative in a quantified way (Remark E.0a); the rate of E.0′ is within 12% of the exact rate above its threshold. Novelty (literature audit): partially anticipated [12], [27]; the strict Lyapunov function is of the Mazenc–Malisoff type.

Proof.

1. *Symmetry.* Relabelling the network, $(e, \vartheta) \mapsto (e, \vartheta\circ\sigma)$, maps solutions to solutions. It fixes e, is an isometry of the matched metric, and conjugates the dynamics near $(0, \vartheta^*\circ\pi)$ to those near $(0, \vartheta^*)$. So it suffices to treat $\pi = \mathrm{id}$.

2. *Domain and linearisation.* Let $r := \min(\rho, \delta)$ and $D := \{z = (e, \tilde{\vartheta}) \in \mathbb{R}^{n+mp} : \|z\|_2 < r\}$.

   - For $z \in D$ we have $\|e\| < \rho$, so $f = \hat{f}^*$.
   - Also $\|\tilde{\vartheta}_i\| < \delta$ for every i, so $\vartheta_i \in \operatorname{int} \Theta_K^\varepsilon$ and $\mathrm{Proj}_K = \mathrm{id}$.

   On $[0, \infty) \times D$ the closed loop is

   $$F(t, z) = \Big(-k_e e + \hat{f}^*(x) - \hat{f}(x),\ \big(G \nabla_\vartheta \Phi(x; \vartheta_i)^\top e\big)_i\Big), \qquad x = x_m(t) + e.$$

   - It is C^1 in (t, z), since $x_m \in C^1$, and C^∞ in z.
   - Its derivatives through order two are bounded uniformly in t, because the Gaussian's derivatives through order three are bounded uniformly in x for $s \geq s_{\min}$.
   - Its Jacobian at 0 is

     $$A(t) = \begin{pmatrix} -k_e I & -\frac{1}{m}\Psi \\ (I_m \otimes G)\Psi^\top & 0 \end{pmatrix},$$

     and the remainder is $O(\|z\|^2)$.
   - The Jacobian and the identity $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$, with $Q = \mathrm{diag}\big(I/2,\ (I \otimes G^{-1})/(2m)\big)$, are checked numerically by `test_theorem_E_linearisation_and_lyapunov_identity`.

3. *LTV stability, with explicit constants.* Put $w := (e, \zeta)$ with $\zeta := m^{-1/2}(I_m \otimes G^{-1/2})\tilde{\vartheta}$, that is $w = Sz$, and $\Psi_G := m^{-1/2}\Psi(I \otimes G^{1/2})$.

   - The system becomes $\dot{e} = -k_e e - \Psi_G \zeta$ and $\dot{\zeta} = \Psi_G^\top e$: this is (LTV) of Lemma E.0 with $k = k_e$ and $\Psi = \Psi_G$, that is, $\dot{w} = [A_0 + B(t)]w$ with $A_0 = \mathrm{diag}(-k_e I, 0)$ and B skew-symmetric, bounded, with bounded derivative.
   - The excitation transfers: $\int \Psi_G^\top \Psi_G \, \mathrm{d}r \succeq (\alpha \lambda_{\min}(G)/m) I = \alpha_G I$, and $\|\Psi_G\| \leq \Psi_{G,M}$, $\|\dot{\Psi}_G\| \leq \Psi_{G,D}$.
   - Lemma E.0 gives the strict Lyapunov function $V(t, w)$ with $p_{\mathrm{lo}}\|w\|^2 \leq V \leq p_{\mathrm{hi}}\|w\|^2$ and $\dot{V} \leq -\gamma\|w\|^2$ along the linear system, hence

     $$\|w(t)\| \leq \Big(\frac{p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} e^{-(\gamma/(2p_{\mathrm{hi}}))(t - t_0)} \|w(t_0)\| :$$

     uniform exponential stability with explicit overshoot and rate.
   - The w-coordinates are the coordinates of the matched Lyapunov function: $\|\zeta\|^2 = \frac{1}{m}\sum_i \|\tilde{\vartheta}_i\|^2_{G^{-1}} = W_\pi^2$, so $\|w\|^2 = \|e\|^2 + W_\pi^2 = 2V_\pi$.
   - The bound on $\dot{\Psi}$ cannot be dropped. $\Psi = \sin t^2$ is bounded and PE, but for $\dot{e} = -e - \Psi\zeta$, $\dot{\zeta} = \Psi e$, from $w(t_0) = (e, \zeta) = (0, 1)$, a numerical computation gives the ratio $\|w(t_0 + 200)\|/\|w(t_0)\|$ as 0.9715, 0.9942 and 0.9980 for $t_0 = 5$, 20 and 50, and part (f) of the proof of Lemma E.0 shows that the decay over a fixed horizon tends to 0 as $t_0 \to \infty$. So that system is not UES. Consistently, the rate of Lemma E.0 tends to 0 as $\Psi_D \to \infty$.

4. *Nonlinear system, with an explicit ball.* On the closure of D (to which the Lipschitz bound extends) the field is $F(t, z) = A(t)z + R(t, z)$ with $\|R(t, z)\| \leq (L_J/2)\|z\|^2$ (Taylor with the Lipschitz Jacobian). In w-coordinates, $\dot{w} = [A_0 + B(t)]w + \tilde{R}(t, w)$ with $\tilde{R}(t, w) := S\,R(t, S^{-1}w)$, so $\|\tilde{R}(t, w)\| \leq \tilde{C}_R\|w\|^2$ on $\{\|w\| < r/\|S^{-1}\|\} \subset S(D)$. Along the nonlinear system, since $\|\partial V/\partial w\| \leq g_V\|w\|$,

   $$\dot{V} \leq -\gamma\|w\|^2 + g_V\tilde{C}_R\|w\|^3 \leq -\frac{\gamma}{2}\|w\|^2 \qquad \text{whenever } \|w\| \leq r_w.$$

The sublevel set $\{V \le p_{\mathrm{lo}} r_w^2\}$ is contained in $\{\|w\| \le r_w\}$, so it is forward invariant, and it contains the ball $\{\|w\| \le r_0\}$, since $V \le p_{\mathrm{hi}}\|w\|^2$. On it $\dot V \le -(\gamma/(2p_{\mathrm{hi}}))V$, so for $\|w(t_0)\| \le r_0$,

$$\|w(t)\| \le \Big(\frac{p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} e^{-\lambda(t-t_0)}\|w(t_0)\|, \qquad \lambda = \frac{\gamma}{4p_{\mathrm{hi}}}.$$

These trajectories stay in D, where the closed loop equals F. (Khalil [40, Thm. 4.13] gives the same conclusion with existence constants; the explicit form is what Lemma E.0 adds.)

5. *Transfer to* $(\|e\|, W)$. Let π be optimal at t_0. Then $\|w(t_0)\|^2 = 2V_\pi(t_0) = 2V_{\mathrm{OT}}(t_0) \le 2c_0 = r_0^2$, so step 4 applies. Moreover $\|e\| + W \le \|e\| + W_\pi = \|e\| + \|\zeta\| \le \sqrt{2}\|w\|$ at every t, and $\|w(t_0)\| \le \|e(t_0)\| + \|\zeta(t_0)\| = \|e(t_0)\| + W(t_0)$. This gives the estimate with $C = (2p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2}$.

   Uniqueness. Suppose $V_\pi, V_{\pi'} \le c_0$ with $\pi \ne \pi'$. Then the state lies in the invariant balls of step 4 around both $(0, \vartheta^* \circ \pi)$ and $(0, \vartheta^* \circ \pi')$. The solution would converge to two distinct equilibria, which is impossible.

   Limits. Hence $z(t) \to 0$, that is, $\vartheta_i(t) \to \vartheta^*_{\pi(i)}$. For $\pi' \ne \pi$,

$$V_{\pi'}(t) \to \frac{1}{2m}\sum_i \|\vartheta^*_{\pi(i)} - \vartheta^*_{\pi'(i)}\|^2_{G^{-1}} \ge \frac{d^2}{2} > 0,$$

   while $V_\pi(t) \to 0$. So π is the unique optimal assignment for all large t, and E_{lab} converges as stated. □

Checked by: `tests/test_transport.py`, `tests/test_excitation.py` and `experiments/exp_pe_convergence.py`.

- Linearisation: the Jacobian of the implemented loop equals $A(t)$, and $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$ (`test_theorem_E_linearisation_and_lyapunov_identity`).
- Explicit constants: on the implemented loop, from a start inside the explicit ball, $\|w(t)\| \le (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-\lambda t}\|w(0)\|$ at every sample (L_J estimated from samples) (`test_theorem_E_explicit_nonlinear_envelope_on_the_implemented_loop`).
- Convergence: the matched error decays and the labelled error does not (qualitative); sliding-window excitation and fitted rates (`test_theorem_E_matched_error_converges_labelled_does_not`, `exp_pe_convergence.py`).
- Not checked in the repository: the sharpness of the explicit rate beyond the tables of Remark E.0a; Corollary E.2 and Remark E.3.

Remark E.1 (the rate)**.**

- **What converges.** "The parameters converge" is false for the labelled error and true for the matched one.
- **The rate is not set by the PE level.** $\lambda > 0$ exists for each admissible Ψ, but it depends on $\Psi(\cdot)$, not only on its PE level. No positive function of $\alpha\lambda_{\min}(G)/m$, k_e, T and $\sup\|\Psi_G\|$ alone bounds it below.
  - For $m = n = 1$, $\vartheta^* = (1.5, 0.3, 0.8)$, $G = I$, $k_e = 3$ and $x_m = 1.5\sin\omega t$ ($T = 2\pi$), these four quantities do not depend on ω. The exact Floquet rate, computed numerically, falls from 4.7×10^{-2} ($\omega = 1$) to 2.9×10^{-5} ($\omega = 64$), like ω^{-2}.
  - A lower bound must also involve $\sup\|\dot\Psi_G\|$ (step 3). Lemma E.0 gives one; it reproduces the ω^{-2} law (Remark E.0a).
  - Exact Floquet rates, computed numerically, are also not monotone in G or in k_e, and they decrease for large G. When k_e is large relative to the reference bandwidth, they approach $\lambda_{\min}\langle\Psi_G^\top\Psi_G\rangle/k_e$.
- **The numbers** (`results/pe_convergence`; $m = 3$, $n = 2$, six runs):

quantity	range
min over sliding 60 s windows of $\lambda_{\min}\big(\int_t^{t+60} \Psi^\top \Psi \,\mathrm{d}r\big)$	5.9×10^{-4} to 2.7×10^{-2}
fitted decay rate of the matched error, second half of the run	2.3×10^{-3} to $4.3 \times 10^{-3}\,\mathrm{s}^{-1}$
rank correlation between the two	0.60

The convergence test is therefore qualitative. The machine-checked part of the proof is the linearisation.

Corollary E.2 (uniform-in-time exponential merging under PE)**.** *Take two solutions of the same closed loop with $V_{\mathrm{OT}}(0) \le c_0$ and $V'_{\mathrm{OT}}(0) \le c_0$. Then*

$$\|x_t - x'_t\| + W(\mu_t, \mu'_t) \le Ce^{-\lambda t} R_0, \qquad R_0 := \|e(0)\| + \|e'(0)\| + W(\mu_0, \mu^*) + W(\mu'_0, \mu^*). \tag{15.2}$$

Let $\delta_0 := \|x_0 - x'_0\| + W(\mu_0, \mu'_0)$. Combine this with the coupling estimate of Theorem D(c) on $[0, T_0]$, with $T_0 := (\Lambda + \lambda)^{-1} \log(CR_0/\delta_0)$.

- *Both solutions stay in D (proof step 4), so $\|x - x_m(t)\| < r \le \rho$.*
- *The segment between x_t and x'_t lies in $B(x_m(t), \rho)$, where $f = \hat{f}^*$, and $\mathrm{Proj}_K = \mathrm{id}$.*
- *Hence Theorem D(c) holds with $M_x := k_e + 2a_{\max} J_c$, with L_x and L_ϑ computed for $\bar{E} = r$ and an inactive projection, and with $L_2 = \lambda_G^{1/2} L_1$. This Λ depends neither on T_0 nor on f outside the tube.*

This gives uniform-in-time Hölder continuity in the basin:

$$\sup_{t \ge 0} \big(\|x_t - x'_t\| + W(\mu_t, \mu'_t) \big) \le (CR_0)^{\Lambda/(\Lambda+\lambda)} \, \delta_0^{\lambda/(\Lambda+\lambda)}. \tag{15.3}$$

Status. Proved, through Theorem E (self-contained since Lemma E.0). Not checked in the repository.

Proof. $\|x_t - x'_t\| = \|e - e'\| \le \|e\| + \|e'\|$, and $W(\mu_t, \mu'_t) \le W(\mu_t, \mu^*) + W(\mu'_t, \mu^*)$. Bound each bracket by Theorem E. On $[0, T_0]$ use $e^{\Lambda T_0} \delta_0$; on $[T_0, \infty)$ use $CR_0 e^{-\lambda T_0}$. Note that $\delta_0 \le R_0 \le CR_0$. □

Remark E.3 (the collapse point)**.** Take the G^{-1}-closest pair (j, k). Set $e = 0$ and $\vartheta_j = \vartheta_k = (\vartheta_j^* + \vartheta_k^*)/2$, with all other particles ideal.

- At this point $V_{\mathrm{OT}} = d^2/8$, and there are two optimal assignments.
- By exchange symmetry, $\vartheta_j \equiv \vartheta_k$ for all t, so $W(\mu_t, \mu^*) \ge d/2$ forever.

The uniqueness argument of step 5 shows $c_0 < d^2/8$ automatically, and $d^2/8$ is sharp along this direction.

16 Theorem F: the transport cost of adaptation

Theorem F. *Assume* (S0)–(S3)*, and take $t_0 \le t_1$ in the maximal interval of existence (all $t \ge 0$ under the hypothesis of Theorem D(b)). μ^*,* (S4) *and* (S5) *are not needed. Along* (L)*:*

(i) $\big(\frac{1}{m} \sum_i \|\dot{\vartheta}_i\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e(t)\|$*;*

(ii) $W(\mu_{t_1}, \mu_{t_0}) \le \lambda_G^{1/2} L_1 \int_{t_0}^{t_1} \|e\| \,\mathrm{d}r$*. Hence the W-length of $r \mapsto \mu_r$ on $[t_0, t_1]$ is at most $\lambda_G^{1/2} L_1 \int_{t_0}^{t_1} \|e\| \,\mathrm{d}r$;*

(iii) $W(\mu_{T_2}, \mu_{T_1}) \le \lambda_G^{1/2} L_1 (T_2 - T_1) \cdot \mathrm{RMS}_{[T_1, T_2]} \|e\|$.

Along (L_σ) *with the OT leak, for a.e. t:*

(i′) $\big(\frac{1}{m}\sum_i \|\dot\vartheta_i\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_t, \mu^0)$;

(ii′) $W(\mu_{t_1}, \mu_{t_0}) \le \int_{t_0}^{t_1} \big(\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0)\big)\,\mathrm{d}r$;

(iii′) $W(\mu_{T_2}, \mu_{T_1}) \le \lambda_G^{1/2} L_1 (T_2 - T_1) \cdot \mathrm{RMS}_{[T_1,T_2]} \|e\| + \sigma_m \int_{T_1}^{T_2} W(\mu_r, \mu^0)\,\mathrm{d}r$.

For the labelled leak, replace $W(\mu_r, \mu^0)$ by $d_{\mathrm{lab}}(r) := \big(\frac{1}{m}\sum_i \|\vartheta_i(r) - \vartheta_i^0\|^2_{G^{-1}}\big)^{1/2} \ge W(\mu_r, \mu^0)$. The term $W(\mu_r, \mu^0)$ does not suffice there.

All bounds also hold with $L_G := \big(g_a + a_{\max}^2 (g_c J_c^2 + g_s J_s^2)\big)^{1/2} \le \lambda_G^{1/2} L_1$ in place of $\lambda_G^{1/2} L_1$. Since $\max_i \|\dot\vartheta_i\|_{G^{-1}} \le L_G \|e\|$ along (L)*, also $W_\infty(\mu_{t_1}, \mu_{t_0}) \le L_G \int_{t_0}^{t_1} \|e\|\,\mathrm{d}r$.*

Status. Proved and checked, including the leak clause, a tight leak instance with $e \equiv 0$ and a tight velocity instance. Novelty (literature audit): elementary.

Proof. (i) By Lemma P.1 and Lemma B.2(b),

$$\|\dot\vartheta_i\|_{G^{-1}} \le \|\nabla g_t(\vartheta_i)\|_G \le \lambda_G^{1/2} \|\nabla g_t(\vartheta_i)\| \le \lambda_G^{1/2} L_1 \|e(t)\|.$$

Square and average over i.

(ii) Use the identity coupling $(\vartheta_i(t_0), \vartheta_i(t_1))$ and Minkowski's inequality in $L^2\big(\frac{1}{m}\sum_i; G^{-1}\big)$. For general measures, use the Lagrangian coupling $(X_{t_0}, X_{t_1})_\# \mu_0$, or Ambrosio–Gigli–Savaré [2, Thm. 8.3.1] after the isometry $\vartheta \mapsto G^{-1/2}\vartheta$. Summing over a partition gives the length statement.

(iii) Cauchy–Schwarz.

(i′) Apply Lemma P.1 with $y = \nabla g_t - \sigma_m G^{-1}(\vartheta_i - \vartheta^0_{\tau(i)})$. Then Minkowski's inequality gives $\|y\|_G \le \|\nabla g_t\|_G + \sigma_m \|\vartheta_i - \vartheta^0_{\tau(i)}\|_{G^{-1}}$. Average over i with τ optimal. At ties, Filippov velocities are convex combinations of fields with τ optimal at the current point. Each such field satisfies (i′), and the norm is convex.

(ii′) and (iii′) follow as (ii) and (iii).

For L_G: $\|\nabla g\|_G^2 = g_a \|\varphi e\|^2 + (a^\top e)^2 \big(g_c \|\nabla_c \varphi\|^2 + g_s (\partial_s \varphi)^2\big) \le \|e\|^2 L_G^2$. □

Reading. Along (L), moving the kernel measure by W costs at least $W/(\lambda_G^{1/2} L_1)$ of integrated tracking error. This is an exchange rate between learning and tracking.

Along a leaky law the proved statement is

$$\int_{t_0}^{t_1} \big(\lambda_G^{1/2} L_1 \|e\| + \sigma_m d_r\big)\,\mathrm{d}r \ge W(\mu_{t_1}, \mu_{t_0}),$$

with $d_r = W(\mu_r, \mu^0)$ for (L_σ) and $d_r = d_{\mathrm{lab}}(r)$ for the labelled leak. This gives no lower bound on $\int \|e\|$, because the leak can pay for the whole motion. Under (L_σ) there are runs with $e \equiv 0$ and $W(\mu_{10}, \mu_0) \approx 1.05$ (found numerically).

Numerical checks. The following are checked at every sample (with the particle path length $\ge$ the Wasserstein length):

- (i) and Lemma P.1, including a tight instance with ratio ≥ 0.99;
- (ii) along (L);
- (ii′) along (L_σ), on a grid of pairs, including an instance with $e \equiv 0$ where it is tight (ratio ≥ 0.99) and the (L) reading fails.

Checked by: `tests/test_transport.py`:
`test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound` ((i) and Lemma P.1);
`test_theorem_F_wasserstein_length_bounded_by_integrated_error` ((ii) along (L));
`test_theorem_F_leak_clause_along_ot_sigma_mod` and
`test_theorem_F_leak_clause_is_tight_when_the_leak_pays_for_all_motion` ((ii′) along (L_σ)).

17 Theorem D′: propagation of chaos for the dithered closed loop on finite horizons

Theorem D covers the deterministic projected law (L). The route to Conjecture H.6 (Section 20) lists the mean-field limit of the dithered law (L_τ) as one of its three missing pieces. This section proves it on finite horizons, with an explicit $m^{-1/2}$ rate that does not depend on the temperature, by a synchronous coupling. The uniform-in-time statement is Section 25, branch 3.

Setting. The design of Theorem H: saturated features $\tilde\Phi$, $\|\tilde\Phi\| \le K_\Phi$, with all ϑ-derivatives bounded uniformly in x, and the m-particle closed loop

$$\begin{aligned} \dot x &= f(x) + \dot x_m - k_e e - \frac{1}{m}\sum_j \tilde\Phi(x;\vartheta_j), \qquad e = x - x_m(t), \\ \mathrm{d}\vartheta_i &= b(x,e,\vartheta_i)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad i = 1,\dots,m, \\ b(x,e,\vartheta) &:= G\nabla_\vartheta\big[e^\top\tilde\Phi(x;\vartheta)\big] - \sigma(\vartheta-\vartheta^0), \end{aligned} \tag{L_τ}$$

with independent Brownian motions B_i, i.i.d. initial particles $\vartheta_i(0) \sim \bar\mu_0 \in \mathcal{P}_2(\mathbb{R}^p)$ independent of the B_i, and a deterministic $x(0) = \bar x_0$. f is C^1 and $x_m \in C^1$ with $\dot x_m$ bounded; no autonomy or periodicity is needed. The **McKean–Vlasov closed loop** is

$$\begin{aligned} \dot{\bar x} &= f(\bar x) + \dot x_m - k_e \bar e - \int \tilde\Phi(\bar x;\vartheta)\,\mathrm{d}\bar\mu_t(\vartheta), \qquad \bar e = \bar x - x_m(t), \\ \mathrm{d}\bar\vartheta &= b(\bar x_t, \bar e_t, \bar\vartheta)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B, \qquad \bar\mu_t := \mathrm{Law}(\bar\vartheta_t), \end{aligned} \tag{MV}$$

from the same $\bar x_0$ and $\bar\vartheta(0) \sim \bar\mu_0$. Here $\bar x_t$ is deterministic, and the particle $\bar\vartheta$ interacts with its own law only through $\bar x_t$.

Standing bound. Fix $T > 0$ and suppose both x^m (for every m and every realisation) and $\bar x$ remain in a compact set X on $[0,T]$. Since $\|\frac{1}{m}\sum\tilde\Phi\| \le K_\Phi$ and $\|\int\tilde\Phi\,\mathrm{d}\bar\mu\| \le K_\Phi$ whatever the parameters, the tube bound of Theorem D(b) (with $a_{\max}$ replaced by K_Φ) gives such an X for every T when its self-consistent growth condition holds, for example for bounded f; otherwise the statement is local in time, as in Theorem D(d). Put $E_X := \sup_{t\le T,\,x\in X}\|x - x_m(t)\|$ and let

$$\begin{gathered} L_f := \mathrm{Lip}_X f, \qquad L_x := \sup\|\partial_x\tilde\Phi\|, \qquad L_\vartheta := \sup\|\nabla_\vartheta\tilde\Phi\|, \\ L_b := \lambda_G\big(L_\vartheta + E_X\sup\|\partial_x\nabla_\vartheta\tilde\Phi\| + E_X\sup\|\nabla_\vartheta^2\tilde\Phi\|\big) + \sigma, \\ \Lambda := \max\{L_f + L_x + 2L_b,\ L_\vartheta + L_b\}, \end{gathered}$$

where L_b is a Lipschitz constant of $(x,e,\vartheta) \mapsto b$ on $X \times \{\|e\| \le E_X\} \times \mathbb{R}^p$ for the norm $\|x\| + \|e\| + \|\vartheta\|$. All of these are finite by saturation and independent of m and τ; the suprema of the $\tilde\Phi$-derivatives are over all $x \in \mathbb{R}^n$ (the saturated features and their derivatives are globally bounded), so the segments between x and $\bar x$ used below may leave X. The constants grow with the output dimension n ($K_\Phi = a_{\max}\sqrt{n}$), not with the lifted dimension p beyond that.

Theorem D′. *Under the standing bound:*

(a) Well-posedness. (MV) *has a unique solution* $(\bar x, \mathrm{Law}(\bar\vartheta))$ *on* $[0,T]$*, and* $\bar x$ *is deterministic.*

(b) Synchronous coupling. *Let* $\bar\vartheta_i$*,* $i = 1,\dots,m$*, be the solutions of* (MV)*'s particle equation driven by the same* B_i *and started at the same* $\vartheta_i(0)$ *as the* m*-particle system. Then, pathwise,*

$$\begin{aligned} \|x_t - \bar x_t\| + \frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\| &\le e^{\Lambda t}\int_0^t \zeta_s\,\mathrm{d}s, \\ \zeta_s := \Big\|\frac{1}{m}\sum_j\tilde\Phi(\bar x_s;\bar\vartheta_j(s)) - \int\tilde\Phi(\bar x_s;\cdot)\,\mathrm{d}\bar\mu_s\Big\|, & \end{aligned}$$

and since the $\bar\vartheta_j(s)$ are i.i.d. with law $\bar\mu_s$ and $\bar x_s$ is deterministic, $\mathbb{E}\zeta_s^2 \le K_\Phi^2/m$. Hence

$$\mathbb{E}\sup_{t\le T}\Big(\|x_t-\bar x_t\|+\frac{1}{m}\sum_i\|\vartheta_i(t)-\bar\vartheta_i(t)\|\Big)\le TK_\Phi e^{\Lambda T}m^{-1/2},$$
$$\mathbb{E}\sup_{t\le T}\Big(\|x_t-\bar x_t\|+\frac{1}{m}\sum_i\|\vartheta_i(t)-\bar\vartheta_i(t)\|\Big)^2\le T^2K_\Phi^2e^{2\Lambda T}m^{-1},$$

with constants independent of $\tau\ge 0$ and of $\bar\mu_0$. The exponent $-1/2$ does not depend on the dimension; the constants do, through K_Φ and the Lipschitz bounds of $\tilde\Phi$ (standing bound).

(c) Consequences.

- *Propagation of chaos: for each fixed k, the law of $(\vartheta_1,\dots,\vartheta_k)$ on $[0,T]$ converges to the k-fold product of the path law of $\bar\vartheta$, with the per-particle bound $\mathbb{E}\sup_{t\le T}\|\vartheta_i(t)-\bar\vartheta_i(t)\|\le 2(e^{L_bT}-1)TK_\Phi e^{\Lambda T}m^{-1/2}$ for every i. (Exchangeability alone gives only the fixed-t bound $\mathbb{E}\|\vartheta_i(t)-\bar\vartheta_i(t)\|=\mathbb{E}\bar\Delta(t)\le TK_\Phi e^{\Lambda T}m^{-1/2}$, since $\sup_t$ of a mean is not the mean of the $\sup_t$; the per-particle sup bound is the pathwise Grönwall estimate of the proof.)*
- *The learned function converges at the rate $m^{-1/2}$: at each fixed t,*

$$\begin{aligned}\mathbb{E}\Big\|\hat f^m(x_t)-\int\tilde\Phi(\bar x_t;\cdot)\,\mathrm{d}\bar\mu_t\Big\|&\le(L_x+L_\vartheta)TK_\Phi e^{\Lambda T}m^{-1/2}+\mathbb{E}\zeta_t\\&\le\big((L_x+L_\vartheta)Te^{\Lambda T}+1\big)K_\Phi m^{-1/2}\end{aligned}$$

  (the supremum over t would need a chaining argument for ζ and is not pursued).
- *The parameter measure converges in W_1 (Euclidean cost on $\mathbb{R}^p$; the G^{-1} metric used elsewhere changes it by a factor $\lambda_G^{\pm1/2}$) at the i.i.d. rate:*

$$\mathbb{E}W_1(\mu_t^m,\bar\mu_t)\le TK_\Phi e^{\Lambda T}m^{-1/2}+\mathbb{E}W_1(\bar\mu_t^{\mathrm{emp},m},\bar\mu_t),$$

  where $\bar\mu_t^{\mathrm{emp},m}$ is the empirical measure of m i.i.d. samples of the time-t marginal $\bar\mu_t$; the last term is $O(m^{-1/p})$ for $p>2$ (Fournier–Guillin [23]; applicable because $\bar\mu_t$ has a bounded second moment on $[0,T]$, the drift being linearly bounded). Only this term carries the dimension.

(d) $\tau=0$. *The statement holds for the deterministic saturated law with i.i.d. initial particles, where it gives, for this law (not the projected law of Theorem D(d)), the explicit $m^{-1/2}$ rate in place of a rate $\delta_m\to 0$ for the state and the paired particles.*

Status. Proved; consistency of the $m^{-1/2}$ rate checked by coupling m-particle systems to a 4096-particle reference (`exp_chaos.py`). Novelty (literature audit): the technique is Sznitman's; new in that the plant is in the loop and the rate for the state is dimension-free.

Proof. (a) Let X' be a compact neighbourhood of X on which f is Lipschitz. For a given curve $\bar x\in C([0,T_1];X')$, the particle equation of (MV) is an SDE with drift Lipschitz in ϑ (bounded $\nabla^2_\vartheta\tilde\Phi$ and the linear leak; the derivative bounds of $\tilde\Phi$ are global in x) and constant diffusion; it has a unique strong solution, whose law $\bar\mu_t[\bar x]$ depends Lipschitz-continuously on $\bar x$: by synchronous coupling of two solutions driven by $\bar x$ and $\bar x'$, $\frac{\mathrm{d}}{\mathrm{d}t}\|\bar\vartheta-\bar\vartheta'\|\le L_b\big(2\|\bar x-\bar x'\|+\|\bar\vartheta-\bar\vartheta'\|\big)$, so $W_1(\bar\mu_t[\bar x],\bar\mu_t[\bar x'])\le 2L_b\int_0^t e^{L_b(t-s)}\|\bar x_s-\bar x'_s\|\,\mathrm{d}s$. The plant input $\dot x_m-k_e\bar e-\int\tilde\Phi\,\mathrm{d}\bar\mu$ is bounded by $\sup\|\dot x_m\|+k_eE_{X'}+K_\Phi$ whatever the measure, so for T_1 small the plant solutions from $\bar x_0$ stay in X', and the map taking $\bar x$ to the solution of the plant ODE with input $\int\tilde\Phi(\cdot\,;\vartheta)\,\mathrm{d}\bar\mu_t[\bar x]$ sends $C([0,T_1];X')$ into itself and is a contraction there, since $\mu\mapsto\int\tilde\Phi(x;\cdot)\,\mathrm{d}\mu$ is W_1-Lipschitz with constant L_ϑ and f is Lipschitz on X'. This gives a unique solution on $[0,T_1]$ in the class of

curves with values in X'; the standing bound keeps it in X, so the local solutions concatenate to $[0,T]$, and uniqueness holds among solutions with $\bar{x} \in C([0,T];X)$. This is the Picard argument of Theorem D(a) with the Dobrushin estimate replaced by the coupling estimate. (Theorem D(b)'s tube bound, used in the standing bound for a law it does not formally cover, rests only on $\|\hat{f}\| \le a_{\max}$ — here K_Φ — and on the pathwise error ODE, so it applies.)

(b) Set $\Delta_i := \vartheta_i - \bar{\vartheta}_i$ and $\delta := x - \bar{x}$. The noise cancels in $\mathrm{d}\Delta_i$, so Δ_i is absolutely continuous with $\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta_i\| \le \|b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i)\| \le L_b\big(\|x-\bar{x}\| + \|e-\bar{e}\| + \|\Delta_i\|\big) = L_b\big(2\|\delta\| + \|\Delta_i\|\big)$, using $e - \bar{e} = \delta$ (same x_m). For δ,

$$\dot{\delta} = f(x) - f(\bar{x}) - k_e\delta - \Big[\frac{1}{m}\sum_j \tilde{\Phi}(x;\vartheta_j) - \int \tilde{\Phi}(\bar{x};\cdot)\,\mathrm{d}\bar{\mu}_t\Big],$$

and the bracket splits as

$$\frac{1}{m}\sum_j\big[\tilde{\Phi}(x;\vartheta_j) - \tilde{\Phi}(\bar{x};\vartheta_j)\big] + \frac{1}{m}\sum_j\big[\tilde{\Phi}(\bar{x};\vartheta_j) - \tilde{\Phi}(\bar{x};\bar{\vartheta}_j)\big] + \Big[\frac{1}{m}\sum_j \tilde{\Phi}(\bar{x};\bar{\vartheta}_j) - \int \tilde{\Phi}(\bar{x};\cdot)\,\mathrm{d}\bar{\mu}_t\Big],$$

of norms at most $L_x\|\delta\|$, $L_\vartheta \frac{1}{m}\sum\|\Delta_j\|$ and ζ_t. Dropping the dissipative $-k_e\delta$, $\frac{\mathrm{d}}{\mathrm{d}t}\|\delta\| \le (L_f + L_x)\|\delta\| + L_\vartheta\bar{\Delta} + \zeta_t$ with $\bar{\Delta} := \frac{1}{m}\sum\|\Delta_j\|$, and $\frac{\mathrm{d}}{\mathrm{d}t}\bar{\Delta} \le L_b\big(2\|\delta\| + \bar{\Delta}\big)$. For $D := \|\delta\| + \bar{\Delta}$ this gives $\dot{D} \le \Lambda D + \zeta_t$ with $D(0) = 0$, and Grönwall gives the pathwise bound, whose right-hand side is non-decreasing in t. For the expectation, $\bar{x}_s$ is deterministic and the $\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_j(s))$ are i.i.d. bounded by K_Φ, so

$$\mathbb{E}\zeta_s^2 = \frac{1}{m}\mathbb{E}\big\|\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_1(s)) - \mathbb{E}\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_1(s))\big\|^2 \le K_\Phi^2/m,$$

and $\mathbb{E}\int_0^T \zeta_s\,\mathrm{d}s \le TK_\Phi m^{-1/2}$; for the second moment, $\mathbb{E}\big(\int_0^T \zeta\big)^2 \le T\int_0^T \mathbb{E}\zeta^2 \le T^2K_\Phi^2/m$.

(c) For the first item, $\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta_i\| \le L_b\big(2\|\delta\| + \|\Delta_i\|\big)$ with $\|\delta(s)\| \le D(s) \le e^{\Lambda s}\int_0^s \zeta$ and $\Delta_i(0) = 0$ give, by Grönwall and $\Lambda \ge L_b$,

$$\|\Delta_i(t)\| \le 2L_b\int_0^t e^{L_b(t-s)}e^{\Lambda s}\Big(\int_0^s \zeta\Big)\,\mathrm{d}s \le 2(e^{L_bt} - 1)e^{\Lambda t}\int_0^t \zeta,$$

a pathwise bound whose right-hand side is non-decreasing in t; take $\sup_t$ and expectations. Convergence of the k-particle law follows from the pathwise coupling bound. For the second, $\|\hat{f}^m(x_t) - \int \tilde{\Phi}(\bar{x}_t;\cdot)\,\mathrm{d}\bar{\mu}_t\|$ is bounded by the same three terms as the bracket in (b), the first two by $(L_x + L_\vartheta)D(t)$ and the third by ζ_t. For the third, couple μ_t^m with the empirical measure of the $\bar{\vartheta}_j(t)$ through the identity pairing (cost $\frac{1}{m}\sum\|\Delta_j\|$) and use the triangle inequality.

(d) Nothing in the argument uses $\tau > 0$: the noise cancels in the coupling and only the i.i.d. sampling of the initial particles enters through ζ. □

Remark D′.1 (what carries the dimension)**.** The plant and the learned function see the particle measure only through $\int \tilde{\Phi}\,\mathrm{d}\mu$, the mean of a bounded function. Their fluctuation is $K_\Phi/\sqrt{m}$: the exponent does not see the dimension p of the lifted parameter, and the constant $K_\Phi = a_{\max}\sqrt{n}$ grows with the output dimension only. The curse of dimensionality of Corollary B.6 (rates $m^{-1/p}$) concerns the Wasserstein distance between the parameter measures, which the closed loop never evaluates. Theorem F, by contrast, bounds the transport of μ by the tracking error and is dimension-free as well.

Remark D′.2 (what is still missing for Conjecture H.6)**.** The constant $e^{\Lambda T}$ grows with the horizon; uniformity in time would follow from a contraction of the coupled system, which the plant coupling and the non-reciprocal averaged interaction (Proposition H.7) obstruct for the general law. For the averaged system, Remark H.5 points to the uniform-in-time result of [8]; with the plant in the loop the question is Section 25, branch 3.

Numerics (`exp_chaos.py` → `results/chaos/table.md`). An $M = 4096$ particle system stands in for (MV); for $m = 16, 32, \dots, 1024$ the m-particle system is driven by the first m initial particles and Brownian motions of the reference — the coupling of (b) — with $n = 1$, $k_e = 10$, $\sigma = 0.02$, $T = 20$, at $\tau = 0$ and $\tau = 2{\cdot}10^{-3}$, and $\sup_t D_m$, $D_m := \|x^m - x^M\| + \frac{1}{m}\sum_{i\le m}\|\vartheta_i^m - \vartheta_i^M\|$, is averaged over 24 seeds. Because the m-system shares its particles with the reference, the sampling fluctuation that drives D_m has variance proportional to $1/m - 1/M$ (the shared particles contribute $(1/m - 1/M)\sum_{i\le m}$, the others $(1/M)\sum_{i>m}$), so the predicted law for the mean is a constant times $(1/m - 1/M)^{1/2}$, and $m^{-1/2}$ for $m \ll M$.

	$\tau = 0$	$\tau = 2 \cdot 10^{-3}$
mean $\sup_t D_m$, $m = 16 \to 1024$	$1.49 \cdot 10^{-1} \to 2.17 \cdot 10^{-2}$	$1.52 \cdot 10^{-1} \to 2.32 \cdot 10^{-2}$
log-log slope of the mean against m over $m \le 256$ (bootstrap s.e. over seeds)	-0.47 ± 0.06	-0.46 ± 0.05
slope of the median	-0.46	-0.45
slope of the mean against $(1/m - 1/M)^{1/2}$, all m (predicted 1)	0.94 ± 0.08	0.92 ± 0.08
mean / $(1/m - 1/M)^{1/2}$, $m = 32 \dots 1024$	0.70–0.80	0.67–0.86

The rate is the $m^{-1/2}$ of (b) within one standard error, the same at both temperatures (the noise cancels in the coupling, (d)), and the two-scale law in $(1/m - 1/M)^{1/2}$ holds up to $m = M/4$. Two features of the data are worth recording. First, single seeds differ by more than an order of magnitude at fixed m (the table lists the minimum and maximum over seeds): $\sup_t D_m$ is governed by the projection of the initial sampling fluctuation on a few slowly amplified directions of the closed loop, so the expectation in (b) needs tens of seeds — a first version of this experiment with three seeds gave slopes -0.25 and -0.21 and was withdrawn (`review/REVIEW_LOG.md`, round 5, D2). Second, the measured ratio $\sup_t D_m/(1/m - 1/M)^{1/2} \approx 0.7$ is astronomically smaller than the constant $TK_\Phi e^{\Lambda T}$ of (b): on this horizon the theorem is a rate statement, not a quantitative bound.

Checked by: `exp_chaos.py`: the experiment only; there is no test for Theorem D′ (its hypotheses are structural and the coupling is the experiment itself).

Theorem D″: uniform-in-time propagation of chaos in the leak-dominated regime

Theorem D″ (uniform-in-time propagation of chaos in the leak-dominated regime)**.** *In the setting and with the constants of Theorem D′, assume the standing bound for all $t \ge 0$ and let $L_2 := \sup\|\nabla_\vartheta^2\tilde\Phi\|$, $L_{x\vartheta} := \sup\|\partial_x\nabla_\vartheta\tilde\Phi\|$ (over all x and ϑ; for $n > 1$ the operator norms are contracted with unit vectors as in (1) of Section 9), and*

$$\kappa_e := k_e - L_f - L_x, \qquad \kappa_\vartheta := \sigma - \lambda_G E_X L_2,$$
$$L_{b'} := \lambda_G(L_\vartheta + E_X L_{x\vartheta}), \qquad \det A := \kappa_e\kappa_\vartheta - L_{b'}L_\vartheta.$$

If $\kappa_e > 0$ and $\det A > 0$ (then $\kappa_\vartheta > 0$ as well) — the leak dominates the loop gain of the adaptive law — then, under the synchronous coupling of Theorem D′(b), for every $t \ge 0$ and every $\tau \ge 0$,

$$\mathbb{E}\|x_t - \bar x_t\| \le K_\Phi\kappa_\vartheta m^{-1/2}/\det A,$$
$$\mathbb{E}\frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\| = \mathbb{E}\|\vartheta_1(t) - \bar\vartheta_1(t)\| \le K_\Phi L_{b'} m^{-1/2}/\det A,$$

with constants independent of t: propagation of chaos is uniform in time, at the rate $m^{-1/2}$. Moreover $\sup_t \mathbb{E}W_1(\mu_t^m, \bar\mu_t) \le K_\Phi L_{b'} m^{-1/2}/\det A + \sup_t \mathbb{E}W_1(\bar\mu_t^{\mathrm{emp},m}, \bar\mu_t)$, and the second term is finite: $\sup_t \int\|\vartheta - \vartheta^0\|^2\,\mathrm{d}\bar\mu_t \le \max\{\int\|\vartheta - \vartheta^0\|^2\,\mathrm{d}\bar\mu_0,\ ((\lambda_G E_X L_\vartheta)^2/\sigma + 2\tau\,\mathrm{tr}\,G)/\sigma\}$.

Status. Proved; checked numerically on one slow-adaptation instance (`exp_chaos_uniform.py`), where it also shows its limit. The contraction condition $\kappa_\vartheta > 0$ makes the particle map a contraction uniformly in t, so in the certified regime the particles of a single system stay together: at $\tau = 0$ they synchronise exponentially and the network collapses to one kernel; at $\tau > 0$ they form the dither cloud of the prior, $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$ to leading order, and the network's output stays at the level of its prior (Remark D″.1). In both cases the leak dominates the learning: the certified regime is the one in which the network does not learn. What the theorem adds to Theorem D′ is the removal of the horizon from the bound in that regime, with explicit constants (non-trivially at $\tau > 0$, where the coupled distance is of order $m^{-1/2}$ and not zero), and a precise statement of where the contraction method stops. Referee round 6 (`review/REVIEW_round6_Dpp.md`) confirmed the proof, corrected a double-counted factor 2 in the interconnection gain $L_{b'}$ (sharp constants below), and found that the run first reported as "learning regime" was a slow collapse to the dead kernel of an odd plant; the instance was changed, and the new long-horizon run in the learning regime shows the coupled distance growing in time — the uniform statement is supported only where the theorem proves it.

Proof. Keep the contracting terms that Theorem D′'s proof dropped. For $\delta := x - \bar{x}$, the term $-k_e\delta$ gives $D^+\|\delta\| \le -(k_e - L_f - L_x)\|\delta\| + L_\vartheta\bar{\Delta} + \zeta_t$ with the same decomposition of the bracket as in D′(b). For $\Delta_i := \vartheta_i - \bar{\vartheta}_i$, split $b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i) = [b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\vartheta_i)] + [b(\bar{x},\bar{e},\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i)]$. Since $e - \bar{e} = \delta$, the first bracket is $G\nabla_\vartheta\tilde{\Phi}(x;\vartheta_i)^\top(e-\bar{e}) + G[\nabla_\vartheta\tilde{\Phi}(x;\vartheta_i) - \nabla_\vartheta\tilde{\Phi}(\bar{x};\vartheta_i)]^\top\bar{e}$, of norm at most $\lambda_G L_\vartheta\|\delta\| + \lambda_G L_{x\vartheta}\|\delta\|E_X = L_{b'}\|\delta\|$ on the tube. The second bracket is $G[\nabla_\vartheta\tilde{\Phi}(\bar{x};\vartheta_i) - \nabla_\vartheta\tilde{\Phi}(\bar{x};\bar{\vartheta}_i)]^\top\bar{e} - \sigma\Delta_i$, and $\langle\Delta_i, \cdot\rangle \le \lambda_G\|\bar{e}\|L_2\|\Delta_i\|^2 - \sigma\|\Delta_i\|^2 \le -\kappa_\vartheta\|\Delta_i\|^2$, so $D^+\|\Delta_i\| \le -\kappa_\vartheta\|\Delta_i\| + L_{b'}\|\delta\|$, and the same for the average $\bar{\Delta}$. The pair $u := (\|\delta\|, \bar{\Delta})$ thus satisfies $D^+u \le Au + (\zeta_t, 0)^\top$ componentwise, $u(0) = 0$, with the Metzler matrix $A := \left[\begin{smallmatrix} -\kappa_e & L_\vartheta \\ L_{b'} & -\kappa_\vartheta \end{smallmatrix}\right]$ (non-negative off-diagonal entries), whose flow e^{As} is entrywise non-negative. The comparison principle for cooperative systems gives $u(t) \le \int_0^t e^{A(t-s)}(\zeta_s, 0)^\top \mathrm{d}s$ entrywise. A is Hurwitz exactly when its trace is negative and its determinant positive; with $\kappa_e > 0$ and $\det A = \kappa_e\kappa_\vartheta - L_{b'}L_\vartheta > 0$ one has $\kappa_\vartheta > L_{b'}L_\vartheta/\kappa_e \ge 0$, so both hold, and then $\int_0^\infty e^{As}\,\mathrm{d}s = -A^{-1} = (\det A)^{-1}\left[\begin{smallmatrix} \kappa_\vartheta & L_\vartheta \\ L_{b'} & \kappa_e \end{smallmatrix}\right]$, entrywise non-negative. Taking expectations, with $\mathbb{E}\zeta_s \le (\mathbb{E}\zeta_s^2)^{1/2} \le K_\Phi m^{-1/2}$ for every s (the $\bar{\vartheta}_j(s)$ are i.i.d. with law $\bar{\mu}_s$ and $\bar{x}_s$ is deterministic, as in D′(b)), $\mathbb{E}u(t) \le K_\Phi m^{-1/2}(-A^{-1})e_1 = K_\Phi m^{-1/2}(\kappa_\vartheta, L_{b'})^\top/\det A$ for every t. The per-particle statement is exchangeability at fixed t (valid here, unlike for $\sup_t$: Theorem D′(c)); the W_1 statement is the triangle inequality as in D′(c). For the second moment of $\bar{\mu}_t$: $z(t) := \mathbb{E}\|\bar{\vartheta}_t - \vartheta^0\|^2$ satisfies, by Itô's formula with $\|G\nabla_\vartheta(\bar{e}^\top\tilde{\Phi})\| \le \lambda_G E_X L_\vartheta =: \beta$, $\dot{z} \le -2\sigma z + 2\beta\sqrt{z} + 2\tau\,\mathrm{tr}\,G \le -\sigma z + \beta^2/\sigma + 2\tau\,\mathrm{tr}\,G$, whence $z(t) \le \max\{z(0), (\beta^2/\sigma + 2\tau\,\mathrm{tr}\,G)/\sigma\}$. The noise cancels in Δ_i, so the coupling estimate does not depend on τ. □

Remark D″.1 (reading)**.**

- *The certified regime is the leak-dominated one.* The condition $\kappa_\vartheta = \sigma - \lambda_G E_X L_2 > 0$ says that $\vartheta \mapsto b(\bar{x}, \bar{e}, \vartheta)$ is a contraction uniformly in t. At $\tau = 0$ the same estimate applied to two particles of the *same* system gives $D^+\|\vartheta_i - \vartheta_j\| \le -\kappa_\vartheta\|\vartheta_i - \vartheta_j\|$: all particles synchronise exponentially, the network collapses to a single kernel, and the m-system and the M-system — which share their first m particles — become identical, so the coupled distance is not just bounded but zero (the numerics: particle spread and coupled distance at the rounding level). At $\tau > 0$ the two Brownian increments do not cancel in $\vartheta_i - \vartheta_j$; the particles form a cloud around the common mean motion whose second moment is bounded as in the proof, $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$ to leading order when the leak dominates (measured: rms spread within 3% of $(\tau\,\mathrm{tr}\,G/\sigma)^{1/2}$ in the referee's runs), the coupled distance is of order $m^{-1/2}$ and the bound is not trivial — but the network's output stays at the level of its prior: $\sup|\hat{f}|$ is a few percent of $\sup|f|$ in the certified runs. Either way, propagation of chaos is uniform in time because the leak has stopped the learning. This is the price of the contraction method, not of the closed loop: the learning regime, in which kernels separate

and the network fits the plant (σ below the critical leak of Proposition K at the collapsed critical point, the leak below the separating force of the residual), is where Theorems H, I and K live, where the averaged interaction is non-reciprocal, and where uniformity in time remains the open part of Section 25, branch 3.

- *What the conditions say in the scalings of the program.* On the a-priori tube of Theorem D(b) with $e(0) = 0$, $k_e E_X = F + K_\Phi$, so the certified leak is $\sigma_{\mathrm{cert}} := \lambda_G E_X L_2 + L_{b'} L_\vartheta / \kappa_e = [\lambda_G (F + K_\Phi) L_2 + \lambda_G L_\vartheta^2]/k_e + O(k_e^{-2})$: every fixed leak gain $\sigma > 0$ certifies uniform-in-time propagation of chaos for k_e large, provided the tracking error starts at zero (or $\|e(0)\| = O(1/k_e)$; otherwise $E_X \geq \|e(0)\|$ and the condition does not improve with k_e) — and stops the learning. In the scaling $\sigma = \sigma_0/k_e$ of Theorem J the condition reads $\sigma_0 > \lambda_G (F + K_\Phi) L_2 + \lambda_G L_\vartheta^2$, which implies that J_m is strongly convex in the G^{-1} metric (its Hessian is bounded below by $\sigma_0 - \lambda_G \sup_\Theta \|r_\Theta\| L_2 \geq \sigma_0 - \lambda_G (F + K_\Phi) L_2$ there) with a unique, collapsed minimiser.

- *Why the constants are conservative.* L_ϑ, L_2, $L_{x\vartheta}$ are suprema over the parameter box and the tube; for sharp kernels (small $s_{\min}$) they are large, so σ_{cert} can exceed the adaptation gains by an order of magnitude; the a-priori E_X is itself several times the measured $\sup\|\bar{e}\|$ (0.23 against 0.07 in the numerics below). The gap to the learning regime is not a matter of constants, however: in the numerics the coupled distance grows over the horizon once the kernels separate.

- *Technique.* The comparison system is the standard contraction argument for McKean–Vlasov systems with a confining potential, with the plant as a second contracting channel (rate κ_e) and the two interconnection gains L_ϑ (particles $\to$ plant) and $L_{b'}$ (plant $\to$ particles); $\det A > 0$ is the small-gain condition of that interconnection.

Numerics (`exp_chaos_uniform.py` $\to$ `results/chaos/uniform_table.md`; reproduced in Table 16). A slow-adaptation instance ($n = 1$, $k_e = 25$, $G = 0.05 \cdot \mathrm{diag}(4, 1, 1)$, saturated features with $a_{\max} = 4$, $s \geq 0.3$, prior $\vartheta^0 = 0$), a two-bump plant that is not odd, $f(x) = 1.6e^{-(x-1)^2/0.4} - 1.2e^{-(x+1)^2/0.4} + 0.1x$, on $x_m = 1.5 \sin t + 0.5 \sin 2t$ (so that the dead kernel is not a critical point; referee round 6), $m = 16$, 64, 256 coupled to an $M = 2048$ reference. The constants are exact: the Lipschitz constants of the saturated features in closed form, $L_x = L_\vartheta = a_{\max} e^{-1/2}/s_{\min} = 8.087$ and $L_2 = L_{x\vartheta} = a_{\max}/s_{\min}^2 = 44.4$ (a numerical search over $1.2 \cdot 10^5$ parameter samples with Nelder–Mead polish reproduces them to 10^{-8}), $L_f = 2.29$, and the a-priori tube radius of Theorem D(b) with $e(0) = 0$, $E_X = (F + K_\Phi)/k_e = 0.228$ ($F = 1.70$; the measured $\sup\|\bar{e}\|$ is 0.070): $\kappa_e = 14.6$, $L_{b'} = 3.64$, $\sigma_{\mathrm{cert}} = 4.04$. Proposition K gives the critical leak of the collapsed critical point (the best single kernel, $a = 0.136$, $c = 0.068$, $s = 1.15$): $\sigma_K = \sigma_{0,\mathrm{crit}}/k_e = 0.00144$ — the certified leak is 2800 times the leak below which the kernels separate. Four runs:

Table 16: Theorem D″ on the slow-adaptation instance: four runs at different leaks (`results/chaos/uniform_table.md`).

σ, τ	regime	T, seeds	reference network at T: rms spread in (a, c, s) (from 1.53), fit error $\|f - \hat{f}\|/\|f\|$ (from 1.00)	max D_m, first $\to$ last window, $m = 16$ and 256	slope against m (per-seed, $\pm$ s.e.)	bound $\sup_t \mathbb{E}D_m$, $m = 16$ / 256
4.85 ($1.2\sigma_{\text{cert}}$), 0.02	certified, with dither	80 s, 8	0.033 (the dither cloud: $(\tau \operatorname{tr} G/\sigma)^{1/2} = 0.035$), 1.000	$6.7 \cdot 10^{-4} \to 7.9 \cdot 10^{-4}$ and $1.6 \cdot 10^{-4} \to 1.9 \cdot 10^{-4}$	-0.51 ± 0.02	0.55 / 0.14
4.85, 0	certified, no dither	80 s, 8	$1.7 \cdot 10^{-14}$, 1.000	$2.5 \cdot 10^{-25} \to 0$ and $1.2 \cdot 10^{-26} \to 0$	—	0.55 / 0.14
0.505 ($\sigma_{\text{cert}}/8$), 0	leak-dominated, below the threshold	80 s, 8	$3.2 \cdot 10^{-14}$, 1.000	$7.2 \cdot 10^{-5} \to 4.3 \cdot 10^{-11}$ and $3.1 \cdot 10^{-5} \to 1.8 \cdot 10^{-11}$	-0.33 ± 0.20	—
0.00048 ($\sigma_K/3$), 0	learning regime	1000 s, 6	1.53, 0.632	$6.1 \cdot 10^{-2} \to 2.0 \cdot 10^{-1}$ and $1.3 \cdot 10^{-2} \to 3.9 \cdot 10^{-2}$	-0.62 ± 0.16	—

(Windows $(10, 20]$, $(20, 40]$, $(40, 80]$ for the 80 s runs and $(100, 200]$, $(200, 500]$, $(500, 1000]$ for the long run; the table reports $\mathbb{E} \max_t D_m$ over a window, which exceeds the theorem's $\sup_t \mathbb{E}D_m$.) At the certified leak with dither the particles stay in the cloud of the prior, the network's output stays at the level of the empty network (fit error 1.000), and the coupled distance is of order $m^{-1/2}$ (slope -0.51 ± 0.02), bounded in time (the last window, twice as long, has maxima 15–18% above the first) and seven hundred times below the bound: the theorem's content at $\tau > 0$. Without dither the reference collapses (spread 10^{-14}) and the coupled distance is zero to rounding; at $\sigma_{\text{cert}}/8$ the particles still synchronise and the distance decays with them. In the learning regime ($\sigma_K/3$, the collapsed configuration unstable) the network stays spread, fits the plant (fit error $1.00 \to 0.63$ at $T = 1000$ s with these slow gains) and the coupled distance follows the $m^{-1/2}$ law at each time but *grows* over the horizon — by a factor 3 between the windows $(100, 200]$ and $(500, 1000]$ for every m: on this horizon propagation of chaos in the learning regime is the finite-horizon statement of Theorem D′, not a uniform one. Whether the growth saturates on longer horizons (the leak's own time scale is $1/\sigma = 2000$ s) is open; the earlier claim that the distance stays bounded in a "spread" regime (version 4, first draft) rested on a run that was in fact a slow collapse, and is withdrawn (referee round 6, DPP-3).

Checked by: `exp_chaos_uniform.py`: the experiment only.

18 Proposition G: the slow geometry

This section concerns joint descent, and variable projection (VarPro) in the induced metric.

Setting.

- $f = f(x)$ is time-invariant and C^2.
- x_m is T_0-periodic, with $\omega_{\text{ref}} := 2\pi/T_0$ and occupation measure $\rho := \frac{1}{T_0} \int_0^{T_0} \delta_{x_m(t)} \, \mathrm{d}t$.
- Law (L) acts in the interior, with gains $(g_a, \delta\gamma_c, \delta\gamma_s)$. Put $\Gamma := \operatorname{diag}(\gamma_c I_n, \gamma_s)$, acting per kernel, and $G_\xi := \delta\Gamma$. Write the geometry as $\xi := (c, s)$.
- $\bar{L}(a, c, s) := \frac{1}{2} \int \|f - \frac{1}{m} \sum_i a_i \varphi(\cdot\,; c_i, s_i)\|^2 \, \mathrm{d}\rho$, and $M(c, s) := \frac{1}{m^2} \int \varphi\varphi^\top \, \mathrm{d}\rho \otimes I_n$ is the Hessian of $\bar{L}$ in a.

- $\mathcal{K}$ is the set of geometries ξ on which (H1) holds, and $\mathcal{K}' \Subset \mathcal{K}$ is compact. (The set $\mathcal{K}$ is not the functional $\mathcal{G}(\mu)$ of Section 19.)
- $a^*(\xi) := \operatorname{argmin}_a \bar{L}$, and $L^*(\xi) := \bar{L}(a^*(\xi), \xi)$ is the profile (variable-projection) loss.
- Three dimensionless parameters:
  - the rate ratio $R := g_a m_0 / \big(\delta \cdot \sup_{\mathcal{K}'} \|\Gamma^{1/2} \nabla^2 L^* \Gamma^{1/2}\|\big)$;
  - the slope $S := (\delta/g_a) \cdot \sup_{\mathcal{K}'} \|D_\xi a^* \cdot \Gamma^{1/2}\|^2$;
  - the averaging parameter $\varepsilon_{\text{avg}} := (m/k_e)\, \lambda_{\max}(G^{1/2} \nabla^2 \bar{L}\, G^{1/2})/\omega_{\text{eff}}$, the second ratio in (H2a′). The result tables in `results/variable_projection` report its inverse, with ω_{ref} in place of ω_{eff}.

Hypotheses.

(H0) On the interval considered, the closed-loop and reduced geometries stay in $\mathcal{K}'$. The parameters stay in $\operatorname{int} \Theta_K^\varepsilon$, and $\sup_{\mathcal{K}'} \max_i \|a_i^*\| < a_{\max} - \varepsilon_K$.

(H1) $M \succeq m_0 I$ on $\mathcal{K}$.

(H2a′) *Two scales.* $k_e \gg \omega_{\text{eff}}$ and $\varepsilon_{\text{avg}} \ll 1$, where

$$\omega_{\text{eff}} := \max\Big\{\omega_{\text{ref}},\ \frac{\sup_t \|\dot{x}_m\|}{s_{\min}},\ \sup_X \|\partial_x (f - \hat{f})\|,\ \frac{\sup_t \|\frac{\mathrm{d}}{\mathrm{d}t} r(x_m(t))\|}{\sup \|r\|}\Big\}.$$

The last entry is the bandwidth of the residual along the reference. Also $\sup\|r\|/k_e \ll s_{\min}$.

(H2b) *Rate separation.* $R \gg 1$: the slowest weight rate $g_a m_0 m/k_e$ is much larger than the geometry rate $(m\delta/k_e)\|\Gamma^{1/2} \nabla^2 L^* \Gamma^{1/2}\|$.

(H2c) *Flat slow manifold.* $S \ll 1$. Since $Da^* = -M^{-1} \partial^2_{a,\xi} \bar{L}$, a checkable sufficient condition is $\delta \cdot \max(\gamma_c, \gamma_s) \cdot \|\partial^2_{a,\xi} \bar{L}\|^2 \ll g_a m_0^2$.

Three remarks on the hypotheses.

- **(H2b) is about $\lambda_{\min}$.** It is a condition on $\lambda_{\min}$ of the feature Gram matrix. Since $0 < \varphi \le 1$, $m_0 \le 1/m^2$. So the slowest weight rate is at most $g_a/(k_e m)$, and overlapping kernels make (H2b) fail.
- **The ratios are gain choices.** R and S contain neither k_e nor ω_{ref}. The separation between weights and geometry is therefore set by the gains.
- **(H2b) does not imply (H2c).** The product

$$R \cdot S = m_0 \cdot \sup\|Da^* \Gamma^{1/2}\|^2 \big/ \sup\|\Gamma^{1/2} \nabla^2 L^* \Gamma^{1/2}\|$$

does not depend on g_a, δ or k_e. It is fixed by the problem and by the shape of Γ: in the slope configuration below, a numerical check gives values from 166 to 1149 over $\gamma_s/\gamma_c \in [10^{-4}, 10^4]$. It is large whenever refitting the weights largely compensates a change of geometry.

Proposition G (formal reduction)**.**

(a) Two scales, under (H0) and (H2a′). *To leading order in ω_{eff}/k_e and ε_{avg}, the period-averaged closed loop is the joint gradient flow*

$$\dot{\vartheta}_i = -\frac{m}{k_e}\, G\, \nabla_{\vartheta_i} \bar{L}(\vartheta_1, \dots, \vartheta_m). \tag{18.1}$$

Its measure form is $-\frac{1}{k_e} G \nabla(\delta \bar{L}/\delta \mu)$.

- *This is the drift of* $(\mathfrak{A})$ *at* $\sigma = \tau = 0$*, with* Φ *in place of* $\tilde{\Phi}$ *and* (a, c, s) *in place of* (α, γ, κ)*, by the same averaging computation as Theorem H(i).*
- *It is not literally a special case of H(i): the saturated coordinates there induce a different, state-dependent metric.*
- $\bar{L}$ *is non-increasing along the averaged flow. Along the closed loop,* $\bar{L}$ *at period-averaged parameters decreases while the averaged dissipation* $\frac{m}{k_e}\|\nabla\bar{L}\|_G^2$ *dominates the* $O(\omega_{\text{eff}}/k_e)$ *corrections of Proposition H.7 and Remark H.7a. Near equilibrium it can increase by* $O((\omega_{\text{eff}}/k_e)^2)$*, as the weights settle at their shifted equilibrium (in a numerical check:* $+8 \cdot 10^{-7}$ *over 35 periods at* $k_e = 20$*, and* $3.7 \cdot 10^{-8}$*, about 22 times less, at* $k_e = 80$*).*

(b′) Three scales, for the averaged system (a), under (H0), (H1), (H2a′) and (H2b). *After a boundary layer, the weights follow the slow manifold:*

$$\begin{aligned} a - a^*(\xi) &= -\frac{k_e}{m g_a} M^{-1} Da^* \cdot \dot{\xi} + \text{h.o.t.} \\ &= \frac{\delta}{g_a} M^{-1} Da^* \Big(\Gamma^{-1} + \frac{\delta}{g_a}(Da^*)^\top Da^*\Big)^{-1} \nabla L^*(\xi) + \text{h.o.t.} \end{aligned} \tag{18.2}$$

This reduces to $\frac{\delta}{g_a} M^{-1} Da^* \, \Gamma \, \nabla L^*(\xi)$ *when* $S \ll 1$*. The geometry follows, to leading order in* $1/R$ *and for every* S*,*

$$\dot{\xi} = -\frac{m\delta}{k_e}\Big(\Gamma^{-1} + \frac{\delta}{g_a}(Da^*)^\top Da^*\Big)^{-1} \nabla L^*(\xi). \tag{18.3}$$

- *This is the gradient flow of* $(m/k_e)L^*$ *in the metric that the* G^{-1} *metric induces on the graph* $\{(a^*(\xi), \xi)\}$*.*
- *Its metric tensor is* $G_\xi^{-1} + g_a^{-1}(Da^*)^\top Da^*$*, and it couples all kernels through* Da^**.*
- *In the closed loop, the weight equilibrium is further shifted by* $O(\omega_{\text{eff}}/k_e)$ *(Proposition H.7). This shift can exceed the offset displayed above: in a numerical check on the* `satisfying` *run it is* $3\text{–}6 \cdot 10^{-3}$*, against an offset of* $4 \cdot 10^{-5}\text{–}3 \cdot 10^{-4}$*.*

(b) Variable projection, under (H2c) in addition. *Let* $(\bar{c}, \bar{s})$ *solve* $\bar{\xi}' = -\frac{m\delta}{k_e}\Gamma\nabla L^*(\bar{\xi})$*, that is,* $\bar{c}_i' = -\frac{m\delta\gamma_c}{k_e}\nabla_{c_i} L^*$ *and* $\bar{s}_i' = -\frac{m\delta\gamma_s}{k_e}\partial_{s_i} L^*$*, with* $(\bar{c}, \bar{s})(0) = (c, s)(0)$*. Then*

$$\sup_{t \le \tau_1 k_e/(m\delta)} \|(c,s)(t) - (\bar{c},\bar{s})(t)\| \to 0 \quad \text{as } (\omega_{\text{eff}}/k_e,\ \varepsilon_{\text{avg}},\ S,\ 1/R) \to 0, \text{ with } \tau_1 \text{ fixed.} \tag{18.4}$$

- *Formally the error is* $O(\omega_{\text{eff}}/k_e + \varepsilon_{\text{avg}} + S + 1/R)$*.*
- L^* *is non-increasing along* $\bar{\xi}$*, and the weights track* a^**.*
- *This flow is the* $S \to 0$ *limit of (b′).*

Status. Formal (a derivation without error control); checked directly, including a counterexample to plain VarPro at $S = O(1)$ (see the numerics below). Novelty (literature audit): a fixed-m, closed-loop analogue of the two-timescale/VarPro training of [48], [5].

No mean-field form. For fixed m, on configurations of m distinct geometries in $\mathcal{K}$, the flow (b) is the gradient flow of L^*/k_e in the metric $(1/m)\sum_i \|\dot{\xi}_i\|^2_{G_\xi^{-1}}$. Locally, that metric is the Wasserstein metric restricted to uniform m-atom measures.

- L^* is the unregularised profile loss. It is the $\lambda \to 0$ limit of $\lambda\mathcal{L}^\lambda$ of [5] (time rescaled by λ). It is not the $\lambda \to 0$ functional $\mathcal{L}^0$ of [5], which is a min-norm interpolation functional whose flow is an ultra-fast diffusion.

- As a functional of the geometry marginal, L^* depends only on the support, because the free weights absorb the masses. So it is not W-continuous, and the statement has no mean-field form. This is consistent with $m_0 \le 1/m^2$.

Derivation (formal). Each step is a standard theorem on its own time scale. Their composition on the $O(k_e/(m\delta))$ horizon is formal.

1. *Error channel (exact identity).* For $w := e - r(x,\vartheta)/k_e$, $\dot w = -k_e w - \dot r/k_e$. So $\|w(t)\| \le e^{-k_e t}\|w(0)\| + \sup\|\dot r\|/k_e^2$, and
$$e = r(x_m)/k_e + O\big(\sup\|\dot r\|/k_e^2 + \sup\|\partial_x r\| \cdot \sup\|r\|/k_e^2\big).$$
This is $O(\omega_{\text{eff}} \cdot \sup\|r\|/k_e^2)$, with the bandwidth term in ω_{eff}.
2. *Period averaging of the joint system* (Khalil [40, Thm. 10.4]; Sastry–Bodson [67, Ch. 4]). This gives (a), on $[0, b/\varepsilon_{\text{avg}}]$.
3. *Weights against geometry.* Use Tikhonov in slow time (Kokotović–Khalil–O'Reilly [41]) with the uniformly exponentially stable averaged weights on $\mathcal{K}'$ (H1).
   - On the slow manifold $a = h(\xi)$, $\dot a = Dh\,\dot\xi = -\frac{mg_a}{k_e} M(h - a^*)$, so $h - a^* = -\frac{k_e}{mg_a} M^{-1} Dh\,\dot\xi$. This must be evaluated with the (b′) velocity, not the VarPro velocity.
   - Since $\bar L$ is quadratic in a and $\partial^2_{\xi a}\bar L = -(Da^*)^\top M$, we get $\nabla_\xi \bar L(h,\xi) = \nabla L^* - (Da^*)^\top M(h - a^*) + O((\delta/g_a)^2)$.
   - Substituting into $\dot\xi = -\frac{m\delta}{k_e}\Gamma\nabla_\xi \bar L(h,\xi)$, with $Dh = Da^* + O(1/R)$, gives (b′). The resummed form solves the invariance equation exactly once $Dh - Da^*$ and the quadratic term are dropped, so the remaining corrections are in $1/R$ (they involve D^2a^* and $\partial_\xi M$), not in S.
4. *Envelope theorem.* $\nabla_\xi \bar L(a^*(\xi),\xi) = \nabla L^*(\xi)$. When $S \ll 1$, the metric correction in (b′) drops out, which gives (b).

The composition requires the averaged system to remain valid up to $t \sim \tau_1 k_e/(m\delta)$, which is much longer than $b/\varepsilon_{\text{avg}}$. Step 2 does not certify this. A proof would combine:

- a near-identity averaging transform;
- uniform exponential stability of the averaged weights on $\mathcal{K}'$;
- Tikhonov in slow time with an $O(\varepsilon_{\text{avg}})$ perturbation;
- Grönwall in ξ (Sastry–Bodson [67, §4.4]; Sanders–Verhulst–Murdock [64, Ch. 5]).

□

Numerics. The results are in `results/variable_projection` of the accompanying repository; Table 17 summarises the comparison.

- The first two rows use centred differences of period means.
- The slope rows use the displacement between the first and last period means. They start on the slow manifold of (b′), so no weight boundary layer enters the comparison.
- (b′) was evaluated only in the slope rows.
- k_e/ω_{eff} is 2.0 in the first two rows with the box's $s_{\min}$, and 2.6–4.8 with the actual widths, so (H2a′) holds only marginally there. In the `satisfying` run the residual bandwidth, about 7.7, then dominates.
- k_e/ω_{eff} is about 16 in the slope rows.

configuration	(H2b) ratio	S	G(a): closed loop against the joint flow (a)	G(b) and the induced-metric flow (b′)
`violating` ($m = 4$, overlapping)	0.10–0.23	5.4–25	joint cosine 0.9994 / 0.9985, size ratio 1.00 / 0.94	(b) fails: cosine -0.12 / 0.39, and L^* *increases* from $3.8 \cdot 10^{-3}$ to $5.9 \cdot 10^{-3}$
`satisfying` ($m = 2$)	79–87	0.006	joint cosine 0.985 / 0.962, size 0.96 / 0.94; geometry block 0.999 / 0.998; the weight block fails (see below)	(b) holds: cosine 0.999, ratio 0.97; L^* decreases from $1.20 \cdot 10^{-2}$ to $4.11 \cdot 10^{-3}$
`slope`, $\delta = 0.47$ ($m = 1$)	81–178	2.0–2.8	cosine 0.9992, size 1.10	(b) fails: ratio 0.42, cosine 0.93, and the centre moves the opposite way; (b′): cosine 0.9986, ratio 1.12
`slope`, $\delta = 0.47$, all gains $\div 4$	81–178	2.0–2.8	cosine 1.0000, size 1.05	(b) fails: ratio 0.40, cosine 0.92; (b′): cosine 0.9999, ratio 1.065
`slope`, $\delta = 0.047$	654–1779	0.20–0.32	cosine 1.0000, size 1.04	(b): ratio 0.85, cosine 0.9986; (b′): ratio 1.05, cosine 1.0000
`slope`, $\delta = 0.0047$	13162–17790	0.020–0.023	cosine 1.0000, size 1.04	(b): ratio 1.02; (b′): ratio 1.04; both cosines 1.0000

Table 17: Proposition G: the closed loop against the averaged predictions. In the first two rows, an entry 'p / q' gives the medians over the first quarter (p) and over the last quarter (q) of the run.

- **Joint descent.** In all runs, $\bar{L}$ at the period-mean parameters is non-increasing, sampled once per period.
- **(H2b) is not enough.** The `slope` rows satisfy (H2b) by a wide margin, yet (b) fails until S is small. Meanwhile (b′) tracks the closed loop at every S. So (H2c) is the hypothesis missing from an earlier formulation of Proposition G(b), and (b′) is the correct leading-order slow flow. The configuration is a counterexample to that earlier formulation, made reproducible in `experiments/exp_slow_manifold_slope.py`.
- **Error decomposition** (`slope`, $\delta = 0.47$, started on the slow manifold):
  - $\|(\mathrm{a})\|/\|(\mathrm{b}')\| = 1.015$. This is the intrinsic $O(1/R)$ error of (b′).
  - $\|\mathrm{closed}\|/\|(\mathrm{a})\| = 1.10$. It falls to 1.05 when all gains are divided by 4, which leaves R and S unchanged and divides $\varepsilon_{\mathrm{avg}} \approx 0.08$ by 4. So this part is the averaging error.
  - The remaining $\approx 4.4\%$ is the $O(1/k_e)$ error-channel factor at $k_e = 20$. It is the same at $\delta = 0.047$ and 0.0047.
- **The size ratio in the `satisfying` run.** The centred ratio of 0.97 is the error-channel factor times a $+1.4\%$ averaging correction from the weights' oscillation within each period, the analogue of Remark H.7a.
  - The frozen-parameter ratio is 0.951–0.956 along the run, and 0.996 at $k_e = 80$.
  - The fastest weight rate over ω_{ref} is about 0.14.
  - In a further numerical check, with the weights running at frozen geometry the ratio is 0.969–0.970; with $g_a/4$ it is 0.956–0.960.
  - The slope correction is below 0.01% here, because ∇L^* is nearly orthogonal to the range of $(Da^*)^\top$. An earlier estimate of 0.94–0.95 contained a 2.5% forward-difference bias.

- **The weight block.** In the `satisfying` run the weights sit at $a^* + O(1/k_e)$, within 0.1% of a^*.
  - Their actual drift is tiny: 0.001–0.008 of the geometry speed in the G^{-1} norm.
  - The prediction $-(m/k_e)g_a\nabla_a\bar{L}$ at the period-mean weights is 0.17–0.30 of the geometry speed, 40–200 times the actual drift. It is the restoring velocity of the $O(1/k_e)$ offset, which (a) does not contain.
  - So the weight block fails (cosine -0.21 / 0.50, carrying 92–95% of the residual), while the geometry block matches.

Checked by: `experiments/exp_variable_projection.py`, which writes `table_violating.md` and `table_satisfying.md`; and `experiments/exp_slow_manifold_slope.py`, which writes `table_slope.md` (all in `results/variable_projection`).

Status.

- **What the loop supplies.** The error-channel and averaging scales, and the data distribution (the occupation measure of the reference), come from the closed loop.
- **What the designer supplies.** The separation between weights and geometry is a gain choice, as in [48], [5]: $\delta \ll g_a \cdot \min\big(m_0/\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|,\ 1/\|Da^*\Gamma^{1/2}\|^2\big)$. Equivalently, $\|G_\xi^{1/2}\nabla^2 L^* G_\xi^{1/2}\| \ll g_a m_0$ and $\|Da^* G_\xi^{1/2}\|^2 \ll g_a$.
- **What is new here.**
  - The learner's data distribution is its own excitation, and at finite k_e it moves with the parameters (Proposition H.7(c)).
  - The slow flow is variable projection in the *induced* metric, valid for every S. The slope S measures the departure from plain VarPro.

19 Theorem H: Langevin adaptive control and global convergence to the regularised optimum

Why something new is needed. Theorems B–G are local, or describe the dynamics. A global statement about kernel placement needs either the Chizat–Bach theory or entropy.

- **Boundary conditions.** The Chizat–Bach theory for noiseless Wasserstein flows (partially 1-homogeneous case, [10]) requires boundary conditions: either Neumann conditions on a bounded convex domain, or C^1-uniform behaviour at infinity. Gaussian kernels with free centres and widths satisfy neither.
- **Separation.** Its Theorem 3.5 also needs the support of μ_0 to separate $\{-r_0\} \times \Theta$ from $\{r_0\} \times \Theta$. A Dirac start violates this, and that is an obstruction distinct from the boundary conditions.
- **The collapsed configuration is invariant.** It remains a moving Dirac mass and never separates without noise, by permutation symmetry and uniqueness of solutions. The test `test_noiseless_symmetric_initialisation_never_separates_but_dither_does` in the accompanying repository checks this on the real closed loop with $m = 6$. The rows stay bit-identical at every snapshot, the common particle moves, and dither separates the rows within 0.1 s. (Whether the collapsed configuration is a strict local minimum of the regularised loss, so that nearby non-collapsed starts return to it, is decided by the leak gain against the critical leak σ_{crit} of Proposition K, Section 22.)

Entropy removes the obstruction: Theorem H(ii) covers Dirac starts after any positive time.

Design. The particles are unconstrained, $\vartheta = (\alpha, \gamma, \kappa) \in \mathbb{R}^p$, and the features are saturated coordinate by coordinate:

$$a = a_{\max}\tanh(\alpha/a_{\max}), \qquad c = c_{\max}\tanh(\gamma/c_{\max}), \qquad s = s_{\min} + (s_{\max} - s_{\min}) \cdot \mathrm{sig}(\kappa).$$

Here $a_{\max}$ bounds each coordinate of a, unlike Section 9, where it bounds $\|a\|$. So $\tilde{\Phi}(x;\vartheta) = a\varphi(x;c,s)$ satisfies $\|\tilde{\Phi}\| \le K_\Phi := a_{\max}\sqrt{n}$, and every ϑ-derivative of $\tilde{\Phi}$ is bounded uniformly in x. The law is

$$\mathrm{d}\vartheta_i = \big[\, G\nabla_\vartheta g_t(\vartheta_i) - \sigma(\vartheta_i - \vartheta^0)\,\big]\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad g_t(\vartheta) := e(t)^\top \tilde{\Phi}(x(t);\vartheta). \tag{L_τ}$$

It has three parts: the classical gradient law, σ-modification written in the G^{-1} metric, and parameter dither of temperature τ.

Averaged mean-field system. Assume $f = f(x)$ is continuous and time-invariant, as in Proposition G (the functions `occupation_loss` and `first_variation` of the accompanying code assume this). Let ρ be the occupation measure of a T_0-periodic reference, and $F_\rho := \sup_{\mathrm{supp}\,\rho}\|f\|$. Define

$$r_\mu := f - \int \tilde{\Phi}(\cdot\,;\vartheta')\,\mathrm{d}\mu(\vartheta'), \qquad \bar{L}(\mu) := \frac{1}{2}\int \|r_\mu\|^2\,\mathrm{d}\rho,$$
$$V_L[\mu](\vartheta) := -\int r_\mu(x)^\top \tilde{\Phi}(x;\vartheta)\,\mathrm{d}\rho(x), \qquad V[\mu] := \frac{1}{k_e}V_L[\mu] + \frac{\sigma}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}},$$

and the averaged system

$$\partial_t\mu = \nabla\cdot\big(\mu G\nabla V[\mu]\big) + \tau\nabla\cdot(G\nabla\mu). \tag{$\mathfrak{A}$}$$

Remark H.0 (from the closed loop to ($\mathfrak{A}$): a formal reduction with a measured error)**.**

- **An algebraic identity.** Substitute $e := r_\mu(x_m)/k_e$ and $x := x_m$ in the drift of (L_τ), and average over one reference period at frozen μ. The result is exactly $-G\nabla V[\mu]$. The test `test_averaged_drift_is_the_mean_field_langevin_drift` checks this identity to 10^{-7}. It does not test averaging.
- **The formal reduction.** ($\mathfrak{A}$) is the leading-order description of the mean-field closed loop under two conditions:
  - *Tikhonov:* $k_e \gg \max\big(\omega_{\mathrm{ref}},\ \mathrm{Lip}(f) + K_\Phi/s_{\min},\ (F_\rho + K_\Phi)/s_{\min}\big)$, so that the shift $x - x_m = O(1/k_e)$ is small on the kernel scale.
  - *Period averaging:* $T_0\cdot\sup_\vartheta\|(1/k_e)G\nabla V_L[\mu](\vartheta)\| \ll \ell$, $\sigma T_0 \ll 1$ and $\tau\lambda_G T_0 \ll \ell^2$, where ℓ is the ϑ-scale of $\tilde{\Phi}$. The left-hand side is at most $(T_0/k_e)\lambda_G(F_\rho + K_\Phi)\cdot\sup\|\nabla_\vartheta\tilde{\Phi}\|$. The confinement drift does not depend on time and needs no averaging.
- **The first corrections.**
  - At frozen μ (and frozen ϑ) they are $O(1/k_e)$ relative to the drift, do not depend on the gains, and are given exactly by Proposition H.7.
  - Along the loop, the within-period oscillation of the parameters adds the second-order averaging term b_{osc} of Remark H.7a. Its size relative to the frozen terms is $O(|G|T_0)$, so it does depend on the gains.
  - In the formal limit below, both are $O(k_e^{-2})$.

  Measured relative errors of the frozen-μ averaged drift against ($\mathfrak{A}$), with b_{osc} excluded:

instance	$k_e = 10$	20	40	80	$k_e \times$ error
$n = 1$	4.2%	2.0%	0.91%	0.48%	≈ 0.4
$n = 2$, ellipse	43%	21%	10%	5.1%	≈ 4.2

Tests: `test_averaging_error_of_the_formal_reduction_decreases_like_one_over_ke` (instance $n = 1$) and `test_averaged_drift_error_decays_like_one_over_ke_for_an_orbit_enclosing_area` (instance $n = 2$; $k_e = 10, 20, 40, 80$).

- $k_e = 10$ is outside the averaging regime for the second instance.
- Its larger error does not come from the enclosed area. Proposition H.7 attributes it to the orientation-even state-shift term: it predicts $k_e \times \text{error} \approx 4.05$, of which U_{odd} alone would give 1.5, and a numerical check on the reversed orbit gives 4.0.
- A numerical check shows that at the gains of these instances b_{osc} is not small. k_e times its relative size is 0.92 ($n = 1$, $G = \text{diag}(4, 1, 1)$) and 4.4 ($n = 2$, $G = I$), against 0.42 and 4.05 for the frozen terms.

- **The exact formal limit** is $k_e \to \infty$ with G, $k_e\sigma$ and $k_e\tau$ fixed, in slow time t/k_e (Conjecture H.6).
- **Not proved here.** The mean-field limit of (L_τ) itself is not proved; Theorem D covers the deterministic projected law (L).

Status. Formal reduction, with a measured error.

Theorem H. *Assume the setting above, and $\sigma > 0$, $\tau > 0$ where stated.*

(i) Identification. *$V[\mu]$ is the first variation of $\mathcal{G}(\mu) := \frac{1}{k_e}\bar{L}(\mu) + \frac{\sigma}{2}\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$. Hence ($\mathfrak{A}$) is the mean-field Langevin dynamics, in the G^{-1} metric, of the free energy $F_\tau := \mathcal{G} + \tau H$, with $H(\mu) = \int \mu \log \mu$. Equivalently, with $\gamma_\tau := \mathcal{N}(\vartheta^0, (\tau/\sigma)G)$,*

$$F_\tau = \frac{1}{k_e}\bar{L} + \tau H(\cdot \,|\, \gamma_\tau) + \text{const.}$$

F_τ is well defined on all of $\mathcal{P}(\mathbb{R}^p)$, and finite exactly when $H(\mu \,|\, \gamma_\tau) < \infty$.

(ii) Global exponential convergence ($\sigma, \tau > 0$). *F_τ has a unique minimiser, $\mu^*_\tau \propto e^{-V[\mu^*_\tau]/\tau}$. For every $\mu_0 \in \mathcal{P}_2(\mathbb{R}^p)$ with $F_\tau(\mu_0) < \infty$,*

$$F_\tau(\mu_t) - F_\tau(\mu^*_\tau) \le e^{-2\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu^*_\tau)\big),$$
$$W_{G^{-1}}(\mu_t, \mu^*_\tau)^2 \le \frac{2}{\tau\rho_\tau}\, e^{-2\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu^*_\tau)\big).$$

Here ρ_τ is a log-Sobolev constant, for the whitened measures, of every proximal Gibbs measure $\hat{\mu} \propto e^{-V[\mu]/\tau}$. In ϑ-coordinates this means Fisher information in the G metric. It satisfies

$$\rho_\tau \ge \frac{\sigma}{\tau} \cdot \exp\Big(-\frac{2K_\Phi(F_\rho + K_\Phi)}{k_e\tau}\Big).$$

Suppose μ_0 is only in $\mathcal{P}_2(\mathbb{R}^p)$, for example a Dirac mass or an empirical measure, so that $H(\mu_0) = +\infty$. Then $F_\tau(\mu_{t_0}) < \infty$ for every $t_0 > 0$, and the estimates hold with (μ_0, t) replaced by $(\mu_{t_0}, t - t_0)$.

(iii) Loss and tracking. *Let $\bar{e}_t := r_{\mu_t}/k_e$. Then $\langle\|\bar{e}_t\|^2\rangle_\rho = 2\bar{L}(\mu_t)/k_e^2$, and*

$$\begin{aligned}|\bar{L}(\mu_t) - \bar{L}(\mu^*_\tau)| &\le (F_\rho + K_\Phi)L_\Phi \cdot W_{G^{-1}}(\mu_t, \mu^*_\tau)\\ &\le (F_\rho + K_\Phi)L_\Phi\Big(\frac{2}{\tau\rho_\tau}\Big)^{1/2} e^{-\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu^*_\tau)\big)^{1/2},\end{aligned}$$

where $L_\Phi := \sup_{x,\vartheta,\|u\|=1} \|G^{1/2}\nabla_\vartheta(u^\top\tilde\Phi)\|$. *The regularisation bias satisfies*

$$\bar L(\mu_\tau^*) \le \inf_\nu \big\{\bar L(\nu) + k_e\tau H(\nu\,|\,\gamma_\tau)\big\}.$$

The link $e \approx \bar e$ *between* $(\mathfrak{A})$ *and the closed loop is formal (Remark H.0).*

(iv) Annealing. *Let* $\mu_0 \in \mathcal{P}_2$ *with* $F_{\tau_{\text{init}}}(\mu_0) < \infty$. *Let* (τ_t) *be smooth and decreasing from* τ_{init}, *with* $\tau_t = \alpha_a/\log t$ *for* $t \ge t_1 > 1$, *where* $\alpha_a > \alpha^* := 2K_\Phi(F_\rho + K_\Phi)/k_e$. *Then* $\mathcal{G}(\mu_t) - \inf\mathcal{G} \le C' \cdot \log\log t/\log t$, *where* $\inf\mathcal{G}$ *is the* σ*-regularised optimum. This is Theorem 4.1 of [11]. Its hypotheses hold:*

- $\mathcal{G} \ge 0$*;*
- *Assumptions 1–3 hold for every* $\tau > 0$*;*
- *for* $\tau \in (0, \tau_{\text{init}}]$, *(ii) gives* $\rho_\tau \ge C_0 e^{-\alpha^*/\tau}$ *with* $C_0 = \sigma/\tau_{\text{init}}$, *which is all its proof uses.*

Status. Proved for $(\mathfrak{A})$; the reduction to $(\mathfrak{A})$ is formal (Remark H.0); real loop: Conjecture H.6. Novelty (literature audit): a bridge to [11], [52]; new in adaptive control.

Checked by: `tests/test_langevin.py`:
`test_first_variation_bound_used_in_holley_stroock` (the oscillation bound in (ii));
`test_lsi_lower_bound_formula` (formula transcription only);
`experiments/exp_langevin_global.py`, `exp_langevin_tradeoff.py` and `exp_gibbs_reference.py` (global behaviour, speed–bias trade-off, reference values).

Proof. (i) We have

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bar L\big((1-\varepsilon)\mu + \varepsilon\nu\big)\Big|_{\varepsilon=0} = -\int r_\mu^\top \Big(\int \tilde\Phi\,\mathrm{d}(\nu-\mu)\Big)\,\mathrm{d}\rho = \int V_L\,\mathrm{d}(\nu-\mu).$$

This is exact, because $\bar L$ is quadratic in μ. The confinement is linear in μ. In the G^{-1} metric, $\nabla\cdot\big(\mu G\nabla(V+\tau\log\mu)\big) = \nabla\cdot(\mu G\nabla V) + \tau\nabla\cdot(G\nabla\mu)$. Since $\log\gamma_\tau = -(\sigma/(2\tau))\|\vartheta-\vartheta^0\|^2_{G^{-1}} + \text{const}$, we get $\tau H(\mu\,|\,\gamma_\tau) = \tau H(\mu) + (\sigma/2)\int\|\vartheta-\vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + \text{const}$.

(ii) Whiten with $\zeta := G^{-1/2}\vartheta$. The metric becomes Euclidean, the confinement becomes $(\sigma/2)\|\zeta-\zeta^0\|^2$, $W_{G^{-1}}$ becomes W_2, and H shifts by $\frac12\log\det G$. $(\mathfrak{A})$ becomes equation (12) of [11]. We verify Assumptions 1–3 of [11].

- *Assumption 1.* $\nabla V[\mu](\zeta)$ is Lipschitz in (ζ,μ). The constants do not depend on μ, because $\|r_\mu\| \le F_\rho + K_\Phi$ on $\operatorname{supp}\rho$ for every $\mu \in \mathcal{P}(\mathbb{R}^p)$.
  - In ζ, this holds because $\nabla^2\tilde\Phi$ is bounded (saturation, and $s \ge s_{\min}$).
  - In μ, it holds because $\int\tilde\Phi\,\mathrm{d}\mu$ is W_1-Lipschitz, $W_1 \le W_2$, and $\nabla\tilde\Phi$ is bounded.
- *Assumption 2.* $\mathcal{G}$ is convex: a linear map followed by a convex quadratic, plus a linear term. F_τ has a minimiser by confinement and lower semicontinuity (proof of [11, Prop. 5.1]).
- *Assumption 3.* $e^{-V[\mu]/\tau} = e^{-(\sigma/(2\tau))\|\zeta-\zeta^0\|^2}\cdot e^{-V_L[\mu]/(k_e\tau)}$.
  - The Gaussian factor satisfies $\mathrm{LSI}(\sigma/\tau)$ (Bakry–Émery [3]).
  - $|V_L[\mu]| \le \int\|r_\mu\|\|\tilde\Phi\|\,\mathrm{d}\rho \le (F_\rho+K_\Phi)K_\Phi$, so the perturbation has oscillation at most $2K_\Phi(F_\rho+K_\Phi)/(k_e\tau)$. Here F_ρ may be replaced by $\int\|f\|\,\mathrm{d}\rho$.
  - Holley–Stroock [35] gives ρ_τ.

Theorem 3.2 and Corollary 3.3 of [11] (the latter with Otto–Villani [53]) give the two inequalities. Uniqueness follows from strict convexity of H.

$\mathcal{P}_2$ *starts.* Work in ζ-coordinates and fix the solution μ_t. Compare the particle SDE with the Ornstein–Uhlenbeck process $\mathrm{d}Y = -\sigma(Y-\zeta^0)\,\mathrm{d}t + \sqrt{2\tau}\,\mathrm{d}B$ started from the same law.

- The two drifts differ by $b_t = -(1/k_e)G^{1/2}\nabla V_L[\mu_t]$, with $\|b_t\| \le B := (1/k_e)\lambda_G^{1/2}(F_\rho + K_\Phi) \cdot \sup\|\nabla_\vartheta \tilde{\Phi}\|$.
- Girsanov and data processing give $H(\mu_{t_0} \,|\, \mathrm{OU}_{t_0}) \le B^2 t_0/(4\tau)$.
- OU_{t_0} has a density at most $\big(2\pi(\tau/\sigma)(1-e^{-2\sigma t_0})\big)^{-p/2}$, so $H(\mu_{t_0}) < \infty$.
- $\mu_{t_0} \in \mathcal{P}_2$, because the drift grows linearly.

Hence $F_\tau(\mu_{t_0}) < \infty$.

(iii) $\bar{L}(\mu) - \bar{L}(\nu) = \frac{1}{2}\int (r_\mu - r_\nu)^\top (r_\mu + r_\nu)\,\mathrm{d}\rho$. Here $\|r_\mu + r_\nu\| \le 2(F_\rho + K_\Phi)$, and $\|r_\mu - r_\nu\| = \|\int \tilde{\Phi}\,\mathrm{d}(\nu - \mu)\| \le L_\Phi \cdot W_{1,G^{-1}} \le L_\Phi \cdot W_{G^{-1}}$. Conclude with (ii). For the bias: $(1/k_e)\bar{L}(\mu_\tau^*) + \tau H(\mu_\tau^* \,|\, \gamma_\tau) \le (1/k_e)\bar{L}(\nu) + \tau H(\nu \,|\, \gamma_\tau)$ for every ν, and $H(\mu_\tau^* \,|\, \gamma_\tau) \ge 0$.

(iv) This is Theorem 4.1 of [11] with the stated hypotheses. □

Remark H.1 (dither adds no output noise in the mean-field limit; its cost is the entropic bias)**.** In the limit, $\hat{f} = \int \tilde{\Phi}\,\mathrm{d}\mu$ is deterministic by the law of large numbers. The only effect of dither is the entropic bias, which Theorem H(iii) quantifies.

In the experiment ($\sigma = 0.02$, $\tau = 0.004$, $G = \mathrm{diag}(4,1,1)$, $k_e = 10$), a grid computation gives $\bar{L}(\mu_\tau^*) = 0.11118$. It is grid-converged (`exp_gibbs_reference.py` → `results/langevin_global/gibbs_reference.md`). At $(\sigma,\tau) = (0.002, 5\cdot 10^{-4})$ the same computation gives 0.0085. The unregularised offline optimum over 32 kernels is about $7.6 \cdot 10^{-6}$ (best of 5 L-BFGS restarts, an upper bound), and $\bar{L}(0) = 0.311$.

At finite m, the fluctuation of $\hat{f}$ is $O(K_\Phi/\sqrt{m})$ at each fixed t (finite-horizon propagation of chaos; Theorem D′ of Section 17 gives the rate $m^{-1/2}$ for the plant and the learned function of the dithered loop). It is uniform in time only under Remark H.5.

Remark H.2 (the control knobs, and what the numbers say)**.**

- **The knobs.** σ is the strong-convexity constant of the confinement, τ is the temperature, and $a_{\max}$ sets K_Φ.
- **The role of k_e.** Since $k_e F_\tau = \bar{L} + (k_e\sigma/2)\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + k_e\tau H$, μ_τ^* and ρ_τ depend on (σ,τ,k_e) only through $(k_e\sigma, k_e\tau)$. Raising k_e at fixed (σ,τ) has three effects:
  - it improves ρ_τ and the guaranteed rate $2\sigma e^{-\alpha^*/\tau}$, since $\alpha^* \propto 1/k_e$;
  - it moves μ_τ^* toward the prior $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$, which increases the regularisation bias;
  - it shrinks the loss-driven drift like $1/k_e$.
- **The guaranteed rate** is exponentially small in $1/\tau$, a known feature of Holley–Stroock bounds [35]:
  - $2\tau\rho_\tau \ge e^{-8401}\,\mathrm{s}^{-1}$ at $(\sigma,\tau) = (0.002, 5\cdot 10^{-4})$;
  - $2\tau\rho_\tau \ge e^{-1053}\,\mathrm{s}^{-1}$ at $(0.02, 0.004)$.

  The experiments do not test this rate.
- **The temperature sweep** (`results/langevin_tradeoff`; $m = 32$, $\sigma = 0.002$, $k_e = 10$, $T = 1500\,\mathrm{s}$, 2 seeds):

τ	escape time of the collapsed start (s), per seed	loss floor of the spread start
0	never, never	$7.0 \cdot 10^{-3}$
$5 \cdot 10^{-4}$	never, 1050	$1.1 \cdot 10^{-2}$
$1 \cdot 10^{-3}$	1480, 1005	$1.5 \cdot 10^{-2}$
$2 \cdot 10^{-3}$	1115, 800	$2.3 \cdot 10^{-2}$
$4 \cdot 10^{-3}$	970, 430	$3.6 \cdot 10^{-2}$
$8 \cdot 10^{-3}$	390, 295	$6.5 \cdot 10^{-2}$
annealed, hot start	990, 530	$3.2 \cdot 10^{-2}$

- **How escape is measured.** Escape is the first sample at which the collapsed run's loss falls to at most twice the spread run's final loss, at the same τ and seed. Two caveats follow:
  - the threshold grows with τ, from 0.022 to 0.13, and the escape time includes convergence after escape;
  - the $\tau = 5 \cdot 10^{-4}$ point is censored and is excluded from the fit.
- **The fit.** The seed-averaged escape time fits $t_{\text{esc}} \approx -1693 + 427 \cdot \log(1/\tau)$ (4.5% relative residual) better than an Arrhenius law (25%). On the eight per-seed values the two laws fit equally poorly (27% and 33%; `langevin_tradeoff/table.md`).
- **No saddle explanation.** The noiseless collapsed end point is a saddle of the averaged landscape, but, as a numerical check shows, a weak one. Its unstable rate is about $5 \cdot 10^{-5}\,\text{s}^{-1}$ for $(\mathfrak{A})$, and about $2 \cdot 10^{-4}\,\text{s}^{-1}$ for the exactly averaged loop at $k_e = 10$. A saddle-escape law would therefore need $\tau \lesssim 10^{-5}$, and would have a $\log(1/\tau)$ slope of 10^3–10^4 s.
- **What the data show.** At the tested temperatures escape is governed by diffusion, by the approach transient and by the τ-dependent threshold. The data do not identify the escape mechanism. An earlier "saddle, not barrier" reading of these data is withdrawn.
- **The annealed runs** use α_a between about $2 \cdot 10^{-3}$ and $2 \cdot 10^{-2}$, far below $\alpha^* = 4.2$. They are not instances of H(iv).

Remark H.3 (why saturation and not projection)**.** The mean-field Langevin results are on $\mathbb{R}^d$. Projecting the drift with the Pomet–Praly operator [63], while keeping additive noise, leaves the noise unconfined. Reflected (Skorokhod) dynamics on the convex body Θ_K would confine it.

- Such dynamics satisfy a log-Sobolev inequality with constant of order $1/\operatorname{diam}^2$, even for $\sigma = 0$.
- The argument is a diameter-based log-Sobolev inequality for the uniform measure on a convex body, followed by Holley–Stroock [35]. The diameter-based inequality comes from $\mathrm{CD}(0, \infty)$ with a convex Neumann boundary plus a diameter bound (F.-Y. Wang 1997 [72]), or from Frieze–Kannan (1999) [24] for log-concave measures on convex bodies. Together these give $\mathrm{LSI}\big(c \cdot \operatorname{diam}^{-2} \cdot e^{-2K_\Phi(F_\rho + K_\Phi)/(k_e \tau)}\big)$.
- Bakry–Émery [3] alone gives nothing at $\sigma = 0$.
- That route connects to [57] and is left open.

Remark H.4 (the role of σ-modification)**.** In the classical theory, σ-modification is a robustness device with a bias. In the saturated design on $\mathbb{R}^p$ it supplies the confinement that makes $e^{-V/\tau}$ integrable and log-Sobolev, so the global result rests on it. Its bias is the regularisation term of the variational problem that μ_τ^* solves. Under reflection on Θ_K, compactness would play that role instead.

Remark H.5 (finite m; conditional)**.** The m-particle averaged system is

$$\mathrm{d}\vartheta_i = -G\nabla V[\mu_t^m](\vartheta_i)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad \mu_t^m := \frac{1}{m}\sum_i \delta_{\vartheta_i}.$$

Uniform-in-time propagation of chaos would follow from [8], under its hypotheses.

- *Available:* convexity of $\mathcal{G}$; a uniform LSI, from (ii); and a bounded, smooth second variation $(1/k_e)\int \tilde{\Phi}(x;\vartheta)^\top \tilde{\Phi}(x;\vartheta')\,\mathrm{d}\rho$.
- *Still to check:* the remaining regularity conditions of [8], verbatim (open question 4 of `LITERATURE_TRANSPORT.md` in the accompanying repository).

Status. Conditional on the hypotheses of [8]; their remaining regularity conditions are still to be checked.

Conjecture H.6 (the real closed loop, in the scaling where $(\mathfrak{A})$ is the limit)**.** *Let $f = f(x)$ satisfy the tube condition of Theorem D(b), with $a_{\max}$ replaced by K_Φ. Fix $G \succ 0$ and $\sigma_0, \tau_0 > 0$, and set $\sigma := \sigma_0/k_e$ and $\tau := \tau_0/k_e$.*

- *Then μ_τ^*, the minimiser of $\bar{L} + (\sigma_0/2)\int \|\vartheta - \vartheta^0\|_{G^{-1}}^2\,\mathrm{d}\mu + \tau_0 H$, does not depend on k_e.*
- *In slow time $s = t/k_e$, $(\mathfrak{A})$ is the formal limit of the closed loop.*

Conjecture: *for the mean-field limit of the closed loop under (L_τ), for every $\mu_0 \in \mathcal{P}_2$ and every $e(0)$, there are $S(\mu_0)$ and $r(k_e) \to 0$ such that $W_{G^{-1}}(\mu_t, \mu_\tau^*) \le r(k_e)$ for all $t \ge k_e \cdot S$.*

Status. Conjectural.

At fixed k_e the analogous statement is false. The limit of small gains is the exactly averaged system, with the true periodic error e_μ and $x = x_m + e_\mu$.

- Its stationary state is offset from μ_τ^*: $\bar{L} = 0.11415$ against $\bar{L}(\mu_\tau^*) = 0.11118$ at $k_e = 10$ (`results/langevin_global/gibbs_reference.md`).
- In the scaling of Conjecture H.6, the $L^2(\rho)$ distance between the two learned functions is $6.8 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$ and $9.7 \cdot 10^{-4}$ at $k_e = 10$, 20 and 40. It is small and decreasing, but a clean $1/k_e$ law is not resolved in this range.
- By Proposition H.7(e) it is not the $W_{G^{-1}}$-gradient flow of any free energy. The cyclic sums of its antisymmetric second variation are non-zero in the checked instances, including the network of the experiments; by Theorem I(iii) (Section 20) they are non-zero at every $n = 1$ measure whose residual is non-zero on the orbit, and by Theorem I(vi) on no open set of measures do they all vanish.

20 Finite gain: semi-gradient and non-reciprocal structure

This section continues Section 19 at finite feedback gain k_e. The notation is that of Section 9 and Section 19: the saturated features $\tilde{\Phi}$, the residual r_μ, the potential $V_L[\mu]$, the averaged system $(\mathfrak{A})$ and the law (L_τ). We write $\tilde{\Phi}_\vartheta := \tilde{\Phi}(\cdot;\vartheta)$.

20.1 Frozen-measure structure of the exactly averaged loop

Proposition H.7 (frozen-measure structure of the exactly averaged closed loop)**.** *Assume the following:*

- *$f \in C^2(\mathbb{R}^n)$ is autonomous;*
- *$x_m \in C^2$ is T_0-periodic, with oriented orbit $\mathcal{O}$;*
- *the features are the saturated features of the design (Section 19);*

- $\mu \in \mathcal{P}(\mathbb{R}^p)$ *is frozen.*

Then for every $R > 0$ there is $k_0(R)$, independent of μ, such that for $k_e \geq k_0(R)$ the following hold.

(a) Periodic error. *The frozen-μ error equation $\dot{e} = r_\mu(x_m + e) - k_e e$ has exactly one T_0-periodic solution e_μ with $\sup_t \|e_\mu\| \leq R$. It attracts exponentially every solution with $\|e(0)\| \leq R$, and*

$$e_\mu = \frac{r_\mu(x_m)}{k_e} + \frac{\nabla r_\mu(x_m)\, r_\mu(x_m) - \frac{\mathrm{d}}{\mathrm{d}t} r_\mu(x_m(t))}{k_e^2} + O(k_e^{-3}), \tag{20.1}$$

uniformly in t and μ.

- *Here ∇r_μ is the Jacobian $(\partial r_k / \partial x_l)$, and $\partial_x \tilde{\Phi} = (\partial \tilde{\Phi}_k / \partial x_l)$.*
- *If f is one-sided Lipschitz, $\langle f(x) - f(y), x - y \rangle \leq L_f \|x - y\|^2$, then for $k_e > L_f + a_{\max}\sqrt{n}\, e^{-1/2}/s_{\min}$ the periodic solution is unique and attracts every solution.*
- *Without such a condition uniqueness is only local. For $f(x) = x^2$ there is a second, repelling periodic solution near $e \approx k_e$, and solutions above it blow up.*

(b) Expansion. *The period average of the drift of (L_τ) along e_μ is $G \nabla_\vartheta P_\mu(\vartheta) - \sigma(\vartheta - \vartheta^0)$, where $P_\mu(\vartheta) := \langle e_\mu^\top \tilde{\Phi}(x_m + e_\mu; \vartheta) \rangle_t$ satisfies*

$$P_\mu(\vartheta) = -\frac{1}{k_e} V_L[\mu](\vartheta) + \frac{1}{k_e^2}\big(U_{\mathrm{even}}[\mu](\vartheta) + U_{\mathrm{odd}}[\mu](\vartheta)\big) + O(k_e^{-3}), \tag{20.2}$$

$$U_{\mathrm{even}}[\mu](\vartheta) := \int (r_\mu \cdot \nabla_x)\big(r_\mu^\top \tilde{\Phi}(\cdot; \vartheta)\big)\, \mathrm{d}\rho, \qquad U_{\mathrm{odd}}[\mu](\vartheta) := -\frac{1}{T_0} \oint_{\mathcal{O}} \tilde{\Phi}(\cdot; \vartheta)^\top \, \mathrm{d}r_\mu, \tag{20.3}$$

uniformly in ϑ and μ, together with ϑ-gradients.

(c) Semi-gradient form.

$$-\frac{1}{k_e} V_L[\mu] + \frac{1}{k_e^2} U_{\mathrm{even}}[\mu] = \frac{1}{k_e} \int r_\mu^\top \tilde{\Phi}(\cdot; \vartheta)\, \mathrm{d}\rho_\mu^{\mathrm{cl}} + O(k_e^{-3}). \tag{20.4}$$

Here ρ_μ^{cl} is the occupation measure of the closed-loop state $x_m + e_\mu$.

- *So, apart from the lag term U_{odd}, the averaged law follows the gradient of the closed-loop loss $\frac{1}{2}\int \|r_\nu\|^2 \, \mathrm{d}\rho_\mu^{\mathrm{cl}}$ at $\nu = \mu$, with the closed-loop state distribution ρ_μ^{cl} held fixed.*
- *This is a semi-gradient: the update freezes part of its own parameter dependence. In TD learning the frozen part is the bootstrapped target, and the data distribution does not depend on the parameters [71, 73].*
- *Here the frozen part is the data distribution ρ_μ^{cl}. This is the structure of performative prediction [59]: steps taken with the distribution held at its current value converge to performatively stable points, not to optima.*
- ***The performative defect carries the even non-reciprocity.*** *With $J^{\mathrm{cl}}(\mu) := \frac{1}{2}\int \|r_\mu\|^2 \, \mathrm{d}\rho_\mu^{\mathrm{cl}}$,*

$$\begin{aligned} &-\frac{1}{k_e} V_L[\mu] + \frac{1}{k_e^2} U_{\mathrm{even}}[\mu] \\ &\qquad = -\frac{1}{k_e} \frac{\delta J^{\mathrm{cl}}}{\delta \mu} - \frac{1}{k_e^2} \int (\tilde{\Phi}_\vartheta \cdot \nabla)\big(\tfrac{1}{2}\|r_\mu\|^2\big)\, \mathrm{d}\rho + O(k_e^{-3}). \end{aligned} \tag{20.5}$$

The first term is a true first variation, hence reciprocal. The second is the defect: it has the same order as U_{even}, and it carries all of A_{even} in (e).

(d) Circulation kernel. $U_{\mathrm{odd}}[\mu](\vartheta) = -\frac{1}{T_0}\oint_{\mathcal{O}} \tilde{\Phi}(\cdot;\vartheta)^\top \,\mathrm{d}f + \int K(\vartheta,\vartheta')\,\mathrm{d}\mu(\vartheta')$, *with*

$$K(\vartheta,\vartheta') := \frac{1}{T_0}\oint_{\mathcal{O}} \tilde{\Phi}(\cdot;\vartheta)^\top \,\mathrm{d}\tilde{\Phi}(\cdot;\vartheta') = \Big\langle \tilde{\Phi}(x_m;\vartheta)^\top \frac{\mathrm{d}}{\mathrm{d}t}\tilde{\Phi}(x_m;\vartheta')\Big\rangle_t. \tag{20.6}$$

- *K is antisymmetric.*
- *By Stokes, $K(\vartheta,\vartheta') = \frac{1}{T_0}\int_\Sigma \sum_k \mathrm{d}\tilde{\Phi}_k(\cdot;\vartheta)\wedge \mathrm{d}\tilde{\Phi}_k(\cdot;\vartheta')$ for any 2-chain Σ with $\partial\Sigma = \mathcal{O}$.*
- *For $n = 2$ and a simple, positively oriented orbit bounding D,*

$$K(\vartheta,\vartheta') = \frac{1}{T_0}\sum_k \int_D \det\big[\nabla_x\tilde{\Phi}_k(\cdot;\vartheta), \nabla_x\tilde{\Phi}_k(\cdot;\vartheta')\big]\,\mathrm{d}A.$$

- *K is an integral over the enclosed region, not the area itself. For Gaussian features, $\sum_k \det[\nabla\tilde{\Phi}_k(\vartheta), \nabla\tilde{\Phi}_k(\vartheta')] = (a^\top a')\cdot\det[\nabla\varphi,\nabla\varphi']$, so $K = 0$ whenever $a^\top a' = 0$ or $c = c'$, whatever the area.*
- *Hence $U_{\mathrm{odd}} \equiv 0$ for $n = 1$.*
- *Reversing the orbit preserves ρ, flips U_{odd}, and leaves U_{even} unchanged.*

(e) Non-reciprocity. *Write $S(\vartheta,\vartheta') := \delta P_\mu(\vartheta)/\delta\mu(\vartheta')$ for the derivative in the direction $\delta_{\vartheta'}$, and call $S(\vartheta,\vartheta') - S(\vartheta',\vartheta)$ its antisymmetric part (without the factor $\frac{1}{2}$). Adding a function of ϑ alone, or a constant $c(\mu)$, to P_μ changes the antisymmetric part by a coboundary, which the criterion below allows.*

- *At order $1/k_e$, S is symmetric: it equals $-\frac{1}{k_e}\int \tilde{\Phi}(\cdot;\vartheta)^\top\tilde{\Phi}(\cdot;\vartheta')\,\mathrm{d}\rho$.*
- *At order $1/k_e^2$ it is not. The second variation of the k_e^{-2} coefficient $U_{\mathrm{even}} + U_{\mathrm{odd}}$ has antisymmetric part $A_{\mathrm{even}} + 2K$, with*

$$A_{\mathrm{even}}(\vartheta,\vartheta') := -\int \Big[(\tilde{\Phi}_{\vartheta'}\cdot\nabla)(r_\mu^\top\tilde{\Phi}_\vartheta) - (\tilde{\Phi}_\vartheta\cdot\nabla)(r_\mu^\top\tilde{\Phi}_{\vartheta'})\Big]\,\mathrm{d}\rho. \tag{20.7}$$

- *P_μ, up to $O(k_e^{-3})$, can be a first variation of a functional of μ only if $A_{\mathrm{even}} + 2K$ is a coboundary $a(\vartheta) - a(\vartheta')$, that is, only if all its cyclic sums vanish.*
- *They do not vanish in the checked instances, for $n = 1$ (from A_{even} alone) and for $n = 2$. Theorem I shows that the cyclic condition is equivalent to $A_{\mathrm{even}} + 2K \equiv 0$, and characterises when that happens: for $n = 1$ exactly when the residual vanishes on the orbit; for $n \geq 2$ at perfect fit exactly when the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit. It also shows (Theorem I(vi)) that no open set of measures satisfies the condition, for any n.*
- *So at finite k_e, the exactly averaged loop is a non-reciprocal McKean–Vlasov dynamics: it is not the $W_{G^{-1}}$-gradient flow of any free energy on any open set of measures. A different, μ-dependent metric is not excluded by this argument.*

Status. The frozen-measure part is proved. Non-reciprocity of the exact second variation is checked to 3% (asserted; 1–1.5% measured) and characterised by Theorem I. The Stokes identity and the parity under orbit reversal hold to machine precision. The expansion is checked through its $O(1/k_e)$ convergence. Novelty (literature audit): no precedent found (targeted search); ingredients known (higher-order averaging; performative prediction [59]; semi-gradient TD [71, 73]).

Proof. (a) Write $e = r_\mu(x_m)/k_e + w$. Then w solves $\dot{w} = -k_e w + g(t,w)$ with $g = \nabla r_\mu(x_m)e - \dot{r}_\mu/k_e + O(|e|^2)$.

- The tube $\{\|e\| \leq R\}$ is forward invariant once $k_e R > \sup_{\|e\|\leq R}\|r_\mu(x_m + e)\|$.

- Inside the tube, solutions contract at rate $k_e - \mathrm{Lip}_R(r_\mu)$, which gives existence, uniqueness and attraction there. A contraction on T_0-periodic functions gives the expansion.
- The bounds on r_μ, ∇r_μ and $\nabla^2 r_\mu$ on the R-neighbourhood of the orbit are uniform in μ, by the saturation and $f \in C^2$.
- The global statement uses L_f in place of the local Lipschitz constant of f.

(b) Taylor-expand $e^\top \tilde\Phi(x_m + e; \vartheta) = e^\top \tilde\Phi(x_m; \vartheta) + e^\top \partial_x \tilde\Phi(x_m; \vartheta) e + O(|e|^3)$, and insert (a).

- The period averages of the time-symmetric terms are integrals against ρ.
- $(\nabla r\, r)^\top \tilde\Phi + r^\top (\partial_x \tilde\Phi) r = (r \cdot \nabla)(r^\top \tilde\Phi)$.
- ϑ-derivatives commute with the average.
- The remainders are uniform, because the x- and ϑ-derivatives of $\tilde\Phi$ of the orders needed are bounded.

(c) We have

$$\int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho_\mu^{\mathrm{cl}} = \big\langle r_\mu(x_m + e_\mu)^\top \tilde\Phi(x_m + e_\mu; \vartheta) \big\rangle = \int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho + \big\langle \nabla_x (r_\mu^\top \tilde\Phi_\vartheta) \cdot e_\mu \big\rangle + O(k_e^{-2}).$$

By (a) this equals $\int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho + (1/k_e) U_{\mathrm{even}} + O(k_e^{-2})$.

(d) $\frac{\mathrm{d}}{\mathrm{d}t} r_\mu(x_m) = \frac{\mathrm{d}}{\mathrm{d}t} f(x_m) - \int \frac{\mathrm{d}}{\mathrm{d}t} \tilde\Phi(x_m; \vartheta') \,\mathrm{d}\mu(\vartheta')$.

- Antisymmetry: $K(\vartheta, \vartheta') + K(\vartheta', \vartheta) = \frac{1}{T_0} \oint \mathrm{d}\big(\tilde\Phi_\vartheta^\top \tilde\Phi_{\vartheta'}\big) = 0$.
- The area form is Stokes, or Green's theorem for $n = 2$.
- For $n = 1$, every 1-form on $\mathbb{R}$ is exact, so its loop integrals vanish.
- Reversal changes the sign of oriented loop integrals and preserves ρ.

(e) The second variation of U_{even} is

$$S_{\mathrm{even}}(\vartheta, \vartheta') = -\int \Big[(\tilde\Phi_{\vartheta'} \cdot \nabla)(r_\mu^\top \tilde\Phi_\vartheta) + (r_\mu \cdot \nabla)(\tilde\Phi_{\vartheta'}^\top \tilde\Phi_\vartheta) \Big] \,\mathrm{d}\rho,$$

whose second term is symmetric. The second variation of U_{odd} is K. A kernel's antisymmetric part A is a coboundary if and only if $A(\vartheta_0, \vartheta_1) + A(\vartheta_1, \vartheta_2) + A(\vartheta_2, \vartheta_0) = 0$ for all triples.

The remainder of (b) is not differentiated in μ. Instead the argument uses a circulation around a small triangle of measures.

1. Fix $\vartheta_0, \vartheta_1, \vartheta_2$ and $\varepsilon \in (0, 1)$. For (a, b) in $T_\varepsilon := \{a, b \geq 0,\ a + b \leq \varepsilon\}$, let $\mu_{a,b} := (1 - \varepsilon)\mu + (\varepsilon - a - b)\delta_{\vartheta_0} + a\delta_{\vartheta_1} + b\delta_{\vartheta_2}$.
2. The circulation $\oint_{\partial T_\varepsilon} \langle Q_{\mu_{a,b}}, \mathrm{d}\mu_{a,b} \rangle$ vanishes when Q_μ is a first variation plus a constant $c(\mu)$, because $\mathrm{d}\mu_{a,b}$ has zero mass. In particular it vanishes for $-(1/k_e)V_L$.
3. $U_{\mathrm{even}} + U_{\mathrm{odd}}$ is quadratic in μ, so its circulation equals $-(\varepsilon^2/2) \cdot \sum_{\mathrm{cyc}} (A_{\mathrm{even}} + 2K)(\vartheta_i, \vartheta_{i+1})$ at the centroid of T_ε.
4. The $O(k_e^{-3})$ remainder of (b) contributes at most $6\varepsilon \cdot \sup|R| = O(\varepsilon k_e^{-3})$.
5. Suppose that, for all large k_e, P_μ were a first variation up to $O(k_e^{-3})$, with constants uniform in k_e. Then $(\varepsilon^2/(2k_e^2))|\mathrm{cyc}| \leq C\varepsilon k_e^{-3}$, that is, $|\mathrm{cyc}| \leq 2C/(\varepsilon k_e)$. The cyclic sum does not depend on k_e, so letting $k_e \to \infty$ at fixed ε gives $\mathrm{cyc} = 0$ at the centroid measure of the triangle; letting $\varepsilon \to 0$, since $A_{\mathrm{even}} + 2K$ is affine in the measure, gives $\mathrm{cyc} = 0$ at μ.

This is checked numerically: for $\varepsilon = 0.05$ and $n = 2$ the circulation equals the centroid formula to all printed digits. □

Checked by: `tests/test_averaging.py` in the accompanying repository, using `src/core/averaging.py`; seven tests: four for Proposition H.7, one for Remark H.0, two for Remark H.7a. The checks of Proposition H.7 are listed in Table 18.

Table 18: Checks of Proposition H.7 (`tests/test_averaging.py`).

check	result
expansion: relative error of $k_e^2(P_\mu - P_0)$ against $U_{\text{even}} + U_{\text{odd}}$	halves when k_e doubles. At $k_e = 40 \to 80$ it goes $0.54 \to 0.25$ for $n = 1$, $0.038 \to 0.019$ for $n = 2$ counter-clockwise, and $0.022 \to 0.011$ clockwise. A Richardson step at $k_e = 80$ and 160 brings $n = 1$ within 3% (asserted)
Stokes: loop-integral K against the area integral over the enclosed ellipse	agreement to $7 \cdot 10^{-15}$ relative (asserted at 10^{-6}); K antisymmetric; $K \equiv 0$ for $n = 1$
parity under orbit reversal (same ρ)	U_{even} unchanged and U_{odd} flipped, to $2 \cdot 10^{-15}$ relative (asserted at 10^{-10})
non-reciprocity: the *exact* second variation S ($k_e = 80$) against the prediction	the full matrix $k_e^2 S + k_e S_0$ is within 5% of $S_{\text{even}} + K$. The cyclic sums of the antisymmetric part agree to 3%: $n = 1$ gives -0.0711 against -0.0704, all from the state shift; $n = 2$ gives -0.0471 against -0.0464 (state shift -0.089, circulation $+0.043$)

20.2 The oscillation correction and the orientation of the reference orbit

Remark H.7a (the full first-order correction, and a test in the real closed loop)**.** Proposition H.7 freezes the measure over a period. In the closed loop the parameters also oscillate within each period, by $O(1/k_e)$.

The parameter equation. In the scaling $\sigma = \sigma_0/k_e$, write

$$\dot{\Theta} = \frac{1}{k_e} H(\Theta, t) + \frac{1}{k_e^2} H_2(\Theta, t) + O(k_e^{-3}), \tag{20.8}$$
$$H_i(\Theta, t) := G \nabla_\vartheta \big[r_\Theta(x_m(t))^\top \tilde{\Phi}(x_m(t); \vartheta_i) \big] - \sigma_0(\vartheta_i - \vartheta^0).$$

Second-order averaging (Sanders–Verhulst–Murdock [64, §2.9]) gives the dynamics of the period mean Y:

$$\dot{Y} = \frac{1}{k_e} \langle H \rangle(Y) + \frac{1}{k_e^2} \Big[\langle H_2 \rangle(Y) + \big\langle D_\Theta H(Y, t) \cdot u(Y, t) \big\rangle \Big] + O(k_e^{-3}). \tag{20.9}$$

- u is the zero-mean antiderivative in t of $H - \langle H \rangle$.
- $\langle H_2 \rangle$ is the frozen-measure correction of Proposition H.7.
- This is formal. The averaging theorem gives it on finite slow-time horizons.
- `averaging.oscillation_correction` computes the oscillation term $b_{\text{osc}} := \langle D_\Theta H \cdot u \rangle$.

Three properties follow.

- **Parity and value.** The oscillation term b_{osc} is odd under reversal of the orbit (`test_oscillation_correction_is_odd_under_orbit_reversal`). Its value matches an independent one-period near-identity flow at $k_e = 80$: the test asserts 3%, and 0.7% is measured (`test_oscillation_correction_matches_a_one_period_flow`).
- **Size.** Relative to the frozen-measure terms it scales like $|G|T_0$, the parameter displacement per period times k_e. It is negligible when adaptation is slow within a period, which is the

small-gain limit in the fixed-k_e discussion of Conjecture H.6. It is not negligible as $k_e \to \infty$ at fixed G. Both terms are $O(k_e^{-2})$ and uniformly bounded, so the order of the route to Conjecture H.6 is unchanged.

- **Non-reciprocity survives.** The only orientation-even part of the full correction is U_{even}, and its antisymmetric second variation is not a coboundary (Proposition H.7(e)). Suppose the full first-order dynamics were reciprocal for both orientations. Then the average of the two antisymmetric parts, A_{even}, would be a coboundary, which it is not. So the full first-order averaged dynamics is non-reciprocal for at least one orientation.

A prediction tested in the real closed loop (`exp_orbit_orientation.py` → `results/orbit_orientation/`).

- **The prediction.** Traverse the same elliptic reference both ways. The occupation measure is the same, so ($\mathfrak{A}$), and any functional of ρ, cannot distinguish the two runs. The first-order theory predicts that the learned kernels differ by $\Theta_+ - \Theta_- \approx -2J^{-1}(b_{\text{odd}} + b_{\text{osc}}) = O(1/k_e)$. Here J is the Jacobian of the leading averaged drift at the equilibrium.
- **The setting.** $n = 2$, $m = 6$, $G = \text{diag}(4, 4, 1, 1, 1)$, $\sigma = 2/k_e$, no dither, and the same initial parameters for both runs. The runs converged to machine precision. The six kernels start spread out (kernel spread 1.05) and converge to a common configuration (spread at most $5.6 \cdot 10^{-17}$ at the end of every run), so the learned network is effectively a single kernel: the test probes the orientation effect at a collapsed equilibrium of the $m = 6$ system.

Table 19: Orientation dependence of the learned kernels in the real closed loop (`results/orbit_orientation/table_sigma2_slow16.md`). The size ratio is measured over predicted.

		frozen measure only (H.7)		full (H.7 + oscillation)		full
k_e	measured $\|\Theta_+ - \Theta_-\|$	size ratio	cosine	size ratio	cosine	vector error
10	$5.7 \cdot 10^{-3}$	2.67	0.76	0.94	0.93	0.36
20	$2.8 \cdot 10^{-3}$	2.41	0.83	0.97	0.98	0.20
40	$1.4 \cdot 10^{-3}$	2.29	0.86	0.97	0.994	0.11

- The measured difference scales like $k_e^{-1.01}$.
- At $k_e = 40$ the full first-order theory predicts it to within 3% in size and 11% as a vector (cosine 0.994).
- The vector error halves with each doubling of k_e: 36%, 20% and 11% at $k_e = 10$, 20 and 40.
- The frozen-measure terms alone account for less than half of it.
- A first attempt with a weaker leak ($\sigma = 0.5/k_e$) and a shorter horizon had not converged: its residual drift was a third of the difference. It is not used.

The direction in which the reference orbit is traversed is therefore a property of the excitation that the learned kernels retain, at order $1/k_e$. The first-order theory predicts its effect.

Status. The second-order averaged equation (20.9) is formal; the averaging theorem gives it on finite slow-time horizons. The oscillation term is checked against an independent one-period near-identity flow (within 3% at $k_e = 80$) and is odd under orbit reversal (to 10^{-6}). The full first-order theory predicts the orientation dependence of the learned kernels in the real closed loop to 3% in size and 11% as a vector (cosine 0.994, $k_e = 40$).

Checked by: `tests/test_averaging.py` (the two tests named above); `exp_orbit_orientation.py` → `results/orbit_orientation/table_sigma2_slow16.md`.

20.3 Reciprocity at order k_e^{-2}: a pairwise criterion and its characterisation

Theorem I (reciprocity at order k_e^{-2}: a pairwise criterion and its characterisation)**.** *Assume the setting of Proposition H.7: saturated Gaussian features $\tilde{\Phi}(x;\vartheta) = a\varphi_\xi(x)$ with $\xi := (c,s)$, $f \in C^2(\mathbb{R}^n)$ autonomous, $x_m \in C^2$ T_0-periodic with oriented orbit $\mathcal{O}$ and occupation measure ρ, and a frozen $\mu \in \mathcal{P}(\mathbb{R}^p)$. Let S be the k_e^{-2} coefficient of the second variation of P_μ, and $A_\mu := S - S^\top = A_{\mathrm{even}} + 2K$ its antisymmetric part, as in H.7(e). Call μ* reciprocal at order k_e^{-2} *if $A_\mu \equiv 0$ (the 1-form that the k_e^{-2} drift defines is closed at μ). Note that A_μ is affine in μ, because r_μ is and K does not depend on μ.*

(i) Pairwise criterion. *A_μ is a coboundary $a(\vartheta) - a(\vartheta')$ if and only if $A_\mu \equiv 0$. Consequently, by Proposition H.7(e), whose hypothesis concerns a neighbourhood of μ (its triangle of measures), P_ν can be a first variation of a functional of ν up to $O(k_e^{-3})$, with constants uniform in k_e and in ν near μ, only if $A_\nu \equiv 0$ for every ν in a neighbourhood of μ.*

(ii) Structure. *With Dr the Jacobian of r_μ, $(Dr)_{kl} = \partial_l r_k$, and $[X,Y] := (X\cdot\nabla)Y - (Y\cdot\nabla)X$ the Lie bracket of vector fields on $\mathbb{R}^n$,*

$$A_{\mathrm{even}}(\vartheta,\vartheta') = -\int \Big\{ \tilde{\Phi}_\vartheta^\top (Dr_\mu - Dr_\mu^\top)\tilde{\Phi}_{\vartheta'} + r_\mu^\top [\tilde{\Phi}_{\vartheta'}, \tilde{\Phi}_\vartheta] \Big\}\, \mathrm{d}\rho. \tag{20.10}$$

For the Gaussian features A_μ is a bilinear form in the weights: $A_\mu(\vartheta,\vartheta') = a^\top B_\mu(\xi,\xi')a'$ with

$$\begin{aligned} B_\mu(\xi,\xi') :=\ & -\int (Dr_\mu - Dr_\mu^\top)\varphi_\xi \varphi_{\xi'}\,\mathrm{d}\rho - \int r_\mu (\nabla\varphi_\xi)^\top \varphi_{\xi'}\,\mathrm{d}\rho \\ & + \int (\nabla\varphi_{\xi'}) r_\mu^\top \varphi_\xi\,\mathrm{d}\rho + 2\kappa(\xi,\xi') I_n, \end{aligned} \tag{20.11}$$

$$\kappa(\xi,\xi') := \Big\langle \varphi_\xi(x_m(t)) \cdot \frac{\mathrm{d}}{\mathrm{d}t}\varphi_{\xi'}(x_m(t)) \Big\rangle_t, \tag{20.12}$$

so $A_\mu \equiv 0$ if and only if $B_\mu(\xi,\xi') = 0$ for all admissible ξ,ξ'. The first term is the curl of the residual paired with the features, the next two are the residual paired with the Lie bracket of the feature fields, and the last is the circulation kernel $K(\vartheta,\vartheta') = (a^\top a')\kappa(\xi,\xi')$. The performative part is bounded by the residual and by the antisymmetric part of its Jacobian along the orbit:

$$|A_{\mathrm{even}}(\vartheta,\vartheta')| \le |a||a'| \Big[\int \|Dr_\mu - Dr_\mu^\top\|\,\mathrm{d}\rho + \frac{2e^{-1/2}}{s_{\min}} \int |r_\mu|\,\mathrm{d}\rho \Big].$$

(iii) $n = 1$. *$A_\mu \equiv 0$ if and only if $r_\mu = 0$ ρ-a.e. on the orbit. So the one-dimensional loop is reciprocal at order k_e^{-2}, at a frozen μ, exactly when μ interpolates f along the orbit, and its non-reciprocity is bounded by the residual: $|A_\mu(\vartheta,\vartheta')| \le (2e^{-1/2}/s_{\min})|a||a'| \int |r_\mu|\,\mathrm{d}\rho$.*

(iv) $n \ge 2$ at perfect fit. *Suppose $r_\mu = 0$ ρ-a.e. Then the Lie-bracket terms vanish and*

$$B_\mu(\xi,\xi') = -\int (Dr_\mu - Dr_\mu^\top)\varphi_\xi\varphi_{\xi'}\,\mathrm{d}\rho + 2\kappa(\xi,\xi')I_n,$$

an antisymmetric matrix plus a multiple of the identity. Hence $A_\mu \equiv 0$ if and only if both

- *(a) $K \equiv 0$, which holds if and only if the 1-current of the oriented orbit vanishes, $T_{\mathcal{O}} := \int_0^{T_0} \delta_{x_m(t)}\, \dot{x}_m(t)\,\mathrm{d}t = 0$ (an $\mathbb{R}^n$-valued measure; $\int v \cdot \mathrm{d}T_{\mathcal{O}} = \oint_{\mathcal{O}} v \cdot \mathrm{d}x$); for $n = 2$, iff the winding-number function $w_{\mathcal{O}}$ of the orbit vanishes a.e.; and*
- *(b) $Dr_\mu = Dr_\mu^\top$ ρ-a.e.: the Jacobian of the residual has no antisymmetric part along the orbit.*

Vanishing of r_μ on the orbit constrains only its tangential derivative, so (b) is a genuine extra condition: for $f := \hat{f}_\mu + he_1$ with $h = 0$ on the orbit and $\partial_\perp h \neq 0$, $A_\mu - 2K = -\int \varphi_\xi \varphi_{\xi'} \partial_\perp h \, \mathrm{d}\rho \cdot (e_1 e_\perp^\top - e_\perp e_1^\top) \neq 0$. When r_μ vanishes on a neighbourhood of the orbit, $Dr_\mu = 0$ there and $A_\mu = 2K$: the semi-gradient defect has vanished and the loop is non-reciprocal exactly when the orbit carries circulation, through the orbit-area kernel alone.

(v) General $n \geq 2$. *In any connected finite-dimensional real-analytic family of measures, for example atomic measures with J atoms parametrised by their weights and lifted positions, the set of reciprocal measures is either the whole family or a closed real-analytic subvariety of measure zero. The instances of `tests/test_averaging.py` ($n = 1$ and the $n = 2$ ellipse) have $A_\mu \not\equiv 0$, with cyclic sums -0.0704 and -0.0464 of the predicted kernel.*

(vi) Nowhere a gradient flow. *For every n, every (f, x_m) and every μ, every neighbourhood of μ contains measures ν with $A_\nu \not\equiv 0$: along the segments $\nu_\delta := (1-\delta)\mu + \delta\delta_{\vartheta'}$ either the zero feature $\vartheta' = \vartheta_z$ or a feature with $a' \neq 0$ gives $A_{\nu_\delta} \not\equiv 0$ for all small $\delta > 0$. Hence, by (i), there is no open set of measures on which P_ν is a first variation up to $O(k_e^{-3})$ with uniform constants: the exactly averaged closed loop is nowhere the $W_{G^{-1}}$-gradient flow of a free energy at order k_e^{-2}. The reciprocal measures of (iii) and (iv) are coincidences without interior.*

Status. Proved; every formula checked against the generic second-variation code and, by the round-5 referee, against the exact second variation; the characterisations checked on perfect-fit, curl and zero-current instances. Novelty (literature audit): new; the analytic tools (injectivity of Gaussian convolution; Stokes with winding numbers) are classical.

Proof. (i) Let $\vartheta_z := (\alpha, \gamma, \kappa)$ with $\alpha = 0$, an admissible point of $\mathbb{R}^p$, so $a = 0$ and $\tilde{\Phi}(\cdot; \vartheta_z) \equiv 0$ as a function of x; its x-derivatives vanish too. Every term of $A_{\mathrm{even}}(\vartheta, \vartheta_z)$ contains $\tilde{\Phi}_{\vartheta_z}$ or $(\tilde{\Phi}_{\vartheta_z} \cdot \nabla)$, and $K(\vartheta, \vartheta_z) = \langle \tilde{\Phi}_\vartheta^\top \frac{\mathrm{d}}{\mathrm{d}t} \tilde{\Phi}_{\vartheta_z} \rangle = 0$; so $A_\mu(\vartheta, \vartheta_z) = 0$ for every ϑ. If $A_\mu(\vartheta, \vartheta') = a(\vartheta) - a(\vartheta')$, then $a(\vartheta) = a(\vartheta_z)$ for every ϑ, so $A_\mu \equiv 0$. The converse is trivial. For the consequence, apply H.7(e) at any ν near μ with $\vartheta_2 := \vartheta_z$ in its triangle of measures: the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ is $A(\vartheta_0, \vartheta_1)$, evaluated at the centroid measure of the triangle, which tends to ν as the triangle's size $\varepsilon \to 0$; A is affine in the measure, so $A_\nu(\vartheta_0, \vartheta_1) = 0$.
(ii) Expand $(\tilde{\Phi}' \cdot \nabla)(r^\top \tilde{\Phi}) = \sum_{k,l} \tilde{\Phi}'_l (\partial_l r_k) \tilde{\Phi}_k + \sum_{k,l} \tilde{\Phi}'_l r_k \partial_l \tilde{\Phi}_k = \tilde{\Phi}^\top (Dr) \tilde{\Phi}' + r^\top (\tilde{\Phi}' \cdot \nabla) \tilde{\Phi}$, and subtract the expression with ϑ and ϑ' exchanged. For $\tilde{\Phi} = a\varphi$, $(\tilde{\Phi}' \cdot \nabla)\tilde{\Phi} = a(a'^\top \nabla\varphi)\varphi'$, so $r^\top[\tilde{\Phi}', \tilde{\Phi}] = (r^\top a)(a'^\top \nabla\varphi)\varphi' - (r^\top a')(a^\top \nabla\varphi')\varphi$, and $K(\vartheta, \vartheta') = (a^\top a')\kappa(\xi, \xi')$. Collect the bilinear form in (a, a'). Since a ranges over the open cube $(-a_{\max}, a_{\max})^n$, $A_\mu \equiv 0$ iff $B_\mu \equiv 0$. The bound uses $\varphi \leq 1$ and $|\nabla\varphi| \leq e^{-1/2}/s_{\min}$.
(iii) For $n = 1$, $Dr - Dr^\top = 0$, and for equal widths s, $\varphi_{c'} D\varphi_c - \varphi_c D\varphi_{c'} = \varphi_c \varphi_{c'} (c - c')/s^2$, so $B_\mu(\xi, \xi') = -((c - c')/s^2) \int r_\mu \varphi_c \varphi_{c'} \, \mathrm{d}\rho$. Now $\varphi_c \varphi_{c'} = e^{-(c-c')^2/(4s^2)} \cdot e^{-(x-m)^2/s^2}$ with $m := (c + c')/2$. Put $\nu := r_\mu \rho$, a finite signed measure with compact support. The function $m \mapsto \int e^{-(x-m)^2/s^2} \, \mathrm{d}\nu(x)$ extends to an entire function. If $B_\mu \equiv 0$ on admissible pairs, it vanishes for all $m \in (-c_{\max}, c_{\max})$ (take $c = m - \delta$, $c' = m + \delta$ with $0 < \delta$ small), hence everywhere; its Fourier transform is the product of the Gaussian's, which has no zeros, with $\hat{\nu}$, so $\nu = 0$, that is $r_\mu = 0$ ρ-a.e. Conversely, if $\nu = 0$ then $A_{\mathrm{even}}(\vartheta, \vartheta') = -\int (\varphi' D\varphi - \varphi D\varphi') aa' \, \mathrm{d}\nu = 0$ for all widths.
(iv) If $r_\mu = 0$ ρ-a.e. the two Lie-bracket integrals vanish, which gives the displayed B_μ; the first term is an antisymmetric matrix and the second a multiple of I_n, so $B_\mu \equiv 0$ iff both vanish (take the trace, which is $2n\kappa$, and the antisymmetric part). The first vanishes identically iff the matrix-valued measure $(Dr_\mu - Dr_\mu^\top)\rho$ is zero, by the argument of (iii) applied entrywise with equal widths (the entries of $\int (Dr - Dr^\top)\varphi_c \varphi_{c'} \, \mathrm{d}\rho$ are Gaussian convolutions of the entries of that measure). For (a): with equal widths, $\varphi_\xi \varphi_{\xi'} = e^{-|c-c'|^2/(4s^2)} e^{-|x-m|^2/s^2}$ and $\nabla\varphi_{\xi'} = -((x - c')/s^2)\varphi_{\xi'}$, so

$$\kappa(\xi, \xi') = -\frac{1}{s^2} e^{-|c-c'|^2/(4s^2)} \big\langle e^{-|x_m - m|^2/s^2} (x_m - c') \cdot \dot{x}_m \big\rangle_t .$$

Write $x_m - c' = (x_m - m) + (c - c')/2$. The first part integrates to zero, being the time derivative of $-(s^2/2)e^{-|x_m - m|^2/s^2}$ along a period. Hence

$$\kappa(\xi, \xi') = -\frac{1}{2s^2} e^{-|c-c'|^2/(4s^2)} (c - c') \cdot F_s(m),$$
$$F_s(m) := \left\langle e^{-|x_m(t)-m|^2/s^2} \dot{x}_m(t) \right\rangle_t = \frac{1}{T_0} \int e^{-|x-m|^2/s^2} \, \mathrm{d}T_{\mathcal{O}}(x),$$

the Gaussian convolution of the vector measure $T_{\mathcal{O}}$. If $\kappa \equiv 0$ on admissible pairs, then $F_s(m) \cdot v = 0$ for every m with $\|m\| < c_{\max}$ and every direction v (take $c = m - \delta v$, $c' = m + \delta v$), so $F_s \equiv 0$ on that ball; F_s is entire, so $F_s \equiv 0$ on $\mathbb{R}^n$, and the Fourier argument gives $T_{\mathcal{O}} = 0$. Conversely, $\kappa(\xi, \xi') = \frac{1}{T_0} \int \varphi_\xi \nabla \varphi_{\xi'} \cdot \mathrm{d}T_{\mathcal{O}} = 0$ for all widths when $T_{\mathcal{O}} = 0$. For $n = 2$, $\oint_{\mathcal{O}} v \cdot \mathrm{d}x = \int w_{\mathcal{O}} \operatorname{curl} v \, \mathrm{d}A$ for every smooth v (Stokes with winding numbers), and every smooth compactly supported g is $\operatorname{curl} v$ for some smooth v, so $T_{\mathcal{O}} = 0$ iff $w_{\mathcal{O}} = 0$ a.e. ($T_{\mathcal{O}}$ is compactly supported, so v need not be). For the example, $r = he_1$ with $h = 0$ on $\mathcal{O}$ gives $(Dr - Dr^\top)_{1l} = \partial_l h$ and the only non-zero entries are the normal derivative; the Lie-bracket terms vanish because $r = 0$ on $\mathcal{O}$.

(v) The saturated features are real-analytic in the lifted coordinates and extend holomorphically to a complex neighbourhood of $\mathbb{R}^p$ (tanh and the logistic function are holomorphic on the strips $|\mathrm{Im}| < \pi a_{\max}/2$, $\pi c_{\max}/2$, π, where s stays away from 0), uniformly for x on the compact orbit; integrals over the orbit of such integrands are real-analytic in the parameters, and r_μ is affine in the weights. The zero set of a real-analytic function on a connected real-analytic manifold is either the whole manifold or a closed set of Lebesgue measure zero. The instances are those of `test_exactly_averaged_dynamics_is_non_reciprocal_for_n1_and_n2`.

(vi) Since r_ν is affine in ν, $B_{\nu_\delta} = B_\mu + \delta B^{\mathrm{even}}[g]$ with $g := \hat{f}_\mu - \tilde{\Phi}_{\vartheta'}$, where $B^{\mathrm{even}}[g]$ denotes the three residual terms of B with g in place of r_μ. The trace of $B^{\mathrm{even}}[g]$ at equal widths is $-(1/s^2)(c - c') \cdot \int g \varphi_c \varphi_{c'} \, \mathrm{d}\rho$ (the curl term is traceless). If $A_{\nu_\delta} \equiv 0$ for all small δ, then $B^{\mathrm{even}}[g] \equiv 0$, so $(c - c') \cdot F[g\rho]((c + c')/2) = 0$ for all admissible c, c', where $F[g\rho]$ is the Gaussian convolution of the vector measure $g\rho$; as in (iv), $g\rho = 0$, that is $g = 0$ ρ-a.e. For $\vartheta' = \vartheta_z$ this says $\hat{f}_\mu = 0$ on the orbit; for ϑ' with $a' \neq 0$ it then says $\tilde{\Phi}_{\vartheta'} = 0$ on the orbit, which is false since a Gaussian has no zeros. So one of the two segments has $A_{\nu_\delta} \not\equiv 0$ for all small $\delta > 0$. □

Checked by: `tests/test_nonreciprocity.py`, using `src/core/nonreciprocity.py` against `src/core/averaging.py`: the bilinear form $a^\top B_\mu a'$ reproduces the antisymmetric part of the generic second-variation coefficient (10^{-10}, $n = 1$ and 2); $A_\mu(\vartheta, \vartheta_z) = 0$ (10^{-14}) and the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ equals $A_\mu(\vartheta_0, \vartheta_1)$; for $n = 1$ with $f := \hat{f}_\mu$ (perfect fit) $A_\mu \equiv 0$ (below 10^{-13}) while a non-zero residual gives $A_\mu \neq 0$, equal to the closed form; the closed form of κ (10^{-10}), the second component of $F_s(m)$ at $m = (0.5, 0)$, $s = 0.7$ is positive on the ellipse, and $\kappa \equiv 0$ on a segment traversed back and forth (below 10^{-12}); with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ (so $Dr_\mu \equiv 0$), $B_\mu = 2\kappa I$ (10^{-13}), $A_\mu \neq 0$ on the ellipse and $A_\mu \equiv 0$ on the segment; with $f := \hat{f}_\mu + he_1$, $h = (x_1/1.2)^2 + (x_2/0.8)^2 - 1$ (zero on the ellipse, non-zero normal derivative), $B_\mu - 2\kappa I$ equals $-\int \varphi\varphi' \partial_2 h \, \mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$ (10^{-12}) and is not small; the positivity witness $f \equiv 0$, $\mu = \delta_{\bar{\vartheta}}$, for which $A_\mu(\vartheta_0, \vartheta_1) = a_0 a_1 \bar{a} \, ((c_0 - c_1)/s^2) \int \varphi_{\bar{c}} \varphi_{c_0} \varphi_{c_1} \, \mathrm{d}\rho$ has a strictly positive integral. The round-5 referee re-derived every formula and checked $a^\top B_\mu a'$ against the exact second variation of the averaged potential (Richardson in k_e) to $2 \cdot 10^{-4}$–$9 \cdot 10^{-4}$ relative, including parameters at the saturation limits.

Remark I.1 (what Theorem I changes in the reading of Proposition H.7)**.**

- The non-reciprocity of H.7(e) is no longer an instance-by-instance check. Pointwise, for $n = 1$ it holds at every frozen μ that does not interpolate f along the orbit; for $n \geq 2$ at perfect fit it holds unless the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit; in general it fails only on a null set of any connected analytic family that contains one non-reciprocal member. Globally, (vi) shows that no open set of measures is reciprocal, for any n: the averaged loop is nowhere a gradient flow at order k_e^{-2}.

- The two defects scale differently. The performative (semi-gradient) defect is bounded by the residual and the antisymmetric part of its Jacobian along the orbit, so it is small at a well-fitting equilibrium whose residual is smooth across the orbit; the circulation defect does not depend on the fit at all.
- For the $n = 1$ Langevin experiments and Conjecture H.6: the non-gradient correction that falsifies H.6 at fixed k_e is, by (iii), bounded by the residual, which at the regularised optimum μ_τ^* is the regularisation bias. Both the leading drift and the correction are proportional to the residual there, so the correction is $O(1/k_e)$ relative to the leading drift and $O(k_e^{-2})$ absolute, as in Remark H.0.
- The pairwise criterion also simplifies the test of reciprocity: a single pair (ϑ, ϑ') with $B_\mu(\xi, \xi') \neq 0$ suffices.

20.4 Input-to-state stability of $(\mathfrak{A})$ and the route to Conjecture H.6

Proposition H.8 (input-to-state stability of $(\mathfrak{A})$)**.** *Assume $\sigma, \tau > 0$ and the setting of Theorem H. Let μ_t solve the perturbed equation*

$$\partial_t \mu = \nabla \cdot \big(\mu(G\nabla V[\mu] - b_t)\big) + \tau \nabla \cdot (G\nabla \mu) \tag{20.13}$$

from $\mu_0 \in \mathcal{P}_2$ with $F_\tau(\mu_0) < \infty$. Here b_t is a velocity perturbation with $\beta(t)^2 := \int \|b_t\|^2_{G^{-1}} \,\mathrm{d}\mu_t \leq \bar{\beta}^2$. Assume the chain rule of [11, Lemma 2.2] holds along this solution:

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\tau(\mu_t) = \int \langle \nabla(\delta F_\tau/\delta\mu)(\mu_t),\, v_t\rangle \,\mathrm{d}\mu_t,$$

in whitened coordinates, with v_t the velocity. It is expected for smooth bounded perturbations (parabolic regularity and the entropy-dissipation estimate), but it is not verified here. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(F_\tau(\mu_t) - F^*\big) \leq -\tau\rho_\tau\big(F_\tau(\mu_t) - F^*\big) + \tfrac{1}{2}\beta(t)^2, \tag{20.14}$$

$$F_\tau(\mu_t) - F^* \leq e^{-\tau\rho_\tau t}\big(F_\tau(\mu_0) - F^*\big) + \frac{\bar{\beta}^2}{2\tau\rho_\tau}, \tag{20.15}$$

$$\limsup_{t\to\infty} W_{G^{-1}}(\mu_t, \mu_\tau^*) \leq \frac{\bar{\beta}}{\tau\rho_\tau}. \tag{20.16}$$

Status. Proved, assuming the chain rule for the perturbed flow. A proof only; not checked in the repository. Novelty (literature audit): elementary given [11].

Proof. Work in $\zeta = G^{-1/2}\vartheta$, and let $\xi_t := \nabla(\delta F_\tau/\delta\mu)(\mu_t) = \nabla\tilde{V}[\mu_t] + \tau\nabla\log\mu_t$. The velocity is $-\xi_t + \tilde{b}_t$, with $\|\tilde{b}_t\| = \|b_t\|_{G^{-1}}$.

1. By the chain rule and Young's inequality,
$$\frac{\mathrm{d}}{\mathrm{d}t} F_\tau(\mu_t) = -\int \|\xi_t\|^2 \,\mathrm{d}\mu_t + \int \langle \xi_t,\, \tilde{b}_t\rangle \,\mathrm{d}\mu_t \leq -\frac{1}{2}\int \|\xi_t\|^2 \,\mathrm{d}\mu_t + \frac{1}{2}\beta^2.$$
2. With $\hat{\mu}_t \propto e^{-\tilde{V}[\mu_t]/\tau}$,
$$\int \|\xi_t\|^2 \,\mathrm{d}\mu_t = \tau^2 I(\mu_t|\hat{\mu}_t) \geq 2\rho_\tau\tau^2 H(\mu_t|\hat{\mu}_t) \geq 2\rho_\tau\tau\big(F_\tau(\mu_t) - F^*\big),$$
by the LSI of Theorem H(ii) and the entropy sandwich [11, Lemma 3.4].
3. Grönwall gives the first two lines.
4. For the third, $F_\tau(\mu) - F^* \geq \tau H(\mu|\mu_\tau^*)$ [11, Lemma 3.4]. μ_τ^* satisfies LSI(ρ_τ), hence Talagrand's inequality $W^2 \leq (2/\rho_\tau)H(\cdot|\mu_\tau^*)$ (Otto–Villani [53]). So $W^2 \leq (2/(\tau\rho_\tau))(F_\tau - F^*)$, and the limsup follows from the second line. □

The gain $1/(\tau\rho_\tau) \leq e^{\alpha^*/\tau}/\sigma$ is exponentially large in $1/\tau$, like the inverse of the rate.

Route to Conjecture H.6. In the scaling of Conjecture H.6:

- $\tau\rho_\tau \geq (\sigma_0/k_e) \cdot e^{-2K_\Phi(F_\rho+K_\Phi)/\tau_0}$, so $1/(\tau\rho_\tau) = O(k_e)$.
  - In this scaling the whitened proximal Gibbs measures do not depend on k_e, so $\tau\rho_\tau = \Theta(1/k_e)$.
  - At the experiment's $\tau_0 = 0.04$ the constant is e^{1049}.
- Define $b := (1/k_e^2)\big[G\nabla(U_{\text{even}} + U_{\text{odd}}) + b_{\text{osc}}\big] + O(k_e^{-3})$. The first part is the frozen-measure correction of Proposition H.7; the second is the oscillation term of Remark H.7a. Both are bounded uniformly in μ and ϑ, so $\bar{\beta} = O(k_e^{-2})$.
- Proposition H.8 then confines every solution of $(\mathfrak{A}) + b$ to an asymptotic W-ball around μ_τ^* of radius $O(1/k_e) \cdot e^{2K_\Phi(F_\rho+K_\Phi)/\tau_0}/\sigma_0$. This needs $F_\tau(\mu_0) < \infty$, or a $\mathcal{P}_2$ start, after the Girsanov regularisation of Theorem H(ii), which applies because b is bounded.

Three things are still missing for Conjecture H.6:

1. control of the frozen-measure approximation along the moving μ_t, that is, a near-identity averaging transformation for the stochastic McKean–Vlasov loop, uniformly in time;
2. the mean-field limit of (L_τ), since Theorem D covers only (L);
3. the chain rule that Proposition H.8 assumes for the perturbed flow.

20.5 Evidence from the real closed loop

The evidence comes from the real closed loop with $m = 32$ and $n = 1$, not from the averaged system.

- **Dither is necessary.** Without dither the collapsed start is trapped forever. Its loss stays at 0.166, against 0.007 from the spread start (`results/langevin_tradeoff`, 1500 s).
- **Equalisation at moderate temperature.** At $\sigma = 0.02$ and $\tau = 0.004$ (3 seeds, `results/langevin_global/table_sigma0.02_tau0.004.md`):
  - collapsed and spread starts reach the same loss (ratio 1.03) and tracking RMS (0.0465 against 0.0452);
  - their final 32-particle configurations, driven by the same noise realisation, are 4.2× closer in W_2 over the physical coordinates (a, c, s) than without dither (0.40 against 1.69);
  - a numerical check shows that two independent 32-point samples of μ_τ^* are typically about 0.6 apart, so 0.40 is at the sampling floor;
  - the common loss of about 0.11 matches the grid value $\bar{L}(\mu_\tau^*) = 0.11118$. It is dominated by the regularisation bias.
- **Metastability at low temperature.** At $\sigma = 0.002$ and $\tau = 5 \cdot 10^{-4}$ (table for these values in `results/langevin_global/`), dither frees the collapsed start ($0.166 \to 0.048$). But the starts have not equalised within 900 s (loss ratio 5.2). The spread start's $9.2 \cdot 10^{-3}$ is close to the regularised optimum $\bar{L}(\mu_\tau^*) = 8.5 \cdot 10^{-3}$.
- **Limits of the evidence.**
  - The Langevin runs have $n = 1$. By Proposition H.7, the averaged dynamics is already non-reciprocal for $n = 1$, through the state shift.
  - The only $n \geq 2$ closed-loop runs are the deterministic orientation runs of Remark H.7a. They confirm the first-order theory quantitatively. No $n \geq 2$ run with dither has been made.

– The runs use fixed k_e, not the scaling of Conjecture H.6.

– All Langevin runs use $k_e = 10$, below the Tikhonov scale of Remark H.0 for this network ($(F_\rho + K_\Phi)/s_{\min} \approx 35$). The measured frozen-$\mu$ drift error at $k_e = 10$ is 4.2% ($n = 1$), so the condition is conservative, but the agreement with ($\mathfrak{A}$) is empirical.

– So the evidence supports Conjecture H.6 qualitatively, but the equalisation time can exceed practical horizons at low τ.

21 Theorem J: the finite-m closed loop as a perturbed gradient flow

This section proves the first step of Section 25, branch 1: at finite m, without dither, in the scaling $\sigma = \sigma_0/k_e$, the real closed loop has, near every hyperbolic critical point of the m-particle regularised occupation loss, a unique periodic orbit with the same stability type, at distance $O(1/k_e)$, whose first-order location is the one Proposition H.7 and Remark H.7a predict. It turns Remark H.7a's "formal" second-order averaging into a theorem for this system.

Setting. The closed loop of Theorem H's design with $\tau = 0$, at finite m, in the scaling $\sigma = \sigma_0/k_e$:

$$\begin{aligned} \dot{x} &= f(x) + \dot{x}_m - k_e e - \hat{f}_\Theta(x), \qquad e := x - x_m(t), \qquad \hat{f}_\Theta := \frac{1}{m}\sum_j \tilde{\Phi}(\cdot\,;\vartheta_j), \\ \dot{\vartheta}_i &= G\nabla_\vartheta\big[e^\top \tilde{\Phi}(x;\vartheta_i)\big] - \frac{\sigma_0}{k_e}(\vartheta_i - \vartheta^0), \qquad i = 1,\dots,m, \end{aligned} \tag{CL}$$

with $\Theta := (\vartheta_1,\dots,\vartheta_m) \in \mathbb{R}^{mp}$, $G_m := I_m \otimes G$, saturated features, $x_m \in C^2$ T_0-periodic with occupation measure ρ, and $f \in C^2$ on a neighbourhood of the tube $\mathcal{T}_R := \{x : \operatorname{dist}(x, x_m([0,T_0])) \le R\}$ for some $R > 0$. Put

$$r_\Theta(x) := f(x) - \hat{f}_\Theta(x), \qquad J_m(\Theta) := m\bar{L}_m(\Theta) + \frac{\sigma_0}{2}\sum_i \|\vartheta_i - \vartheta^0\|^2_{G^{-1}},$$

$$\bar{L}_m(\Theta) := \frac{1}{2}\int \|r_\Theta\|^2 \,\mathrm{d}\rho,$$

$$H_i(\Theta,t) := G\nabla_\vartheta\big[r_\Theta(x_m(t))^\top \tilde{\Phi}(x_m(t);\vartheta)\big]\Big|_{\vartheta=\vartheta_i} - \sigma_0(\vartheta_i - \vartheta^0),$$

$$\langle H\rangle(\Theta) = -G_m \nabla J_m(\Theta).$$

The averaged system of (CL) is $\dot{\Theta} = \frac{1}{k_e}\langle H\rangle(\Theta) = -\frac{1}{k_e}G_m\nabla J_m(\Theta)$: the gradient flow of J_m in the G^{-1} metric, which is ($\mathfrak{A}$) at $\tau = 0$ for the m-atom measure (Remark H.0); in slow time $s := t/k_e$ it does not depend on k_e. Its equilibria are the critical points of J_m. A critical point Θ^* is *hyperbolic* if $\nabla^2 J_m(\Theta^*)$ is non-singular; its index is the number of negative eigenvalues; it is a *non-degenerate minimum* if $\nabla^2 J_m(\Theta^*) \succ 0$. Write $\nu_1 \le \cdots \le \nu_{mp}$ for the eigenvalues of $G_m^{1/2}\nabla^2 J_m(\Theta^*)G_m^{1/2}$ (those of $G_m\nabla^2 J_m(\Theta^*)$), and $\lambda_* := \nu_1$ when Θ^* is a minimum.

Let $u(\Theta,t)$ be the zero-mean antiderivative in t of $H(\Theta,t) - \langle H\rangle(\Theta)$ (T_0-periodic), and define, as in Proposition H.7 and Remark H.7a, at the atomic measure $\mu^* := \frac{1}{m}\sum_j \delta_{\vartheta_j^*}$:

$$J := D_\Theta\big[\tfrac{1}{k_e}\langle H\rangle\big](\Theta^*) = -\frac{1}{k_e}G_m\nabla^2 J_m(\Theta^*),$$

$$b_{\mathrm{frozen},i} := \frac{1}{k_e^2}G\nabla_\vartheta\big(U_{\mathrm{even}} + U_{\mathrm{odd}}\big)[\mu^*](\vartheta)\Big|_{\vartheta=\vartheta_i^*}, \qquad b_{\mathrm{osc}} := \frac{1}{k_e^2}\big\langle D_\Theta H(\Theta^*,t)\cdot u(\Theta^*,t)\big\rangle_t.$$

Theorem J. *Let Θ^* be a hyperbolic critical point of J_m. There are $k_0, \rho_0, C > 0$, depending on (f, x_m, G, σ_0, R, the feature bounds, Θ^*) but not on k_e, such that for every $k_e \ge k_0$:*

(a) Existence, uniqueness, location. (CL) *has a T_0-periodic solution (x_p, Θ_p), unique among the T_0-periodic solutions with $\sup_t \|\Theta_p(t) - \Theta^*\|_{G^{-1}} \le \rho_0$ and $\sup_t \|x_p(t) - x_m(t)\| \le R$. It satisfies*

$$\sup_t \|x_p(t) - x_m(t)\| \le C/k_e, \qquad \sup_t \|\Theta_p(t) - \Theta^*\|_{G^{-1}} \le C/k_e,$$

$$\Theta_p(t) - \bar{\Theta}_p = \frac{1}{k_e} u(\Theta^*, t) + O(k_e^{-2}),$$

with $\bar{\Theta}_p$ the period mean of Θ_p.

(b) Stability type. *The monodromy matrix of (x_p, Θ_p) has n eigenvalues of modulus at most $e^{-(k_e - L)T_0}$, with L independent of k_e (the error channel), and mp eigenvalues*

$$\lambda_j = \exp\bigl(-(T_0/k_e)\nu_j + O(k_e^{-2})\bigr), \qquad j = 1, \dots, mp.$$

Hence the orbit is hyperbolic with the index of Θ^*: exponentially stable if $\nabla^2 J_m(\Theta^*) \succ 0$, with parameter Floquet exponents $(1/k_e)\nu_j + O(k_e^{-2})$, and unstable otherwise.

(c) Basin (non-degenerate minimum). *If $\nabla^2 J_m(\Theta^*) \succ 0$, every solution of* (CL) *with $\|\Theta(0) - \Theta^*\|_{G^{-1}} \le \rho_0$ and $\|x(0) - x_m(0)\| \le R$ converges to the periodic orbit, and*

$$\|\Theta(t) - \Theta_p(t)\|_{G^{-1}} + \|x(t) - x_p(t)\| \le C e^{-\mu t}\bigl(\|\Theta(0) - \Theta_p(0)\|_{G^{-1}} + \|x(0) - x_p(0)\|\bigr),$$

$$\mu := \frac{\lambda_*}{k_e}(1 - \kappa),$$

with $\kappa = \kappa(\rho_0, 1/k_e) \to 0$ as $(\rho_0, 1/k_e) \to 0$. The basin radius ρ_0 does not depend on k_e; the rate is λ_*/k_e to leading order, that is, λ_* in slow time.

(d) First-order location. *The period mean satisfies*

$$\begin{aligned}\bar{\Theta}_p &= \Theta^* - J^{-1}(b_{\text{frozen}} + b_{\text{osc}}) + O(k_e^{-2}) \\ &= \Theta^* + \frac{1}{k_e}\bigl(G_m \nabla^2 J_m(\Theta^*)\bigr)^{-1} \cdot k_e^2 (b_{\text{frozen}} + b_{\text{osc}}) + O(k_e^{-2}).\end{aligned}$$

The offset is $O(1/k_e)$. Its orientation-odd part is the quantity measured in Remark H.7a, so the prediction $\Theta_+ - \Theta_- = -2J^{-1}(b_{\text{odd}} + b_{\text{osc}}) + O(k_e^{-2})$ of that remark is now a statement with error control.

Status. Proved (tube and slaving, periodic Riccati block-triangularisation, second-order averaging, Poincaré map); checked by Newton shooting for the exact orbit, its monodromy matrix and its period mean against the averaged Hessian and the first-order prediction, at $k_e = 10$–160 (`tests/test_periodic.py`, `exp_periodic_orbit.py`). Novelty (literature audit): the first step of §8 branch 1; the tools (two-time-scale averaging under attraction) are classical.

Proof. Write $\varepsilon := 1/k_e$. All constants below are uniform in $\varepsilon \le 1/k_0$ and in Θ in the ball $\mathcal{K} := \{\|\Theta - \Theta^*\| \le 2\rho_0\}$, because the saturated features and their derivatives are bounded uniformly in Θ and $f \in C^2$ on the compact tube $\mathcal{T}_R$.

Step 1 (the tube and the slaved error). Let $M_r := \sup\{\|r_\Theta(x)\| : x \in \mathcal{T}_R,\ \Theta \in \mathcal{K}\}$ and $L_r := \sup\|\partial_x r_\Theta\|$ on the same set. For $k_e > M_r/R$ the tube $\{\|e\| \le R\}$ is forward invariant: on $\|e\| = R$, $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|e\|^2 = e^\top r_\Theta(x) - k_e\|e\|^2 < 0$. Put $w := e - \varepsilon r_\Theta(x)$ with x the actual state. Since $r_\Theta(x) - k_e e = -k_e w$ and $\dot{x} = \dot{x}_m - k_e w$,

$$\dot{w} = -\bigl(k_e I - \partial_x r_\Theta(x)\bigr)w - \varepsilon\, \partial_x r_\Theta(x)\dot{x}_m - \varepsilon\, \partial_\Theta r_\Theta(x) \cdot \dot{\Theta},$$

and $\dot{\Theta} = F(e, \Theta, t) := \bigl(G\nabla_\vartheta[e^\top \tilde{\Phi}(x; \vartheta_i)] - \varepsilon\sigma_0(\vartheta_i - \vartheta^0)\bigr)_i$ is $O(\|e\| + \varepsilon) = O(\|w\| + \varepsilon)$. So for $k_e \ge 2L_r + C$, $\frac{\mathrm{d}}{\mathrm{d}t}\|w\| \le -(k_e/2)\|w\| + C_1\varepsilon$ on $\{\|w\| \le 1\} \cap \{\|e\| \le R\}$, whence $\|w(t)\| \le$

$e^{-k_e t/2}\|w(0)\| + 2C_1\varepsilon^2$. The set $W_\varepsilon := \{\|w\| \le 3C_1\varepsilon^2\}$ is forward invariant and absorbs every solution of the tube within $t_1 = O(\varepsilon \log(1/\varepsilon))$, during which Θ moves by $O(\varepsilon\|e(0)\| + \varepsilon^2 \log(1/\varepsilon))$ ($\dot\Theta = O(\|e\| + \varepsilon)$ and $\|e\|$ decays at rate $k_e/2$). On W_ε, $e = \varepsilon r_\Theta(x_m(t)) + \varepsilon[r_\Theta(x_m + e) - r_\Theta(x_m)] + w = \varepsilon r_\Theta(x_m(t)) + O(\varepsilon^2)$.

Step 2 (the slow equation and the fast variable). Let $e_\Theta(t)$ be the frozen-Θ periodic error of Proposition H.7(a): the unique T_0-periodic solution of $\dot e = r_\Theta(x_m(t) + e) - k_e e$, exponentially stable with rate at least $k_e/2$, with $e_\Theta = \varepsilon r_\Theta(x_m) + \varepsilon^2(\partial_x r_\Theta\, r_\Theta - \dot r_\Theta)(x_m) + O(\varepsilon^3)$ in sup norm (Taylor expansion of r_Θ in x to second order; $f \in C^2$). Because Θ enters $r_\Theta = f - \hat f_\Theta$ only through the smooth $\hat f_\Theta$, $e_\Theta(t)$ is C^2 in Θ as soon as $f \in C^1$, and differentiating the frozen equation (gain $1/k_e$) gives $\partial_\Theta e_\Theta = \varepsilon\, \partial_\Theta r_\Theta(x_m) + O(\varepsilon^2)$ and $\partial_\Theta^2 e_\Theta = O(\varepsilon)$, uniformly on $\mathcal{K}$. Let $\zeta := e - e_{\Theta(t)}(t)$ be the fast variable. Then

$$\dot\zeta = \big[r_\Theta(x_m + e) - r_\Theta(x_m + e_\Theta)\big] - k_e\zeta - \partial_\Theta e_\Theta(t)\cdot\dot\Theta,$$

where the bracket is Lipschitz in ζ with constant L_r and the last term is $O(\varepsilon)\cdot O(\varepsilon) = O(\varepsilon^2)$ ($\dot\Theta = F = O(\|e\| + \varepsilon)$). So for $k_e \ge 2L_r + C$ the set $Z_\varepsilon := \{\|\zeta\| \le C_2\varepsilon^3\}$ is forward invariant and absorbs every solution of the tube within $O(\varepsilon\log(1/\varepsilon))$; on W_ε of Step 1 one has $\|\zeta\| = O(\varepsilon^2)$ at once and $O(\varepsilon^3)$ after this further transient. On Z_ε, since F is C^1 in e with $\partial_e F = O(1)$,

$$\dot\Theta = F(e_\Theta(t), \Theta, t) + R(\Theta, \zeta, t) = \varepsilon H(\Theta, t) + \varepsilon^2 H_2(\Theta, t) + O(\varepsilon^3), \qquad \|R\| \le C\|\zeta\| = O(\varepsilon^3), \tag{S}$$

where $\varepsilon H + \varepsilon^2 H_2$ is the expansion of the frozen-Θ drift along the frozen periodic error, and by Proposition H.7(b), $\langle H_2\rangle(\Theta)_i = G\nabla_\vartheta(U_{\text{even}} + U_{\text{odd}})[\mu_\Theta](\vartheta_i)$, that is, $\varepsilon^2\langle H_2\rangle(\Theta^*) = b_{\text{frozen}}$. H and H_2 are T_0-periodic and C^2 in Θ on $\mathcal{K}$, with bounds uniform in ε.

Regularity of the remainders (used in Steps 4 and 6). The second-order terms of (S), and of (N) below, are used in sup norm only, for the location (d). The implicit-function and contraction arguments of Steps 4 and 6 need first-order C^1 closeness, which holds: $F(e_\Theta(t), \Theta, t)$ is C^2 in Θ with $\partial_\Theta[F(e_\Theta(t), \Theta, t)] = \varepsilon\, \partial_\Theta H(\Theta, t) + O(\varepsilon^2)$ (from $\partial_\Theta e_\Theta = \varepsilon\, \partial_\Theta r_\Theta(x_m) + O(\varepsilon^2)$ and the smooth dependence of F on (e, Θ) through $\tilde\Phi$); the remainder R is Lipschitz in ζ with constant $\sup\|\partial_e F\| = O(1)$ and Lipschitz in Θ with constant $O(\|\zeta\|) = O(\varepsilon^3)$; and the forcing $-\partial_\Theta e_\Theta\cdot\dot\Theta$ of the ζ-equation is Lipschitz in Θ with constant $O(\varepsilon^2)$ (from $\partial_\Theta^2 e_\Theta = O(\varepsilon)$ and $\partial_\Theta F = O(\varepsilon)$). None of this uses more than $f \in C^2$.

Step 3 (second-order averaging normal form). Let u be the zero-mean antiderivative of $H - \langle H\rangle$ and $\Theta = Y + \varepsilon u(Y, t)$. Then $\dot Y = \varepsilon\langle H\rangle(Y) + \varepsilon^2\big[H_2(Y, t) + D_\Theta H(Y, t)u(Y, t) - D_Y u(Y, t)\langle H\rangle(Y)\big] + O(\varepsilon^3)$. Let u_2 be the zero-mean antiderivative of the fluctuating part of the bracket and $Y = Y_2 + \varepsilon^2 u_2(Y_2, t)$. Since $\langle D_Y u\rangle = D_Y\langle u\rangle = 0$,

$$\dot Y_2 = \varepsilon\langle H\rangle(Y_2) + \varepsilon^2\Phi_2(Y_2) + O(\varepsilon^3), \qquad \Phi_2 := \langle H_2\rangle + \langle D_\Theta H\cdot u\rangle, \tag{N}$$

the second-order averaged equation of Remark H.7a (Sanders–Verhulst–Murdock [64, §2.9]), now with a remainder that is uniformly $O(\varepsilon^3)$ on $Z_\varepsilon \times \mathcal{K}$ because (S) holds there. The transformation $\Theta \mapsto Y_2$ is a T_0-periodic near-identity diffeomorphism from $\mathcal{K}$ onto its image, $O(\varepsilon)$-close to the identity with $O(\varepsilon)$ derivative, and the period mean of Θ along any curve $Y_2(t) = Y_2^0 + O(\varepsilon^2)$ equals the period mean of Y_2 up to $O(\varepsilon^3)$, because u and u_2 have zero mean.

Step 4 (the Poincaré map in the slow coordinates). Let $P = (P_Y, P_\zeta)$ be the time-T_0 map of (CL) from $t = 0$, written in the coordinates (Y_2, ζ) on $\mathcal{K}\times Z_\varepsilon$. By (N) and Step 2,

$$P_Y(Y_2, \zeta) = Y_2 + \varepsilon T_0\tilde A(Y_2) + O(\varepsilon^3), \qquad \tilde A := \langle H\rangle + \varepsilon\big[\Phi_2 + (T_0/2)D\langle H\rangle\cdot\langle H\rangle\big],$$
$$\|P_\zeta(Y_2, \zeta)\| \le e^{-k_e T_0/2}\|\zeta\| + O(\varepsilon^3),$$

the $\varepsilon^2(T_0^2/2)D\langle H\rangle\langle H\rangle$ term being the second-order term of the time-T_0 flow of (N). The Lipschitz constants of the coupling are $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ (a perturbation of ζ acts on $\dot\Theta$ through $\partial_e F = O(1)$

while it decays at rate $k_e/2$, so its integrated effect over a period is $O(\varepsilon)\|\zeta\|$) and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$ (the forcing of the ζ-equation is $O(\varepsilon^2)$-Lipschitz in Θ and is integrated against the fast decay). Hence $\zeta \mapsto P_\zeta(Y_2, \zeta)$ has a unique fixed point $\zeta^*(Y_2) = O(\varepsilon^3)$, Lipschitz in Y_2 with constant $O(\varepsilon^3)$, and the fixed points of P are the zeros of

$$\tilde{A}_\varepsilon(Y) := \frac{P_Y(Y, \zeta^*(Y)) - Y}{\varepsilon T_0} = \tilde{A}(Y) + O(\varepsilon^2).$$

By the regularity paragraph, $D_Y P_Y = I + \varepsilon T_0 D\langle H\rangle(Y) + O(\varepsilon^2)$ (the variational equation of the slow flow over one period, whose coefficient is $\varepsilon\,\partial_\Theta H + O(\varepsilon^2)$), and the ζ^*-dependence contributes $\mathrm{Lip}_\zeta P_Y \cdot \mathrm{Lip}_Y \zeta^* = O(\varepsilon^4)$, so $\tilde{A}_\varepsilon$ is Lipschitz with $\mathrm{Lip}(\tilde{A}_\varepsilon - \langle H\rangle) = O(\varepsilon)$ on $\mathcal{K}$. Since $D\langle H\rangle(\Theta^*) = -G_m \nabla^2 J_m(\Theta^*)$ is invertible, $Y \mapsto Y - D\langle H\rangle(\Theta^*)^{-1}\tilde{A}_\varepsilon(Y)$ maps a ball of ε-independent radius ρ_0 around Θ^* into itself and is a contraction there for ε small (the Lipschitz form of the implicit function theorem), which gives a unique zero $Y_2^* = \Theta^* + O(\varepsilon)$, hence a unique fixed point of P with Θ-component within $O(\varepsilon)$ of Θ^*, that is, a unique T_0-periodic solution in the stated set; the bound on $x_p - x_m$ is Step 1. Expanding $\tilde{A}_\varepsilon(Y_2^*) = 0$ to first order, $D\langle H\rangle(\Theta^*)(Y_2^* - \Theta^*) + \varepsilon\Phi_2(\Theta^*) = O(\varepsilon^2)$ (the term $(T_0/2)D\langle H\rangle\langle H\rangle(Y_2^*)$ is $O(\varepsilon)$ because $\langle H\rangle(Y_2^*) = O(\varepsilon)$), so

$$Y_2^* = \Theta^* - \varepsilon D\langle H\rangle(\Theta^*)^{-1}\Phi_2(\Theta^*) + O(\varepsilon^2) = \Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}}) + O(\varepsilon^2),$$

using $J = \varepsilon D\langle H\rangle(\Theta^*)$ and $\varepsilon^2\Phi_2(\Theta^*) = b_{\mathrm{frozen}} + b_{\mathrm{osc}}$. Along the periodic orbit $Y_2(t) = Y_2^* + O(\varepsilon^2)$ (since $\langle H\rangle(Y_2^*) = O(\varepsilon)$), so by Step 3 the period mean of Θ_p is $Y_2^* + O(\varepsilon^2)$: this is (d), and $\Theta_p(t) - \bar{\Theta}_p = \varepsilon u(\Theta^*, t) + O(\varepsilon^2)$ is (a).

Step 5 (the monodromy matrix). Linearise (CL) along the periodic orbit: with δe and $\delta\Theta$,

$$\delta\dot{e} = (\partial_x r - k_e I)\delta e + \partial_\Theta r \cdot \delta\Theta, \qquad \delta\dot{\Theta} = \partial_e F \cdot \delta e + \partial_\Theta F \cdot \delta\Theta,$$

all coefficients T_0-periodic and evaluated along the orbit; $\partial_x r$, $\partial_\Theta r$, $\partial_e F$ are $O(1)$ and $\partial_\Theta F = O(\varepsilon)$ (F is linear in e with coefficients smooth in Θ, plus the leak $-\varepsilon\sigma_0(\vartheta_i - \vartheta^0)$, so $\partial_\Theta F = O(\|e\| + \varepsilon)$, and $e = O(\varepsilon)$ on the orbit). Let $S(t)$ be a T_0-periodic solution of the Riccati equation

$$\dot{S} = (\partial_x r - k_e I)S + \partial_\Theta r - S\,\partial_e F\,S - S\,\partial_\Theta F.$$

For k_e large it exists, is unique with $\|S\| \le C/k_e$, and depends smoothly on the data (contraction of the periodic solution operator of the dominant part $\dot{S} = -k_e S + g$; cf. Proposition H.7(a)). In the coordinates $\eta := \delta e - S\delta\Theta$ the linearisation is block triangular:

$$\dot{\eta} = \big[(\partial_x r - k_e I) - S\,\partial_e F\big]\eta, \qquad \delta\dot{\Theta} = \partial_e F \cdot \eta + \big[\partial_e F\,S + \partial_\Theta F\big]\delta\Theta,$$

so the monodromy matrix is similar to a block-triangular matrix whose diagonal blocks are the monodromies of the two diagonal equations. The η-block has norm at most $e^{-(k_e - L)T_0}$ with $L := L_r + \sup\|S\,\partial_e F\| = O(1)$ (Grönwall): these are the n error-channel multipliers. The $\delta\Theta$-block is the linearisation along the orbit of the slow equation (S) in Θ-coordinates, $\varepsilon D_\Theta H(\Theta_p(t), t) + O(\varepsilon^2) = \varepsilon D_\Theta H(\Theta^*, t) + O(\varepsilon^2)$, whose monodromy is $I + \varepsilon T_0 D\langle H\rangle(\Theta^*) + O(\varepsilon^2) = I - \varepsilon T_0 G_m \nabla^2 J_m(\Theta^*) + O(\varepsilon^2)$. Its eigenvalues are $1 - \varepsilon T_0\nu_j + O(\varepsilon^2) = \exp(-\varepsilon T_0\nu_j + O(\varepsilon^2))$, those of $G_m\nabla^2 J_m(\Theta^*)$ being the ν_j by similarity with $G_m^{1/2}\nabla^2 J_m(\Theta^*)G_m^{1/2}$; the $O(\varepsilon^2)$ perturbation moves them by $O(\varepsilon^2)$ because $I - \varepsilon T_0 G_m \nabla^2 J_m(\Theta^*)$ is diagonalisable (similar to a symmetric matrix through $G_m^{1/2}$), so the Bauer–Fike theorem applies with a condition number independent of ε — this covers the repeated eigenvalues of a collapsed configuration. This is (b).

Step 6 (the basin). Let $\nabla^2 J_m(\Theta^*) \succ 0$. In the norm $\|v\|_{G^{-1}} := \|G_m^{-1/2} v\|$, the matrix $I - \varepsilon T_0 G_m \nabla^2 J_m(\Theta^*)$ is similar to a symmetric contraction with norm exactly $1 - \varepsilon T_0 \lambda_*$ once $\varepsilon T_0 \nu_{mp} < 1$. By Step 4, $D_Y P_Y = I - \varepsilon T_0 G_m \nabla^2 J_m(Y) + O(\varepsilon^2)$ on $\mathcal{K}$, and $\nabla^2 J_m$ is Lipschitz, so

on the ball $\{\|Y_2 - Y_2^*\|_{G^{-1}} \le \rho_0\}$ the Lipschitz constant of P_Y in Y_2 is at most $1 - \varepsilon T_0 \lambda_*(1-\kappa)$ with $\kappa = O(\rho_0 + \varepsilon)$; the coupling constants are $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$, and P_ζ contracts ζ by $e^{-k_e T_0/2}$ (Step 4). The 2×2 matrix of Lipschitz constants

$$\begin{pmatrix} 1 - \varepsilon T_0 \lambda_*(1-\kappa) & C\varepsilon \\ C\varepsilon^3 & e^{-k_e T_0/2} \end{pmatrix}$$

has spectral radius $1 - \varepsilon T_0 \lambda_*(1-\kappa')$ with $\kappa' = O(\rho_0 + \varepsilon)$, so on $\{\|Y_2 - Y_2^*\|_{G^{-1}} \le \rho_0\} \times Z_\varepsilon$, with ρ_0 small but independent of ε, P is a contraction in an adapted product norm with that factor: its iterates converge to the fixed point geometrically, uniformly in ε. Solutions started in the tube with $\|e(0) - e_p(0)\| \le 1$ enter Z_ε within $O(\varepsilon \log(1/\varepsilon))$, during which Θ moves by $O(\varepsilon)\|e(0) - e_p(0)\|$ (Step 1), which is absorbed by halving ρ_0. Between the sampling times the solutions stay within a factor C of each other (Grönwall on one period, with the fast variable absorbing $\|e(0) - e_p(0)\|$ into an $O(\varepsilon)\|e(0) - e_p(0)\|$ displacement of Θ, Step 1). Transporting back through the near-identity transformations gives (c). □

Numerics (`exp_periodic_orbit.py` → `results/periodic_orbit/table.md`; `tests/test_periodic.py`). Two instances, $k_e = 10$, 20, 40, 80, 160. For each k_e the periodic orbit of (CL) is found by Newton shooting on the Poincaré map (DOP853 at relative tolerance 10^{-11}; residual below 10^{-11} in every case), its monodromy matrix is the finite-difference Jacobian of the map, and its period mean is compared with Θ^* and with the first-order prediction $\Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ of (d). Distances and cosines are in the labelled G^{-1} metric.

- **Instance A** ($n = 2$, $m = 3$, the plant and elliptic reference of Remark H.7a, $\sigma_0 = 0.5$, $G = \mathrm{diag}(4, 4, 1, 1, 1)$). The minimiser of J_m is a collapsed configuration (kernel spread $4 \cdot 10^{-14}$) with $\lambda_{\min}(\nabla^2 J_m) = 0.0148$: a non-degenerate minimum with permutation symmetry.

k_e	$\|\bar\Theta_p - \Theta^*\|$	full prediction: size ratio, cosine, vector error	frozen-measure part only: vector error	parameter Floquet exponents against $(1/k_e)\nu_j$: max, median relative error
10	$4.09 \cdot 10^{-2}$	1.149, 0.9972, 0.169	0.398	$8.3 \cdot 10^{-2}$, $1.2 \cdot 10^{-2}$
20	$1.91 \cdot 10^{-2}$	1.074, 0.9993, 0.083	0.360	$7.3 \cdot 10^{-2}$, $7.3 \cdot 10^{-3}$
40	$9.22 \cdot 10^{-3}$	1.036, 0.9998, 0.041	0.348	$4.5 \cdot 10^{-2}$, $4.0 \cdot 10^{-3}$
80	$4.53 \cdot 10^{-3}$	1.018, 1.0000, 0.020	0.343	$2.5 \cdot 10^{-2}$, $2.1 \cdot 10^{-3}$
160	$2.24 \cdot 10^{-3}$	1.009, 1.0000, 0.010	0.342	$1.3 \cdot 10^{-2}$, $1.1 \cdot 10^{-3}$

  The offset scales like $k_e^{-1.05}$ (fitted exponent; (a) predicts -1), the vector error of the full prediction halves with every doubling of k_e (the $O(k_e^{-2})$ remainder of (d) against an $O(1/k_e)$ offset), and the frozen-measure part alone has a 35% vector error at every k_e: the oscillation term b_{osc} of Remark H.7a is an $O(1)$ fraction of the first-order location. The median relative error of the Floquet exponents is $O(1/k_e)$, as (b) states; the largest relative error sits on the $(m-1)$-fold degenerate separation pair with the smallest exponent (0.0018 at $k_e = 20$), where an $O(k_e^{-2})$ absolute error is a large fraction of a small exponent. The two error-channel multipliers are below the resolution of the finite-difference monodromy (exponents of order k_e); their product is recovered from Liouville's identity $\sum \log|\lambda| = \int_0^{T_0} \mathrm{tr}\ \partial F/\partial z \, \mathrm{d}t$ along the orbit, which gives a mean error-channel exponent of $k_e - 0.16$, $k_e - 0.11$, $k_e - 0.086$, $k_e - 0.074$, $k_e - 0.068$ at $k_e = 10 \ldots 160$ (instance B: $k_e + 0.05 \ldots k_e + 0.09$): the multipliers are $e^{-(k_e - L)T_0}$ with $|L| < 0.2$, as (b) states.

- **Instance B** ($n = 1$, $m = 2$, a two-bump plant $f(x) = 1.6e^{-(x-1)^2/0.4} - 1.2e^{-(x+1)^2/0.4} + 0.1x$ on $x_m = 2\sin t$, $\sigma_0 = 0.05$, $G = \mathrm{diag}(2, 1, 1)$). The minimiser has two distinct kernels

of opposite sign (spread 2.1, $\lambda_{\min}(\nabla^2 J_m) = 0.066$). Here $b_{\mathrm{osc}} = 0$ exactly: for a one-dimensional sinusoidal sweep the reflection $t \mapsto \pi - t$ maps x_m to itself and u to $-u$, so $\langle D_\Theta H \cdot u \rangle = 0$, and the full and frozen predictions coincide. The reference is fast on the kernel scale ($\sup\|\dot{x}_m\|/s$ is $2/s$: between 1.3 and 10 over the admissible widths, about 2.6 at the minimiser's width; the residual bandwidth is of the same order), so $k_e = 10$–20 is outside the averaging regime:

k_e	$\|\bar{\Theta}_p - \Theta^*\|$	prediction: size ratio, cosine, vector error	Floquet: max, median relative error
10	$6.44 \cdot 10^{-2}$	2.818, 0.880, 1.995	$9.5 \cdot 10^{-2}$, $4.0 \cdot 10^{-2}$
20	$2.22 \cdot 10^{-2}$	1.942, 0.929, 1.078	$3.2 \cdot 10^{-2}$, $1.4 \cdot 10^{-2}$
40	$8.37 \cdot 10^{-3}$	1.464, 0.970, 0.551	$1.2 \cdot 10^{-2}$, $5.4 \cdot 10^{-3}$
80	$3.51 \cdot 10^{-3}$	1.227, 0.990, 0.277	$4.6 \cdot 10^{-3}$, $2.2 \cdot 10^{-3}$
160	$1.59 \cdot 10^{-3}$	1.111, 0.9969, 0.139	$2.0 \cdot 10^{-3}$, $9.5 \cdot 10^{-4}$

The vector error and the Floquet error halve with each doubling of k_e, as the theorem's $O(k_e^{-2})$ remainders predict, but the constants are large: the asymptotic regime starts around $k_e \approx 80$ here. This instance shows what the theorem does not claim: the $O(1/k_e)$ location is sharp only once k_e exceeds the effective bandwidth ω_{eff} of Proposition G's (H2a′) (Section 18).

Checked by: `tests/test_periodic.py`: ∇J_m against finite differences (10^{-7}) and the non-degenerate two-kernel minimum of instance B; at $k_e = 20$ and 40 the shooting residual is below 10^{-10}, $\|\bar{\Theta}_p - \Theta^*\|$ and $\max\|e\|$ fall by more than 35% from 20 to 40, the vector error of the first-order location falls by more than 40% and is below 0.6 at $k_e = 40$, and the maximal relative Floquet error falls by more than 40% and is below 2% at $k_e = 40$.

Remark J.1 (what Theorem J says about the program of Section 25)**.**

- It is the deterministic, finite-m first step of branch 1: near every hyperbolic critical point of the m-particle regularised loss $k_e \cdot \mathcal{G}$ at $\tau = 0$ the real closed loop has exactly one stationary state (a periodic orbit), an $O(1/k_e)$-perturbation of it with the same stability type, and the convergence rate is the one of the averaged gradient flow in slow time. (The theorem does not say that every periodic orbit of (CL) arises this way.) The two missing pieces of Conjecture H.6 remain: dither ($\tau > 0$, where the stationary state is a measure) and uniformity in m.
- It makes the stationary (period-mean) prediction of Remark H.7a a theorem for this system: the first-order location of the orbit, hence the orientation difference of the period-mean kernels, is $\Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ with an $O(k_e^{-2})$ error. When the leading offset is itself small, the $O(k_e^{-2})$ remainder can dominate its *direction* at moderate k_e: a collapsed saddle instance examined in the round-5 review (`review/REVIEW_round5_J.md`) has a first-order location nearly orthogonal to the true offset at $k_e = 20$ and aligned with it at $k_e = 80$, while the sizes obey the theorem's orders throughout.
- The leak is not needed for the structure, only for the hyperbolicity of Θ^*. At a collapsed critical point the separation Hessian is $\sigma_0 G^{-1} - R_2$ (Proposition K, Section 22); for $n = 1$ and $\sigma_0 = 0$ the amplitude row of R_2 vanishes at a leak-free critical point ($\langle r\varphi \rangle = \partial_c \langle r\varphi \rangle = \partial_s \langle r\varphi \rangle = 0$ there), so the $m - 1$ amplitude-redistribution directions are exactly flat and the theorem does not apply, while for $n \geq 2$ only the amplitude block of R_2 vanishes and a collapsed critical point is generically non-degenerate even without a leak. Where $\nabla^2 J_m$ is singular the theorem does not apply.

- At a collapsed critical point (all ϑ_i^* equal) the permutation symmetry makes the separation modes $(m-1)$-fold degenerate; the Floquet exponents come in such multiplets (`results/periodic_orbit/table.md`); Proposition K (Section 22) works out this case.

22 Proposition K: collapsed configurations, the critical leak, and the orientation of the orbit as a design input

Branch 2 of Section 25 asks whether the circulation of the reference orbit can be designed. Theorem J (Section 21) makes the question precise for the collapsed configuration, the trap that Theorem H's dither was introduced to escape: its stability is decided by the eigenvalues of a Hessian that the orientation shifts at order k_e^{-2}.

Setting. As in Theorem J. A configuration is *collapsed* if $\vartheta_i = \bar\vartheta$ for all i. Then $\hat f_\Theta = \tilde\Phi(\cdot\,;\bar\vartheta)$, so Θ is a critical point of J_m if and only if $\bar\vartheta$ is a critical point of the single-kernel objective $J_1(\vartheta) := \frac12 \int \|f - \tilde\Phi(\cdot\,;\vartheta)\|^2 \,\mathrm{d}\rho + \frac{\sigma_0}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}}$. Write $\bar r := f - \tilde\Phi(\cdot\,;\bar\vartheta)$ and

$$M_2 := \int \nabla_\vartheta \tilde\Phi(\cdot\,;\bar\vartheta)^\top \nabla_\vartheta \tilde\Phi(\cdot\,;\bar\vartheta)\,\mathrm{d}\rho \succeq 0,$$

$$R_2 := \int \sum_k \bar r_k \nabla^2_\vartheta \tilde\Phi_k(\cdot\,;\bar\vartheta)\,\mathrm{d}\rho = \nabla^2_\vartheta \big\langle \bar r^\top \tilde\Phi(\cdot\,;\vartheta)\big\rangle_\rho\Big|_{\bar\vartheta} \quad \text{(residual frozen)}.$$

R_2 is the Hessian of $-V_L[\bar\mu](\vartheta)$ at the atom: the second-order cost of moving a small amount of mass off the atom at frozen residual.

Proposition K (collapsed configurations, the critical leak, and the orientation as a design input)**.** *Let Θ^* be a collapsed critical point of J_m with atom $\bar\vartheta$.*

(i) Symmetry splitting. $\nabla^2 J_m(\Theta^*) = I_m \otimes (\sigma_0 G^{-1} - R_2) + \frac1m(\mathbf{1}\mathbf{1}^\top)\otimes M_2$. *On the separation subspace $\{(u_1,\dots,u_m) : \sum_i u_i = 0\}$, of dimension $(m-1)p$, it acts as $\sigma_0 G^{-1} - R_2$; on the collective subspace $\{u_i$ all equal$\}$ as $\nabla^2 J_1(\bar\vartheta) = \sigma_0 G^{-1} - R_2 + M_2$. The same splitting holds for the Jacobian of any permutation-equivariant particle drift at a collapsed configuration, in particular for the second-order averaged drift of Remark H.7a (Section 20): it has the form $I_m \otimes D + \frac1m(\mathbf{1}\mathbf{1}^\top)\otimes E$.*

(ii) Separation stability at leading order. *Θ^* is a non-degenerate local minimum of J_m iff $\nabla^2 J_1(\bar\vartheta) \succ 0$ and $\sigma_0 G^{-1} \succ R_2$, that is, iff $\bar\vartheta$ is a non-degenerate minimiser of J_1 and $\sigma_0 > \sigma_{\mathrm{crit}}(\bar\vartheta) := \lambda_{\max}(G^{1/2} R_2 G^{1/2})$. The separation Floquet exponents of the collapsed periodic orbit of* (CL) *are then, by Theorem J(b), $(1/k_e)\lambda_j(\sigma_0 I - G^{1/2} R_2 G^{1/2}) + O(k_e^{-2})$, each with multiplicity $m-1$, and they do not depend on m. The condition $\sigma_0 G^{-1} \succ R_2$ is the statement that the atom is a strict local minimiser of the first-variation potential $V_L[\bar\mu] + \frac{\sigma_0}{2}\|\cdot\|^2_{G^{-1}}$ on its support, the local form of the Chizat–Bach optimality condition for measures.*

(iii) The orientation-odd shift. *Let $\mathcal{O}$ be traversed in either direction (the same ρ, hence the same Θ^* and the same leading exponents). Write the second-order normal form of Theorem J's proof as*

$$\dot\Theta = -\frac{1}{k_e} G_m \nabla J_m(\Theta) + b(\Theta) + O(k_e^{-3}), \qquad b := b_{\mathrm{frozen}} + b_{\mathrm{osc}} = O(k_e^{-2}), \tag{N}$$

with $b_{\mathrm{frozen}} = \frac{1}{k_e^2} G_m \nabla(U_{\mathrm{even}} + U_{\mathrm{odd}})$ (Proposition H.7) and b_{osc} of Remark H.7a. (N) *has a unique equilibrium $\bar\Theta_2$ near Θ^*, which is collapsed, with $\bar\Theta_2 - \Theta^* = -J_1^{-1} b(\Theta^*) + O(k_e^{-2})$ for $J_1 := -\frac{1}{k_e} G_m \nabla^2 J_m(\Theta^*)$ (the first-order location of Theorem J(d)); let J_2 be the Jacobian*

of (N) *at* $\bar\Theta_2$*. The parameter Floquet exponents of the collapsed orbit are the eigenvalues of* $-J_2$ *up to* $O(k_e^{-3})$*, and*

$$J_2 = J_1 + \partial b(\Theta^*) + \partial J_1(\Theta^*)[\bar\Theta_2 - \Theta^*] + O(k_e^{-3}),$$

where the third term — the third derivative of J_m *contracted with the first-order offset — is* $O(k_e^{-2})$*, of the same order as* ∂b*, and must be kept: the Jacobian of* (N) *at* Θ^* *is not the right linearisation. Under orbit reversal* ρ*,* Θ^**,* J_1 *and* U_{even} *are unchanged while* U_{odd} *and* b_{osc} *change sign (Proposition H.7(d), Remark H.7a), so with* $b_{\mathrm{even}} := \frac{1}{k_e^2} G_m \nabla U_{\mathrm{even}}$ *and* $b_{\mathrm{odd}} := \frac{1}{k_e^2} G_m \nabla U_{\mathrm{odd}} + b_{\mathrm{osc}}$*,*

$$J_2 = J_1 + E \pm O,$$
$$E := \partial b_{\mathrm{even}}(\Theta^*) - \partial J_1(\Theta^*)\big[J_1^{-1} b_{\mathrm{even}}(\Theta^*)\big], \qquad O := \partial b_{\mathrm{odd}}(\Theta^*) - \partial J_1(\Theta^*)\big[J_1^{-1} b_{\mathrm{odd}}(\Theta^*)\big],$$

for the two orientations. J_1*,* E *and* O *are permutation-equivariant and preserve the separation subspace; by* (i) *their restrictions to one separation block are* $p \times p$ *matrices, written* $J_1|_{\mathrm{sep}} = -\frac{1}{k_e}(\sigma_0 I - G R_2)$*,* D_{even}*,* D_{odd}*. The separation Floquet exponents of the two orientations differ by*

$$\lambda_j^{(+)} - \lambda_j^{(-)} = -2\,\frac{w_j^\top D_{\mathrm{odd}} v_j}{w_j^\top v_j} + O(k_e^{-3})$$

(*simple eigenvalue* λ_j *of* $-J_1|_{\mathrm{sep}}$*, right and left eigenvectors* v_j*,* w_j*), an* $O(k_e^{-2})$ *quantity that scales like* k_e^{-2} *at fixed* (G, σ_0)*, while the exponents themselves are* $O(1/k_e)$*. The difference is a property of the excitation's orientation alone: the occupation measure, the averaged system* ($\mathfrak{A}$) *and all of Theorems B–H are blind to it.*

(iv) The critical leak as a design point. *At* $\sigma_0 = \sigma_{\mathrm{crit}}(\bar\vartheta)$ *the softest separation exponent vanishes at leading order and equals the* k_e^{-2} *value* $-(w^\top D_{\mathrm{even}} v)/(w^\top v) \mp (w^\top D_{\mathrm{odd}} v)/(w^\top v) + O(k_e^{-3})$*. If* $|w^\top D_{\mathrm{odd}} v| > |w^\top D_{\mathrm{even}} v|$*, the collapsed configuration is stable for one orientation and unstable for the other: the direction in which the reference is traversed decides whether the kernels separate. More generally, for* σ_0 *in a window of width* $O(1/k_e)$ *around* σ_{crit} *— the window where the leading exponent* $(\sigma_0 - \sigma_{\mathrm{crit}})(1 + o(1))/k_e$ *lies between the two* k_e^{-2} *values — the orientation decides the sign, and the centre of the window,* $\sigma_0 = \sigma_{\mathrm{crit}} + k_e (w^\top D_{\mathrm{even}} v)/(w^\top v) + O(k_e^{-1})$*, can be computed from the averaged data alone.*

Status. (i) and (ii) are proved (linear algebra at the collapsed point and Theorem J(b)). (iii) and (iv) are proved given the second-order normal form (N) with an $O(k_e^{-3})$ remainder, which is Theorem J's proof carried one order further (periodic averaging to second order, Sanders–Verhulst–Murdock [64, Ch. 2–3]; the first-order step is written out in the proof of Theorem J, the second-order step is Remark H.7a's derivation and is not written out here). The orientation-odd shift, its k_e^{-2} scaling, the prediction and the design window are checked numerically in one instance (`exp_orientation_stability.py`).

Proof. (i) $\partial^2 J_m/\partial\vartheta_i\partial\vartheta_j = \frac{1}{m}\int \nabla\tilde\Phi_i^\top \nabla\tilde\Phi_j \,\mathrm{d}\rho - \delta_{ij}\int \sum_k r_k \nabla^2\tilde\Phi_{i,k}\,\mathrm{d}\rho + \delta_{ij}\sigma_0 G^{-1}$ (the first term from the quadratic dependence of $\frac{1}{2}\int\|r\|^2\,\mathrm{d}\rho$ on the m features with weights $1/m$, the second from the ϑ_i-curvature of the residual term: with $r = f - \frac{1}{m}\sum\tilde\Phi_j$, $\partial_{\vartheta_i}(m\bar L_m) = -\int r^\top\nabla\tilde\Phi_i\,\mathrm{d}\rho$ and $\partial_{\vartheta_j}\partial_{\vartheta_i}(m\bar L_m) = \frac{1}{m}\int\nabla\tilde\Phi_j^\top\nabla\tilde\Phi_i\,\mathrm{d}\rho - \delta_{ij}\int\sum_k r_k\nabla^2\tilde\Phi_{i,k}\,\mathrm{d}\rho$). At a collapsed point all blocks coincide, giving the Kronecker form; the two subspaces are the isotypic components of the permutation action, on which $\frac{1}{m}(\mathbf{1}\mathbf{1}^\top)$ acts as 0 and 1 respectively. The second-order averaged drift is permutation-equivariant (each particle moves by a function of its own position and of the measure), so its Jacobian at a collapsed point has the same form with D the self-derivative and E the derivative through the measure.

(ii) Positive definiteness of a block-diagonal form; the exponents are Theorem J(b) applied to the separation block, whose eigenvalues of $G_m^{1/2}\nabla^2 J_m G_m^{1/2}$ restricted to the separation subspace are those of $\sigma_0 I - G^{1/2}R_2G^{1/2}$, each repeated $m-1$ times. For the last sentence, the first variation of $\mathcal{G} := \bar{L} + \frac{\sigma_0}{2}\int\|\cdot-\vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$ at $\bar{\mu}=\delta_{\bar{\vartheta}}$ is $V_L[\bar{\mu}](\vartheta) + \frac{\sigma_0}{2}\|\vartheta-\vartheta^0\|^2_{G^{-1}} = -\langle \bar{r}^\top\tilde{\Phi}(\cdot\,;\vartheta)\rangle_\rho + \frac{\sigma_0}{2}\|\vartheta-\vartheta^0\|^2_{G^{-1}}$, whose Hessian at $\bar{\vartheta}$ is $-R_2+\sigma_0G^{-1}$.

(iii) The proof of Theorem J carries one order further: the near-identity change of variables of its Step 3 removes the oscillating terms through order k_e^{-2}, which turns (CL) into (N) with an $O(k_e^{-3})$ remainder (this is the normal form of Remark H.7a's derivation, with b_{osc} the term that the frozen-measure expansion misses). (N) is a regular perturbation of the averaged system near the hyperbolic point Θ^*, so it has a unique equilibrium $\bar{\Theta}_2$ within $O(1/k_e)$ of Θ^*, obtained by the implicit function theorem; by the first-order expansion, $\bar{\Theta}_2-\Theta^* = -J_1^{-1}b(\Theta^*)+O(k_e^{-2})$ ($J_1^{-1}=O(k_e)$, $b=O(k_e^{-2})$). (N) is permutation-equivariant and Θ^* is collapsed, so the equivariant Newton iterates stay collapsed and $\bar{\Theta}_2$ is collapsed. Theorem J(b) applied to (N) (same proof, remainder $O(k_e^{-3})$) gives the Floquet exponents as the eigenvalues of $-J_2+O(k_e^{-3})$. Expanding the Jacobian of (N) about Θ^*: $J_2 = J_1 + \partial b(\Theta^*) + \partial J_1(\Theta^*)[\bar{\Theta}_2-\Theta^*] + O(\|\bar{\Theta}_2-\Theta^*\|^2/k_e) + O(k_e^{-2}\|\bar{\Theta}_2-\Theta^*\|)$, and both error terms are $O(k_e^{-3})$; substituting the offset gives E and O, which are the even and odd parts under reversal because ∂J_1 is even and b_{even}, b_{odd} have the parities named. All three operators are permutation-equivariant, hence preserve the separation subspace, and by the Kronecker structure of (i) each acts on it as $I_{m-1}\otimes$ (a $p\times p$ block). First-order eigenvalue perturbation of the simple eigenvalue λ_j of $-J_1|_{\mathrm{sep}}$ gives

$$\lambda_j^{(\pm)} = \lambda_j - \frac{w_j^\top D_{\mathrm{even}} v_j}{w_j^\top v_j} \mp \frac{w_j^\top D_{\mathrm{odd}} v_j}{w_j^\top v_j} + O(k_e^{-3}),$$

and the difference as stated. (For the $(m-1)$-fold degenerate separation modes the degeneracy is the trivial one of the Kronecker structure, which the restriction to one p-dimensional block removes; a further degeneracy within the block would call for degenerate perturbation theory.)

(iv) is (iii) at a vanishing leading eigenvalue; the window follows because the leading exponent is $(\sigma_0-\sigma_{\mathrm{crit}})/k_e\cdot(1+o(1))$ near σ_{crit} ($\bar{\vartheta}$, R_2 and the eigenvectors depend smoothly on σ_0) while the shifts are $O(k_e^{-2})$; its centre is where the leading exponent cancels the even shift. □

Remark K.1 (reading)**.** Three things.

- The collapsed configuration's stability is a single-kernel question at leading order (the Hessian $\sigma_0G^{-1}-R_2$ does not depend on m), which explains why the $m=6$ and $m=32$ experiments of Section 19 collapsed alike: a leak above σ_{crit} traps every network size.
- Dither (Theorem H) and the orientation (this proposition) are two different ways to leave the trap: the first adds entropy, the second tilts the Hessian by $O(k_e^{-2})$. The second is free but small; it matters only near σ_{crit}, that is, when the leak is tuned to the edge of trapping, or for nearly flat separation modes such as the weak saddle of Remark H.2.
- The sign of the shift depends on the data $(f,\mathcal{O},\bar{\vartheta})$ through D_{odd} and has no general sign: the design rule is to compute D_{odd} (from the averaged data: U_{odd}, b_{osc} and the third derivative of J_m) and choose the orientation with the smaller softest separation exponent (or the negative one); the window of (iv) is where the choice matters.

Numerics (`exp_orientation_stability.py` → `results/orientation_stability/table.md`). Instance A of Theorem J's numerics (Section 21; $n=2$, $m=3$, the elliptic reference traversed in both directions, $G=\mathrm{diag}(4,4,1,1,1)$). For each orientation the collapsed periodic orbit is found by Newton shooting (residual $\le 4\cdot10^{-13}$), the parameter block of its monodromy matrix is restricted to the separation and collective subspaces, and the exponents are compared

with the leading order $(\sigma_0 I - G^{1/2} R_2 G^{1/2})/k_e$ and with the second-order prediction, the eigenvalues of $-J_2$ at the equilibrium $\bar{\Theta}_2$ of (N) (Newton from Θ^*; every $\bar{\Theta}_2$ found is collapsed to 10^{-8}). The second-order equilibrium is $\|\bar{\Theta}_2 - \Theta^*\|_{G^{-1}} = 3.7 \cdot 10^{-2}$, $1.8 \cdot 10^{-2}$, $9.0 \cdot 10^{-3}$, $4.5 \cdot 10^{-3}$ away from Θ^* at $k_e = 10$, 20, 40, 80: the $O(1/k_e)$ offset of Theorem J(d).

- *The shift and its prediction* ($\sigma_0 = 0.5$, a stable collapse). The softest separation exponents of the two orientations differ by $9.0 \cdot 10^{-4}$, $1.9 \cdot 10^{-4}$, $4.1 \cdot 10^{-5}$, $9.7 \cdot 10^{-6}$ at $k_e = 10$, 20, 40, 80: fitted exponent -2.18 against k_e ((iii): -2). Scaled by k_e^2, the measured corrections to the leading order are -0.032, -0.057, -0.070, -0.077 for (+) and -0.123, -0.132, -0.136, -0.139 for (−), approaching the predicted -0.083 and -0.143 with a gap that halves at every doubling of k_e (the $O(k_e^{-3})$ remainder); the predicted difference at $\bar{\Theta}_2$ — $7.8 \cdot 10^{-4}$, $1.7 \cdot 10^{-4}$, $3.9 \cdot 10^{-5}$, $9.4 \cdot 10^{-6}$ — is within 14%, 9%, 5%, 3% of the measured one. The prediction with the Jacobian taken at Θ^* instead of $\bar{\Theta}_2$ (the third-derivative term omitted) reads $9.0 \cdot 10^{-4}$, $2.1 \cdot 10^{-4}$, $5.0 \cdot 10^{-5}$, $1.2 \cdot 10^{-5}$: it agrees at $k_e = 10$ by accident and its k_e^2-scaled value converges to 0.078 while the measured one converges to 0.060, the 30% error that the numerics exposed (`review/REVIEW_LOG.md`, round 5, K-1). The separation exponents come in exact pairs (multiplicity $m - 1 = 2$) and the collective exponents differ between the orientations by 2% at $k_e = 10$ and 0.2% at $k_e = 80$, as (i)–(iii) describe.
- *At the critical leak* ($\sigma_0 = \sigma_{\mathrm{crit}} = 0.4606$). The leading exponent vanishes and the k_e^{-2} terms decide: the softest separation exponent is $-4.1 \cdot 10^{-4}$ for (+) and $-1.27 \cdot 10^{-3}$ for (−) at $k_e = 10$, $-4.8 \cdot 10^{-5}$ and $-8.7 \cdot 10^{-5}$ at $k_e = 40$ — unstable for both orientations in this instance, the even part dominating the odd one by a factor 2–3; the predictions at $\bar{\Theta}_2$ are within 14% and 6% of the measured values at $k_e = 40$.
- *The orientation decides* (σ_0 designed from the averaged data). Solving $\mathrm{pred}_+(\sigma_0) + \mathrm{pred}_-(\sigma_0) = 0$ for the predicted softest exponents gives $\sigma_0 - \sigma_{\mathrm{crit}} = 0.0125$ at $k_e = 10$ and 0.0061 at $k_e = 20$, the centre of the window of (iv); there the measured softest separation exponents are $+8.5 \cdot 10^{-4}$ and $-2.5 \cdot 10^{-5}$ ($k_e = 10$), $+1.4 \cdot 10^{-4}$ and $-3.7 \cdot 10^{-5}$ ($k_e = 20$): the same reference orbit, traversed one way, keeps the collapsed kernels together, and traversed the other way makes them separate. The window is narrow — the leading exponent at these σ_0 is $1.2 \cdot 10^{-3}$ and $3.0 \cdot 10^{-4}$ — and the design lands inside it although the k_e^{-3} terms move both measured exponents by about $+4.7 \cdot 10^{-4}$ and $+6 \cdot 10^{-5}$ against the prediction.

Checked by: `tests/test_periodic.py`: the Hessian splitting (i) against finite differences (10^{-5} relative) at the collapsed minimiser of instance A, the isotypic eigenvalue structure, the gap equation of σ_{crit} (10^{-8}), and the two-fold zero of the separation block at σ_{crit} with a positive definite collective block (`test_proposition_K_hessian_splitting_and_critical_leak`); the experiment above for (iii)–(iv) (`exp_orientation_stability.py` → `results/orientation_stability/table.md`).

23 Theorem J′: the stationary law of the dithered finite-m closed loop

This section proves the second step of Section 25, branch 1 — the finite-m closed loop *with dither* — and answers the first "next step" of branch 2 (the orientation effect when the stationary state is a measure). It also opens a new angle: the stationary state of the dithered loop is a non-equilibrium steady state, with a stationary probability current of order $1/k_e$ that vanishes exactly when the first-order averaged drift is reciprocal (Theorem I), and the stationary parameter process is not reversible in time.

Setting. The closed loop of Theorem H's design at finite m, in the scaling $\sigma = \sigma_0/k_e$, $\tau = \tau_0/k_e$ (the formal limit of Remark H.0: $k_e \to \infty$ with G, $k_e\sigma$, $k_e\tau$ fixed), $\varepsilon := 1/k_e$:

$$\begin{aligned} \dot{x} &= f(x) + \dot{x}_m - k_e e - \hat{f}_\Theta(x), \qquad e := x - x_m(t), \\ \mathrm{d}\vartheta_i &= \big[G\nabla_\vartheta\big(e^\top \tilde{\Phi}(x;\vartheta_i)\big) - (\sigma_0/k_e)(\vartheta_i - \vartheta^0)\big]\,\mathrm{d}t + \sqrt{2\tau_0/k_e}\,G^{1/2}\,\mathrm{d}B_i, \qquad i = 1..m, \end{aligned} \tag{CL_τ}$$

with the notation of Section 21: saturated features, $\Theta \in \mathbb{R}^{mp}$, $G_m := I_m \otimes G$, $x_m \in C^\infty$ T_0-periodic with occupation measure ρ, $f \in C^\infty$ on a neighbourhood of the tube $\mathcal{T}_R$, $F_R := \sup_{\mathcal{T}_R}\|f\|$, and r_Θ, J_m, H, u, $\langle H\rangle = -G_m\nabla J_m$ as there. Assume $k_e \geq k_T := (F_R + K_\Phi)/R$, so that the ball $\{\|e\| \leq R\}$ is forward invariant for every parameter path ($\|\dot{e}\| \leq F_R + K_\Phi - k_e\|e\|$ on it): the plant never sees f outside the tube, and f may be modified outside a neighbourhood of $\mathcal{T}_R$ to be smooth with bounded derivatives on $\mathbb{R}^n$ without changing the law of the loop.

Slow time and the three scales. Put $s := \varepsilon t$, $\theta := t \bmod T_0 \in \mathbb{T} := \mathbb{R}/T_0\mathbb{Z}$, and $\eta := e/\varepsilon = k_e e$. Then $\mathrm{d}W_s := \sqrt{\varepsilon}\,\mathrm{d}B_t$ is a Brownian motion in slow time and (CL_τ) reads

$$\begin{aligned} \frac{\mathrm{d}\eta}{\mathrm{d}s} &= \varepsilon^{-2}\big[r_\Theta(x_m(\theta) + \varepsilon\eta) - \eta\big], \\ \mathrm{d}\vartheta_i &= \big[G\nabla_\vartheta\big(\eta^\top\tilde{\Phi}(x_m(\theta) + \varepsilon\eta;\vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]\,\mathrm{d}s + \sqrt{2\tau_0}\,G^{1/2}\,\mathrm{d}W_i, \qquad \frac{\mathrm{d}\theta}{\mathrm{d}s} = \varepsilon^{-1}: \end{aligned}$$

the error relaxes at rate ε^{-2} to the slaved value $\eta^*(\Theta,\theta) := r_\Theta(x_m(\theta))$, the phase turns at rate ε^{-1}, and the parameters move at rate one. The generator of the Markov process $Z := (\Theta, \eta, \theta)$ on $\mathcal{S} := \mathbb{R}^{mp} \times \{\|\eta\| \leq F_R + K_\Phi\} \times \mathbb{T}$ is

$$\begin{aligned} L_\varepsilon &= \varepsilon^{-2}A_\varepsilon + \varepsilon^{-1}\partial_\theta + B_\varepsilon, \qquad A_\varepsilon := \big(r_\Theta(x_m(\theta) + \varepsilon\eta) - \eta\big)\cdot\nabla_\eta, \\ B_\varepsilon &:= \sum_i \big[G\nabla_\vartheta\big(\eta^\top\tilde{\Phi}(x_m(\theta) + \varepsilon\eta;\vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]\cdot\nabla_{\vartheta_i} + \tau_0\sum_i \mathrm{tr}\big(G\nabla^2_{\vartheta_i}\big), \end{aligned}$$

with $A_\varepsilon = A_0 + \varepsilon A_1 + \ldots$, $A_0 = (\eta^* - \eta)\cdot\nabla_\eta$, $A_1 = (D_x r_\Theta(x_m)\eta)\cdot\nabla_\eta$, and $B_\varepsilon = B_0 + \varepsilon B_1 + \ldots$, where B_0 has the drift $\sum_i[G\nabla_\vartheta(\eta^\top\tilde{\Phi}(x_m;\vartheta_i)) - \sigma_0(\vartheta_i - \vartheta^0)]$ (equal to $H(\Theta,\theta)$ at $\eta = \eta^*$) and $B_1 = \sum_i G\nabla_\vartheta\big(\eta^\top D_x\tilde{\Phi}(x_m;\vartheta_i)\eta\big)\cdot\nabla_{\vartheta_i}$.

The averaged objects. $\bar{L}_0 := -G_m\nabla J_m\cdot\nabla + \tau_0\,\mathrm{tr}(G_m\nabla^2)$ is the Langevin generator in the G^{-1} metric with invariant measure the Gibbs measure $\pi := Z^{-1}\exp(-J_m/\tau_0)$ (Theorem H(ii) at finite m; π has Gaussian tails because $J_m \geq (\sigma_0/2)\sum\|\vartheta_i - \vartheta^0\|^2_{G^{-1}} - C$). The first-order drift is the vector field on $\mathbb{R}^{mp}$

$$\begin{aligned} b(\Theta) &:= b_{\mathrm{fr}}(\Theta) + b_{\mathrm{osc}}(\Theta), \qquad b_{\mathrm{fr},i} := G\nabla_\vartheta(U_{\mathrm{even}} + U_{\mathrm{odd}})[\mu_\Theta](\vartheta)\big|_{\vartheta=\vartheta_i}, \\ b_{\mathrm{osc}} &:= \big\langle (D_\Theta H)(\Theta,\theta)\cdot u(\Theta,\theta)\big\rangle_\theta, \end{aligned}$$

with $\mu_\Theta := \frac{1}{m}\sum_j \delta_{\vartheta_j}$ and U_{even}, U_{odd} the functionals of Proposition H.7 (so $b = k_e^2(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ of Theorem J, now at every Θ and not only at critical points; `periodic.second_order_drift` and `dither.SingleKernelInstance.averaged` compute it). The *first-order correction* $g\colon \mathbb{R}^{mp} \to \mathbb{R}$ is the solution with $\pi(g) = 0$ of the weighted Poisson equation

$$\tau_0\nabla\cdot(\pi G_m\nabla g) = \nabla\cdot(\pi b)$$

(equivalently $\bar{L}_0 g = -\tau_0\pi^{-1}\nabla\cdot(\pi b) = \nabla J_m\cdot b - \tau_0\nabla\cdot b$, which has zero π-mean),

the *first-order current* is $j_1 := \pi(b - \tau_0 G_m\nabla g) = \pi b_\perp$ with $b_\perp := b - \tau_0 G_m\nabla g$, so that $\nabla\cdot j_1 = 0$ and $b = G_m\nabla(\tau_0 g) + b_\perp$ is the Hodge decomposition of b in $L^2(\pi; G_m^{-1})$ into a G_m-gradient and a π-divergence-free field. For an observable $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ with polynomially bounded derivatives, ψ_φ denotes the solution of $\bar{L}_0\psi_\varphi = \varphi - \pi(\varphi)$, $\pi(\psi_\varphi) = 0$.

Theorem J′ (the stationary law of the dithered finite-m closed loop)**.** *Assume the setting above, with G diagonal (as in every instance of this document; for a general symmetric G replace $G_l\partial_{\vartheta_l}$ by the contraction with G below). There are $\varepsilon_0 > 0$ and, for every observable $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ (three derivatives, the third Hölder continuous) with polynomially bounded derivatives up to order three, a constant C_φ (depending on φ, f, x_m, G, σ_0, τ_0, R, m, the feature bounds, and not on ε) such that for every $\varepsilon = 1/k_e \in (0, \varepsilon_0]$:*

(a) Invariant law. *The process Z has an invariant probability measure m_ε on $\mathcal{S}$, every invariant measure is supported in $\{\|\eta\| \le F_R + K_\Phi\}$ and has moments of all orders in Θ bounded uniformly in ε, and its θ-marginal is uniform. If moreover $a_{\max} > F_R$ componentwise and the tube lies in the box of the saturated centres, the invariant measure is unique, so the law of Z_t converges to m_ε and time averages along a single trajectory converge to m_ε-expectations.*

(b) The Gibbs measure at leading order. *For every invariant measure m_ε, $|m_\varepsilon(\varphi) - \pi(\varphi)| \le C_\varphi \varepsilon$.*

(c) First-order correction. $m_\varepsilon(\varphi) = \pi(\varphi) - \varepsilon\,\pi(b\cdot\nabla\psi_\varphi) + O(\varepsilon^2) = \pi(\varphi) + \varepsilon\int \varphi g\,\mathrm{d}\pi + O(\varepsilon^2)$, *with $|O(\varepsilon^2)| \le C_\varphi\varepsilon^2$. In particular the stationary parameter means are $\pi(\Theta) + \varepsilon\int \Theta g\,\mathrm{d}\pi + O(\varepsilon^2)$.*

(d) Observables of the full state. *For $\varphi(\Theta, \eta, \theta)$ smooth, T_0-periodic in θ, with polynomially bounded derivatives in Θ ($\bar\varphi$ below in the class of* (b)*), $m_\varepsilon(\varphi) = \pi(\bar\varphi) + O(\varepsilon)$, where $\bar\varphi(\Theta) := \langle \varphi(\Theta, r_\Theta(x_m(\theta)), \theta)\rangle_\theta$. In particular the stationary mean-square tracking error is*

$$\mathbb{E}_{m_\varepsilon}\|e\|^2 = \frac{2}{k_e^2}\,\pi(\bar L_m) + O(k_e^{-3}), \qquad \bar L_m(\Theta) = \tfrac{1}{2}\int \|r_\Theta\|^2\,\mathrm{d}\rho :$$

the Gibbs average of the occupation loss, divided by $k_e^2/2$.

(e) Circulation and irreversibility. *Let*

$$V(\Theta, \eta, \theta) := k_e \cdot (\textit{drift of } \Theta \textit{ in physical time}) = \sum_i \big[G\nabla_\vartheta\big(\eta^\top \tilde\Phi(x_m(\theta) + \varepsilon\eta; \vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]$$

(so $\dot\Theta = \varepsilon V + \text{noise}$). For every pair of coordinates (k, l) of $\mathbb{R}^{mp}$,

$$\Lambda_\varepsilon^{kl} := \mathbb{E}_{m_\varepsilon}[\Theta_k V_l - \Theta_l V_k] = \varepsilon\Big[\int (\Theta_k j_{1,l} - \Theta_l j_{1,k})\,\mathrm{d}\Theta + \pi\big(\langle u_k \dot u_l - u_l \dot u_k\rangle_\theta\big)\Big] + O(\varepsilon^2),$$

where $\dot u := \partial_\theta u = H - \langle H\rangle$. If the bracket is non-zero for some (k, l), then for ε small the stationary parameter process $(\Theta_t)_{t\ge 0}$ under m_ε is not reversible in time: $\mathbb{E}[\Theta_{k,t}\Theta_{l,t+h}] \ne \mathbb{E}[\Theta_{l,t}\Theta_{k,t+h}]$ for small $h > 0$.

Status. Proved. The proof imports three standard theorems, named where they are used: the existence, growth and regularity theory of the Poisson equation for the Langevin generator with a confining potential (Pardoux–Veretennikov [54], with interior Schauder estimates [28] for the higher derivatives), and, for the uniqueness in (a) only, Hörmander's theorem for the smoothness of the transition densities of a hypoelliptic time-periodic diffusion [36], [7], the Stroock–Varadhan support theorem [69] and the Doob–Khasminskii and Harris theorems [17], [33], [34]. Parts (b)–(e) hold for every invariant measure and do not use the uniqueness. The first-order drift b is the one of Theorem J, so (c) is the thermal version of Theorem J(d): at low temperature it reduces to the $O(1/k_e)$ offset of the periodic orbit (Remark J′.1). The orientation effect of Remark H.7a and Proposition K appears in (c) as the orientation-odd part of $\int \Theta g\,\mathrm{d}\pi$, and in the numerics it is measured to within 7–8% at $k_e = 10$ and within the Monte Carlo resolution (10–30%) at $k_e = 20$–80. Checked numerically on the single-kernel instance below (`exp_dither_stationary.py`, `tests/test_dither.py`), where the first-order prediction removes the $O(1/k_e)$ discrepancy between the real loop and the Gibbs measure; the circulation of (e) is below the Monte Carlo resolution of the real loop at the affordable cost, and the current machinery behind it is checked on the first-order averaged SDE instead (Proposition J′.2).

Proof. Throughout, C denotes a constant independent of ε, and "polynomially bounded" means bounded by $C(1+\|\Theta\|^q)$ on $\mathcal{S}$ for some q.

Step 0 (invariance, moments, existence). By the choice $k_e \geq k_T$, $\{\|e\| \leq R\}$ is forward invariant and $\|e(t)\| \leq e^{-k_e t}\|e(0)\| + (F_R + K_\Phi)/k_e$, so the set $\{\|\eta\| > F_R + K_\Phi + \delta\}$ is left in finite time and never re-entered: every invariant measure is supported in $\{\|\eta\| \leq F_R + K_\Phi\}$, and on this set the Θ-drift of (CL_τ) is bounded by $\lambda_G(F_R+K_\Phi)L_\vartheta + \sigma\|\Theta - \Theta^0\|$. For $W := \|\Theta - \Theta^0\|^2$ (slow time) $L_\varepsilon W \leq -2\sigma_0 W + 2\lambda_G(F_R + K_\Phi)L_\vartheta\sqrt{mW} + 2\tau_0 \operatorname{tr} G_m \leq -\sigma_0 W + C$, and likewise $L_\varepsilon W^q \leq -c_q W^q + C_q$ for every $q \geq 1$ (Itô's formula and Young's inequality). Hence $\mathbb{E}_z W^q(Z_s) \leq e^{-c_q s} W^q(z) + C_q/c_q$: the Cesàro averages of the laws of Z_s are tight (W^q is coercive in Θ, η and θ live in compact sets), the semigroup is Feller (the coefficients are smooth with bounded derivatives in η and θ and of linear growth in Θ), and Krylov–Bogoliubov gives an invariant probability measure m_ε. For any invariant measure, $0 = m_\varepsilon(L_\varepsilon W^q) \leq -c_q\, m_\varepsilon(W^q) + C_q$ gives the uniform moment bounds; the θ-marginal is uniform because θ moves at constant speed. Dynkin's formula with these moment bounds shows $m_\varepsilon(L_\varepsilon \psi) = 0$ for every $\psi \in C^2(\mathcal{S})$, T_0-periodic in θ, with polynomially bounded derivatives up to order two: the local martingale in Itô's formula is a martingale, and the stationary expectation of $\psi(Z_s) - \psi(Z_0)$ vanishes.

Step 1 (uniqueness). The noise acts on all of Θ; the x-equation carries no noise. Write the SDE for (x, Θ) in physical time with the vector fields X_0 (drift, time-periodic) and $X_{i,k} := \sqrt{2\tau_0/k_e}\,(G^{1/2})_{\cdot k} \cdot \partial_{\vartheta_i}$. The bracket $[X_{i,\alpha_k}, X_0]$ contains, in the x-direction, $-\frac{1}{m}(\partial_{\alpha_k}\tilde{\Phi})(x;\vartheta_i)\cdot \partial_{x_k} = -\frac{1}{m}(\,\mathrm{d}a_k/\,\mathrm{d}\alpha_k)\varphi(x;c_i,s_i)\partial_{x_k}$ with $\mathrm{d}a_k/\,\mathrm{d}\alpha_k > 0$ and $\varphi > 0$: together with the $X_{i,k}$ these brackets span $\mathbb{R}^n \times \mathbb{R}^{mp}$ at every point and every time, so the parabolic Hörmander condition holds everywhere. By Hörmander's theorem for time-dependent hypoelliptic diffusions with smooth coefficients of bounded derivatives ([36]; the non-homogeneous case is [7]), the transition probabilities $P_{s,t}(z,\cdot)$ have densities that are smooth in z, so the period map $P := P_{0,T_0}$ is strong Feller. Accessibility: fix Θ_1 and let $z^* := (e^*, \Theta_1)$ be the state at time T_0 of the deterministic skeleton that follows, on $[0, T_0/2]$, a C^1 parameter path along which $\hat{f}_{\Theta(t)}(x(t)) = f(x(t)) - w(t)$ with $w(t) := \lambda \cdot \beta(t)$ for a fixed bump β supported in $(0, T_0/2)$ and $\lambda \in \mathbb{R}$ chosen so that $e(T_0/2) = 0$ (possible since $\dot{e} = -k_e e + w$ is controllable and the required λ is $O(e^{-k_e T_0/2})$; such parameter paths exist because $a_{\max} > F_R$: m kernels centred at $x(t)$ with amplitude $f(x(t)) - w(t)$ realise $\hat{f} = f - w$, and the saturated centre box contains the tube), and on $[T_0/2, T_0]$ any C^1 path from that configuration to Θ_1. From every initial (e_0, Θ_0) in the invariant set the same construction (with λ depending on e_0) reaches $e(T_0/2) = 0$ at the same parameter configuration, hence z^* at time T_0. By the Stroock–Varadhan support theorem [69] (the skeleton paths are realised by controls $h \in L^2$: the Θ-noise is non-degenerate and the skeleton's Θ-path is C^1), $z^* \in \operatorname{supp} P(z,\cdot)$ for every z. The supports of two distinct ergodic invariant measures of a strong Feller kernel are disjoint, and both contain z^* ([17, Prop. 4.1.1 and Thm 4.2.1]; [33, Cor. 1.14–Thm 1.16]), so P has at most one invariant measure. The periodic family $P^*_{0,t} m_\varepsilon$ is then the unique invariant law of the augmented process; the strong Feller property with the accessible point gives a minorisation of P^2 on the sublevel sets of the Lyapunov function W ([33, Lemma 1.13]), and Harris' theorem ([34]) gives the convergence of the law of Z_{nT_0} to m_ε in a W-weighted total variation norm, hence the convergence of time averages along a single trajectory.

Step 2 (the fast operator). At frozen (Θ, θ), A_0 is the generator of the flow $\dot{\eta} = \eta^* - \eta$, which contracts to η^* at rate one. For g smooth with $g(\Theta, \eta^*, \theta) = 0$ the function

$$(A_0^{-1} g)(\Theta, \eta, \theta) := -\int_0^\infty g\big(\Theta, \eta^* + e^{-s}(\eta - \eta^*), \theta\big)\,\mathrm{d}s$$

is smooth, satisfies $A_0(A_0^{-1}g) = g$, and its derivatives in (Θ, θ) of any order are bounded by those of g; if g is linear in $\eta - \eta^*$, $g = (\eta - \eta^*)^\top w(\Theta, \theta)$, then $A_0^{-1} g = -(\eta - \eta^*)^\top w$. Conversely $A_0 \psi = 0$ with ψ bounded in η forces $\psi = \psi(\Theta, \theta)$.

Step 3 (the expansion). Let $\varphi \in C^5(\mathbb{R}^{mp})$ with polynomially bounded derivatives up to order five. Define

$$\psi_1 := -u \cdot \nabla\varphi, \qquad w_\varphi := \sum_{i,l} G_l \, \partial_{\vartheta_l}\tilde{\Phi}(x_m(\theta);\vartheta_i)\, \partial_{(i,l)}\varphi \in \mathbb{R}^n$$

$$\text{(for a symmetric } G\text{: } (w_\varphi)_k = \sum_i \big(\nabla_\vartheta \tilde{\Phi}_k(x_m;\vartheta_i)\big)^\top G \nabla_{\vartheta_i}\varphi),$$

$$\psi_2 := (\eta - \eta^*)^\top w_\varphi + \psi_2'(\Theta,\theta),$$

where ψ_2' is the zero-mean antiderivative in θ of $b\cdot\nabla\varphi - [-\dot r^\top w_\varphi + (D_x r_\Theta\, r)^\top w_\varphi + (B_0\psi_1)|_{\eta^*} + (B_1\varphi)|_{\eta^*}]$ with $\dot r := \partial_\theta[r_\Theta(x_m(\theta))]$ and $r := r_\Theta(x_m(\theta))$, and $\psi_3 := A_0^{-1}$ of the η-dependent remainder defined below. Then $\psi_\varepsilon := \varphi + \varepsilon\psi_1 + \varepsilon^2\psi_2 + \varepsilon^3\psi_3$ satisfies, order by order:

- ε^{-2}: $A_0\varphi = 0$.
- ε^{-1}: $A_0\psi_1 + \partial_\theta\varphi + A_1\varphi = 0$, since ψ_1 does not depend on η.
- ε^0: $A_0\psi_2 + \partial_\theta\psi_1 + A_1\psi_1 + (B_0 + A_2)\varphi$. Here $A_1\psi_1 = A_2\varphi = 0$ and $A_0\psi_2 = -(\eta-\eta^*)^\top w_\varphi = (B_0\varphi)|_{\eta^*} - B_0\varphi$, so the sum equals $\partial_\theta\psi_1 + (B_0\varphi)|_{\eta^*} = -\dot u\cdot\nabla\varphi + H\cdot\nabla\varphi + \tau_0\,\mathrm{tr}(G_m\nabla^2\varphi) = \langle H\rangle\cdot\nabla\varphi + \tau_0\,\mathrm{tr}(G_m\nabla^2\varphi) = \bar L_0\varphi$, because $\dot u = H - \langle H\rangle$. This is Remark H.0's algebraic identity: the slaved drift averages to the Langevin drift.
- ε^1: $A_0\psi_3 + \partial_\theta\psi_2 + A_1\psi_2 + (B_0 + A_2)\psi_1 + (B_1 + A_3)\varphi$. At $\eta = \eta^*$: $\partial_\theta\psi_2 = -\dot r^\top w_\varphi + \partial_\theta\psi_2'$ (the lag of the error behind its slaved value), $A_1\psi_2 = (D_x r_\Theta(x_m)r)^\top w_\varphi$ (the state shift in the error equation), $B_0\psi_1 = -H\cdot\nabla(u\cdot\nabla\varphi) - \tau_0\,\mathrm{tr}(G_m\nabla^2(u\cdot\nabla\varphi))$, $A_2\psi_1 = A_3\varphi = 0$, and $B_1\varphi = \sum_i G\nabla_\vartheta\big(r^\top D_x\tilde\Phi(x_m;\vartheta_i)r\big)\cdot\nabla_{\vartheta_i}\varphi$ (the state shift in the feature). By the choice of ψ_2' the sum at $\eta = \eta^*$ equals $b\cdot\nabla\varphi$ pointwise in θ, once the θ-averages are identified: $\langle -\dot r^\top w_\varphi\rangle = G\nabla_\vartheta\langle -\dot r^\top\tilde\Phi(x_m;\vartheta)\rangle\cdot\nabla\varphi = (G_m\nabla U_{\mathrm{odd}})\cdot\nabla\varphi$ at the particles, $\langle (D_x r\, r)^\top w_\varphi + (B_1\varphi)|_{\eta^*}\rangle = \langle (r\cdot\nabla_x)(r^\top\tilde\Phi)\rangle$-gradient $\cdot\nabla\varphi = (G_m\nabla U_{\mathrm{even}})\cdot\nabla\varphi$ (the two state shifts combine into U_{even}), so these three give $b_{\mathrm{fr}}\cdot\nabla\varphi$; and $\langle B_0\psi_1|_{\eta^*}\rangle = -\langle (D_\Theta u)H\rangle\cdot\nabla\varphi - \langle H\otimes u\rangle : \nabla^2\varphi - \tau_0\,\mathrm{tr}(G_m\nabla^2(\langle u\rangle\cdot\nabla\varphi)) = \langle (D_\Theta H)u\rangle\cdot\nabla\varphi = b_{\mathrm{osc}}\cdot\nabla\varphi$, because $\langle u\rangle = 0$, $\langle (D_\Theta u)\dot u\rangle = -\langle (D_\Theta\dot u)u\rangle = -\langle (D_\Theta H)u\rangle$ (integration by parts in θ, $\langle D_\Theta u\rangle\langle H\rangle = 0$), and $\langle H_j u_k\rangle + \langle H_k u_j\rangle = \langle\partial_\theta(u_j u_k)\rangle = 0$ so the symmetric contraction with $\nabla^2\varphi$ vanishes (the same cancellation as $N_2 = 0$ in Lemma E.0′). Writing $T_1(\Theta,\eta,\theta)$ for the sum of the order-ε terms other than $A_0\psi_3$, one has $T_1(\Theta,\eta^*,\theta) = b\cdot\nabla\varphi$, so $\psi_3 := A_0^{-1}(b\cdot\nabla\varphi - T_1)$ is well defined (Step 2) and $A_0\psi_3 + T_1 = b\cdot\nabla\varphi$: the order-$\varepsilon$ term is exactly $\bar L_1\varphi := b\cdot\nabla\varphi$.
- ε^2: $R_\varepsilon := \varepsilon^{-2}[L_\varepsilon\psi_\varepsilon - \bar L_0\varphi - \varepsilon\bar L_1\varphi]$ is a finite sum of products of derivatives of f on the tube (orders ≤ 3), of $\tilde\Phi$ in (x,ϑ) (bounded), of x_m, of u, w_φ, ψ_2', ψ_3 and of derivatives of φ: ψ_2' contains $\nabla^3\varphi$ (through $\tau_0\,\mathrm{tr}(G_m\nabla^2(u\cdot\nabla\varphi))$ in $(B_0\psi_1)|_{\eta^*}$) and ψ_3 contains $\nabla^2\varphi$, so $B_0\psi_2 \ni \tau_0\,\mathrm{tr}(G_m\nabla^2\psi_2')$ carries $\nabla^5\varphi$ and $B_0\psi_3$ carries $\nabla^4\varphi$; using the exact operators A_ε, B_ε and Taylor's theorem in ε rather than their formal series, $|R_\varepsilon| \le C(1 + \|\Theta\|^q)$ on $\mathcal{S}$ uniformly in $\varepsilon \in (0,\varepsilon_0]$ for $\varphi \in C^5$ with polynomially bounded derivatives up to order five. (For the zeroth-order statement (b) alone ψ_2' can be dropped — it only serves to make the order-ε term exactly $b\cdot\nabla\varphi$ — and C^3 suffices.)

Step 4 (the Poisson equation). $\bar L_0$ is uniformly elliptic with smooth coefficients, ∇J_m has linear growth with bounded derivatives of all orders, and $\langle -G_m\nabla J_m(\Theta),\, \Theta - \Theta^0\rangle \le -\sigma_0\|\Theta - \Theta^0\|^2 + C\|\Theta - \Theta^0\|$: the recurrence condition of Pardoux–Veretennikov [54, Thm 1] holds with every polynomial Lyapunov function. For $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ with polynomially bounded derivatives up to order three there is a unique $\psi_\varphi \in C^2(\mathbb{R}^{mp})$ with $\bar L_0\psi_\varphi = \varphi - \pi(\varphi)$, $\pi(\psi_\varphi) = 0$, and ψ_φ, $\nabla\psi_\varphi$ are polynomially bounded ([54, Thm 1 and 2]); interior Schauder estimates on unit balls [28, Thm 6.2 and 6.17], which gain two derivatives and whose constants depend polynomially on $\|\Theta\|$ through the local Hölder norms of the coefficients, give $\psi_\varphi \in C^{5,\alpha}$ with polynomially bounded derivatives up to order five.

Step 5 ((b) and (c)). Apply Step 3 to $\psi := \psi_\varphi$ and Step 0's identity to ψ_ε: $0 = m_\varepsilon(L_\varepsilon\psi_\varepsilon) = m_\varepsilon(\bar L_0\psi) + \varepsilon m_\varepsilon(b\cdot\nabla\psi) + \varepsilon^2 m_\varepsilon(R_\varepsilon) = m_\varepsilon(\varphi) - \pi(\varphi) + \varepsilon m_\varepsilon(b\cdot\nabla\psi) + O(\varepsilon^2)$, the $O(\varepsilon^2)$ by the moment bounds of Step 0. Since $b\cdot\nabla\psi$ is polynomially bounded, $|m_\varepsilon(\varphi) - \pi(\varphi)| \le C\varepsilon$, which is (b). Now $b\cdot\nabla\psi \in C^4(\mathbb{R}^{mp})$ with polynomially bounded derivatives (b is smooth with bounded derivatives in Θ, $\psi \in C^{5,\alpha}$), so (b) applies to it (its Poisson solution is in C^3, which the zeroth-order expansion needs): $m_\varepsilon(b\cdot\nabla\psi) = \pi(b\cdot\nabla\psi) + O(\varepsilon)$, and $m_\varepsilon(\varphi) = \pi(\varphi) - \varepsilon\pi(b\cdot\nabla\psi_\varphi) + O(\varepsilon^2)$. The second form of (c) is Green's identity for the weighted Poisson equation:

$$\pi(b\cdot\nabla\psi) = \int b\cdot\nabla\psi\,\pi = -\int \psi\nabla\cdot(\pi b) = -\tau_0\int \psi\nabla\cdot(\pi G_m\nabla g) = -\tau_0\int g\nabla\cdot(\pi G_m\nabla\psi)$$
$$= -\int g\pi\bar L_0\psi = -\int g(\varphi - \pi(\varphi))\pi = -\int \varphi g\,\mathrm{d}\pi,$$

using $\tau_0\pi^{-1}\nabla\cdot(\pi G_m\nabla\psi) = \bar L_0\psi$ and $\pi(g) = 0$; the boundary terms vanish by the Gaussian tails of π and the polynomial growth of g, ψ and their gradients (the same regularity for g as for ψ_φ, with φ replaced by the polynomially bounded $\nabla J_m\cdot b - \tau_0\nabla\cdot b$).

Step 6 ((d)). Write $\varphi = \bar\varphi + \varphi_\theta + \varphi_\eta$ with $\varphi_\theta(\Theta,\theta) := \varphi(\Theta,\eta^*,\theta) - \bar\varphi(\Theta)$ (zero θ-mean) and $\varphi_\eta := \varphi(\Theta,\eta,\theta) - \varphi(\Theta,\eta^*,\theta)$ (zero at η^*). With $\chi_1 := -$(zero-mean antiderivative in θ of φ_θ) and $\chi_2 := -A_0^{-1}\varphi_\eta$ (Step 2), $L_\varepsilon(\varepsilon\chi_1 + \varepsilon^2\chi_2) = \partial_\theta\chi_1 + A_0\chi_2 + \varepsilon[B_0\chi_1 + (\partial_\theta + A_1)\chi_2] + O(\varepsilon^2) = -(\varphi_\theta + \varphi_\eta) + \varepsilon[\dots] + O(\varepsilon^2)$, all terms polynomially bounded. Step 0's identity gives $m_\varepsilon(\varphi) = m_\varepsilon(\bar\varphi) + \varepsilon m_\varepsilon(B_0\chi_1 + (\partial_\theta + A_1)\chi_2) + O(\varepsilon^2) = \pi(\bar\varphi) + O(\varepsilon)$ by (b). For $\varphi = \|\eta\|^2$, $\bar\varphi = \langle\|r_\Theta(x_m(\theta))\|^2\rangle_\theta = 2\bar L_m(\Theta)$, and $\mathbb{E}\|e\|^2 = \varepsilon^2\mathbb{E}\|\eta\|^2$.

Step 7 ((e)). Expand $V = V^0 + \varepsilon V^1 + O(\varepsilon^2)$ with V^0 the B_0-drift and V^1 the B_1-drift, and apply Step 6 to $\varphi := \Theta_k V_l - \Theta_l V_k$ keeping the order-ε terms. For $\varphi^0 := \Theta\wedge V^0$ ($\wedge$ on the pair (k,l)): $\bar\varphi^0 = \Theta\wedge\langle H\rangle = \Theta\wedge(-G_m\nabla J_m)$, and $\pi(\bar\varphi^0) = 0$ because $-\pi G_m\nabla J_m = \tau_0 G_m\nabla\pi$ and $\int\Theta\wedge(G_m\nabla\pi) = 0$ for a symmetric G_m (integration by parts: the integrand is a divergence plus $\pi\cdot(G_{lk} - G_{kl})$). Next, $\varphi^0_\theta = \Theta\wedge(H - \langle H\rangle) = \Theta\wedge\dot u$, so $\chi_1 = -\Theta\wedge u$ and $\langle(B_0\chi_1)|_{\eta^*}\rangle = -\langle H\cdot\nabla_\Theta(\Theta\wedge u)\rangle - \tau_0\langle\mathrm{tr}\,G_m\nabla^2(\Theta\wedge u)\rangle = -\langle H\wedge u\rangle - \langle\Theta\wedge(D_\Theta u\cdot H)\rangle = -\langle\dot u\wedge u\rangle + \Theta\wedge\langle(D_\Theta H)u\rangle = \langle u\wedge\dot u\rangle + \Theta\wedge b_{\mathrm{osc}}$ (the τ_0 term and $\langle H\rangle\wedge\langle u\rangle$ vanish since $\langle u\rangle = 0$, and $\langle(D_\Theta u)H\rangle = -\langle(D_\Theta H)u\rangle$ as in Step 3). Further, $\varphi^0_\eta = \Theta\wedge(V^0(\eta) - V^0(\eta^*))$ is linear in $\eta - \eta^*$, so $\chi_2 = \varphi^0_\eta$ and $\langle((\partial_\theta + A_1)\chi_2)|_{\eta^*}\rangle = \Theta\wedge\big\langle\sum_i G\nabla_\vartheta\big((-\dot r + D_x r\,r)^\top\tilde\Phi(x_m;\vartheta_i)\big)\big\rangle = \Theta\wedge$(lag and error-shift parts of b_{fr}); with $\bar\varphi^1 = \Theta\wedge\langle V^1|_{\eta^*}\rangle = \Theta\wedge$(feature-shift part of b_{fr}) these complete $\Theta\wedge b_{\mathrm{fr}}$. Collecting, and using (c) for $\bar\varphi^0$ ($m_\varepsilon(\bar\varphi^0) = \varepsilon\int\Theta\wedge(-G_m\nabla J_m)g\,\mathrm{d}\pi + O(\varepsilon^2)$) and (b) for the order-$\varepsilon$ observables,

$$\Lambda_\varepsilon = \varepsilon\Big[\int\Theta\wedge(b - gG_m\nabla J_m)\,\mathrm{d}\pi + \pi\big(\langle u\wedge\dot u\rangle\big)\Big] + O(\varepsilon^2),$$

and $b - gG_m\nabla J_m = (b - \tau_0 G_m\nabla g) + (\tau_0 G_m\nabla g - gG_m\nabla J_m) = b_\perp + \tau_0\pi^{-1}G_m\nabla(g\pi)$, whose second term integrates to zero against $\Theta\wedge$ as above. This is the formula of (e). For the irreversibility: under m_ε the Θ-process is stationary; reversibility would give $\mathbb{E}[\Theta_{k,t}\Theta_{l,t+h}] = \mathbb{E}[\Theta_{l,t}\Theta_{k,t+h}]$ for all h, i.e. $\mathbb{E}[\Theta_t\wedge\Theta_{t+h}] = 0$, whereas $\mathbb{E}[\Theta_t\wedge(\Theta_{t+h} - \Theta_t)] = \mathbb{E}[\Theta_t\wedge\int_t^{t+h}\varepsilon V\,\mathrm{d}s] + 0$ (the martingale term has mean zero) $= h\varepsilon\Lambda_\varepsilon + o(h)$ by stationarity and the continuity of the integrand; so $\Lambda_\varepsilon \neq 0$ excludes reversibility. □

Remark J′.1 (reading)**.**

- *From the periodic orbit to the thermal state.* As $\tau_0 \to 0$ with Θ^* the non-degenerate global minimiser of J_m, $\pi \to \delta_{\Theta^*}$, $\psi_{\Theta_k} \to$ the linear function $v_k^\top(\Theta - \Theta^*)$ with $-(G_m\nabla^2 J_m)^\top v_k = e_k$, and (c) gives $m_\varepsilon(\Theta) \to \Theta^* + \varepsilon(G_m\nabla^2 J_m(\Theta^*))^{-1}b(\Theta^*)$: Theorem J(d)'s period mean of the periodic orbit. Theorem J′(c) is the finite-temperature version: the correction is a Gibbs average of the same first-order drift, weighted by the Poisson solution instead of the inverse Hessian.

- *The orientation effect with dither (branch 2).* Reversing the direction of traversal of the reference flips U_{odd} and b_{osc} and leaves U_{even}, J_m and π unchanged, so the stationary means of the two traversals differ at order ε by $2\varepsilon\int\Theta g_{\mathrm{odd}}\,\mathrm{d}\pi$ with g_{odd} the correction driven by the odd part of b: the quantity measured in Remark H.7a for the periodic orbit is now a statement about stationary measures, and it is checked below (to within 7–8% at $k_e = 10$, and within the resolution of the runs at larger k_e) in the single-kernel instance (where, in one state dimension, $U_{\mathrm{odd}} \equiv 0$ — a 1-form integrated over a path that retraces itself — and the whole odd effect comes from b_{osc}; a reference with the time-reversal symmetry $x_m(\theta_0+\theta) = x_m(\theta_0-\theta)$, such as a sinusoid, has no odd effect at all, since its reversal is a time shift). As a design rule: a time-symmetric reference (one whose Fourier series carries only cosines after a time shift, such as $x_m = X[\cos\theta + r_2\cos 2\theta]$) removes b_{osc} and U_{odd} identically, so its first-order correction is the performative term $G_m\nabla U_{\mathrm{even}}$ alone, which is reciprocal exactly when Theorem I's $A_{\mathrm{even}} \equiv 0$; a reference traversed one way carries both, and the odd part can be used (Proposition K) or cancelled by reversing the traversal. Checked: for the cosine reference the code's b_{osc} and $\langle u \wedge \dot{u}\rangle$ are 10^{-17} at every point tested.

- *Non-equilibrium.* (e) says that the dithered learning loop, at stationarity, is not in detailed balance even to first order: the periodic forcing alone (the term $\pi\langle u \wedge \dot{u}\rangle$, the signed area swept by the fast parameter loop) breaks the time-reversal symmetry, and the slow dynamics carries in addition the current j_1 of Proposition J′.2, nonzero exactly when the first-order drift is not reciprocal. The first is a property of any periodically forced system; the second is the signature of closed-loop learning (Theorem I).

- *Hypotheses.* Smoothness of f on the tube and of the features is used through the expansion (three x-derivatives of f, five Θ-derivatives of the Poisson solution, i.e. three Hölder derivatives of the observable) and, for the uniqueness only, through Hörmander's theorem; $k_e \geq k_T$ keeps the plant in the tube for every parameter path. The constants are not explicit: the result is an asymptotic expansion of the invariant measure, not a quantitative bound at a given k_e, and the numerics show where it applies.

Proposition J′.2 (the first-order current, entropy production, and reciprocity)**.** *Consider the first-order averaged dynamics* $\mathrm{d}\Theta = (-G_m\nabla J_m + \varepsilon b)\,\mathrm{d}s + \sqrt{2\tau_0}\,G_m^{1/2}\,\mathrm{d}W$ *on* $\mathbb{R}^{mp}$*, with* b *as above.*

(i) *Its stationary law* $m_\varepsilon^{(1)}$ *satisfies* $m_\varepsilon^{(1)}(\varphi) = \pi(\varphi) + \varepsilon\int\varphi g\,\mathrm{d}\pi + O(\varepsilon^2)$ *for every* φ *as in Theorem J′ (the expansion* $m_\varepsilon^{(1)} = \pi(1+\varepsilon g) + O(\varepsilon^2)$ *in the weak sense; on the grid it holds in* L^1 *with an* ε^2 *remainder, Numerics below), and its stationary current* $j := m_\varepsilon^{(1)}(-G_m\nabla J_m + \varepsilon b) - \tau_0 G_m\nabla m_\varepsilon^{(1)}$ *equals* $\varepsilon j_1 + O(\varepsilon^2)$ *in the same sense, with* $j_1 = \pi b_\perp$*; its entropy production rate (in slow time) is*

$$\mathrm{EPR}_\varepsilon := \int \frac{\|j\|^2_{G_m^{-1}}}{\tau_0 m_\varepsilon^{(1)}}\,\mathrm{d}\Theta = \varepsilon^2\tau_0^{-1}\pi\big(\|b_\perp\|^2_{G_m^{-1}}\big) + O(\varepsilon^3)$$

(formal in the remainder: the ε^2 *term uses only the first-order current, the* $O(\varepsilon^3)$ *presumes the expansion of* j *in* $L^2(1/\pi)$*).*

(ii) *The following are equivalent:* $j_1 \equiv 0$*;* $b_\perp \equiv 0$*;* b *is a* G_m*-gradient field,* $b = -G_m\nabla J_1$*;* $D_\Theta(G_m^{-1}b)$ *is a symmetric matrix at every* Θ*. In that case the first-order averaged dynamics* is *the Langevin dynamics of* $J_m + \varepsilon J_1$*, with stationary density exactly* $\propto \exp(-(J_m+\varepsilon J_1)/\tau_0)$*, in detailed balance.*

(iii) *The antisymmetric part of* $D_\Theta(G_m^{-1}b)$ *is, block by block (*(i,j) *blocks of size* $p\times p$*),*

$$\tfrac{1}{2}\big[D(G_m^{-1}b) - D(G_m^{-1}b)^\top\big]_{ij} = \frac{1}{2m}\nabla_\vartheta\nabla_{\vartheta'}A_{\mu_\Theta}(\vartheta_i,\vartheta_j) + \tfrac{1}{2}\big\langle[\nabla^2_\Theta\mathcal{J},\nabla^2_\Theta\mathcal{U}]_{G_m}\big\rangle_{\theta,ij},$$

where $A_\mu = S - S^\top$ is the antisymmetric kernel of Theorem I at the atomic measure μ_Θ (derivatives in the kernel arguments, μ_Θ frozen), $\mathcal{J}(\Theta,\theta) := \frac{m}{2}\|r_\Theta(x_m(\theta))\|^2 + \frac{\sigma_0}{2}\sum_i \|\vartheta_i - \vartheta^0\|^2_{G^{-1}}$ is the instantaneous loss (so $H = -G_m\nabla\mathcal{J}$, $\langle\mathcal{J}\rangle = J_m$), $\mathcal{U}$ its zero-mean antiderivative in θ, and $[X,Y]_G := XGY - YGX$. The first term is the finite-m form of the reciprocity defect of Theorem I; the second is the failure of the instantaneous Hessians at different phases to commute, the non-gradient part of second-order averaging of a time-periodic gradient flow.

(iv) *In the setting of Theorem I (saturated Gaussian features, $f \in C^2$, $x_m \in C^2$ periodic), for $m \geq 3$, b_{fr} is never a G_m-gradient on $\mathbb{R}^{mp}$ (b_{fr} does not depend on σ_0 and depends on G through the factor G only); consequently, along the gain scale $t \mapsto tG$, the full drift $b = b_{\mathrm{fr}} + b_{\mathrm{osc}}$ is a $(tG)_m$-gradient for at most one value of $t > 0$. For $m = 1, 2$ the condition of* (ii) *is the explicit vanishing of* (iii)*, checked to fail in the single-kernel instance below (non-gradient fraction $\pi(\|b_\perp\|^2)/\pi(\|b\|^2) \approx 0.3$–$0.5$).*

Proof. (i) Stationary Fokker–Planck: $\nabla\cdot(mG_m\nabla J_m) - \varepsilon\nabla\cdot(mb) + \tau_0\nabla\cdot(G_m\nabla m) = 0$; at order ε^0 the Gibbs measure, at order ε^1 with $\pi_1 = g\pi$: $\nabla\cdot(g\pi G_m\nabla J_m) + \tau_0\nabla\cdot(G_m\nabla(g\pi)) = \tau_0\nabla\cdot(\pi G_m\nabla g)$ (using $\tau_0\nabla\pi = -\pi\nabla J_m$), so the equation is $\tau_0\nabla\cdot(\pi G_m\nabla g) = \nabla\cdot(\pi b)$, the defining equation of g; the expansion is justified by the same Poisson-equation argument as in Theorem J′ (the generator of this SDE is $\bar{L}_0 + \varepsilon b\cdot\nabla$, and Steps 4–5 apply verbatim without the fast variables). The current: $j = \pi(-G_m\nabla J_m) - \tau_0 G_m\nabla\pi + \varepsilon[\pi b - g\pi G_m\nabla J_m - \tau_0 G_m\nabla(g\pi)] + O(\varepsilon^2) = \varepsilon\,\pi(b - \tau_0 G_m\nabla g) + O(\varepsilon^2)$. The entropy production rate of a diffusion with mobility G_m and temperature τ_0 in a stationary state m with current j is $\int j^\top(\tau_0 G_m)^{-1} j/m$ [68], which gives the formula.

(ii) $j_1 \equiv 0$ iff $b = \tau_0 G_m\nabla g$ iff b is a G_m-gradient (then $\tau_0 g = -J_1 + \text{const}$ and the equation for g is satisfied); on the simply connected $\mathbb{R}^{mp}$ a smooth field $G_m^{-1}b$ is a gradient iff its Jacobian is symmetric (Poincaré lemma). When $b = -G_m\nabla J_1$ the drift $-G_m\nabla(J_m + \varepsilon J_1)$ is a gradient drift, so the SDE is the Langevin dynamics of $J_m + \varepsilon J_1$ with the exact Gibbs law $\propto \exp(-(J_m + \varepsilon J_1)/\tau_0)$ and $j \equiv 0$; consistently, $g = -J_1/\tau_0 + \text{const}$ solves the defining equation.

(iii) $G^{-1}b_{\mathrm{fr},i} = \nabla_\vartheta P_2[\mu_\Theta](\vartheta)|_{\vartheta_i}$ with $P_2 := U_{\mathrm{even}} + U_{\mathrm{odd}}$, a functional of μ (quadratic in μ through r_μ) and of the evaluation point. Differentiating with respect to ϑ_j moves the atom ϑ_j of μ_Θ, which gives $\frac{1}{m}\nabla_\vartheta\nabla_{\vartheta'}S(\vartheta_i,\vartheta_j)$ with $S(\vartheta,\vartheta') := \delta P_2[\mu](\vartheta)/\delta\mu(\vartheta')$ (Theorem I's k_e^{-2} second-variation kernel), and, for $j = i$, additionally the symmetric $\nabla^2_\vartheta P_2(\vartheta_i)$. The antisymmetric part of the (i,j) block is therefore $\frac{1}{2m}\nabla\nabla'[S(\vartheta_i,\vartheta_j) - S(\vartheta_j,\vartheta_i)] = \frac{1}{2m}\nabla\nabla' A_\mu(\vartheta_i,\vartheta_j)$. For b_{osc}: $H = -G_m\nabla\mathcal{J}$ and $u = -G_m\nabla\mathcal{U}$, so $G_m^{-1}b_{\mathrm{osc}} = \langle\nabla^2\mathcal{J}\,G_m\nabla\mathcal{U}\rangle$ and $\partial_j(G_m^{-1}b_{\mathrm{osc}})_k = \langle\partial_j\partial_k\partial_l\mathcal{J}\,G_l\partial_l\mathcal{U}\rangle + \langle(\nabla^2\mathcal{J}G_m\nabla^2\mathcal{U})_{kj}\rangle$; the first is symmetric in (j,k), the antisymmetric part of the second is $\frac{1}{2}\langle\nabla^2\mathcal{J}G_m\nabla^2\mathcal{U} - \nabla^2\mathcal{U}G_m\nabla^2\mathcal{J}\rangle$.

(iv) Suppose b_{fr} is a G_m-gradient for some $m \geq 3$. By (iii), $\nabla_\vartheta\nabla_{\vartheta'}A_{\mu_\Theta}(\vartheta_i,\vartheta_j) = 0$ for all Θ and all i, j. A_μ is affine in μ (Theorem I): $A_\mu = A_f + B(\int\tilde{\Phi}\,\mathrm{d}\mu)$ with B linear in the function $\int\tilde{\Phi}(\cdot\,;\vartheta)\,\mathrm{d}\mu(\vartheta)$, so for $\Theta = (\vartheta,\vartheta',\vartheta_3,\dots)$, $\nabla_\vartheta\nabla_{\vartheta'}[A_f + \frac{1}{m}\sum_j B(\tilde{\Phi}(\cdot\,;\vartheta_j))](\vartheta,\vartheta') = 0$ with the atom positions inside B frozen. Varying ϑ_3 alone shows that $\nabla_\vartheta\nabla_{\vartheta'}B(\tilde{\Phi}(\cdot\,;\vartheta_3))(\vartheta,\vartheta')$ does not depend on ϑ_3; at the zero feature $\vartheta_3 = \vartheta_z$ it vanishes ($B(0) = 0$), so it vanishes for all ϑ_3, and then $\nabla\nabla' A_f(\vartheta,\vartheta') = 0$ as well: $\nabla_\vartheta\nabla_{\vartheta'}A_\mu(\vartheta,\vartheta') = 0$ for every μ and all (ϑ,ϑ'). A smooth antisymmetric kernel with vanishing mixed derivative on the connected $\mathbb{R}^p\times\mathbb{R}^p$ has the form $a(\vartheta) - a(\vartheta')$, a coboundary, hence $A_\mu \equiv 0$ by Theorem I(i) — for every μ, contradicting Theorem I(vi). For the gain scale: under $G \to tG$, $b_{\mathrm{fr}} \to t\,b_{\mathrm{fr}}$ and $b_{\mathrm{osc}} \to t^2 b_{\mathrm{osc}}$, so the antisymmetric part of $D((tG)^{-1}b)$ is $a_{\mathrm{fr}} + t\,a_{\mathrm{osc}}$ with $a_{\mathrm{fr}} \not\equiv 0$; it vanishes identically in Θ for at most one t. □

Corollary J′.3 (the stationary law in the iterated limit $k_e \to \infty$, $m \to \infty$)**.** *Let π_m be the Gibbs measure of Theorem J′ on $\mathbb{R}^{mp}$, and μ_τ^* the unique minimiser of Theorem H's free energy F_τ at $\sigma = \sigma_0/k_e$, $\tau = \tau_0/k_e$; since $k_e F_\tau = \bar{L} + \frac{\sigma_0}{2}\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + \tau_0 H(\mu) = \bar{L} + \tau_0 H(\cdot\,|\,\gamma_\tau) + \text{const}$ with $\gamma_\tau = \mathcal{N}(\vartheta^0, (\tau_0/\sigma_0)G)$, μ_τ^* does not depend on k_e. Then:*

(i) Chaoticity of the Gibbs measures. *Under π_m the empirical measure $\mu_\Theta = \frac{1}{m}\sum_i \delta_{\vartheta_i}$ converges in probability to μ_τ^* as $m \to \infty$, and for every k the k-particle marginal of π_m converges weakly to $(\mu_\tau^*)^{\otimes k}$.*

(ii) The iterated limit of the real loop. *For every $\varphi \in C^2(\mathbb{R}^p)$ bounded with bounded derivatives up to order two,*

$$\lim_{m\to\infty}\lim_{k_e\to\infty} \mathbb{E}_{m_\varepsilon}\Big[\frac{1}{m}\sum_i \varphi(\vartheta_i)\Big] = \int \varphi \,\mathrm{d}\mu_\tau^*,$$

and the inner limit is attained at the rate $O(1/k_e)$ of Theorem J′(b), with a constant that depends on m.

Status. Proved (Sanov's theorem and Varadhan's lemma, [19, Thms 6.2.10 and 4.3.1]; Theorem H(i)–(ii); Theorem J′(b)). The joint limit, with k_e and m related, is Conjecture H.6 and remains open; the corollary places the stationary law of the finite loop between Theorem J′ and Theorem H. Numerics: `exp_gibbs_m.py` → `results/dither/gibbs_m.md` (below).

Proof. $J_m(\Theta) = m[\bar L(\mu_\Theta) + \frac{\sigma_0}{2}\int\|\vartheta-\vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu_\Theta]$, so $\pi_m(\,\mathrm{d}\Theta) \propto \exp(-m\bar L(\mu_\Theta)/\tau_0)\,\gamma_\tau^{\otimes m}(\,\mathrm{d}\Theta)$. The map $\mu \mapsto \bar L(\mu) = \frac12\int\|f - \int\tilde\Phi(\cdot\,;\vartheta)\,\mathrm{d}\mu(\vartheta)\|^2\,\mathrm{d}\rho$ is bounded and continuous for the weak topology of $\mathcal{P}(\mathbb{R}^p)$ (the features are bounded and continuous in ϑ, and ρ is a probability measure). Under $\gamma_\tau^{\otimes m}$ the empirical measure satisfies Sanov's large-deviation principle with rate function $H(\cdot\,|\,\gamma_\tau)$; by Varadhan's lemma (tilting by the bounded continuous $-m\bar L/\tau_0$), under π_m it satisfies a large-deviation principle with rate function $I(\mu) = \bar L(\mu)/\tau_0 + H(\mu\,|\,\gamma_\tau) - \inf = (k_e/\tau_0)(F_\tau(\mu) - F_\tau(\mu_\tau^*))$, by Theorem H(i). F_τ is strictly convex ($\bar L$ is convex, the entropy strictly convex) with the unique minimiser μ_τ^* (Theorem H(ii)), so I has the unique zero μ_τ^* and $\mu_\Theta \to \mu_\tau^*$ in probability. Exchangeability of π_m then gives the convergence of the k-marginals to $(\mu_\tau^*)^{\otimes k}$ (Sznitman's equivalence of chaoticity and the convergence of the empirical measure). For (ii), Theorem J′(b) applied to the observable $\frac1m\sum_i\varphi(\vartheta_i) \in C^2(\mathbb{R}^{mp})$ gives $\lim_{k_e\to\infty}\mathbb{E}_{m_\varepsilon}[\frac1m\sum\varphi(\vartheta_i)] = \pi_m(\frac1m\sum\varphi(\vartheta_i)) = \pi_m^{(1)}(\varphi)$, the one-particle marginal, which converges to $\int\varphi\,\mathrm{d}\mu_\tau^*$ by (i). □

Remark J′.4 (what the corollary says about Conjecture H.6)**.** The conjecture asks for the limit of the real loop's law when $k_e \to \infty$ with m fixed or growing, uniformly in time. Theorem J′ settles the stationary part at fixed m, and the corollary shows that the resulting Gibbs measures are chaotic with limit μ_τ^*, the stationary state of ($\mathfrak{A}$): the stationary law of the real dithered loop is the minimiser of Theorem H's free energy in the iterated limit. What is missing for the conjecture is the uniformity in m of Theorem J′'s $O(1/k_e)$ (its constant grows with m through the Poisson equation on $\mathbb{R}^{mp}$) and the time-dependent statement.

Numerics (`exp_dither_stationary.py` → `results/dither/table.md`). The single-kernel instance: $n=1$, $m=1$, plant $f(x) = e^{-(x-0.5)^2/0.5} + 0.4x$, reference $x_m(t) = 1.5[\sin t' + \frac13\sin 2t']$ with $t' = \pm t$ (the two orientations), saturated amplitude and centre $\vartheta = (\alpha,\gamma)$ ($a_{\max} = 3$, $c_{\max} = 2.5$), frozen width $s = 0.6$, $G = \mathrm{diag}(2,2)$, $\sigma_0 = 0.4$, $\vartheta^0 = 0$, $\tau_0 = 0.1$, $k_e \in \{10,20,40,80\}$. The averaged data on a 201^2 grid: π, g, j_1 (rms $b_{\mathrm{fr}} = 0.143$, rms $b_{\mathrm{osc}} = 0.225$; non-gradient fraction $\pi(\|b_\perp\|^2)/\pi(\|b\|^2) = 0.32$ and 0.46 for the two orientations; EPR coefficient $\tau_0^{-1}\pi(\|b_\perp\|^2_{G^{-1}}) = 0.144$ and 0.118; circulation coefficient $\int\Theta\wedge j_1 + \pi\langle u\wedge\dot u\rangle = +0.0117 - 0.0167 = -0.0051$ and $-0.0103 + 0.0167 = +0.0065$). The real loop: $2\cdot10^4$ independent replicas per (k_e, orientation), started from the first-order law $\pi(1+\varepsilon g)$, Strang splitting with $k_e h = 1/2$, burn-in ≥ 10 and averaging ≥ 40 slow-time units (whole numbers of reference periods), standard errors from the spread of the per-replica time averages (the slowest mode of $\bar L_0$ relaxes in 3.5 slow-time units on this instance, so batch means in time would be correlated; referee round 6).

The first-order law (b), (c). The stationary means against the Gibbs value and the first-order prediction, orientation $+1$ (ε-coefficients $\int\varphi g\,\mathrm{d}\pi$: -0.255 for α, -0.426 for γ); Table 20.

Table 20: Theorem J′(b), (c) on the single-kernel instance: the stationary means of the real loop against the Gibbs value and the first-order prediction (`results/dither/table.md`).

k_e	$m_\varepsilon(\alpha)-\pi(\alpha)$ (s.e. 0.0013)	$\times k_e$	$m_\varepsilon(\alpha)-$ first order	$m_\varepsilon(\gamma)-\pi(\gamma)$ (s.e. 0.0018)	$\times k_e$	$m_\varepsilon(\gamma)-$ first order
10	-0.0276	-0.28	-0.0021	-0.0441	-0.44	-0.0015
20	-0.0179	-0.36	-0.0051	-0.0269	-0.54	-0.0056
40	-0.0089	-0.36	-0.0025	-0.0127	-0.51	-0.0020
80	-0.0046	-0.37	-0.0015	-0.0083	-0.67	-0.0030

The discrepancy between the real loop and the Gibbs measure falls like $k_e^{-0.87}$ (α), $k_e^{-0.83}$ (γ), $k_e^{-0.72}$ (α^2) and $k_e^{-1.09}$ ($\alpha\gamma$) (γ^2 has a first-order coefficient 0.03, below resolution), and the first-order prediction removes most of it: 93% and 97% of the α and γ discrepancies at $k_e = 10$, 70–85% at $k_e = 20$–40; at $k_e = 80$ the residuals (-0.0015 ± 0.0013 and -0.0030 ± 0.0018) are at the Monte Carlo floor. The second-order decay of the residual is *not* resolved: the residuals at $k_e = 20$–80 are 1–3 standard errors with a common sign, compatible with an $O(\varepsilon^2)$ coefficient of order -1 to -2 and with noise; at $k_e = 10$ the even part of the α^2 and γ^2 shifts exceeds its prediction by a factor 1.5–1.8 (an $O(\varepsilon^2)$ term of the size seen in Theorem J's instance A, where $c_2/c_1 \approx 1.4$).

The orientation-odd part (Remark J′.1). Measured $(m_+ - m_-)/2$ over predicted $(p_{1+} - p_{1-})/(2k_e)$; Table 21.

Table 21: Theorem J′ on the single-kernel instance: the orientation-odd part of the stationary means, measured over predicted (`results/dither/table.md`).

k_e	α	γ	$\alpha\gamma$
10	0.92 ± 0.04	0.93 ± 0.04	0.89 ± 0.06
20	1.23 ± 0.09	1.10 ± 0.07	1.28 ± 0.13
40	1.28 ± 0.17	1.15 ± 0.15	1.03 ± 0.26
80	1.25 ± 0.34	1.38 ± 0.29	1.11 ± 0.51

The dither version of Remark H.7a's test: the direction in which the reference is traversed shifts the stationary kernel by $2\varepsilon \int \Theta g_{\mathrm{odd}}\, \mathrm{d}\pi$ (here entirely from b_{osc}), measured to within 7–8% at $k_e = 10$ (where the odd shift is 0.02–0.03 against standard errors of 0.001) and consistent with the prediction within the 10–30% resolution of the runs at $k_e = 20$–80 (the ratios sit 10–30% above one with error bars of the same size: a second-order odd term of relative size $O(\varepsilon)$ is not excluded).

The tracking error (d). $k_e^2 \mathbb{E}\|e\|^2/(2\pi(\bar{L})) = 1.054$, 1.033, 1.017, 1.009 (orientation $+1$) and 1.023, 1.014, 1.008, 1.004 (-1) at $k_e = 10$, 20, 40, 80, with standard errors 0.001: the deviation from one is 0.54–0.74/k_e and 0.23–0.35/k_e, the $O(1/k_e)$ of (d).

The circulation (e) and Proposition J′.2. The predicted first-order circulation $\varepsilon[\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle]$ is $-5 \cdot 10^{-4}$ ($+1$) and $+6.5 \cdot 10^{-4}$ (-1) at $k_e = 10$ and scales like $1/k_e$, against a Monte Carlo standard error of $5 \cdot 10^{-4}$ at every k_e: the real loop's circulation is not resolved at first order for $k_e \geq 20$ (measured $-4 \cdot 10^{-4} \cdots + 1.2 \cdot 10^{-3}$, all within 1.7 standard errors of the prediction), and at $k_e = 10$ the measurement ($+3.7 \cdot 10^{-3} \pm 0.5 \cdot 10^{-3}$ for $+1$, $-0.6 \cdot 10^{-3} \pm 0.5 \cdot 10^{-3}$ for -1) is dominated by higher orders (an even $O(\varepsilon^2)$ term of coefficient $\approx +0.2$ and an odd one of the opposite sign to the first-order odd part). The current machinery is therefore checked where the signal is clean, on the first-order averaged SDE itself: its stationary density m_ε and current j_ε, computed on the grid (the discrete current at $\varepsilon = 0$, $-6.6 \cdot 10^{-5}$, a residual of the central scheme, subtracted), give $\int \Theta \wedge (j_\varepsilon - j_0)/(\varepsilon \int \Theta \wedge j_1) = 1.47$, 1.21, 1.09, 1.03 at $\varepsilon = 0.2$, 0.1, 0.05, 0.025 — one plus a term linear in ε — and $\|m_\varepsilon - m_0(1 + \varepsilon g)\|_1/\varepsilon^2 = 0.307$, 0.308, 0.307, 0.305: the first-order expansion of Proposition J′.2(i) with a clean ε^2 remainder.

Corollary J′.3 (`exp_gibbs_m.py` → `results/dither/gibbs_m.md`). On the same instance, μ_τ^* is computed on the 201^2 grid as the fixed point of the first-variation condition (59 damped iterations, residual 10^{-11}): $\int \alpha \, \mathrm{d}\mu^* = 0.4963$, $\int \gamma \, \mathrm{d}\mu^* = 0.1873$, $\int \alpha^2 \, \mathrm{d}\mu^* = 0.9353$, $\bar{L}(\mu_\tau^*) = 0.1394$. The Gibbs measures π_m for $m = 1$, 2, 4, 8, 16, 32 are sampled by MALA (4000 replicas, acceptance 0.98–0.996, standard errors from the independent replicas; the $m = 1$ row agrees with the quadrature of π_1 within one standard error); Table 22.

Table 22: Corollary J′.3 on the single-kernel instance: the one-particle marginals of the Gibbs measures π_m against the mean-field minimiser μ_τ^*, with the rescaled deviations (`results/dither/gibbs_m.md`).

m	$\mathbb{E}_{\pi_m}[\alpha_1^2]$	$\mathbb{E}_{\pi_m}[\gamma_1]$	$\mathbb{E}_{\pi_m}[\bar{L}(\mu_\Theta)]$	$(\mathbb{E}[\alpha_1^2] - \int \alpha^2 \, \mathrm{d}\mu^*) \cdot m$	$(\mathbb{E}[\gamma_1] - \int \gamma \, \mathrm{d}\mu^*) \cdot m$	$(\mathbb{E}[\bar{L}] - \bar{L}(\mu^*)) \cdot m$
1	0.536	0.251	0.1960	−0.40	+0.064	+0.057
2	0.729	0.225	0.1681	−0.41	+0.075	+0.057
4	0.832	0.204	0.1540	−0.41	+0.067	+0.058
8	0.881	0.194	0.1470	−0.43	+0.050	+0.061
16	0.909	0.191	0.1432	−0.42	+0.051	+0.060
32	0.922	0.190	0.1413	−0.42	+0.084	+0.059

The one-particle marginals converge to μ_τ^* with a clean $1/m$ law (slopes -1.01 for γ and -0.99 for the Gibbs-averaged occupation loss, whose rescaled deviation is 0.058 ± 0.002 at every m; the second moment of α moves from 0.54 at $m = 1$ to 0.92 at $m = 32$ against 0.935, with rescaled deviation -0.41 ± 0.01 throughout: the single kernel sits in a narrow well, while in the mean field each particle sees the wide one-body potential $V[\mu_\tau^*]$); the mean of α is within 0.01 of its limit already at $m = 1$ (a small $1/m$ coefficient, its slope not resolved). The stationary law of the finite dithered loop thus approaches the minimiser of Theorem H's free energy in the iterated limit at the rate $1/m$, as the corollary states.

Checked by: `tests/test_dither.py`: the single-kernel averaged data (J, ∇J, b_{fr}, b_{osc}) against the general implementation of Proposition H.7 and Remark H.7a (`averaging.expansion_terms`, `averaging.oscillation_correction`, `periodic.ClosedLoop`) at three parameter points ($2 \cdot 10^{-5}$ relative); the weighted Poisson solver against the Ornstein–Uhlenbeck closed forms ($g = G^{-1} b \cdot \Theta/\tau_0$, mean shift $(G\lambda)^{-1} b$, the Green identity $\int \varphi g \, \mathrm{d}\pi = -\pi(b \cdot \nabla \psi_\varphi)$, zero current and entropy production for a gradient drift); the stationary Fokker–Planck solver of the first-order SDE (OU exact; on the instance the current ratio $\to 1$ and $\|m - \pi(1 + \varepsilon g)\|_1 \propto \varepsilon^2$); the real loop at $k_e = 20$ (8000 replicas: the stationary means of α and γ closer to the first-order prediction than to π, the γ shift resolved at ≥ 3 standard errors, the tracking error within 10% of $2\pi(\bar{L})$); and the Corollary J′.3 numerics (the mean-field fixed point to 10^{-9}, the Gibbs sampler against the quadrature of π_1)
(`test_single_kernel_averaged_data_matches_the_general_H7_implementation`,
`test_poisson_solver_reproduces_the_ornstein_uhlenbeck_closed_forms`,
`test_first_order_stationary_density_and_current_of_the_averaged_sde`,
`test_real_loop_means_are_closer_to_the_first_order_prediction_than_to_gibbs`,
`test_corollary_J3_mean_field_minimiser_and_gibbs_sampler_consistency`).

Part III
Evidence and outlook

24 Numerical evidence

Every number in this section is produced by the named script from raw files, and no table is edited by hand. The scripts are in the directory `experiments/` of the accompanying repository and write their output to subdirectories of `results/`; "`script.py` → `dir/`" below names a script and its output directory or file.

- The test files check the inequalities at every sample of simulated trajectories, or on sampled states: `tests/test_transport.py`, `tests/test_langevin.py`, `tests/test_averaging.py`, `tests/test_excitation.py`, `tests/test_nonreciprocity.py`, `tests/test_periodic.py` and `tests/test_barycentric.py`.
- They are run with `venv\Scripts\python -m pytest tests`. The current number of tests is given in the repository's `README.md`.

24.1 Theorem B: the label-free Lyapunov function

Script: `exp_transport_permutation.py` → `transport_permutation/`.

Each network starts as a random relabelling of the ideal configuration plus noise 0.05, with mean-field gains and 3 seeds per m. "Excess" is $|(1/m)\sum$ Taylor remainders$|$, the part that actually enters $\dot{V}$. Table 23 and Figure 4 show the results.

Table 23: Theorem B, permuted-ideal start, mean-field gains; means over 3 seeds. The two bounds are the same bound, evaluated with the matched (OT) error and with the labelled error, along the same trajectory.

m	labelled error ($t = 0$)	OT error W ($t = 0$)	RMS$\|e\|$	actual excess	bound, matched error	same bound, labelled error
8	2.675	0.1070	0.0113	$1.96 \cdot 10^{-5}$	$8.27 \cdot 10^{-2}$	2.93
16	2.532	0.1044	0.0065	$9.58 \cdot 10^{-6}$	$4.37 \cdot 10^{-2}$	1.86
32	2.684	0.1122	0.0079	$2.69 \cdot 10^{-5}$	$7.07 \cdot 10^{-2}$	2.11
64	2.764	0.1136	0.0094	$1.23 \cdot 10^{-5}$	$6.80 \cdot 10^{-2}$	2.43

- The bound is at least the actual excess at every sample of every run.
- The matched error is about 1/25 of the labelled one.
- The bound is 30–45× smaller when evaluated with the matched error.
- Nothing grows with m.
- The bounds remain far above the actual excess, because the Hessian supremum is global.

Corollary B.5 along (L_σ)**.** This is checked by the test `test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`. The test asserts every hypothesis of C(iv) and measures $\sup W \le \bar{\eta}$. There, the plain B.5 bound is exceeded by more than 3×, while the leak variant with $\sigma_m b^2/4$ holds.

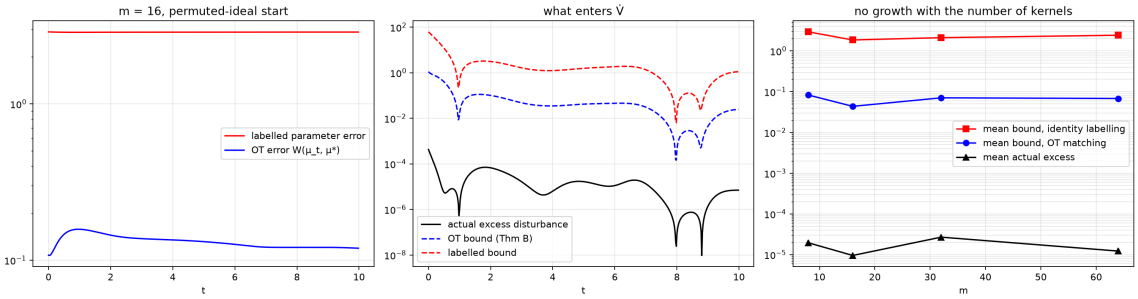


Figure 4: Theorem B, permuted-ideal start. Left: labelled parameter error and OT error $W(\mu_t, \mu^*)$ for $m = 16$. Centre: the actual excess disturbance that enters $\dot{V}$, the OT bound of Theorem B and the labelled bound, for $m = 16$. Right: the mean bound with identity labelling, the mean bound with OT matching and the mean actual excess, against m.

24.2 Theorem C: OT σ-modification

Scripts: `exp_ot_sigma_mod.py` and `exp_sigma_mod_transient.py` → `ot_sigma_mod/`.

The two starts share plant, prior, geometry and x_0. They differ only in particle labels (Table 24, Figure 5). Table 25 shows the reshuffling transient from the relabelled start (5 seeds).

Table 24: Theorem C: OT σ-modification against labelled σ-modification (oracle plant, $m = 6$, 5 seeds, $\sigma_m = 0.3$). RMS$\|e\|$ is taken over 20–40 s and reported as mean ± 95% confidence interval over seeds; W_{end} is the OT error at the end.

start	mechanism	RMS$\|e\|$ (20–40 s)	W_{end}	labelled error at end
aligned	projection	0.0051 ± 0.0010	0.2953	0.2953
aligned	labelled σ-mod.	0.0091 ± 0.0018	0.3040	0.3040
aligned	OT σ-mod.	0.0091 ± 0.0018	0.3040	0.3040
relabelled	projection	0.0051 ± 0.0010	0.2953	0.2953
relabelled	labelled σ-mod.	0.0091 ± 0.0018	0.3040	**2.0590**
relabelled	OT σ-mod.	0.0091 ± 0.0018	0.3040	0.3040

Table 25: Reshuffling transient from the relabelled start ($m = 6$, 5 seeds, $\sigma_m = 0.3$): RMS$\|e\|$ over time windows, and the particle path length.

leak	$[0, 5]$ s	$[5, 10]$ s	$[10, 20]$ s	$[20, 40]$ s	particle path length
labelled	0.0771	0.0086	0.0093	0.0091	2.2585
OT	0.0743	0.0072	0.0093	0.0091	0.3917

- The labelled leak's longer path costs only about 4% of early RMS$\|e\|$. Theorem F does not price leak-driven transport (Proposition C.5).
- The C(iv) instance in the tests satisfies every hypothesis, asserted. V_{OT} stays below the envelope $e^{-\alpha t}V(0) + \beta/\alpha$ at every sample and contracts by more than 5×.
- The C(ii) test asserts $C \le b^2 + 4\eta R$, $\sqrt{C} \le b + 2\eta$, and $C = b^2$ under both gap conditions, on 300 configurations.

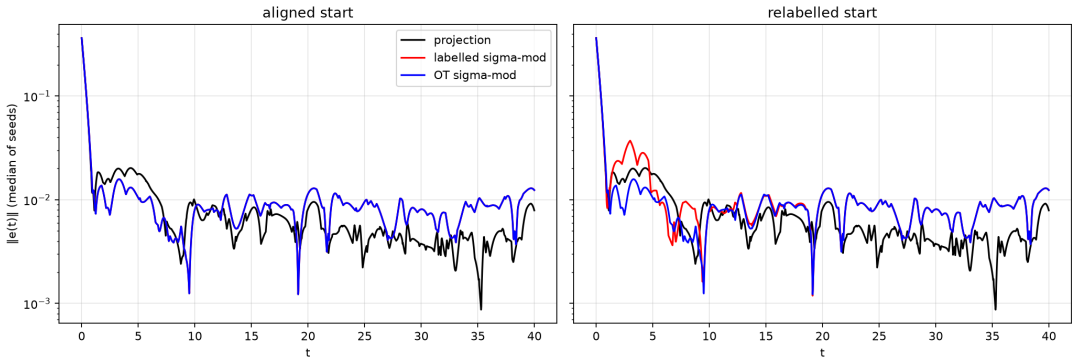


Figure 5: Theorem C: $\|e(t)\|$ (median of seeds) for projection, labelled σ-modification and OT σ-modification, from the aligned start (left) and from the relabelled start (right).

24.3 Theorem C′: the barycentric OT leak

Script: `exp_barycentric_leak.py` → `barycentric_leak/`. Table 26 shows the pure leak toward a prior with more atoms than the network, and Table 27 the closed loop with three leaks of equal gain toward three priors.

Table 26: Theorem C′: pure barycentric leak toward a 48-atom prior ($m = 6$, $n = 1$, lifted dimension 3, $G = \mathrm{diag}(2, 1, 0.5)$, $\sigma_m = 1$, $\mathrm{d}t = 0.01$, 3000 steps). The last column is the best of 20 equal-mass Lloyd runs.

seed	$W^2(\mu_0, \nu)$	$W^2(\mu_T, \nu)$	V^2_{bar} at T	Var_γ at T	W^2 non-increasing	plan switches	best Lloyd
0	2.5526	1.1642	$1.1 \cdot 10^{-26}$	1.1642	yes	3	1.1708
1	2.3032	1.2128	$4.8 \cdot 10^{-26}$	1.2128	yes	2	1.1708
2	2.5807	1.2030	$2.2 \cdot 10^{-25}$	1.2030	yes	6	1.1708

Table 27: Theorem C′: closed loop with three leaks of equal gain ($\sigma_m = 0.5$) toward three priors (representable plant, $m = 4$, $k_e = 4$): (A) the OT leak to the ideal, (B) the barycentric leak to a 32-atom cloud, (C) the labelled leak to a random 4-subset. Each entry of (A)–(C) is RMS$\|e\|$ on $[20, 40]$ s, $W(\mu_T, \mu^*)$. Here $b = W(\mu^*, \mathrm{cloud})$, b_{bar} is the centroid defect of the ideal with respect to the cloud, and "labelled" is the labelled bias to the subset.

seed	b	b_{bar}	$W(\mu^*, \mathrm{subset})$	labelled	(A)	(B)	(C)
0	0.446	0.168	0.647	1.987	0.0000, 0.000	0.0035, 0.167	0.0431, 0.614
1	0.328	0.097	1.027	1.479	0.0000, 0.000	0.0024, 0.097	0.0430, 1.016
2	0.401	0.159	0.395	1.765	0.0000, 0.000	0.0069, 0.157	0.0293, 0.352

- The pure leak stops exactly where $W^2 = \mathrm{Var}_\gamma$ (every particle at the barycentre of its cell; $V^2_{\mathrm{bar}} \le 2.2 \cdot 10^{-25}$): an equal-mass centroidal configuration of the prior. Like Lloyd's algorithm it finds a local minimiser of the quantisation energy, not always the global one (compare the last two columns of Table 26).
- With $m_0 = m$ the same leak gives the exact decay of Theorem C(i): the maximal relative deviation of $W(\mu_t, \mu^0)$ from $(1 - \sigma_m \, \mathrm{d}t)^k W_0$ over 500 Euler steps is $5.1 \cdot 10^{-16}$.
- In the closed loop, the bias of the barycentric leak is governed by b_{bar}, the distance of the ideal from the centroids of the prior's cells: $W(\mu_T, \mu^*) = 0.167, 0.097, 0.157$ against $b_{\mathrm{bar}} = 0.168, 0.097, 0.159$, not by $b = W(\mu^*, \nu) = 0.45, 0.33, 0.40$ (Theorem C′(ii)–(iii)).

The labelled leak to the subset ends at $W(\mu_T, \mu^*) \approx W(\mu^*, \text{subset})$. A rich prior whose centroidal quantisation is close to the ideal is a better anchor than a crude prior with the right number of atoms.

- Tests (`tests/test_barycentric.py`): the plan reduces to the assignment for $m_0 = m$ and $W^2 = V_{\text{bar}}^2 + \text{Var}_\gamma$; the pure leak dissipates at every step with $D^+ Z \le -\sigma_m V_{\text{bar}}^2$ and ends at a Lloyd fixed point; $\sqrt{C_{\text{bar}}} \le b + 2\eta$ on 90 configurations, with $C_{\text{bar}} = b_{\text{bar}}^2$ at small η; the Lyapunov inequality (iii) along a closed loop with a 24-atom prior.

24.4 Theorem D: convergence in m

Script: `exp_mean_field_m.py` → `mean_field_m/`. Table 28 and Figure 6 show the results.

Table 28: Theorem D: mean-field consistency in m (Van der Pol plant, mean-field gains, smooth initial measure quantised at m points). The ratio is $W(\mu_T^m, \mu_T^{2m}) / W(\mu_0^m, \mu_0^{2m})$.

m	$\sup_t \|e^m - e^{2m}\|$	$W(\mu_0^m, \mu_0^{2m})$	$W(\mu_T^m, \mu_T^{2m})$	ratio
16 vs 32	0.0046	0.2179	0.3658	1.68
32 vs 64	0.0009	0.1091	0.1788	1.64
64 vs 128	0.0001	0.0545	0.0920	1.69
128 vs 256	0.0000	0.0273	0.0459	1.68

This is a consistency check in m, not a check of the constant.

- The guaranteed factor $e^{\Lambda T}$ is at least e^{496} on this box.
- The Van der Pol plant fails the tube condition of Theorem D(b), so the run is covered by the localised D(d).

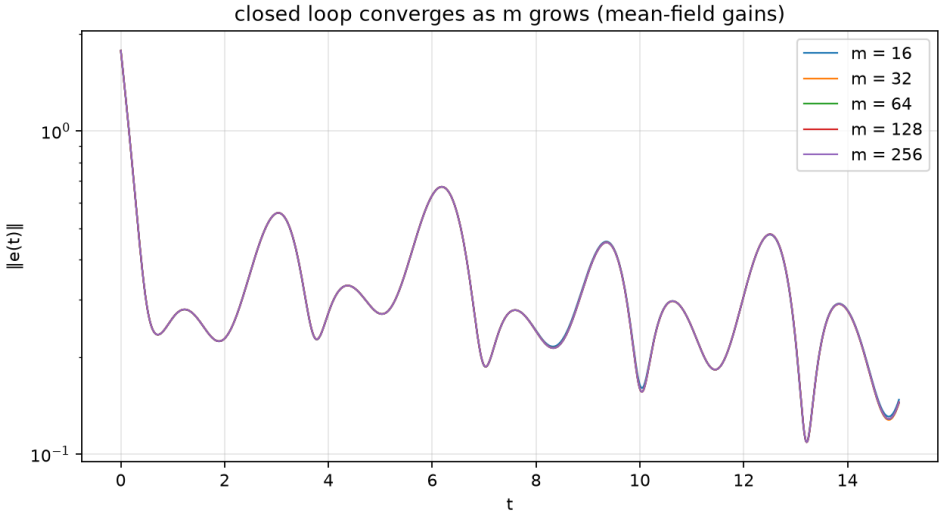


Figure 6: Theorem D: tracking error $\|e(t)\|$ of the closed loop (Van der Pol plant, mean-field gains) for $m = 16$, 32, 64, 128 and 256.

24.5 Theorem D′: propagation of chaos for the dithered loop

Script: `exp_chaos.py` → `chaos/` (raw per-seed values in `raw_sup_D.npz`). The m-particle loop ($n = 1$, $k_e = 10$, $\sigma = 0.02$, $G = \text{diag}(4, 1, 1)$, $T = 20\,\text{s}$, $dt = 0.001$; Euler–Maruyama for the particles and RK4 for the plant) is coupled synchronously to an $M = 4096$ reference that shares its

first m particles and Brownian motions, and $\sup_t D_m$ with $D_m := \|x^m - x^M\| + \frac{1}{m}\sum_i \|\vartheta_i^m - \vartheta_i^M\|$ is averaged over 24 seeds (Table 29).

Table 29: Theorem D′: synchronous coupling of the m-particle dithered loop to an $M = 4096$ reference, 24 seeds. Slopes are log-log fits, with bootstrap standard errors over seeds.

	$\tau = 0$	$\tau = 2 \cdot 10^{-3}$
mean $\sup_t D_m$, $m = 16 \to 1024$	$1.49 \cdot 10^{-1} \to 2.17 \cdot 10^{-2}$	$1.52 \cdot 10^{-1} \to 2.32 \cdot 10^{-2}$
slope of the mean against m ($m \le 256$)	-0.47 ± 0.06	-0.46 ± 0.05
slope of the median	-0.46	-0.45
slope against $(1/m - 1/M)^{1/2}$ (predicted 1)	0.94 ± 0.08	0.92 ± 0.08

- The $m^{-1/2}$ rate of Theorem D′(b) holds within one standard error at both temperatures; the shared-particle coupling predicts the mean $\propto (1/m - 1/M)^{1/2}$, which the data follow up to $m = M/4$.
- Single seeds differ by more than an order of magnitude at fixed m; a three-seed version of the experiment gave slopes -0.25 and -0.21 and was withdrawn (review round 5). The constant of the theorem ($TK_\Phi e^{\Lambda T}$) is not checked.
- There is no test: the experiment is the check.
- Theorem D″ (`exp_chaos_uniform.py` → `chaos/uniform_table.md`; $n = 1$, $k_e = 25$, $G = 0.05 \cdot \mathrm{diag}(4, 1, 1)$, a two-bump plant that is not odd, $M = 2048$): exact Lipschitz constants $L_x = L_\vartheta = 8.087$, $L_2 = L_{x\vartheta} = 44.4$ and the a-priori tube $E_X = 0.228$ give $\sigma_{\mathrm{cert}} = 4.04$, against Proposition K's critical leak $\sigma_K = 0.00144$ of the collapsed critical point (ratio 2800). At $1.2\sigma_{\mathrm{cert}}$ with dither $\tau = 0.02$ (80 s, 8 seeds) the coupled distance is $6.7 \cdot 10^{-4} \to 7.9 \cdot 10^{-4}$ ($m = 16$) and $1.6 \cdot 10^{-4} \to 1.9 \cdot 10^{-4}$ ($m = 256$) between the first and the last window, slope -0.51 ± 0.02, bound 0.55 and 0.14; the network stays at its prior (fit error 1.000, rms spread 0.033 = the dither cloud). Without dither the reference collapses (spread 10^{-14}) and the distance is zero; at $\sigma_{\mathrm{cert}}/8$ it decays to 10^{-11}. In the learning regime ($\sigma_K/3$, 1000 s, 6 seeds) the network stays spread (rms 1.53) and fits the plant (fit error 0.632), and the distance follows $m^{-1/2}$ (slope -0.62 ± 0.16) but grows threefold between the windows $(100, 200]$ and $(500, 1000]$: no empirical uniformity there (the first draft's "spread regime" run was a slow collapse; review round 6).

24.6 Theorem E: label-free convergence under excitation

Script: `exp_pe_convergence.py` → `pe_convergence/`. Table 30 and Figure 7 show the results.

Table 30: Theorem E: tangential excitation and label-free parameter convergence ($m = 3$, $n = 2$, permuted start, $T = 200$ s). $\lambda_{\min}$ is that of the excitation Gramian.

reference	seed	$\lambda_{\min}$, first 60 s window	min over sliding 60 s windows	$W(T)/W(0)$	fitted rate of W, $t \ge 100$ s (1/s)	labelled(T) / labelled(0)
rich (6 tones)	0	$8.60 \cdot 10^{-3}$	$4.12 \cdot 10^{-3}$	0.544	$2.31 \cdot 10^{-3}$	0.992
rich (6 tones)	1	$5.52 \cdot 10^{-3}$	$1.02 \cdot 10^{-3}$	0.409	$2.32 \cdot 10^{-3}$	0.962
rich (6 tones)	2	$1.08 \cdot 10^{-3}$	$5.91 \cdot 10^{-4}$	0.369	$3.07 \cdot 10^{-3}$	1.007
single sine	0	$2.54 \cdot 10^{-2}$	$2.41 \cdot 10^{-2}$	0.274	$4.05 \cdot 10^{-3}$	0.988
single sine	1	$2.96 \cdot 10^{-2}$	$2.71 \cdot 10^{-2}$	0.314	$4.26 \cdot 10^{-3}$	0.967
single sine	2	$6.74 \cdot 10^{-3}$	$6.61 \cdot 10^{-3}$	0.226	$2.91 \cdot 10^{-3}$	1.008

- The matched error decays in all six runs. The labelled error stays within 4% of its initial value.
- The rank correlation between the uniform excitation level and the fitted rate is 0.60 over the six runs. The PE level is a hypothesis of Theorem E, not a rate predictor (Remark E.1).
- The proof's linearisation is checked exactly: the Jacobian equals $A(t)$, and $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$.

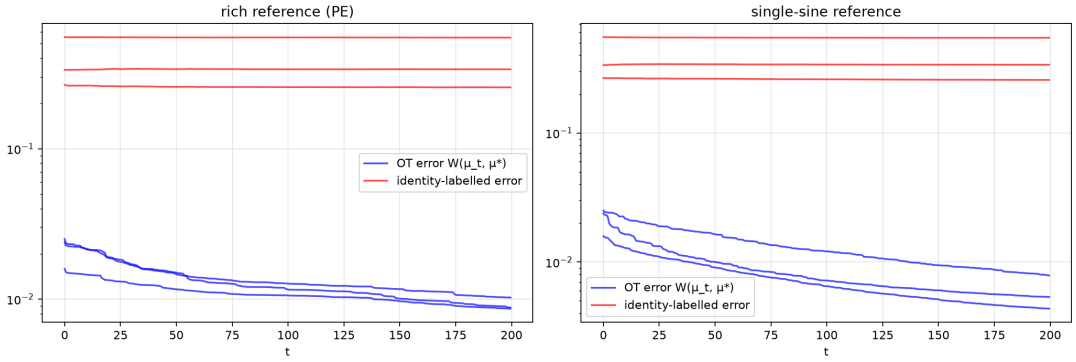


Figure 7: Theorem E: OT error $W(\mu_t, \mu^*)$ and identity-labelled error for the three seeds, with the rich reference (left) and the single-sine reference (right).

24.7 Lemma E.0 and Theorem E with explicit constants

Script: `exp_explicit_rate.py` → `explicit_rate/`. The explicit rate of Lemma E.0 is compared with the exact Floquet rate of the linearised loop (Tables 31 and 32). It lies below the exact rate in every row; the ordering is numerical (Ψ_M, Ψ_D are grid maxima, α a converged quadrature and ε a grid optimum), with margins of at least 17×.

Table 31: Lemma E.0: the explicit rate against the exact Floquet rate of the linearised loop (`explicit_rate/table.md`).

instance	regime	exact / explicit
Remark E.1 ($m = n = 1$, $G = I$, $k_e = 3$), $\omega = 1, \dots, 64$	$\Psi_M^2/k_e \approx 1$	$954 \to 17.6$ (decreasing in ω)
the ideal of the Theorem E tests ($m = 3$, $n = 2$, $k_e = 6$), two-tone reference of period 20π, $G = g \cdot \mathrm{diag}(2, 2, 1)$, $g = 30, \dots, 0.01$	fast → slow	$6.56 \cdot 10^{10} \to 4.22 \cdot 10^5$ (optimised ε; $8.74 \cdot 10^{10} \to 5.63 \cdot 10^5$ with the closed-form ε)

- In Remark E.1's instance the exact rate falls like ω^{-2} (log-log slope -1.97 over $\omega \geq 4$); the explicit rate approaches that law (slope -1.56 over $\omega \geq 4$, -1.92 over the last octave $\omega = 32 \to 64$). Lemma E.0(iii) derives the ω^{-2} law from $\Psi_D \propto \omega$.
- In the slow-adaptation limit ($T\Psi_M^2 \ll k_e$) the ratio decreases towards a constant of order $\kappa_\Psi(1 + \Psi_D/(k_e\Psi_M))^2$ for the optimised ε ($3\kappa_\Psi(\cdots)^2$ for the closed-form bound (ii)), with κ_Ψ the condition number of the averaged Gramian ($1.27 \cdot 10^5$ for this ideal; formula values $4.9 \cdot 10^5$ and $1.47 \cdot 10^6$; measured $4.22 \cdot 10^5$ with the optimised ε and $5.63 \cdot 10^5$ with the closed-form ε at $g = 0.01$). On this instance the exact rate is 0.66 of $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k_e$, because $k_e = 6$ is not large against the reference's bandwidth and the $T^2\Psi_M^3$ term is still

Table 32: Lemma E.0 in Remark E.1's instance, $x_m = 1.5 \sin \omega t$ ($m = n = 1$, $G = I$, $k_e = 3$): PE window T, level $\alpha(T)$, Ψ_D ($\Psi_M = 1.78$ throughout), the exact Floquet rate and the explicit rate with the closed-form and with the optimised ε.

ω	T	$\alpha(T)$	Ψ_D	exact rate	explicit, closed-form ε	explicit, optimised ε
1	6.28	1.13	3.69	$4.660 \cdot 10^{-2}$	$3.688 \cdot 10^{-5}$	$4.886 \cdot 10^{-5}$
2	3.14	0.566	7.38	$2.158 \cdot 10^{-2}$	$7.858 \cdot 10^{-5}$	$1.040 \cdot 10^{-4}$
4	1.57	0.283	14.8	$6.834 \cdot 10^{-3}$	$8.696 \cdot 10^{-5}$	$1.152 \cdot 10^{-4}$
8	0.785	0.142	29.5	$1.831 \cdot 10^{-3}$	$4.736 \cdot 10^{-5}$	$6.301 \cdot 10^{-5}$
16	0.393	0.0708	59	$4.662 \cdot 10^{-4}$	$1.650 \cdot 10^{-5}$	$2.199 \cdot 10^{-5}$
32	0.196	0.0354	118	$1.171 \cdot 10^{-4}$	$4.720 \cdot 10^{-6}$	$6.286 \cdot 10^{-6}$
64	0.0982	0.0177	236	$2.931 \cdot 10^{-5}$	$1.249 \cdot 10^{-6}$	$1.665 \cdot 10^{-6}$

13% of c_1; the PE level is evaluated with 800 Gauss–Legendre nodes, converged over the 20π window. The exact rate pays for ill-conditioning once, the bound twice (Remark E.0a).

- Lemma E.0′ (the slow regime; table 4 of the same file): on the $g = 0.01$ ideal ($\kappa(\bar{M}) = 7.8 \cdot 10^3$; peak factors $\rho = 69.9$, $\rho_D = 3.7 \cdot 10^3$, $\rho_{DD} = 3.7 \cdot 10^5$; threshold $k_0 = 226$ for $\delta = 0.2$, 321 for $\delta = 0.1$) the guaranteed rate is within a factor 1.26 ($\delta = 0.2$) and 1.12 ($\delta = 0.1$) of the exact Floquet rate at $k = 400$ and 600, where Lemma E.0's optimised rate is $1.3 \cdot 10^5$ times smaller; the certificate $\sup \dot{V}/V \le -\gamma$ holds on the grid wherever the conditions hold; exact/$(\lambda_{\min}/k) = 0.9885$, 0.9971, 0.9993, 0.9997, 0.9998, 0.9999 at $k = 50, \dots, 600$ (the k^{-2} law of $N_2 = 0$).

- Corollary E.0′.2 (Theorem E in the slow regime; table 5): on the single-kernel instance at $k_e = 40$ the nonlinear rate from Lemma E.0′'s V is $4.53 \cdot 10^{-4}$ ($\delta = 0.2$) against $2.63 \cdot 10^{-5}$ from Lemma E.0 and an exact linear rate of $1.35 \cdot 10^{-3}$; the explicit ball radius is $1.53 \cdot 10^{-6}$ against $6.15 \cdot 10^{-6}$ (the anisotropy of the quadratic form), the overshoot 11.8 against 1.43.

- Tests (`tests/test_excitation.py`): the pointwise inequalities of Lemma E.0 on random states for four regressors, the envelope along trajectories, the ordering explicit $\le$ exact, the ω^{-2} law, the no-growth of the ratio as the gains shrink, and the explicit nonlinear envelope of Theorem E on the implemented closed loop (single-kernel instance with $G = 0.3\,I$, where $r_0 \approx 10^{-5}$ in the w-coordinates and the explicit nonlinear rate is $3.7 \cdot 10^{-5}$: the inequality holds at every sample over 8000 s, about 0.3 e-folds of the explicit envelope, while the state itself decays by far more). Lemma E.0′: the certificate and the sharpness on $\Psi = (\sin t, \cos t)$ ($k = 20$, 50, 200; ratio $\to 1/(1-\delta)$) and on the slow-adaptation ideal at $k = 400$ (ratio ≤ 1.5; more than 10^4 above Lemma E.0's rate). Corollary E.0′.2: the explicit envelope on the implemented single-kernel loop at $k_e = 40$ over 1.5 e-folds (d$t = 0.005$, below the fixed-step discretisation floor).

- Corollary E.0′.3 (table 6): the explicit bound on the stationary parameter spread of the dithered (LTV), with $\Sigma = (2 \cdot 10^{-3})^{1/2} I$, against the exact periodic stationary covariance: on $\Psi = (\sin t, \cos t)$ the ratio bound/exact is 2.69, 1.58, 1.31 ($\delta = 0.2$) and 2.52, 1.41, 1.17 ($\delta = 0.1$) at $k = 20$, 50, 200, falling toward $1/(1-\delta)$; on the Theorem E ideal ($\kappa = 7.8 \cdot 10^3$) it is 15.4–17.5, the price of $q = 15$ directions charged at the worst excitation; the exact value equals the averaged Langevin covariance $k \cdot \operatorname{tr} X$ within $3 \cdot 10^{-4}$ (ideal) and $1.25/k^2$ (sinusoid), and $\mathbb{E}\|e\|^2 = \|\Sigma\|_F^2/(2k) = \operatorname{tr}(\bar{M}X)/k$ to five digits.

- Lemma E.0″ (`exp_pe_general.py` → `explicit_rate/general_pe.md`): the exponential-average construction on two non-periodic regressors. Quasi-periodic $\Psi = (\sin t, \cos \sqrt{2}\, t)$: $\lambda_T = 0.297$, 0.391, 0.444 for windows 2π, 4π, 8π; conditions from $k = 20$; exact/guaranteed ($\delta = 0.2$ / 0.1) 2.49/2.13 $\to$ 1.63/1.45 at $T = 4\pi$ and 2.10/1.81 $\to$ 1.43/1.27 at $T = 8\pi$

from $k = 20$ to 200; the exact rate is $0.5/k$ to 0.5%. Chaotic $\Psi = (x_1/20, (x_3 - 25)/20)$ along a Lorenz orbit traversed five times slower than standard time: $\lambda_T = 0.015$, 0.024, 0.051 for windows 5, 10, 20 against an effective excitation 0.15 (exact rates $0.00759/k \cdot 20$ ...); conditions from $k = 100$; exact/guaranteed 5.0/4.3 ($k = 100$) $\to$ 4.2/3.7 ($k = 200$) at $T = 20$, dominated by the window loss $\lambda_\infty/\lambda_T \approx 3$; the certificate holds at every admissible row.

24.8 Theorem F: transport cost

Theorem F is checked by tests.

- Along (L), the particle path length (which is at least the Wasserstein length) stays below $\lambda_G^{1/2} L_1 \int \|e\|$ at every sample.
- Along (L_σ) with the OT leak, $W(\mu_{t_1}, \mu_{t_0}) \le \int (\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0))$ on a grid of pairs.
- Lemma P.1 holds on 2000 adversarial states, and a tight instance of F(i) reaches ratio ≥ 0.99.

24.9 Proposition G: slow geometry

Scripts: `exp_variable_projection.py` and `exp_slow_manifold_slope.py`, both $\to$ `variable_projection/`.

In Table 33, ranges are along the run (5 samples), and "a / b" means first-quarter / last-quarter medians. Velocities are centred differences of period means.

Table 33: Proposition G: the two configurations, `violating` ($m = 4$, $k_e = 20$, $g_a = 2$, $\delta = 0.1$, $T = 800$ s) and `satisfying` ($m = 2$, $k_e = 20$, $g_a = 13$, $\delta = 0.004$, $T = 3000$ s). In the rows for L^* and $\bar{L}$, "$x \to y$" is first sample $\to$ last sample.

quantity	violating ($m = 4$)	satisfying ($m = 2$)
(H2a) ω_{ref} / joint fastest	15.3–25.6	6.53–7.27
(H2b) slowest weight / geometry	0.0993–0.231	78.8–87.2
(H2c) S	5.44–25	0.00601–0.00602
G(a) direct check (cosine, G^{-1} metric)	0.9994 / 0.9985	0.9848 / 0.9624 (geometry block 0.9992 / 0.9982)
G(b) cosine	-0.118 / 0.385	0.999 / 0.999
G(b) size ratio	4.207 / 0.472	0.969 / 0.970
L^*	**increases** ($3.8136 \cdot 10^{-3} \to 5.8866 \cdot 10^{-3}$)	decreases ($1.1987 \cdot 10^{-2} \to 4.1073 \cdot 10^{-3}$)
$\bar{L}$ at period-mean weights	non-increasing (0.21892 $\to$ 0.0080127)	non-increasing

Slope counterexample. Table 34 (`table_slope.md`). Here $m = 1$, $k_e = 20$, gains (g_a, δ, δ), and (H2b) holds in every row. The closed loop and the joint flow start on the slow manifold of (b′).

- (a) / (b′) isolates the $O(1/R)$ error of the induced-metric flow.
- closed / (a) isolates the averaging error, which shrinks with the gains, plus a 4.4% $O(1/k_e)$ error-channel factor.

Table 34: Proposition G(b): closed loop against the averaged predictions. "closed / (b)" compares the closed loop with plain variable projection (VarPro) G(b), and "closed / (b′)" with the induced-metric flow G(b′); size is the ratio of displacement norms.

case	(H2b)	S	closed / (b)		closed / (b′)		(a) /	closed /
	ratio		size	cosine	size	cosine	(b′)	(a)
$\delta = 0.47$, $g_a = 20$	81–178	2–2.81	0.422	0.9303†	1.121	0.9986	1.015	1.104
$\delta = 0.47/4$, $g_a = 5$ (all gains ÷4)	81–178	2–2.81	0.401	0.9156	1.065	0.9999	1.015	1.049
$\delta = 0.047$, $g_a = 20$	654–1779	0.2–0.319	0.850	0.9986	1.047	1.0000	1.003	1.044
$\delta = 0.0047$, $g_a = 20$	13162–17790	0.02–0.0229	1.023	1.0000	1.044	1.0000	1.000	1.044

†The centre moves the opposite way.

What the tables show:

- G(b) holds in the satisfying configuration, where (H2b) and $S \ll 1$ both hold. The violating configuration fails both.
- The slope rows show that (H2b) alone is not sufficient.
- The induced-metric flow G(b′) tracks the closed loop at every S.
- In the satisfying configuration, the G(a) weight block is dominated by the $O(1/k_e)$ shift of the weight equilibrium: cosine -0.21 / 0.50, carrying 92–95% of the residual. The geometry block matches.

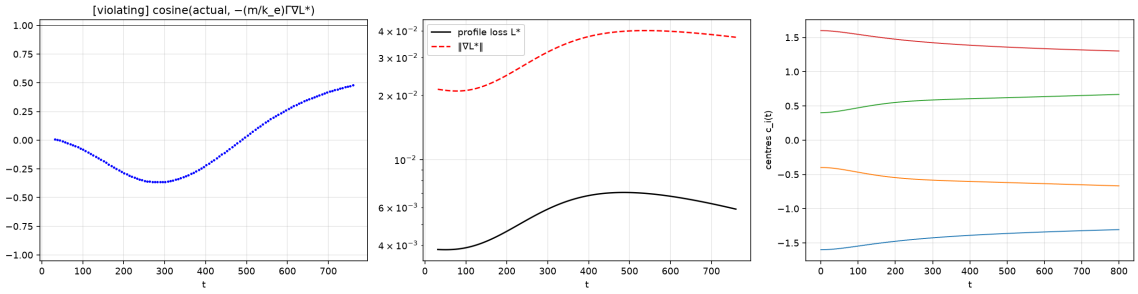


Figure 8: Proposition G, `violating` configuration. Left: the G(b) cosine between the actual geometry velocity and the VarPro direction $-(m\delta/k_e)\Gamma\nabla L^*$, along the run. Centre: the profile loss L^* and $\|\nabla L^*\|$. Right: the centres $c_i(t)$.

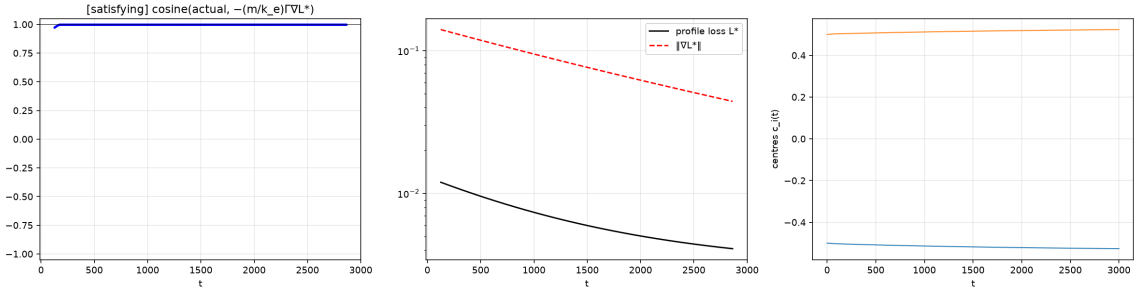


Figure 9: Proposition G, `satisfying` configuration; panels as in Figure 8.

24.10 Theorem H: Langevin adaptive control

Scripts: `exp_langevin_global.py` → `langevin_global/`, `exp_langevin_tradeoff.py` → `langevin_tradeoff/`, and `exp_gibbs_reference.py` → `langevin_global/gibbs_reference.md`.

The setting is the real closed loop with $m = 32$, a scalar plant and a periodic reference.

- The unregularised offline optimum of the occupation loss is about $7.6 \cdot 10^{-6}$ (best of 5 L-BFGS restarts, an upper bound). One run gave $7.0 \cdot 10^{-6}$, because of floating-point summation order.
- The regularised optimum $\bar{L}(\mu_\tau^*)$ is 0.11118 at $(\sigma, \tau) = (0.02, 0.004)$ and 0.00850 at $(0.002, 5 \cdot 10^{-4})$, grid-converged (`exp_gibbs_reference.py` → `langevin_global/gibbs_reference.md`).
- In the scaling $\sigma = 0.2/k_e$, $\tau = 0.04/k_e$ of Conjecture H.6, $\bar{L}(\mu_\tau^*) = 0.11118$ for every k_e. The exactly averaged stationary state gives $\bar{L} = 0.11415$, 0.11148 and 0.11101 at $k_e = 10$, 20 and 40. The $L^2(\rho)$ distance between the two learned functions is $6.81 \cdot 10^{-3}$, $1.43 \cdot 10^{-3}$ and $9.7 \cdot 10^{-4}$.

Speed–bias trade-off. Table 35 and Figure 10 ($\sigma = 0.002$, 2 seeds, `langevin_tradeoff/table.md`).

Table 35: Theorem H: speed–bias trade-off of parameter dither ($m = 32$, $\sigma = 0.002$, $k_e = 10$, $T = 1500\,\mathrm{s}$, 2 seeds). The escape time is the first sample at which the collapsed run's loss is at most twice the spread run's final loss (same seed); per-seed values are listed, and ∞ means not within the horizon. The losses are the final values of $\bar{L}$.

τ	escape of the collapsed start (s)	loss, collapsed	loss, spread
0	∞, ∞	0.166	0.00696
0.0005	∞, 1050	0.0249	0.0109
0.001	1480, 1005	0.0216	0.0148
0.002	1115, 800	0.0288	0.0233
0.004	970, 430	0.0392	0.0362
0.008	390, 295	0.0603	0.0653
annealed (hot start)	990, 530	0.0317	0.0320

- **Fit.** The seed-averaged escape time fits $A + B \log(1/\tau)$ with an RMS relative residual of 0.045, against 0.253 for an Arrhenius fit. On the eight per-seed values both laws fit poorly (0.274 and 0.330). The data do not identify the escape mechanism, and the "saddle" reading of an earlier version of this analysis is withdrawn (Remark H.2).
- **Caveats.** The threshold grows with τ, and the censored $\tau = 5 \cdot 10^{-4}$ point is excluded from the fit.
- **Guaranteed rate.** The Holley–Stroock [35] rate $2\tau\rho_\tau$ is at least $e^{-8401}\,\mathrm{s}^{-1}$ at $(0.002, 5 \cdot 10^{-4})$ and at least $e^{-1053}\,\mathrm{s}^{-1}$ at $(0.02, 0.004)$.

Two fixed settings. Three seeds (`langevin_global/table_*.md`; Figure 11).

- $\sigma = 0.02$**,** $\tau = 0.004$**.** Both starts equalise: the loss ratio is 1.03, and the final measures are 4.2× closer in W than without dither. The common loss is about 0.11, which matches the grid value $\bar{L}(\mu_\tau^*) = 0.111$. It is dominated by the regularisation bias.

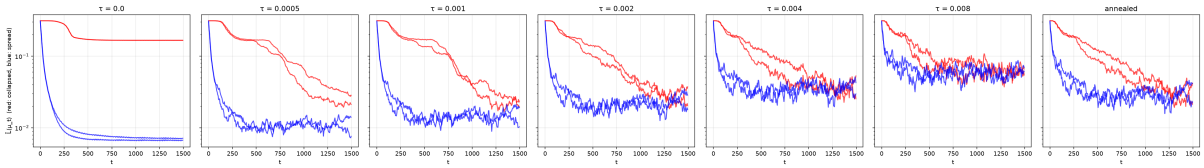
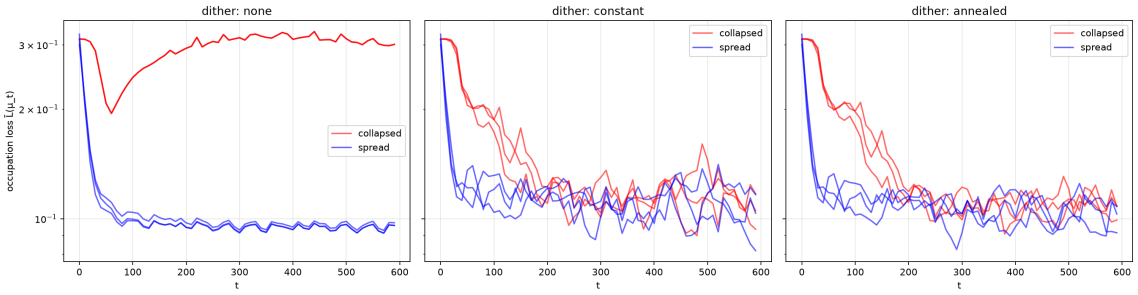
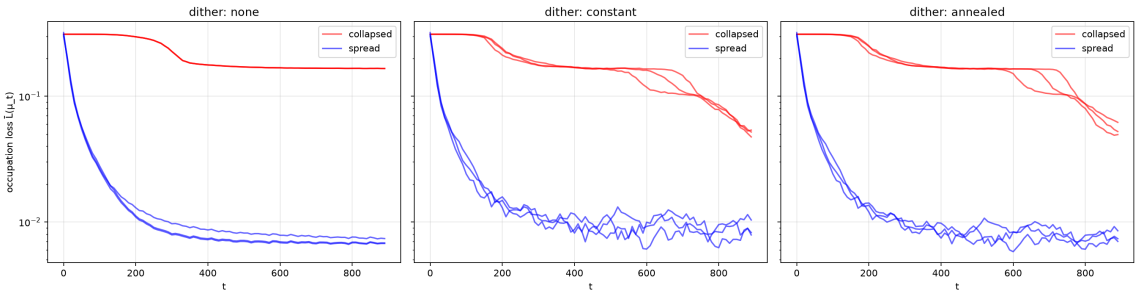


Figure 10: Theorem H, speed–bias trade-off: $\bar{L}(\mu_t)$ of the collapsed start (red) and of the spread start (blue), 2 seeds, for $\tau = 0$, 0.0005, 0.001, 0.002, 0.004, 0.008 and the annealed (hot-start) schedule.

- $\sigma = 0.002$, $\tau = 5 \cdot 10^{-4}$. Dither frees the collapsed start ($0.166 \to 0.0480$), but the starts do not equalise within 900 s (loss ratio 5.23).



(a) $\sigma = 0.02$, $\tau = 0.004$



(b) $\sigma = 0.002$, $\tau = 5 \cdot 10^{-4}$

Figure 11: Theorem H, global kernel placement ($m = 32$, scalar plant, $k_e = 10$, 3 seeds): the occupation loss $\bar{L}(\mu_t)$ of the collapsed and spread starts without dither, with constant dither and with annealed dither.

24.11 Remark H.0 and Proposition H.7: the exactly averaged closed loop

These are checked by tests. Table 36 gives the relative error of the frozen-μ averaged drift against $(\mathfrak{A})$, and Table 37 the structure of the first correction (`test_averaging.py`).

Table 36: Remark H.0: relative error of the frozen-μ averaged drift against $(\mathfrak{A})$.

instance	$k_e = 10$	20	40	80	$k_e \times$ error
$n = 1$ (`test_langevin.py`)	4.2%	2.0%	0.91%	0.48%	≈ 0.4
$n = 2$, ellipse (`test_averaging.py`)	43%	21%	10%	5.1%	≈ 4.2

Table 37: Proposition H.7: structure of the first correction (`test_averaging.py`).

check	$n = 1$	$n = 2$ (ellipse)
expansion error of $k_e^2(P_\mu - P_0)$ against $U_{\text{even}} + U_{\text{odd}}$, $k_e = 40 \to 80$	$0.54 \to 0.25$	$0.038 \to 0.019$
circulation kernel K	$\equiv 0$ (asserted at 10^{-12})	antisymmetric; equals the enclosed-area integral to $7 \cdot 10^{-15}$ (asserted at 10^{-6})
orbit reversal (same ρ)	$U_{\text{odd}} \equiv 0$	U_{even} unchanged, U_{odd} flipped, to $2 \cdot 10^{-15}$ (asserted at 10^{-10})
$k_e^2 \cdot$ cyclic sum of the antisymmetric second variation, exact ($k_e = 80$) against predicted (asserted at 3%; the full matrix at 5%)	-0.0711 against -0.0704	-0.0471 against -0.0464

24.12 Remark H.7a: the orientation of the reference orbit

Script: `exp_orbit_orientation.py` → `orbit_orientation/`.

The setup is deterministic: $n = 2$, $m = 6$, $G = \text{diag}(4, 4, 1, 1, 1)$, $\sigma = 2/k_e$, no dither. The same elliptic reference is traversed both ways, so the occupation measure is the same, and so is the initial configuration. The runs converged to machine precision: the residual drift over the last 10 periods is about 10^{-16} (between $1.0 \cdot 10^{-16}$ and $5.0 \cdot 10^{-16}$). The six kernels converge to a common configuration (kernel spread 1.05 at the start, at most $5.6 \cdot 10^{-17}$ at the end of every run), so the learned network is effectively a single kernel and the test probes the effect at a collapsed equilibrium. Predictions are linear responses at the (+) equilibrium (Table 38).

Table 38: Remark H.7a: orientation dependence of the learned kernels in the real closed loop. Size ratios are measured / predicted.

		frozen-measure prediction (H.7)		full prediction (H.7 + oscillation term)	
k_e	measured $\|\Theta_+ - \Theta_-\|$	size ratio	cosine	size ratio	cosine
10	$5.729 \cdot 10^{-3}$	2.670	0.7577	0.937	0.9314
20	$2.843 \cdot 10^{-3}$	2.414	0.8302	0.965	0.9809
40	$1.403 \cdot 10^{-3}$	2.294	0.8631	0.969	0.9944

- The vector error $\|\text{measured} - \text{predicted}\|/\|\text{predicted}\|$ of the full prediction is 0.364, 0.195 and 0.109 at $k_e = 10$, 20 and 40.
- The measured difference scales like $k_e^{-1.01}$, against a predicted -1.
- The full first-order theory accounts for it. The frozen-measure terms alone account for less than half.
- A first attempt with $\sigma = 0.5/k_e$ and half the horizon had not converged (residual drift about a third of the difference). It was discarded and rerun.

24.13 Theorem I: reciprocity at order k_e^{-2}

There is no new experiment. Every statement of Theorem I is checked against the generic second-variation code of `src/core/averaging.py` by the tests of `tests/test_nonreciprocity.py` (Table 39).

Table 39: Theorem I: the checks of `tests/test_nonreciprocity.py`.

check	result
$A(\vartheta,\vartheta') = a^\top B(\xi,\xi')a'$ against the antisymmetric part of $S_{\text{even}} + K$, $n = 1$ and 2	agreement to 10^{-10}
$A(\vartheta,\vartheta_z) = 0$ for the zero feature; the cyclic sum over $(\vartheta_0,\vartheta_1,\vartheta_z)$ equals $A(\vartheta_0,\vartheta_1)$	10^{-14}
$n = 1$, $f := \hat{f}_\mu$ (perfect fit on the orbit)	$\max\|A\| < 10^{-13}$ over 20 random pairs
$n = 1$, non-zero residual	$\|A\| > 10^{-3}$ found; equals the closed form $-aa' \frac{c-c'}{s^2} \int r\varphi_c\varphi_{c'}\,\mathrm{d}\rho$ to 10^{-12}
$\kappa(\xi,\xi')$ against its closed form for $n = 2$: the Gaussian factor $-\frac{1}{2s^2}e^{-\lvert c-c'\rvert^2/(4s^2)}$ times $(c-c')\cdot F_s$ at the midpoint of c and c'	10^{-10}
$F_2((0.5,0)) > 0$ on the ellipse; $\kappa \equiv 0$ on a segment traversed back and forth	$F_2 > 10^{-3}$; $\|\kappa\| < 10^{-12}$
$n = 2$ with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ (residual and its Jacobian vanish on the orbit): $B = 2\kappa I$	10^{-13}; $A \neq 0$ on the ellipse, $A \equiv 0$ ($< 10^{-13}$) on the segment
$n = 2$ with $f = \hat{f}_\mu + he_1$, h zero on the ellipse (residual zero on the orbit, Jacobian not symmetric): $B - 2\kappa I = -\int \varphi\varphi'\partial_2 h\,\mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$	10^{-12}, and not small (the round-5 correction of Theorem I(iv))
positivity witness $f \equiv 0$, $\mu = \delta_{\bar{\vartheta}}$	integral > 0; the generic code equals the closed form to 10^{-12}

24.14 Theorem J: periodic orbits of the finite-m closed loop

Script: `exp_periodic_orbit.py` → `periodic_orbit/`. Newton shooting on the Poincaré map finds the exact periodic orbit of the real closed loop near the critical point Θ^* of J_m (residual $< 10^{-11}$); its period mean $\bar{\Theta}_p$ and its monodromy matrix are compared with the averaged system (Table 40). Instance A has $n = 2$, $m = 3$, $\sigma_0 = 0.5$, $G = \mathrm{diag}(4,4,1,1,1)$ and a collapsed minimiser ($J_m(\Theta^*) = 0.762775$, $\lambda_{\min}(\nabla^2 J_m) = 0.0148$); instance B has $n = 1$, $m = 2$, $\sigma_0 = 0.05$, $G = \mathrm{diag}(2,1,1)$, two distinct kernels ($J_m(\Theta^*) = 0.258249$, $\lambda_{\min}(\nabla^2 J_m) = 0.0661$) and a fast sweep. Distances and cosines are in the labelled G^{-1} metric.

Table 40: Theorem J: the exact periodic orbit against the averaged system. "vec. error" is the vector error of the first-order location $\Theta^* - J^{-1}(b_{\text{frozen}} + b_{\text{osc}})$ of Proposition H.7 and Remark H.7a; "frozen only" omits b_{osc} (in instance B, $b_{\text{osc}} = 0$ by symmetry). "Floquet" is the maximal relative error of the parameter Floquet exponents against $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})$.

	instance A (collapsed minimiser)				instance B (two distinct kernels)		
k_e	$\|\bar{\Theta}_p - \Theta^*\|$	vec. error	frozen only	Floquet	$\|\bar{\Theta}_p - \Theta^*\|$	vec. error	Floquet
10	$4.09 \cdot 10^{-2}$	0.169	0.398	$8.31 \cdot 10^{-2}$	$6.44 \cdot 10^{-2}$	1.995	$9.48 \cdot 10^{-2}$
20	$1.91 \cdot 10^{-2}$	0.083	0.360	$7.34 \cdot 10^{-2}$	$2.22 \cdot 10^{-2}$	1.078	$3.21 \cdot 10^{-2}$
40	$9.22 \cdot 10^{-3}$	0.041	0.348	$4.50 \cdot 10^{-2}$	$8.37 \cdot 10^{-3}$	0.551	$1.16 \cdot 10^{-2}$
80	$4.53 \cdot 10^{-3}$	0.020	0.343	$2.46 \cdot 10^{-2}$	$3.51 \cdot 10^{-3}$	0.277	$4.63 \cdot 10^{-3}$
160	$2.24 \cdot 10^{-3}$	0.010	0.342	$1.28 \cdot 10^{-2}$	$1.59 \cdot 10^{-3}$	0.139	$2.03 \cdot 10^{-3}$

- Fitted exponent of the offset $\|\bar{\Theta}_p - \Theta^*\|$ against k_e: -1.05 (A) and -1.33 (B; not yet asymptotic below $k_e \approx 80$); Theorem J gives -1.
- The vector error of the first-order location halves with every doubling of k_e in both instances. In A the frozen-measure terms alone leave a 35% error at every k_e, so the oscillation term of Remark H.7a is an $O(1)$ fraction of the offset.

- The median relative error of the parameter Floquet exponents falls from $1.2 \cdot 10^{-2}$ to $1.1 \cdot 10^{-3}$ (A) and from $4.0 \cdot 10^{-2}$ to $9.5 \cdot 10^{-4}$ (B) between $k_e = 10$ and 160.
- Error channel: the error-channel multipliers are below the resolution of the finite-difference monodromy (exponents of order k_e), so their product is recovered from Liouville's identity $\sum \log|\lambda| = \int_0^{T_0} \operatorname{tr} \partial F/\partial z \, \mathrm{d}t$ along the orbit. The mean error-channel exponent is $k_e - L$ with $L = 0.160$, 0.111, 0.086, 0.074, 0.068 (A) and $L = -0.054$, -0.074, -0.084, -0.089, -0.092 (B) at $k_e = 10$, 20, 40, 80, 160, as Theorem J(b) states ($L = O(1)$).
- Tests (`tests/test_periodic.py`): ∇J_m against finite differences; shooting residual $< 10^{-10}$; the offset, the vector error and the Floquet error all fall by the required factors from $k_e = 20$ to 40 on instance B.

24.15 Proposition K: collapsed configurations and the orientation of the orbit

Script: `exp_orientation_stability.py` → `orientation_stability/`. The setting is instance A of Theorem J ($n = 2$, $m = 3$, $G = \operatorname{diag}(4, 4, 1, 1, 1)$) with the ellipse $(1.2\cos t, \pm 0.8 \sin t)$ traversed both ways. The Floquet exponents of the collapsed periodic orbit of the real closed loop are computed by Newton shooting and restricted to the separation subspace $\sum_i u_i = 0$; an exponent is $-\log|\lambda|/T_0$, so positive means stable. The leading-order separation exponents are the eigenvalues of $(\sigma_0 I - G^{1/2} R_2 G^{1/2})/k_e$, identical for both orientations. The second-order prediction is the linearisation of $\frac{1}{k_e}\langle H \rangle + b_{\text{frozen}} + b_{\text{osc}}$ at its own equilibrium $\bar{\Theta}_2$ (every $\bar{\Theta}_2$ found is collapsed, kernel spread below $1.2 \cdot 10^{-8}$); the prediction "at Θ^*" uses the Jacobian at the collapsed critical point instead, which omits the third-derivative term (Table 41).

Table 41: Proposition K: the softest separation Floquet exponent of the collapsed orbit for the two orientations at $\sigma_0 = 0.5$, and its orientation difference, measured against the second-order prediction (`orientation_stability/table.md`).

	softest separation exponent		difference (+) − (−)			$k_e^2 \times$ difference	
k_e	(+)	(−)	measured	pred. at $\bar{\Theta}_2$	pred. at Θ^*	measured	pred. at $\bar{\Theta}_2$
10	$3.563 \cdot 10^{-3}$	$2.660 \cdot 10^{-3}$	$9.03 \cdot 10^{-4}$	$7.77 \cdot 10^{-4}$	$9.01 \cdot 10^{-4}$	0.090	0.078
20	$1.800 \cdot 10^{-3}$	$1.614 \cdot 10^{-3}$	$1.86 \cdot 10^{-4}$	$1.69 \cdot 10^{-4}$	$2.08 \cdot 10^{-4}$	0.075	0.068
40	$9.277 \cdot 10^{-4}$	$8.863 \cdot 10^{-4}$	$4.14 \cdot 10^{-5}$	$3.93 \cdot 10^{-5}$	$4.98 \cdot 10^{-5}$	0.066	0.063
80	$4.737 \cdot 10^{-4}$	$4.640 \cdot 10^{-4}$	$9.71 \cdot 10^{-6}$	$9.44 \cdot 10^{-6}$	$1.22 \cdot 10^{-5}$	0.062	0.060

- Fitted exponent of the difference against k_e: -2.18 (Proposition K(iii): -2). The prediction at $\bar{\Theta}_2$ converges to the measured k_e^2-scaled limit; the prediction at Θ^*, which omits the third-derivative term, converges to a value 30% too large.
- At $\sigma_{\text{crit}} = 0.4606$ both orientations are unstable ($-4.1 \cdot 10^{-4}$ and $-1.27 \cdot 10^{-3}$ at $k_e = 10$); the even part dominates. The two separation exponents coincide (multiplicity $m - 1 = 2$, Proposition K(ii)).
- Designed σ_0, chosen so that the predicted softest exponents of the two orientations are opposite ($\text{pred}_+ + \text{pred}_- = 0$): $\sigma_0 - \sigma_{\text{crit}} = 0.0125$ ($k_e = 10$) and 0.0061 ($k_e = 20$). The measured softest exponents are $+8.5 \cdot 10^{-4} / -2.5 \cdot 10^{-5}$ and $+1.4 \cdot 10^{-4} / -3.7 \cdot 10^{-5}$: one orientation of the same reference orbit (same occupation measure, same averaged system) leaves the collapsed configuration stable and the other makes it unstable.
- Test `test_proposition_K_hessian_splitting_and_critical_leak` in `tests/test_periodic.py`: the Hessian splitting at the collapsed point (10^{-5}), the gap equation of σ_{crit} (10^{-8}), the two-fold zero separation mode at σ_{crit}.

24.16 Theorem J′: the stationary law of the dithered single-kernel loop

Script: `exp_dither_stationary.py` → `dither/`. Single-kernel instance ($n = 1$, $m = 1$, $\vartheta = (\alpha, \gamma)$, frozen width, $G = \mathrm{diag}(2, 2)$, $\sigma_0 = 0.4$, $\tau_0 = 0.1$), reference traversed both ways; $2 \cdot 10^4$ independent replicas per (k_e, orientation) started from the first-order law, $k_e = 10$, 20, 40, 80; the Gibbs measure, the first-order correction g, the current j_1 and the circulation coefficient on a 201^2 grid. Full tables in `results/dither/table.md` (raw in `raw.npz`); Table 42.

Table 42: Theorem J′: the stationary law of the dithered single-kernel loop against the Gibbs measure and the first-order prediction (`dither/table.md`).

	$k_e = 10$	20	40	80
$(m_\varepsilon(\alpha) - \pi(\alpha)) \cdot k_e$, prediction -0.255	-0.28	-0.36	-0.36	-0.37
$(m_\varepsilon(\gamma) - \pi(\gamma)) \cdot k_e$, prediction -0.426	-0.44	-0.54	-0.51	-0.67
residual $m_\varepsilon(\alpha)$ − first order (s.e. 0.0013)	-0.0021	-0.0051	-0.0025	-0.0015
residual $m_\varepsilon(\gamma)$ − first order (s.e. 0.0018)	-0.0015	-0.0056	-0.0020	-0.0030
orientation-odd shift of γ, measured/predicted	0.93 ± 0.04	1.10 ± 0.07	1.15 ± 0.15	1.38 ± 0.29
$k_e^2 \mathbb{E}e^2/(2\pi(\bar{L}))$, orientation +1	1.054	1.033	1.017	1.009
circulation Λ_ε (+1), measured ± 0.0005 / predicted	$+0.0037$ / -0.0005	$+0.0006$ / -0.0003	-0.0004 / -0.0001	$+0.0003$ / -0.0001

- The Gibbs discrepancy decays like $k_e^{-0.87}$ (α), $k_e^{-0.83}$ (γ), $k_e^{-1.09}$ ($\alpha\gamma$), and the first-order prediction removes 93–97% of it at $k_e = 10$ and 70–85% at $k_e = 20$–40; the residual is at the Monte Carlo floor at $k_e = 80$ and its second-order decay is not resolved.
- The orientation-odd shift (the dither version of Remark H.7a) is predicted to 7–8% at $k_e = 10$ and within the 10–30% resolution at larger k_e.
- The tracking error is the Gibbs-averaged occupation loss over $k_e^2/2$ with an $O(1/k_e)$ correction of 0.5–$0.7/k_e$.
- The first-order circulation is below the Monte Carlo resolution for $k_e \geq 20$ and dominated by higher orders at $k_e = 10$; the current machinery of Proposition J′.2 is checked on the first-order averaged SDE on the grid: $\int \Theta \wedge (j_\varepsilon - j_0)/(\varepsilon \int \Theta \wedge j_1) = 1.47$, 1.21, 1.09, 1.03 at $\varepsilon = 0.2$, 0.1, 0.05, 0.025 and $\|m_\varepsilon - m_0(1 + \varepsilon g)\|_1/\varepsilon^2 = 0.31$ constant.
- Referee round 6 (`review/REVIEW_round6_Jprime.md`): the first run's batch standard errors were biased low by 1.4–1.9 and its burn-in too short (JP-4–JP-6); the run reported here is the corrected one.
- Corollary J′.3 (`exp_gibbs_m.py` → `dither/gibbs_m.md`): μ_τ^* on the grid ($\int \alpha \, \mathrm{d}\mu^* = 0.4963$, $\int \gamma \, \mathrm{d}\mu^* = 0.1873$, $\int \alpha^2 \, \mathrm{d}\mu^* = 0.9353$, $\bar{L}(\mu^*) = 0.1394$); the Gibbs measures π_m for $m = 1$–32 by MALA (4000 replicas): $\mathbb{E}[\bar{L}(\mu_\Theta)] = 0.1960$, 0.1681, 0.1540, 0.1470, 0.1432, 0.1413 with $(\mathbb{E}[\bar{L}] - \bar{L}(\mu^*)) \cdot m = 0.057$–$0.061$ (slope -0.99), $\mathbb{E}[\gamma_1]$ slope -1.01, $\mathbb{E}[\alpha_1^2]$ from 0.536 to 0.922 with rescaled deviation -0.41 ± 0.01; the $m = 1$ row matches the quadrature of π_1 within one standard error.

25 Research program

Each branch states what is proved, what is open, and the first concrete step.

1. **Infinite-horizon averaging for McKean–Vlasov closed loops (Conjecture H.6).**
   - *Proved:* the averaged system converges globally and exponentially (H(ii)); the exact first correction at frozen measure (H.7); input-to-state stability of the averaged system (H.8).
   - *Open:* a near-identity averaging transformation for the stochastic McKean–Vlasov loop that keeps the frozen-measure correction valid along the moving measure, uniformly in time.
   - *Done (Theorem J, Section 21):* the deterministic case ($\tau = 0$, $\sigma = \sigma_0/k_e$) at fixed m: near every hyperbolic critical point of J_m the real loop has a unique periodic orbit with the same stability type, a k_e-uniform basin, and the first-order location of Proposition H.7 and Remark H.7a.
   - *Done (Theorem J′, Section 23):* the dithered case at fixed m. The invariant law of the loop has the Gibbs measure $\exp(-J_m/\tau_0)$ as its parameter marginal up to $O(1/k_e)$, with the first-order correction driven by the same drift b as Theorem J; the fast error and the phase are eliminated by a Poisson-equation expansion of the generator rather than a transformation of the Fokker–Planck equation.
   - *Next steps:* uniformity in m — the mean-field limit of the stationary law (the Gibbs measures of J_m converge to the minimiser of the mean-field free energy F_τ of Theorem H, a propagation-of-chaos question for Gibbs measures which is where Conjecture H.6 now lives), the second-order correction, and quantitative constants in place of the $O(\varepsilon^2)$.
2. **Non-reciprocal mean-field dynamics from closed-loop learning (new; Proposition H.7).**
   - *Proved at a frozen measure:* the averaged adaptive dynamics is a semi-gradient flow plus a lag term (Proposition H.7). Along the real loop a second-order averaging term is added (Remark H.7a).
   - Its interaction has two antisymmetric parts:
     - the performative defect A_{even} of the state shift, which is present for every n and is the only part for $n = 1$;
     - the circulation kernel K, an integral over the region that the reference orbit encloses ($n \geq 2$).
   - *Proved (Theorem I, Section 20):* reciprocity at order k_e^{-2} is equivalent to $A_\mu \equiv 0$; for $n = 1$ it holds exactly at perfect fit on the orbit, and for $n \geq 2$ at perfect fit exactly when the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit; no open set of measures is reciprocal, for any n. The performative defect is bounded by the residual and the antisymmetric part of its Jacobian; the circulation defect depends on neither.
   - *Open questions:*
     - Are stationary states unique beyond the perturbative regime?
     - Can circulation be *designed*, choosing the orbit's orientation and area, to break symmetric traps or to speed escape from the collapsed configuration? (Partly answered by Proposition K, below: the orientation decides the stability of the collapsed configuration in a window around the critical leak.)
     - Is there a Lyapunov functional other than F_τ for the non-reciprocal flow?
   - *Context:* the closest analogues are performative prediction [59], semi-gradient TD learning [73], [71] and non-reciprocal phase transitions [25].
   - *Done (Remark H.7a):* deterministic $n = 2$ closed-loop runs with the reference traversed both ways (same ρ). The full first-order theory predicts the $O(1/k_e)$ difference of the

learned kernels to 3% in size and 11% as a vector.

- *Done (Proposition K, Section 22):* at a collapsed configuration the orientation-odd k_e^{-2} terms shift the separation Floquet exponents of the real loop by $O(k_e^{-2})$, with a closed-form prediction from the averaged data; at the critical leak σ_{crit} the direction of traversal decides whether the kernels separate, and the σ_0 at which it does can be computed in advance.
- *Done (Theorem J′):* the test with dither. The orientation-odd part of the stationary parameter means is $2\varepsilon \int \Theta g_{\mathrm{odd}}\, \mathrm{d}\pi$, with g_{odd} driven by the odd part of b; measured on the single-kernel instance of Section 23 (where, in one state dimension, the whole odd effect is b_{osc}).
- *New (Proposition J′.2): the stationary state is a non-equilibrium steady state.* The first-order averaged dynamics carries a stationary current $j_1 = \pi(b - \tau_0 G \nabla g)$ exactly when b is not a G-gradient, which for $m \geq 3$ is always the case; its entropy production rate is $\varepsilon^2 \tau_0^{-1} \pi(\|b_\perp\|^2_{G^{-1}})$; and the real loop's stationary parameter process is not reversible. The Lyapunov-functional question for the non-reciprocal flow is therefore a question about a non-equilibrium steady state (its free energy is not a Lyapunov functional of the real loop at order ε).
- *Next steps:* the design of the orbit's area (not only its orientation) as a control input; the orientation effect at non-collapsed configurations with several kernels; measuring the current of the real loop (below the Monte Carlo resolution in Section 23's instance; a variance-reduced estimator or a stronger instance is needed); the mean-field limit of the current.

3. **Uniform-in-time propagation of chaos with a plant in the loop.**

- Theorems D and D′ are finite-horizon; D′ gives the $m^{-1/2}$ rate for the dithered loop by synchronous coupling, with constants independent of the temperature and an exponent independent of the dimension for the state and the learned function. Uniform-in-time results exist for mean-field Langevin dynamics [8] and for shallow networks [29].
- *Done (Theorem D″, Section 17):* uniform in time when the leak dominates the loop gain of the adaptive law (a small-gain condition between the plant channel and the particle channel, explicit in Lipschitz constants). The certified regime is leak-dominated: the network does not learn (without dither it collapses to one kernel; with dither it stays in the cloud of the prior, where the bound has content: the coupled distance is of order $m^{-1/2}$ and bounded in time). The result marks where the contraction method stops rather than settling the question.
- *Open, and now with a negative indication:* the learning regime (kernels separated, leak below the critical value of Proposition K), where the plant couples all particles through e and (by H.7) the averaged interaction is not reciprocal. The long-horizon run of `exp_chaos_uniform.py` ($\sigma = \sigma_K/3$, 1000 s) shows the coupled distance growing threefold between the windows $(100, 200]$ and $(500, 1000]$ while keeping the $m^{-1/2}$ law at each time: on that horizon only the finite-horizon Theorem D′ is supported. Whether the growth saturates (the leak's time scale is $1/\sigma = 2000$ s), and whether a uniform statement holds at all in the learning regime, is the open question; the first draft's claim of empirical uniformity was based on a run that was in fact a slow collapse and is withdrawn (review round 6).

4. **Geometry of two-time-scale adaptation (Proposition G(b′)).**

- *Proved formally, and checked:* the slow geometry follows the gradient flow of the profile loss in the metric induced on the slow manifold, and the slope S measures the departure from plain variable projection.

- *Open:* a rigorous long-horizon statement; the $O(1/R)$ corrections, which involve D^2a^* and $\partial_\xi M$; gain designs that shape the induced metric.

5. **Noiseless global convergence.**

   - *Done (Proposition K):* the local criterion for the collapsed trap: a collapsed configuration is a strict local minimum of the regularised loss iff $\sigma_0 > \sigma_{\mathrm{crit}}(\bar{\vartheta}) = \lambda_{\max}(G^{1/2}R_2G^{1/2})$, where R_2 is the Hessian of the first-variation potential at the atom; below σ_{crit} the noiseless flow leaves the trap along the softest separation mode, without dither, and the residual at the atom is what drives the separation (R_2 is linear in it).
   - *Open:* a closed-loop separation condition under which the noiseless flow avoids spurious stationary points, despite the failure of the Chizat–Bach [10] boundary conditions. Candidates are a compactified width coordinate and reflecting boundaries (Remark H.3).

6. **Unbalanced transport, meaning online structure adaptation.**

   - Treat the weights as masses (Hellinger–Kantorovich / Wasserstein–Fisher–Rao) to obtain birth and death of kernels with Lyapunov guarantees. The conic particle descent of [12] and [18] is the static analogue.

7. **Excitation theory in measure space.**

   - *Done (Lemma E.0):* a quantitative, uniform UES estimate for the skew-symmetric class, explicit in $(k, \alpha, T, \Psi_M, \Psi_D)$, which reproduces the ω^{-2} law of Remark E.1.
   - *Done (Lemma E.0′):* a strict Lyapunov function whose rate is the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, for k above a threshold set by the peak factors of Ψ, $\dot{\Psi}$, $\ddot{\Psi}$ (second-order slaving of the error to its slow manifold and a second-order averaging correction, whose mean part vanishes identically); the ill-conditioning of the averaged Gramian is paid once, as in the exact rate.
   - *Done (Corollary E.0′.3):* the same Lyapunov function bounds the stationary parameter spread of the dithered linearised loop, $\limsup \mathbb{E}\|\zeta\|^2 \le (1+O(\varepsilon))k\|\Sigma\|_F^2/(2(1-\delta)\lambda_{\min})$, within $1/(1-\delta)$ of the exact (averaged Langevin) value on isotropic regressors and paying the condition number in the trace otherwise: an explicit design formula for how much dither a given tracking budget tolerates.
   - *Done (Lemma E.0″):* the non-periodic case. The period mean is replaced by a forward exponential average over a window T (whose derivative identity $T\dot{\bar{M}} = \bar{M} - M$ gives the antiderivatives in closed form), the peak factors become time-local, and one residual second-order term $\bar{N}_2$ appears, costing $\delta_2 = O(1/k)$ in the rate; the guaranteed rate is $(1-\delta-\delta_2)\lambda_T/k$ with λ_T the exponentially weighted PE level of the window. Checked on a quasi-periodic regressor and on a regressor driven by a Lorenz orbit, where no periodic construction applies.
   - *Open:* the continuum form of (PE); the trajectory kernel

     $$K_T(\vartheta,\vartheta') = \int \Phi(x(s);\vartheta)\Phi(x(s);\vartheta')\,\mathrm{d}s;$$

     identifiability of measures up to symmetry; the relation to [32]. Remark E.1 shows that the PE level alone does not predict the rate. For Lemmas E.0′ and E.0″: lowering the threshold k_0 (a third-order slaving trades ρ_{DD} for the peak factor of $\dddot{\Psi}$ one order higher); the choice of the window T as an optimisation; the nonlinear Theorem E with the constants of E.0″ (Corollary E.0′.2's steps carry over verbatim, with λ_T in place of $\lambda_{\min}(\bar{M})$).

8. **Transport-regularised adaptive laws.**

- Wasserstein proximal (JKO) discrete-time adaptive laws.
- *Done (Theorem C′, Section 13):* the OT leak toward priors with a different number of atoms, using optimal plans with barycentric projection; it is Lloyd's algorithm in continuous time, and its bias is the centroid defect of the ideal.
- *Open:* its mean-field form (a general prior ν and a general μ, where the barycentric projection is the superdifferential of $\frac{1}{2}W^2(\cdot,\nu)$; Ambrosio–Gigli–Savaré [2, Thm. 10.4.12]), and adaptive priors (ν itself learned).
- Transfer: transporting a learned measure to a new plant.

9. **Quantisation and the curse of dimensionality (Corollary B.6).**
   - *Done (Theorem C′):* an adaptive law whose leak tracks the equal-mass centroidal quantisers of a prior measure (centroidal Voronoi tessellations [47]; coverage control [15], [37]), and Theorem D′'s observation that the state and the learned function converge at the rate $m^{-1/2}$, whose exponent is dimension-free: the curse of dimensionality of Corollary B.6 lives in the W-distance of the parameter measures, which the loop never evaluates.
   - *Open:* second-order (centroid) quantisation rates for the ideal measure on the lifted space; approximation rates that do not go through W (Maurey-type).
10. **Symmetric systems beyond kernels.** The OT quotient applies verbatim to swarms of identical agents, mixtures and multi-head models ([20] and [37] are the control-side precedents).
11. **Deep approximators.** Hierarchical (multi-layer) mean-field measures, with a layer-wise OT Lyapunov function.

A Verification map

Table 43 lists the tests and experiments of the accompanying repository that check each statement; statements not checked are listed after it. Test files are abbreviated: `T` is `tests/test_transport.py`, `Lg` is `tests/test_langevin.py`, `Av` is `tests/test_averaging.py`, `Ex` is `tests/test_excitation.py`, `NR` is `tests/test_nonreciprocity.py`, `Pe` is `tests/test_periodic.py` and `Ba` is `tests/test_barycentric.py`.

Table 43: Verification map.

statement	what is checked	where
constants (1) of Section 9, closed forms	$\nabla^2_{(c,s)}\varphi \le H_\varphi$ by brute force	`T::test_gaussian_constants_are_suprema`
Lemma P.1	$\|\mathrm{Proj}_K(\vartheta, Gy)\|_{G^{-1}} \le \|y\|_G$ on 2000 adversarial states, with equality when inactive; a tight instance of F(i) with ratio ≥ 0.99	`T::test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound`
A(i) raw rates	lifted raw rates equal classical raw rates under the gain map (10^{-10})	`T::test_identification_with_classical_law`
A(i) projected rates, only-if	per-kernel projection with rescaled layer matches to 10^{-10} with active projections; an unrescaled layer does not; five joint balls are each distinguished (two-case construction: four active against an inactive per-kernel field, one inactive against an active one)	`T::test_identification_projected_rates_per_kernel`

continued on the next page

Table 43, continued

statement	what is checked	where
B with constant ideal and $\varepsilon = 0$	the identity to floating-point precision; cross term ≤ 0; remainder $\leq$ Lemma B.2; $\dot{V}_\pi \leq$ (B) at every sample	`T::test_theorem_B_inequality_along_trajectories`
B with a moving ideal, $\varepsilon \neq 0$, switches and active projection	the same plus $V_* W$; at least one switch; projection active at ≥ 10 samples; Dini quotient $\leq$ (B)	`T::test_theorem_B_moving_ideal_nonrepresentable_switches_and_active_projection`
B, Dini form	forward-difference quotient of $V_{\mathrm{OT}} \leq$ (B)	`T::test_dini_derivative_of_V_OT_bounded_by_active_branch`
B against labelling	matched error $\ll$ labelled error; the bound is 30–45$\times$ smaller	`T::test_permuted_ideal_labelled_vs_OT`, `exp_transport_permutation.py`
B.5: failure of the plain bound along (L_σ); the (L_σ) variant holds	with every C(iv) hypothesis asserted, the plain bound fails by $> 3\times$ and the leak variant holds	`T::test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`
C(i), Lemma C.3 (unprojected pure leak)	zero switches and exact $e^{-\sigma t}$ decay (explicit Euler, pure leak)	`T::test_theorem_C_i_exact_dissipation_pure_leak`
C(ii)	$C \leq b^2 + 4\eta R$; $\sqrt{C} \leq b + 2\eta$; $C = b^2$ under both gap conditions (300 configurations)	`T::test_lemma_C_ii_induced_coupling_bound`
C(iv)	an instance satisfying **every** hypothesis, asserted, including the gap condition $4\bar{\eta}(b + \bar{\eta}) < \Delta_{\mathrm{gap}}$; the envelope holds at every sample, with actual contraction	`T::test_theorem_C_iv_exponential_estimate`
C(v), Proposition C.5	label invariance of the OT leak; the reshuffling transient of the labelled leak	`exp_ot_sigma_mod.py`, `exp_sigma_mod_transient.py`
C′(i)	the plan reduces to the assignment for $m_0 = m$ (10^{-10}) and $W^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma$ (10^{-10}); along the pure leak W^2 is non-increasing at every step and $D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2$ holds at every state; the flow reaches $V_{\mathrm{bar}}^2 < 10^{-8}$ and ends at a fixed point of the equal-mass Lloyd iteration	`Ba::test_plan_reduces_to_assignment_and_pythagorean_split`, `Ba::test_pure_barycentric_leak_dissipates_to_a_centroidal_configuration`, `exp_barycentric_leak.py` → `results/barycentric_leak/table.md`
C′(ii)	$\sqrt{C_{\mathrm{bar}}} \leq b + 2\eta$ on 90 random configurations; $C_{\mathrm{bar}} = b_{\mathrm{bar}}^2$ at small η (plan stability)	`Ba::test_induced_coupling_bound_and_plan_stability`
C′(iii)	the Lyapunov inequality at every recorded state of a closed loop with the barycentric leak toward a 24-atom prior	`Ba::test_lyapunov_inequality_along_the_barycentric_leak_closed_loop`
D	consistency in m (nearly constant amplification 1.64–1.69; not the constant $e^{\Lambda T}$)	`exp_mean_field_m.py`
D′(b), (d)	the $m^{-1/2}$ rate of the synchronous coupling to a 4096-particle reference: slopes -0.47 ± 0.06 ($\tau = 0$) and -0.46 ± 0.05 ($\tau = 2 \cdot 10^{-3}$) over 24 seeds, 0.94 ± 0.08 and 0.92 ± 0.08 against $(1/m - 1/M)^{1/2}$; not the constant	`exp_chaos.py` → `results/chaos/table.md`

continued on the next page

Table 43, continued

statement	what is checked	where
D″	with exact Lipschitz constants and the a-priori tube, $\sigma_{\text{cert}} = 4.04$ against Proposition K's $\sigma_K = 0.00144$; at $1.2\sigma_{\text{cert}}$ with dither the coupled distance is of order $m^{-1/2}$ (slope -0.51 ± 0.02), bounded in time and 700× below the bound, with the network at its prior (fit error 1.000); without dither the reference collapses (spread 10^{-14}) and the distance is zero; at $\sigma_{\text{cert}}/8$ it decays to 10^{-11}; in the learning regime ($\sigma_K/3$, 1000 s) the network stays spread and fits the plant (fit error 0.63) and the distance follows $m^{-1/2}$ (-0.62 ± 0.16) but grows threefold between the windows $(100, 200]$ and $(500, 1000]$	`exp_chaos_uniform.py` → `results/chaos/uniform_table.md`
F(ii) along (L)	particle path length ($\geq$ Wasserstein length) $\leq \lambda_G^{1/2} L_1 \int \|e\|$ at every sample	`T::test_theorem_F_wasserstein_length_bounded_by_integrated_error`
F(ii′) along (L_σ)	$W(\mu_{t_1}, \mu_{t_0}) \leq \int(\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0))$ on a grid of pairs; a tight instance with $e \equiv 0$ (ratio ≥ 0.99), where the motion is $W \approx 1.05$	`T::test_theorem_F_leak_clause_along_ot_sigma_mod`, `T::test_theorem_F_leak_clause_is_tight_when_the_leak_pays_for_all_motion`
E linearisation	Jacobian of the implemented loop equals $A(t)$; $QA + A^\top Q = \text{diag}(-k_e I, 0)$	`T::test_theorem_E_linearisation_and_lyapunov_identity`
E convergence	the matched error decays and the labelled error does not (qualitative); sliding-window excitation and fitted rates	`T::test_theorem_E_matched_error_converges_labelled_does_not`, `exp_pe_convergence.py`
Lemma E.0	$p_{\text{lo}}\|w\|^2 \leq V \leq p_{\text{hi}}\|w\|^2$ and $\dot{V} \leq -\gamma\|w\|^2$ at random (t, w) for four regressors, closed-form and optimised ε; the envelope along trajectories; explicit rate $\leq$ exact Floquet rate	`Ex::test_lemma_E0_pointwise_inequalities_hold_on_random_states`, `Ex::test_lemma_E0_envelope_and_floquet_ordering`
Lemma E.0(iii), Remark E.0a	the ω^{-2} law of the explicit rate and of the exact rate; the saturation of exact/explicit in the slow-adaptation limit; the tables	`Ex::test_remark_E1_omega_scaling_of_the_explicit_rate`, `Ex::test_slow_adaptation_limit_ratio_is_bounded`, `exp_explicit_rate.py` → `results/explicit_rate/table.md`
E, explicit constants	on the implemented loop, from a start inside the explicit ball, $\|w(t)\| \leq (p_{\text{hi}}/p_{\text{lo}})^{1/2} e^{-\lambda t}\|w(0)\|$ at every sample (L_J estimated from samples)	`Ex::test_theorem_E_explicit_nonlinear_envelope_on_the_implemented_loop`
Corollary E.0′.2	the explicit envelope of Theorem E with Lemma E.0′'s constants on the implemented single-kernel loop at $k_e = 40$: 1.5 e-folds inside the envelope from a start in the explicit ball ($\text{d}t = 0.005$, below the discretisation floor); $\lambda' > 10 \times$ Theorem E's rate from Lemma E.0 and $< 4 \times$ the exact linear rate; the constants table	`Ex::test_corollary_E0prime_2_envelope_on_the_implemented_loop`, `exp_explicit_rate.py` (table 5)
Corollary E.0′.3	the explicit bound on the stationary parameter spread of the dithered (LTV) dominates the exact periodic covariance (sinusoid: ratios $2.7 \to 1.3$ ($\delta = 0.2$) and $2.5 \to 1.2$ ($\delta = 0.1$) at $k = 20 \to 200$, falling toward $1/(1-\delta)$; the Theorem E ideal: 15–18, the price of $q = 15$ directions at the worst excitation); the exact value equals the averaged $k \cdot \text{tr}\, X$ within $2/k^2$	`Ex::test_corollary_E0prime_3_dither_covariance_bound_against_the_periodic_lyapunov_solution`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` (table 6)

continued on the next page

Table 43, continued

statement	what is checked	where
Lemma E.0″	on $\Psi = (\sin t, \cos\sqrt{2}\,t)$ (not periodic) with window 4π: the exponential-average identities on the grid (10^{-4}), the conditions at $k = 20$, 50, 200 with $\delta_2 \le 5\%$, the certificate on 400 times, the guaranteed rate below the exact decay rate (top Lyapunov exponent) within a factor 2.6 and improving with k; the experiment: exact/guaranteed $2.5 \to 1.3$ on the quasi-periodic regressor and $5.0 \to 3.7$ on a slowed Lorenz-driven regressor ($k = 100$–400), the certificate holding at every admissible row	`Ex::test_lemma_E0dprime_quasi_periodic_regressor_certificate_and_rate`, `exp_pe_general.py` → `results/explicit_rate/general_pe.md`
Lemma E.0′	the certificate $\sup_{t,x} \dot{V}/V \le -\gamma$ (a generalised eigenvalue problem on a time grid) on $\Psi = (\sin t, \cos t)$ at $k = 20$, 50, 200 and on the slow-adaptation ideal at $g = 0.01$, $k = 400$; $N_2 = 0$ numerically; the guaranteed rate below the exact Floquet rate and within a factor 2 (sinusoid) and 1.5 (ideal) of it, above Lemma E.0's by more than 10^4 on the ideal; the table of k and δ	`Ex::test_lemma_E0prime_sinusoidal_regressor_certificate_and_sharpness`, `Ex::test_lemma_E0prime_pays_for_ill_conditioning_once_on_the_theorem_E_ideal`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` (table 4)
G(a), two scales	direct check (cosine ≥ 0.998 in the G^{-1} metric, `violating`; the weight block fails in `satisfying`, as explained in Section 18); $\bar{L}$ at period-mean parameters non-increasing	`exp_variable_projection.py` → `table_violating.md`, `table_satisfying.md`
G(b), (b′), (H2c)	VarPro holds when (H2b) and $S \ll 1$ hold; it fails when $S = O(1)$ although (H2b) holds; the induced-metric flow tracks the closed loop at every S (started on the slow manifold; gains $\div 4$ isolates the averaging error)	`exp_variable_projection.py` → `table_satisfying.md`; `exp_slow_manifold_slope.py` → `table_slope.md`
H(i), algebraic identity	with $e := r/k_e$ and $x := x_m$, the averaged drift equals $-G\nabla V$ (code gradients against finite differences, 10^{-7}); does not test averaging	`Lg::test_averaged_drift_is_the_mean_field_langevin_drift`
Remark H.0	the frozen-μ averaging error decays like $1/k_e$, $n = 1$ and $n = 2$	`Lg::test_averaging_error_of_the_formal_reduction_decreases_like_one_over_ke`, `Av::test_averaged_drift_error_decays_like_one_over_ke_for_an_orbit_enclosing_area`
H, need for entropy	finite-m real loop: a noiseless symmetric start stays bit-identical (and moves); dither separates it	`Lg::test_noiseless_symmetric_initialisation_never_separates_but_dither_does`
H(ii), oscillation bound	$\lvert V_L[\mu](\vartheta)\rvert \le K_\Phi(F_\rho + K_\Phi)$ for random and adversarial μ, ϑ	`Lg::test_first_variation_bound_used_in_holley_stroock`
H(ii) constant	formula transcription only	`Lg::test_lsi_lower_bound_formula`
H, global behaviour	collapsed against spread starts, with and without dither, two (σ, τ) settings	`exp_langevin_global.py`
H, speed–bias trade-off	escape time and loss floor against τ; $\log(1/\tau)$ and Arrhenius fits, seed-averaged and per seed (neither identifies the mechanism)	`exp_langevin_tradeoff.py`
H, reference values	$\bar{L}(\mu^*_\tau)$ by a concave dual on a grid; the stationary state of the exactly averaged loop; their offset in the scaling of H.6	`exp_gibbs_reference.py` → `results/langevin_global/gibbs_reference.md`

continued on the next page

Table 43, continued

statement	what is checked	where
H.7	expansion, Stokes area identity, parity under reversal, non-reciprocity of the exact second variation	`Av::test_frozen_measure_expansion_converges_at_rate_one_over_ke`, `Av::test_circulation_kernel_antisymmetric_vanishes_in_1d_and_equals_the_enclosed_area_integral`, `Av::test_shift_correction_is_even_and_lag_correction_is_odd_under_orbit_reversal`, `Av::test_exactly_averaged_dynamics_is_non_reciprocal_for_n1_and_n2`
H.7a, oscillation term	value against an independent one-period near-identity flow (within 3% at $k_e = 80$); odd under orbit reversal (10^{-6})	`Av::test_oscillation_correction_matches_a_one_period_flow`, `Av::test_oscillation_correction_is_odd_under_orbit_reversal`
I(i), (ii)	$a^\top B(\xi, \xi')a'$ equals the antisymmetric part of $S_{\text{even}} + K$ (10^{-10}); $A(\vartheta, \vartheta_z) = 0$ and the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ equals $A(\vartheta_0, \vartheta_1)$ (10^{-14})	`NR::test_bilinear_form_B_reproduces_the_antisymmetric_second_variation`, `NR::test_zero_feature_collapses_the_cyclic_criterion_to_a_pairwise_one`
I(iii)	$n = 1$: $A \equiv 0$ at perfect fit ($< 10^{-13}$); $A \neq 0$ with a residual, equal to the closed form; the positivity witness	`NR::test_n1_reciprocal_iff_residual_vanishes_on_the_orbit`, `NR::test_positivity_witness_n1`
I(iv)	the closed form of κ (10^{-10}); the second component of $F_s((0.5, 0))$ is positive on the ellipse; $\kappa \equiv 0$ on a zero-current orbit; with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ ($Dr_\mu \equiv 0$) $B = 2\kappa I$, $A \neq 0$ on the ellipse and $A \equiv 0$ on the segment; with $f = \hat{f}_\mu + he_1$, h zero on the ellipse, the curl term survives and equals $-\int \varphi\varphi' \partial_2 h \, \mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$	`NR::test_circulation_kernel_closed_form_and_vanishing_for_a_zero_current_orbit`, `NR::test_perfect_fit_n2_leaves_only_the_circulation_kernel`, `NR::test_perfect_fit_on_the_orbit_alone_keeps_the_curl_term`
J, averaged objective	∇J_m against finite differences (10^{-7}); the critical point is a non-degenerate minimum with two distinct kernels	`Pe::test_gradient_of_J_m_matches_finite_differences_and_critical_point_is_nondegenerate`
J(a), (b), (d)	Newton shooting finds the periodic orbit (residual $< 10^{-10}$); $\|\bar{\Theta}_p - \Theta^*\|$ and $\max\|e\|$ halve from $k_e = 20$ to 40; the vector error of the first-order prediction of the offset halves as well (it is 0.55 at $k_e = 40$ in the fast-sweep test instance, and 0.041 at $k_e = 40$ in instance A of the experiment); the parameter Floquet exponents match $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})$ with a maximal relative error below 2% at $k_e = 40$ that halves with k_e; both instances at $k_e = 10$–160 in the experiment; the error-channel exponent recovered from Liouville's trace identity is $k_e - L$ with $\|L\| < 0.2$	`Pe::test_theorem_J_periodic_orbit_offset_and_floquet_exponents`, `exp_periodic_orbit.py` → `results/periodic_orbit/table.md`
K(i), (ii)	the Hessian of J_m at a collapsed configuration equals $I_m \otimes (\sigma_0 G^{-1} - R_2) + \frac{1}{m}(\mathbf{1}\mathbf{1}^\top) \otimes M_2$ (finite differences, 10^{-5} relative); σ_{crit} solves $\sigma = \lambda_{\max}(G^{1/2} R_2(\bar{\vartheta}(\sigma)) G^{1/2})$ (10^{-8}) and the separation block is singular there; the separation Floquet exponents of the collapsed orbit come in $(m-1)$-fold multiplets	`Pe::test_proposition_K_hessian_splitting_and_critical_leak`, `exp_orientation_stability.py`

continued on the next page

Table 43, continued

statement	what is checked	where
K(iii), (iv)	the orientation difference of the softest separation exponent at $\sigma_0 = 0.5$: slope -2.18 against k_e, within 14% $\to$ 3% of the prediction at $\bar{\Theta}_2$ for $k_e = 10 \to 80$, while the prediction at Θ^* converges to a value 30% off; at σ_{crit} both orientations are unstable, predictions within 14% at $k_e = 40$; at the designed σ_0 the signs differ for $k_e = 10$ and 20	`exp_orientation_stability.py` $\to$ `results/orientation_stability/table.md`
J′ averaged data, solver	the single-kernel J, ∇J, b_{fr}, b_{osc} against the general H.7/H.7a code ($2 \cdot 10^{-5}$); the weighted Poisson solver against the Ornstein–Uhlenbeck closed forms (g, the mean shift $(G\lambda)^{-1}b$, the Green identity, zero current for a gradient drift); the stationary Fokker–Planck solver of the first-order SDE: current ratio $\to 1$ linearly in ε, $\|m - \pi(1+\varepsilon g)\|_1 \propto \varepsilon^2$	`tests/test_dither.py` (3 tests)
J′(b), (c), (d)	real loop, $2 \cdot 10^4$ replicas, $k_e = 10$–80, both orientations: the stationary means differ from the Gibbs values by $k_e^{-0.87}$ (α), $k_e^{-0.83}$ (γ), $k_e^{-1.09}$ ($\alpha\gamma$); the first-order prediction removes 93–97% of the discrepancy at $k_e = 10$ and 70–85% at $k_e = 20$–40, the residual at the Monte Carlo floor at $k_e = 80$ (second order not resolved); the orientation-odd shift matches its prediction to 7–8% at $k_e = 10$ and within the 10–30% resolution at $k_e = 20$–80; $k_e^2 \mathbb{E}e^2/(2\pi(\bar{L})) - 1 = 0.5$–$0.7/k_e$	`tests/test_dither.py::test_real_loop_means_are_closer_to_the_first_order_prediction_than_to_gibbs` ($k_e = 20$), `exp_dither_stationary.py` $\to$ `results/dither/table.md`
J′(e), J′.2	the circulation of the real loop is below the Monte Carlo resolution at first order ($k_e \geq 20$) and dominated by higher orders at $k_e = 10$; the current and density expansion of the first-order averaged SDE checked on the grid (ratios 1.47 $\to$ 1.03, ε^2 remainder 0.31); the antisymmetric part of $D(G^{-1}b)$ against the formula of J′.2(iii) to six digits (referee round 6)	`exp_dither_stationary.py` $\to$ `results/dither/table.md`; `review/REVIEW_round6_Jprime.md`
Corollary J′.3	the Gibbs measures π_m (MALA, 4000 replicas, $m = 1$–32) converge to the mean-field minimiser μ_τ^* (fixed point to 10^{-11}) with a $1/m$ law: slopes -1.01 ($\mathbb{E}[\gamma_1]$) and -0.99 ($\mathbb{E}[\bar{L}(\mu_\Theta)]$, rescaled deviation 0.058 ± 0.002 at every m), $\mathbb{E}[\alpha_1^2]$ from 0.54 to 0.92 against 0.935 with rescaled deviation -0.41 ± 0.01; the $m = 1$ row agrees with the quadrature of π_1	`exp_gibbs_m.py` $\to$ `results/dither/gibbs_m.md`; `tests/test_dither.py::test_corollary_J3_mean_field_minimiser_and_gibbs_sampler_consistency`
H.7 + H.7a in the real closed loop	the orientation dependence of the learned kernels (same ρ) is predicted to within 3% in size and 11% as a vector, cosine 0.994 at $k_e = 40$; the measured difference scales like $k_e^{-1.01}$	`exp_orbit_orientation.py` $\to$ `results/orbit_orientation/table_sigma2_slow16.md`

Not checked in the repository:

- B.5 along (L), and B.6 and B.7;
- the Filippov form of C(i), the inequality of C(v), and Proposition C.5's $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2)$;
- D(b), D's constant, and the one-sided constant of Remark D.1;
- the sharpness of the explicit rates of E beyond the tables of Remark E.0a and Lemma E.0′; Lemma E.0′ below its threshold k_0 (where it makes no claim) and for non-periodic PE;

Corollary E.2 and Remark E.3;

- Theorem I(v) beyond the two instances (the analytic dichotomy is a proof only);
- Theorem J(c) (the basin; a single basin run is reported in `review/REVIEW_round5_J.md`);
- Proposition K(iii)–(iv) beyond the instance of `exp_orientation_stability.py`, and the second-order normal form it rests on;
- Theorem C′(iv) (its hypotheses are not met on any tested instance) and the convergence of the pure leak to a critical point beyond the tested instances; Theorem D′'s constant, its per-particle bound and its k-particle convergence (only the rate of (b) is checked);
- H.8, which is a proof only;
- Conjecture H.6.

References

[1] S. K. Ainsworth, J. Hayase, S. Srinivasa. *Git Re-Basin: merging models modulo permutation symmetries*. International Conference on Learning Representations (ICLR), 2023. arXiv:2209.04836.

[2] L. Ambrosio, N. Gigli, G. Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. 2nd edition, Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2008. https://doi.org/10.1007/978-3-7643-8722-8.

[3] D. Bakry, M. Émery. *Diffusions hypercontractives*. In J. Azéma, M. Yor (eds.), Séminaire de Probabilités XIX, 1983/84, Lecture Notes in Mathematics 1123, pp. 177–206. Springer, Berlin, 1985. https://doi.org/10.1007/BFb0075847.

[4] M. Balzer, T. Schlömer, O. Deussen. *Capacity-constrained point distributions*. ACM SIGGRAPH 2009; Q. Mérigot, *A multiscale approach to optimal transport*, Computer Graphics Forum, 30(5), 2011; F. de Goes, K. Breeden, V. Ostromoukhov, M. Desbrun, *Blue noise through optimal transport*, ACM Transactions on Graphics, 31(6), 2012.

[5] R. Barboni, G. Peyré, F.-X. Vialard. *Ultra-fast feature learning for the training of two-layer neural networks in the two-timescale regime*. arXiv:2504.18208, 2025.

[6] E. A. Carlen, K. Craig. *Contraction of the proximal map and generalized convexity of the Moreau–Yosida regularization in the 2-Wasserstein metric*. Mathematics and Mechanics of Complex Systems, 1(1):33–65, 2013. https://doi.org/10.2140/memocs.2013.1.33. arXiv:1205.6565.

[7] P. Cattiaux, L. Mesnager. *Hypoelliptic non-homogeneous diffusions*. Probability Theory and Related Fields, 123:453–483, 2002.

[8] F. Chen, Z. Ren, S. Wang. *Uniform-in-time propagation of chaos for mean field Langevin dynamics*. Annales de l'Institut Henri Poincaré, Probabilités et Statistiques, 61(4):2357–2404, 2025. https://doi.org/10.1214/24-AIHP1499. arXiv:2212.03050.

[9] J. Chevallier. *Uniform decomposition of probability measures: quantization, clustering and rate of convergence*. Journal of Applied Probability, 55(4):1037–1045, 2018. https://doi.org/10.1017/jpr.2018.69.

[10] L. Chizat, F. Bach. *On the global convergence of gradient descent for over-parameterized models using optimal transport*. Advances in Neural Information Processing Systems 31 (NeurIPS 2018), pp. 3036–3046, 2018. arXiv:1805.09545.

[11] L. Chizat. *Mean-field Langevin dynamics: exponential convergence and annealing*. Transactions on Machine Learning Research, 2022. arXiv:2202.01009.

[12] L. Chizat. *Sparse optimization on measures with over-parameterized gradient descent*. Mathematical Programming, 194(1–2):487–532, 2022. https://doi.org/10.1007/s10107-021-01636-z. arXiv:1907.10300.

[13] G. Chowdhary, E. Johnson. *Concurrent learning for convergence in adaptive control without persistency of excitation*. In Proc. 49th IEEE Conference on Decision and Control (CDC), pp. 3674–3679, 2010. https://doi.org/10.1109/CDC.2010.5717148.

[14] G. V. Chowdhary, E. N. Johnson. *Theory and flight-test validation of a concurrent-learning adaptive controller*. Journal of Guidance, Control, and Dynamics, 34(2):592–607, 2011. https://doi.org/10.2514/1.46866.

[15] J. Cortés, S. Martínez, T. Karatas, F. Bullo. *Coverage control for mobile sensing networks*. IEEE Transactions on Robotics and Automation, 20(2):243–255, 2004.

[16] K. Craig, K. Elamvazhuthi, M. Haberland, O. Turanova. *A blob method for inhomogeneous diffusion with applications to multi-agent control and sampling*. Mathematics of Computation, 92(344):2575–2654, 2023. https://doi.org/10.1090/mcom/3841. arXiv:2202.12927.

[17] G. Da Prato, J. Zabczyk. *Ergodicity for Infinite Dimensional Systems*. London Mathematical Society Lecture Note Series 229. Cambridge University Press, Cambridge, 1996.

[18] Y. De Castro, S. Gadat, C. Marteau. *Fast Spawn&Prune (FS&P): global convergence of stochastic conic particle gradient descent via birth/death process*. arXiv:2605.19784, 2026.

[19] A. Dembo, O. Zeitouni. *Large Deviations Techniques and Applications*. 2nd edition. Springer, 1998.

[20] M. Emerick, B. Bamieh. *Continuum swarm tracking control: a geometric perspective in Wasserstein space*. In Proc. 62nd IEEE Conference on Decision and Control (CDC), pp. 1367–1374, 2023. https://doi.org/10.1109/CDC49753.2023.10384000. arXiv:2303.15638.

[21] S. Fabri, V. Kadirkamanathan. *Dynamic structure neural networks for stable adaptive control of nonlinear systems*. IEEE Transactions on Neural Networks, 7(5):1151–1167, 1996. https://doi.org/10.1109/72.536311.

[22] M. Fornasier, B. Piccoli, F. Rossi. *Mean-field sparse optimal control*. Philosophical Transactions of the Royal Society A, 372(2028):20130400, 2014. https://doi.org/10.1098/rsta.2013.0400. arXiv:1402.5657.

[23] N. Fournier, A. Guillin. *On the rate of convergence in Wasserstein distance of the empirical measure*. Probability Theory and Related Fields, 162(3–4):707–738, 2015. https://doi.org/10.1007/s00440-014-0583-7.

[24] A. Frieze, R. Kannan. *Log-Sobolev inequalities and sampling from log-concave distributions*. The Annals of Applied Probability, 9(1):14–26, 1999. https://doi.org/10.1214/aoap/1029962595.

[25] M. Fruchart, R. Hanai, P. B. Littlewood, V. Vitelli. *Non-reciprocal phase transitions*. Nature, 592(7854):363–369, 2021. https://doi.org/10.1038/s41586-021-03375-9.

[26] S. S. Ge, C. Wang. *Direct adaptive NN control of a class of nonlinear systems*. IEEE Transactions on Neural Networks, 13(1):214–221, 2002. https://doi.org/10.1109/72.977306.

[27] R. Giard, Y. De Castro, R. Denis, C. Marteau. *Riemannian gradient descent for Gaussian mixture models with unknown diagonal covariances*. arXiv:2609.30220, 2026.

[28] D. Gilbarg, N. S. Trudinger. *Elliptic Partial Differential Equations of Second Order*. 2nd edition. Springer, 1983 (reprinted 2001).

[29] M. Glasgow, J. Bruna. *Uniform-in-time weak propagation of chaos in shallow neural networks*. Advances in Neural Information Processing Systems 39 (NeurIPS 2026), to appear. arXiv:2605.22010, 2026.

[30] F. Golse. *On the dynamics of large particle systems in the mean field limit*. Lecture notes, arXiv:1301.5494, 2013. Published in A. Muntean, J. Rademacher, A. Zagaris (eds.), Macroscopic and Large Scale Phenomena: Coarse Graining, Mean Field Limits and Ergodicity, Lecture Notes in Applied Mathematics and Mechanics 3, pp. 1–144. Springer, 2016. https://doi.org/10.1007/978-3-319-26883-5_1.

[31] S. Graf, H. Luschgy. *Foundations of Quantization for Probability Distributions*. Lecture Notes in Mathematics 1730. Springer, Berlin, 2000. https://doi.org/10.1007/BFb0103945.

[32] J. Guo, S. T. Paruchuri, A. J. Kurdila. *Partial persistence of excitation in RKHS embedded adaptive estimation*. IEEE Transactions on Automatic Control, 68(10):5850–5861, 2023. https://doi.org/10.1109/TAC.2022.3226710. arXiv:2002.07963.

[33] M. Hairer. *Ergodic properties of Markov processes*. Lecture notes, University of Warwick, 2006.

[34] M. Hairer, J. C. Mattingly. *Yet another look at Harris' ergodic theorem for Markov chains*. In Seminar on Stochastic Analysis, Random Fields and Applications VI, Progress in Probability 63, pp. 109–117. Birkhäuser, 2011.

[35] R. Holley, D. Stroock. *Logarithmic Sobolev inequalities and stochastic Ising models*. Journal of Statistical Physics, 46(5–6):1159–1194, 1987. https://doi.org/10.1007/BF01011161.

[36] L. Hörmander. *Hypoelliptic second order differential equations*. Acta Mathematica, 119:147–171, 1967; M. Hairer, *On Malliavin's proof of Hörmander's theorem*, Bulletin des Sciences Mathématiques, 135(6–7):650–666, 2011.

[37] D. Inoue, Y. Ito, H. Yoshida. *Optimal transport-based coverage control for swarm robot systems: generalization of the Voronoi tessellation-based method*. IEEE Control Systems Letters, 5(4):1483–1488, 2021. https://doi.org/10.1109/LCSYS.2020.3039008. arXiv:2011.08337.

[38] P. A. Ioannou, P. V. Kokotović. *Adaptive Systems with Reduced Models*. Lecture Notes in Control and Information Sciences 47. Springer, Berlin, 1983. https://doi.org/10.1007/BFb0006357.

[39] P. A. Ioannou, J. Sun. *Robust Adaptive Control*. Prentice Hall, Upper Saddle River, NJ, 1996.

[40] H. K. Khalil. *Nonlinear Systems*. 3rd edition. Prentice Hall, Upper Saddle River, NJ, 2002.

[41] P. V. Kokotović, H. K. Khalil, J. O'Reilly. *Singular Perturbation Methods in Control: Analysis and Design*. Academic Press, London, 1986. Reprinted as Classics in Applied Mathematics 25, SIAM, 1999, https://doi.org/10.1137/1.9781611971118.

[42] N. Lanzetti, E. C. Balta, D. Liao-McPherson, F. Dörfler. *Stochastic Wasserstein gradient flows using streaming data with an application in predictive maintenance*. IFAC-PapersOnLine (IFAC World Congress 2023), 56(2):3954–3959, 2023. https://doi.org/10.1016/j.ifacol.2023.10.1334. arXiv:2301.12461.

[43] E. Lavretsky, T. E. Gibson. *Projection operator in adaptive systems*. arXiv:1112.4232, 2011.

[44] E. Lavretsky, K. A. Wise. *Robust and Adaptive Control: With Aerospace Applications*. Advanced Textbooks in Control and Signal Processing. Springer, London, 2013. https://doi.org/10.1007/978-1-4471-4396-3.

[45] F. L. Lewis, A. Yeşildirek, K. Liu. *Multilayer neural-net robot controller with guaranteed tracking performance*. IEEE Transactions on Neural Networks, 7(2):388–399, 1996. https://doi.org/10.1109/72.485674.

[46] F. L. Lewis, S. Jagannathan, A. Yeşildirek. *Neural Network Control of Robot Manipulators and Nonlinear Systems*. Taylor & Francis, London, 1999.

[47] S. P. Lloyd. *Least squares quantization in PCM*. IEEE Transactions on Information Theory, 28(2):129–137, 1982; Q. Du, V. Faber, M. Gunzburger, *Centroidal Voronoi tessellations: applications and algorithms*, SIAM Review, 41(4):637–676, 1999.

[48] P. Marion, R. Berthier. *Leveraging the two-timescale regime to demonstrate convergence of neural networks*. Advances in Neural Information Processing Systems 36 (NeurIPS 2023), pp. 64996–65029, 2023. arXiv:2304.09576.

[49] J. C. Mattingly, A. M. Stuart, M. V. Tretyakov. *Convergence of numerical time-averaging and stationary measures via Poisson equations*. SIAM Journal on Numerical Analysis, 48(2):552–577, 2010.

[50] A. P. Morgan, K. S. Narendra. *On the stability of nonautonomous differential equations* $\dot{x} = [A + B(t)]x$*, with skew symmetric matrix* $B(t)$. SIAM Journal on Control and Optimization, 15(1):163–176, 1977. https://doi.org/10.1137/0315013.

[51] K. S. Narendra, A. M. Annaswamy. *A new adaptive law for robust adaptation without persistent excitation*. IEEE Transactions on Automatic Control, 32(2):134–145, 1987. https://doi.org/10.1109/TAC.1987.1104543.

[52] A. Nitanda, D. Wu, T. Suzuki. *Convex analysis of the mean field Langevin dynamics*. Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 151:9741–9757, 2022. arXiv:2201.10469.

[53] F. Otto, C. Villani. *Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality*. Journal of Functional Analysis, 173(2):361–400, 2000. https://doi.org/10.1006/jfan.1999.3557.

[54] E. Pardoux, A. Yu. Veretennikov. *On the Poisson equation and diffusion approximation. I*. The Annals of Probability, 29(3):1061–1085, 2001 (and *II*, The Annals of Probability, 31(3):1166–1192, 2003).

[55] S. T. Paruchuri, J. Guo, A. J. Kurdila. *Kernel center adaptation in the reproducing kernel Hilbert space embedding method*. International Journal of Adaptive Control and Signal Processing, 36(7):1562–1583, 2022. https://doi.org/10.1002/acs.3407. arXiv:2009.02867.

[56] O. S. Patil, E. J. Griffis, W. A. Makumi, W. E. Dixon. *Simultaneous online system identification and control using composite adaptive Lyapunov-based deep neural networks*. arXiv:2311.13056, 2023 (revised 2026).

[57] O. S. Patil. *Adaptive control as an information gradient flow: excitation, constraints, energy, and stochastic learning*. arXiv:2610.04822, 2026.

[58] G. A. Pavliotis, A. M. Stuart. *Multiscale Methods: Averaging and Homogenization*. Texts in Applied Mathematics 53. Springer, New York, 2008.

[59] J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, M. Hardt. *Performative prediction*. Proceedings of the 37th International Conference on Machine Learning (ICML), PMLR 119:7599–7609, 2020. arXiv:2002.06673.

[60] S. Plunder, B. Simeon. *Coupled systems of linear differential-algebraic and kinetic equations with application to the mathematical modelling of muscle tissue*. In T. Reis, S. Grundel, S. Schöps (eds.), Progress in Differential-Algebraic Equations II, Differential-Algebraic Equations Forum, pp. 357–395. Springer, 2020. https://doi.org/10.1007/978-3-030-53905-4_12. arXiv:1911.05468 (2019).

[61] S. Plunder, B. Simeon. *The mean-field limit for particle systems with uniform full-rank constraints*. Kinetic and Related Models, 16(6):884–912, 2023. https://doi.org/10.3934/krm.2023012. arXiv:2203.07249 (2022).

[62] M. M. Polycarpou. *Stable adaptive neural control scheme for nonlinear systems*. IEEE Transactions on Automatic Control, 41(3):447–451, 1996. https://doi.org/10.1109/9.486648.

[63] J.-B. Pomet, L. Praly. *Adaptive nonlinear regulation: estimation from the Lyapunov equation*. IEEE Transactions on Automatic Control, 37(6):729–740, 1992. https://doi.org/10.1109/9.256328.

[64] J. A. Sanders, F. Verhulst, J. Murdock. *Averaging Methods in Nonlinear Dynamical Systems*. 2nd edition, Applied Mathematical Sciences 59. Springer, New York, 2007. https://doi.org/10.1007/978-0-387-48918-6.

[65] R. M. Sanner, J.-J. E. Slotine. *Gaussian networks for direct adaptive control*. IEEE Transactions on Neural Networks, 3(6):837–863, 1992. https://doi.org/10.1109/72.165588.

[66] F. Santambrogio. {*Euclidean, metric, and Wasserstein*} *gradient flows: an overview*. Bulletin of Mathematical Sciences, 7(1):87–154, 2017. https://doi.org/10.1007/s13373-017-0101-1. arXiv:1609.03890.

[67] S. Sastry, M. Bodson. *Adaptive Control: Stability, Convergence, and Robustness*. Prentice Hall, Englewood Cliffs, NJ, 1989.

[68] U. Seifert. *Stochastic thermodynamics, fluctuation theorems and molecular machines*. Reports on Progress in Physics, 75:126001, 2012; C. Maes, K. Netočný, *Time-reversal and entropy*, Journal of Statistical Physics, 110:269–310, 2003.

[69] D. W. Stroock, S. R. S. Varadhan. *On the support of diffusion processes with applications to the strong maximum principle*. Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, III:333–359, 1972.

[70] N. J. Tatro, P.-Y. Chen, P. Das, I. Melnyk, P. Sattigeri, R. Lai. *Optimizing mode connectivity via neuron alignment*. Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 15300–15311, 2020. arXiv:2009.02439.

[71] J. N. Tsitsiklis, B. Van Roy. *An analysis of temporal-difference learning with function approximation*. IEEE Transactions on Automatic Control, 42(5):674–690, 1997. https://doi.org/10.1109/9.580874.

[72] F.-Y. Wang. *On estimation of the logarithmic Sobolev constant and gradient estimates of heat semigroups*. Probability Theory and Related Fields, 108(1):87–101, 1997. https://doi.org/10.1007/s004400050102.

[73] Y. Zhang, Q. Cai, Z. Yang, Y. Chen, Z. Wang. *Can temporal-difference and Q-learning learn representation? A mean-field theory*. Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 19680–19692, 2020. arXiv:2006.04761.